# Feeding the City: How Rural Productivity Shapes Urban Structural Change*

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*Job Market Paper* (most recent version here)

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**Abstract**

Agricultural productivity in the countryside can promote structural transformation in the cities it supplies with food. Using Indonesia's Transmigration Program, I exploit as-if-random variation in settlers' familiarity with farming conditions at their destinations, which affected how productive the transmigration villages were. I show that more productive rural hinterlands shift employment in nearby cities from agriculture toward manufacturing and services. Urban households consume more food while devoting a smaller share of their budgets to it, consistent with improved food availability supporting greater spending on manufactured goods and services. The gains in food consumption are accompanied by higher incomes and household spending, suggesting improvements in urban living standards. These employment gains are larger in cities with weaker access to international food markets, in line with their greater reliance on local food supply. I use a calibrated spatial model to examine this mechanism, showing how food trade supports urban structural transformation that rural-to-urban migration alone would not explain. The urban effects of rural productivity therefore depend on the cost of trading food with the hinterland and with outside markets.

*JEL codes:* O18, R12, R23.

*Keywords:* structural transformation, agricultural productivity, rural-urban spillovers, migration, urbanization.

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# 1 Introduction

Development programs of governments and intergovernmental organizations often target rural areas: raising farm productivity, expanding cultivation, and improving rural livelihoods. In agriculture-based countries, governments directed about 4% of their public spending to agriculture (World Bank 2007). These policies are typically evaluated based on the rural outcomes they achieve. Yet, when goods and labor markets are linked across space, rural productivity gains can also transform nearby towns and cities. Higher agricultural earnings can retain workers in villages and relax financial constraints on migration, changing the supply of urban labor (Imbert et al. 2022). Greater food supply can lower urban food costs, while rising rural incomes can increase demand for manufactured goods and services. Higher rural productivity can therefore support urban structural transformation through food supply even when it strengthens incentives to remain in agriculture, with most existing research estimating productivity gains only within the region that received them, not being able to distinguish the spatial dimension of productivity effects.

In this paper, I study rural-to-urban spillovers using a policy that generated a large and spatially dispersed rural productivity shock that can be linked to urban outcomes over several decades. Indonesia's Transmigration Program provides this setting. The program resettled over 2 million people from the islands of Java and Bali to the Outer Islands. Although it was designed as a rural resettlement and agricultural development policy, it also created sharp, localized changes in rural population and agricultural production in the hinterlands of many towns. Transmigrant households did not choose their destination villages, and those who were assigned to destination villages more similar to their places of origin were more productive, as their farming skills transferred better (Bazzi et al. 2016). These villages can supply food to the cities around them.

I estimate rural-to-urban spillovers by comparing cities exposed to a more or less productive rural hinterland, measured as the average agro-climatic similarity of the transmigration villages within a five-hour journey of the city. I use three decades of Indonesian census microdata and village and household surveys along with the satellite-derived urban footprints of the Global Human Settlement Layer to construct a consistent-boundary sample of urban centers on the Outer Islands. I compare city population and employment in 2010, controlling for baseline population, employment, and urban structure in 1980, the number of settlers placed in the hinterland, and the characteristics of settlement destinations. The identifying assumption is that city exposure is exogenous conditional on these characteristics. The balance tests do not detect an association between exposure and the baseline values of the main urban outcomes.

Cities surrounded by more productive villages change their sectoral composition more than their size: employment moves out of urban agriculture into manufacturing and services with a positive but imprecisely estimated population response. Per standard deviation of hinterland agro-climatic similarity, the agricultural employment share falls by about 5.1 percentage points, against a 2010 mean of 37.4% and about a quarter of the average city's shift out of agriculture over the three decades, and

modern-sector employment rises by about 17%. Comparing the growth in the workforce to the growth in modern-sector employment, I show using back-of-the-envelope calculations from the main specification that additional workers could fill only part of the modern-sector expansion with the incumbent workforce fixed and without offsetting departures or labor-force entry and exit. Much of the fall in the agricultural share is within birthplace groups. Repeated cross-sections do not identify individual moves or occupational transitions.

I interpret these patterns as reflecting trade in food. Greater food supply can reduce the return to urban farming and support demand for non-agricultural goods and services where households spend half their budgets on agricultural goods. Urban residents in the cities surrounded by more productive villages spend relatively less on food while overall expenditures rise, and they eat more calories per person. Household income per worker is also suggestively higher. I show that transmigration villages with higher agro-climatic similarity produce more food, while their agricultural employment-share response is positive and imprecisely estimated. Rural subdistricts without settlements also reallocate employment toward non-agricultural activity. Changes in food-market competition and urban labor demand also create incentives for sectoral reallocation and migration in other rural areas.

The migration evidence is consistent with transmigrants remaining in productive villages, even long after their initial placement, providing little support for later reallocation of resettled households from villages to cities as the source of the urban outcomes. Cities with more productive rural hinterlands have, if anything, a smaller share of first-generation transmigrants. These cities have larger shares of migrants from other nearby districts, but data constraints do not allow me to establish whether they migrate from other rural areas with or without transmigrants or from other cities. Notably, in the more productive villages, agricultural employment is more prevalent, and I find no evidence of differential settler retention by village productivity. This points to other internal migration not driven by transmigrant settlers. Food trade could support this migration and urban adjustment by alleviating the subsistence constraints and raising non-agricultural labor demand.

To examine this mechanism further, I study how urban responses vary with access to external food markets. Trade openness determines the sectoral response to agricultural productivity in Matsuyama (1992) and the urbanization response in Glaeser (2014). Where a city can import food cheaply, the productivity of its rural hinterland matters less. Where it has access to less food, the hinterland matters more. I measure access using each city's travel time by land and sea to the Thai and Vietnamese hubs that ship most of Indonesia's rice imports (Bazzi 2017). Using detailed household expenditure surveys, I show that in more closed cities, with weaker access to international trade, higher hinterland productivity lowers food prices. There is no price response in more open cities, where the food supply from the rural hinterland likely accounts for a smaller fraction of urban consumption, so the average effect across cities is small and not statistically significant. The modern-sector employment response is also larger in the cities with weaker access to international food markets.

To quantify the trade and migration channels, I use a spatial general equilibrium model with two sectors and multiple locations. Workers choose a location and a sector. All locations have both

sectors, consistent with the empirical setting. A subsistence term in the utility function makes the food expenditure share fall with income. Non-agricultural production includes local services and output sold across locations, allowing rural food supplies to affect urban jobs through household spending and access to wider markets. The counterfactuals ask how much of the shift out of agriculture in cities can come from trade and how much from migration, by making trade costly while migration remains available, and then migration costly while trade remains available. At the selected parameters and cost settings, with trade made costly migration accounts for only a fraction of the agricultural contraction in cities, and with migration made costly trade accounts for most of it. So much sectoral reallocation would not be possible in the absence of trade. For modern-sector employment, both restricted responses are smaller than the full response, in which trade and migration adjust together.

I make two contributions. First, agricultural productivity gains promote structural transformation beyond the regions where productivity rises, shifting employment toward manufacturing and services in nearby cities. Bustos et al. (2020) show that agricultural productivity gains can finance non-agricultural activity in other regions through the banking system. They use variation in productivity at agricultural origins and financial connections to trace its effects in receiving regions and firms. I extend this question to cities' connections with their food-producing hinterlands, linking the productivity variation in transmigration villages to city outcomes through land-and-sea travel times. This analysis also relates to studies of agricultural development and non-agricultural employment in nearby towns and villages (Emran & Shilpi 2018; Edwards 2024).

Local income and employment shocks increase demand for goods and services, with effects on employment in the surrounding economy (Egger et al. 2022; Feyrer et al. 2017; Allcott & Keniston 2018; Moretti 2010). Agricultural development also affects local non-agricultural activity through demand, factor costs and market access (Santangelo 2019; Hornbeck & Keskin 2015; Kraus et al. 2024). Agricultural productivity can also expand the food supply available to cities, changing the cost of meeting food needs and the incentives to work in urban agriculture. Studying the urban sectoral response complements research on sectoral adjustment within agricultural regions (Bustos et al. 2016; Emerick 2018; Foster & Rosenzweig 2004; Moscona 2019). In Indonesia, the related evidence concerns sectoral reallocation within rural areas (Budí-Ors 2026) and market access in transmigration villages (Mitchell 2026). The urban employment effects I find imply that evaluations of rural interventions need to account for structural transformation in nearby cities.[1]

Second, I combine exogenous variation in agricultural productivity with empirical and quantitative evidence to examine the trade and migration channels of rural-to-urban spillovers. Agricultural productivity plays a central role in accounts of aggregate productivity and structural transformation (Restuccia et al. 2008; Gollin et al. 2014; Herrendorf et al. 2014; McArthur & McCord 2017; Gollin et al. 2021). One mechanism is the food problem faced by low-income countries with low agricultural productivity. Meeting basic food needs absorbs a large share of labor and household budgets, restricting

[1] In related work on India, my co-authors and I study the reverse direction: urban employment shocks reduce rural population and increase non-agricultural employment in surrounding villages (Asher et al. 2026).

the movement of labor out of agriculture and spending on non-agricultural goods and services (Gollin et al. 2007). Food trade links this constraint to productivity beyond a location's own farms (Tombe 2015; Nath 2025). I examine this connection at the subnational level, where the food-price and consumption patterns provide suggestive evidence on how rural productivity affects urban households. The source of identification is the productivity shock. The price, consumption, and migration evidence and the spatial model assess how its effects are transmitted. Comparing the urban response when trade or migration is costly helps assess how each channel contributes to structural transformation. In Matsuyama (1992), agricultural productivity gains move labor toward manufacturing in a closed economy and toward agriculture in a small open economy.

The empirical tests of that prediction (Bustos et al. 2016) and the within-country tests that interact an agricultural shock with regional openness (Moscona 2019; Liu et al. 2023; Marden 2016) measure the response on the shocked side. I examine the spillovers in cities, which can be interpreted through the insights of Matsuyama (1992) and Glaeser (2014) on how trade shapes the response to agricultural productivity. Glaeser's city-and-hinterland model shows how access to outside food changes the urban response. The stronger modern-sector employment response in cities with weaker external food access is consistent with local food supply being more consequential there. Research on internal trade and migration quantifies how goods-market and labor-market integration affect aggregate income (Tombe & Zhu 2019; Bryan & Morten 2019; Morten & Oliveira 2024; Lagakos et al. 2023). Assessing the urban effects of rural productivity therefore requires accounting for both the food markets cities can draw on and the movements of workers between locations.

The rest of the paper is organized as follows. Section 2 describes the program and the urban setting, Section 3 the data and the construction of the exposure measure, Section 4 the framework and the empirical strategy, Section 5 the reduced-form results, and Section 6 the model and its quantification. Section 7 concludes.

## 2 Setting

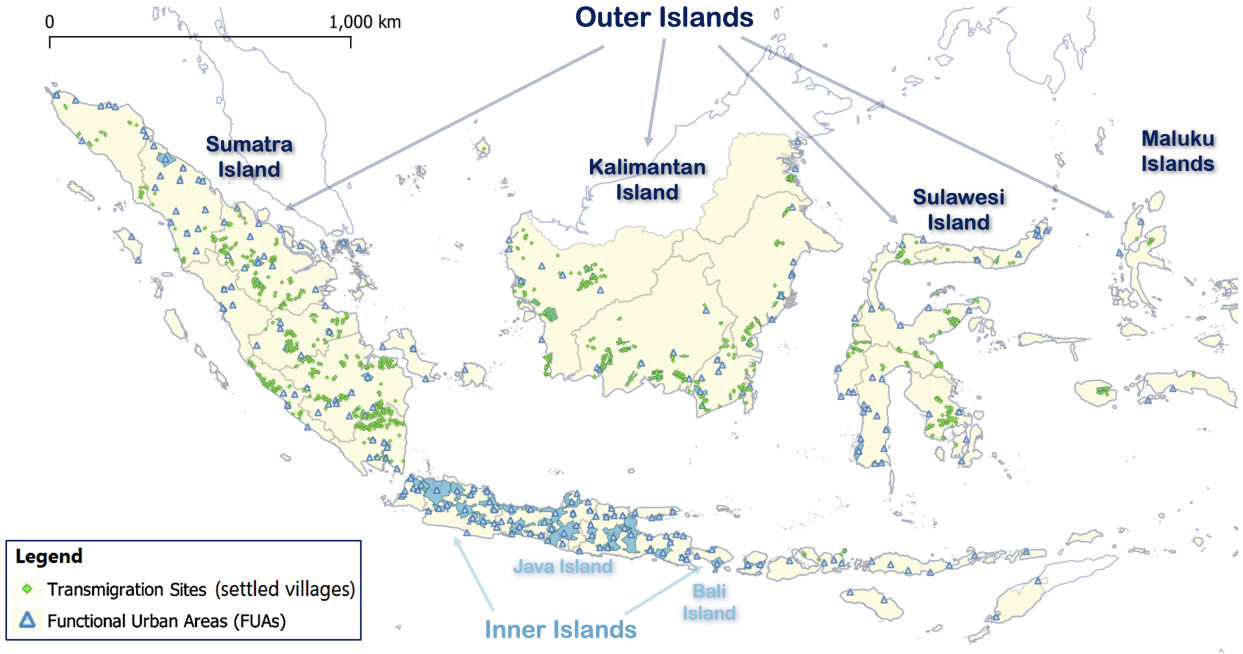
Indonesia's Transmigration Program generated variation in agricultural productivity across newly settled villages on the Outer Islands. I describe the program and its links to nearby cities, then define the urban sample. I discuss what urban agriculture means in this setting and how it appears within city boundaries in Section 2.2.2.

### 2.1 The Transmigration Program and Rural Productivity

The Transmigration Program aimed to alleviate population pressure on Java and Bali by resettling households in less densely populated parts of Indonesia, including Sumatra, Kalimantan, Sulawesi, and Papua. During its peak period, from 1979 to 1988, it moved more than 2 million volunteer migrants to newly established villages on the Outer Islands. Participating households received a land plot, a house, and free transportation to the destination. The program expanded agricultural settlement, largely on previously unused land, with the aim of increasing food production in the receiving regions. Its

expansion in the early 1980s was followed by a sharp contraction as falling oil revenues reduced the funds available for resettlement (Bazzi et al. 2016). The transmigration villages were geographically dispersed across the Outer Islands (Figure 1).

Figure 1: Transmigration Villages and Functional Urban Areas




*Notes:* Diamonds mark the settled transmigration villages, from the Transmigration Census (Bazzi et al. 2016). Triangles and shaded areas mark functional urban areas of the Global Human Settlement Layer, which include commuting zones around urban centers (Moreno-Monroy et al. 2021). Papua is not shown: it is excluded from the analysis throughout (Section 2.2.1).

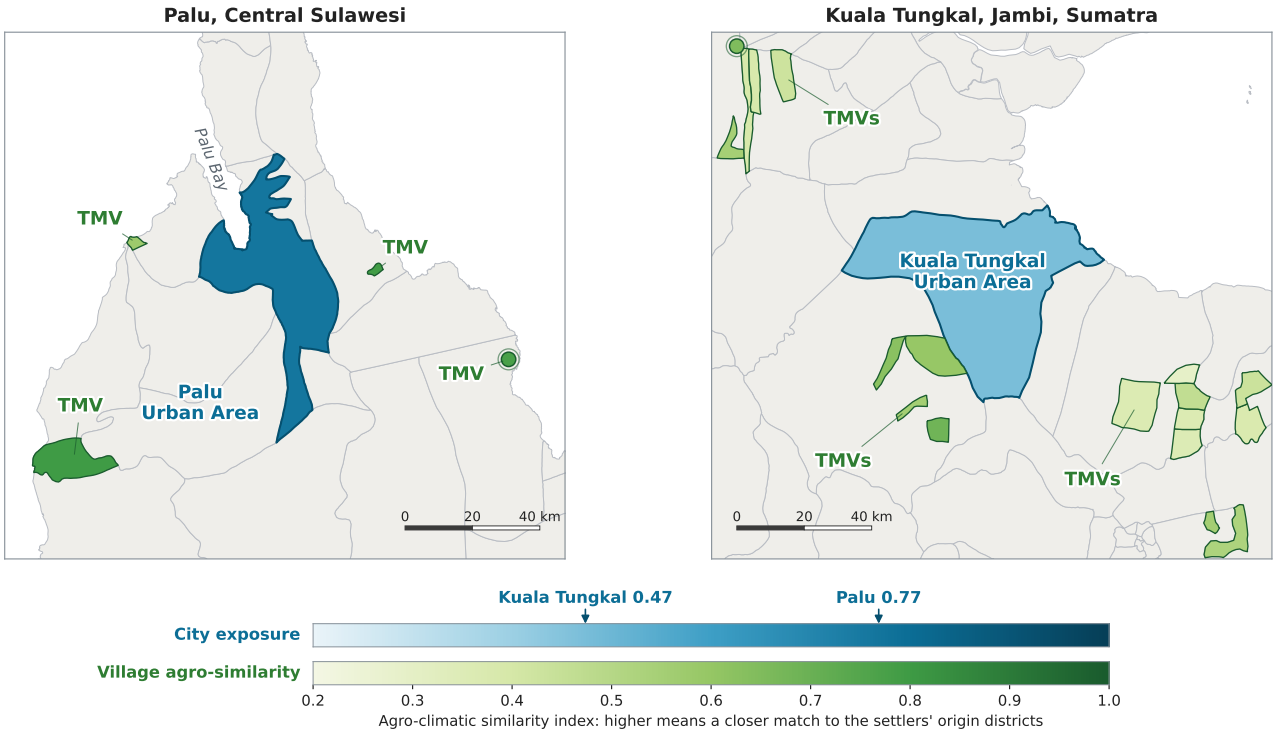
Transmigrant households did not choose their destination villages. Their assignment depended on the timing of arrival at resettlement centers and the availability of newly opened settlements. Officials did not systematically match households to destinations on the basis of their farming skills. Some settlers arrived in places with soils and climates similar to those at home, while others encountered different growing conditions. Conditional on the government's choice of settlement sites, this assignment process generated plausibly exogenous variation in the agro-climatic match between settlers' origins and destinations. Bazzi et al. (2016) assess this interpretation by comparing characteristics across destinations with different agro-climatic similarity. Their balance tests find little systematic association after conditioning on island fixed effects and destination geography. The checks include potential crop yields and destination-district development measures, with the latter constructed from 1980 census records of residents already present in 1978.

The agro-climatic similarity measure developed by Bazzi et al. (2016) captures how closely a destination's growing conditions resemble those in the migrants' regions of origin, and hence how readily their farming skills transfer. They estimate that a one-standard-deviation increase in similarity raises village rice productivity by roughly 20%. I use data from the Transmigration Census for 814 transmi-

gration villages where roughly 1.5 million transmigrants were initially placed, shared in the replication package of Bazzi et al. (2016) (Section 3.3 and Appendix Table A3).

To relate this village-level productivity variation to cities, I estimate travel times between cities and transmigration villages along land and sea routes. For each city, I then average the agro-climatic similarity of the villages within 300 travel minutes.[2] The villages around Palu in Central Sulawesi have higher average similarity than those around Kuala Tungkal in Sumatra, illustrating the differences in rural productivity exposure across cities (Figure 2).

Figure 2: Sample Areas of Urban Centers and Nearby Transmigration Villages




*Notes:* Transmigration villages are shaded by agro-climatic similarity between settlers' origins and destinations. Urban subdistricts are shaded by their city's mean similarity across villages within 300 travel minutes, used as a proxy for rural productivity exposure. Palu, Central Sulawesi (left), and Kuala Tungkal, Jambi, Sumatra (right), are drawn at a common scale. Mean similarity, on the 0 to 1 index, is 0.77 across 9 villages for Palu and 0.47 across 14 for Kuala Tungkal. A village whose boundary is unavailable is drawn as a circle at its recorded location.

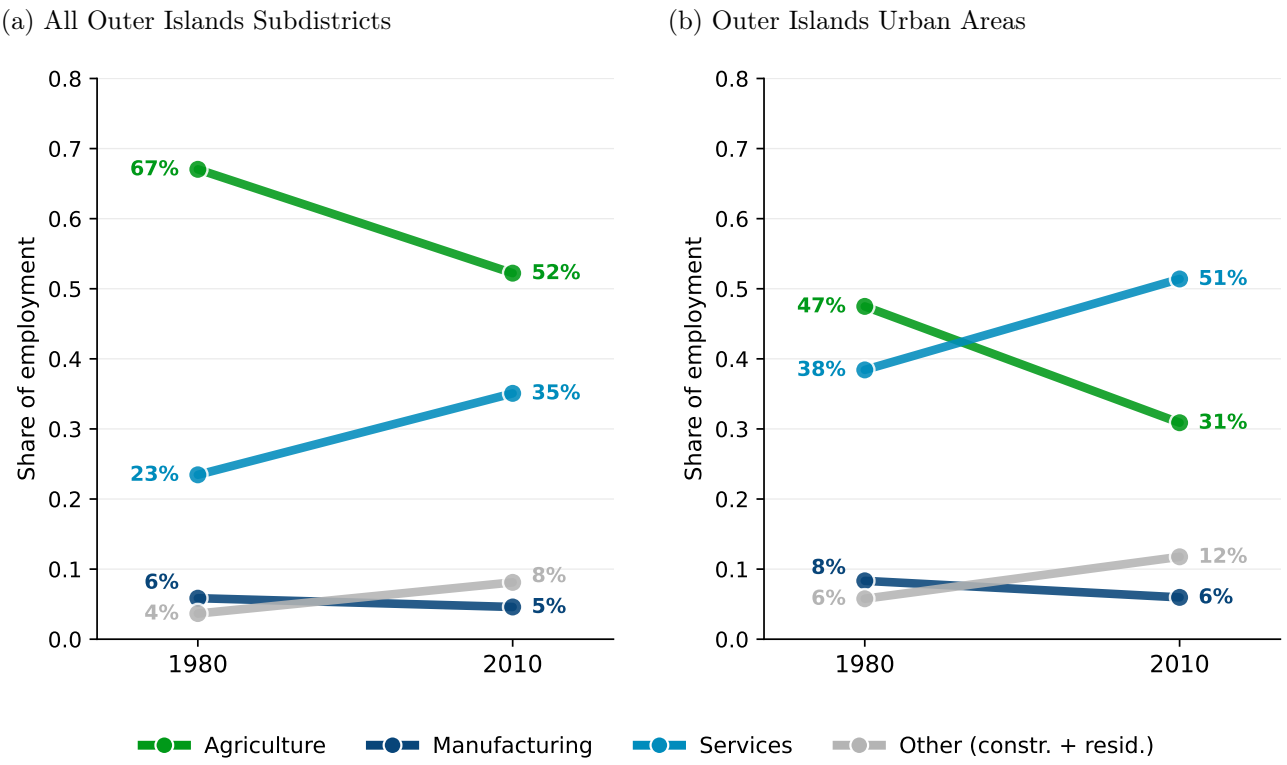
To compare resettlement with the size of the receiving countryside, I count settlers and the 1980 population in rural subdistricts outside every urban center and within 100 km of each city. This fixed straight-line radius describes the local scale of resettlement. Productivity exposure is measured using the travel-time window defined above. Across the 134 cities, settlers placed in these areas amount to about 7% of the rural population on average, and 22 cities have shares of at least 15%. The 1980 census precedes the settlement of 84% of the program's villages, so the denominator includes some early arrivals.

[2] I add maritime connections to the land-travel estimates and calculate travel times between each city and transmigration village (Appendix B.1).

## 2.2 Urbanization and Structural Transformation on the Outer Islands

The Outer Islands were less urbanized and more agricultural than Java and Bali in both 1980 and 2010. During this period, their urban population share rose from 17.9% to 37.5%, while the agricultural employment share fell from 67.0% to 52.2%.[3] Services expanded from 23.5% to 35.1% of employment, while manufacturing's share fell from 5.9% to 4.6%. The same shift from agriculture to services is evident within urban areas, where agriculture nevertheless remained a substantial source of employment (Figure 3).

Figure 3: Sectoral Employment on the Outer Islands, 1980 to 2010




*Notes:* Employment shares pool workers across the subdistricts or urban areas in each panel. The first panel uses subdistricts with measured sectoral employment in each census year. The second uses 183 urban areas with measured sectoral employment in both years, including peripheral rural areas. These are broader geographic samples than the main city sample. Other combines construction, mining, utilities, and undefined industry codes. Papua is excluded. Source: 1980 and 2010 population censuses.

### 2.2.1 The Urban Sample

The analysis follows urban centers on the Outer Islands. Java and Bali do not have transmigration villages in the program sample. I exclude Papua because the 1980 census data needed for baseline controls are unavailable and early spatial coverage is sparse.

[3]The urban population share follows Badan Pusat Statistik (BPS), Indonesia's national statistical agency, which classifies villages using population density, the share of farm households, and access to urban facilities. All Outer Islands aggregates exclude Papua.

To link cities to the census data, I use Indonesia's administrative hierarchy of provinces, districts (*kabupaten* and *kota*), subdistricts (*kecamatan*), and villages (*desa* and *kelurahan*). I aggregate subdistricts into cities to match the geographic detail available consistently across the data sources, and I use city and urban center interchangeably.

Similar to the urban areas plotted in Figure 1, I use a finer set of spatially consistent urban boundaries for Indonesia from the Global Human Settlement (GHS) Urban Center Database (Melchiorri et al. 2024). I hold the database's 2025 boundaries fixed across census years so that changes in outcomes are measured over the same areas.[4] I overlay the urban-center boundaries with subdistrict boundaries and aggregate the overlapping subdistricts into cities, including their peripheral rural areas (Appendix A.1 and Section 3.2).

All baseline population, employment, and urbanization measures used to define the urban sample and in the analysis come from census microdata. Satellite-derived measures of settlement and population extend back to the 1970s, including the GHS population grid for 1975 (European Commission, Joint Research Centre 2024). However, the early imagery provides incomplete coverage of Indonesia: Civelli et al. (2023) document missing 1975 built-up data for all of Sumatra and parts of West Java in an earlier GHS release. The input-observation record for the release used here also shows extensive gaps in the imagery underlying its earliest built-up estimates. The early grids therefore do not provide a directly observed baseline for much of the study region. Coverage is substantially more complete from 1990 onward, and I use later census-based population grids to check the census population patterns.

One challenge of working on rural-to-urban spillovers is a consistent urban definition. Instead of relying on an administrative definition, of which Indonesia has several, I define a comprehensive set of cities that share the characteristics usually thought of as urban, such as density. The urban areas are therefore defined by their footprints of population density and built-up area in the Global Human Settlement database. Recent work shows that the global layers are useful and comprehensive, but they need to be adjusted to local economies depending on what they miss, as the Development Data Lab's work on India illustrates (Development Data Lab 2026). I therefore use the baseline data from the population census and additionally classify areas as urban when they pass thresholds of population density and a low agricultural share. The first rule adds or reattaches subdistricts whose 1980 population density exceeds 500 persons per square kilometer and whose agricultural employment share is below one half, and a second rule includes the remaining subdistricts of a district when their combined 1980 urban population share exceeds one half. Both rules use 1980 census measures to identify urban places that the satellite definition may omit (Appendix Figure A1 and Appendix A.1). The estimated decline in the agricultural employment share and rise in the modern-sector share remain when the reclassified areas are excluded and when stricter criteria are applied to a smaller sample (Appendix Table A1).

I focus on cities close enough to transmigration villages to be exposed to their agricultural productivity, and compare their outcomes conditional on economic conditions observed in 1980. The main

[4] These spatial boundaries are represented as polygons. Because they are defined after resettlement, they can include areas that urbanized during the study period (Section 4.3).

city sample consists of 134 cities with at least one transmigration village within 300 travel minutes and directly observed 1980 census baselines for population, employment, and the urban population share (Section 3.3). These cities had substantial agricultural employment at baseline, leaving scope for structural transformation through changes in the occupations of city residents (Table 1 and Figure 4). Their measured 1980 population, employment, and agricultural employment share do not differ detectably from those of other Outer Islands cities (Appendix Table A4).

Table 1: Summary Statistics: The Main City Sample

**Panel A: Levels (counts and the travel time to the nearest transmigration village)**

| | Mean | SD | Median | Min | Max | N |
|---|---|---|---|---|---|---|
| Total population 1980 | 118,652 | 220,180 | 67,979 | 5,785 | 2,078,951 | 134 |
| Total population 2010 | 229,366 | 404,467 | 115,447 | 28,284 | 3,803,645 | 134 |
| Total employed 1980 | 32,841 | 57,395 | 18,874 | 2,226 | 537,303 | 134 |
| Total employed 2010 | 90,259 | 152,082 | 48,181 | 10,952 | 1,419,054 | 134 |
| Agricultural employment 1980 | 17,376 | 21,328 | 10,622 | 261 | 148,091 | 134 |
| Agricultural employment 2010 | 27,455 | 29,925 | 18,530 | 1,298 | 161,274 | 134 |
| Modern employment 2010 | 53,037 | 110,425 | 23,658 | 1,533 | 1,061,804 | 134 |
| Minutes to nearest transmigration village | 109 | 86 | 78 | 1 | 300 | 134 |

**Panel B: Shares, the unemployment rate, and the exposure measures**

| | Mean | SD | Median | Min | Max | N |
|---|---|---|---|---|---|---|
| Urban population share 1980 | 0.260 | 0.228 | 0.239 | 0.000 | 0.807 | 134 |
| Agricultural employment share 1980 | 0.560 | 0.219 | 0.579 | 0.045 | 0.981 | 134 |
| Agricultural employment share 2010 | 0.374 | 0.183 | 0.362 | 0.050 | 0.899 | 134 |
| Modern employment share 2010 | 0.516 | 0.156 | 0.523 | 0.094 | 0.827 | 134 |
| Unemployment rate 2010 | 0.122 | 0.040 | 0.116 | 0.038 | 0.223 | 134 |
| Settler count (thousands per travel minute) | 0.466 | 0.684 | 0.130 | 0.001 | 4.125 | 134 |
| Agro-similarity | 0.008 | 0.996 | -0.094 | -2.930 | 2.692 | 134 |

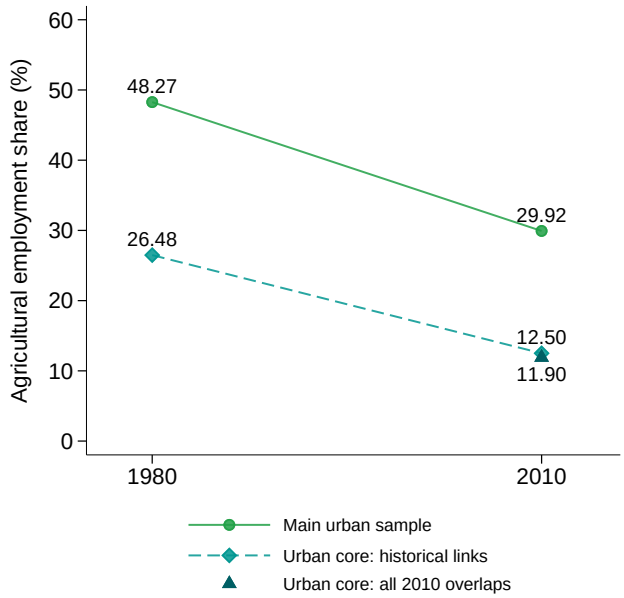
*Notes:* The main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline); every row is measured on the whole sample. The rows are the seven main 2010 outcomes, the exposure measures, the four 1980 baselines of the main specification, and the 1980 agricultural employment share. The settler count sums, over the transmigration villages within 300 travel minutes, the number of settlers placed in each divided by its travel minutes to the city, in thousands, so a village of 1,000 settlers 100 minutes away adds 0.010; agro-similarity is the standardized mean agro-climatic similarity of those villages. The modern sector is manufacturing plus services. Sources: 1980 and 2010 population censuses and the Transmigration Census. Comparison with the other outer-island urban centers: Appendix Table A4.

### 2.2.2 Urban Agriculture

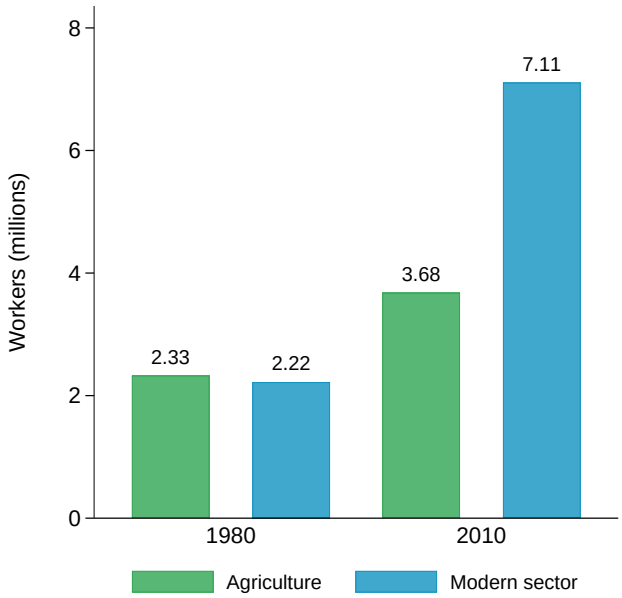
I use urban agriculture to denote agricultural employment among city residents. The census records where workers live, and therefore does not identify the location of their jobs or farm plots. The mean agricultural employment share fell from 56.0% in 1980 to 37.4% in 2010, with declines across most cities (Appendix Figure A2). City residents are nevertheless less likely to work in agriculture and more likely to work in services than residents of rural subdistricts, whether or not those subdistricts host a transmigration village (Appendix Table A5). Agricultural activity within cities is also documented elsewhere. In Nairobi, 20% of households grew crops within city limits in the 1984-1985 survey (Foeken & Mwangi 1998), and Gallaher et al. (2015) document vegetable cultivation in sacks in densely populated

Figure 4: The Agricultural Share Fell as Agricultural Employment Rose

(a) Agricultural Employment Share




(b) Employment by Sector




*Notes:* The main urban series and employment bars use the 134 cities in the main sample. Mapped subdistrict membership is fixed; some historical member codes lack separate census records. The urban-core line pools historically name-linked village groups and is exploratory; the separate 2010 point includes all villages with positive-area urban-center overlap. Each village is counted once, and historical links do not establish unchanged boundaries. Native footprints are available for 132 main-sample cities. Shares pool agricultural workers over industry-reported employment; the summary-statistics table reports averages across cities. Employment is in millions. Modern sector combines manufacturing and services; other industries remain in the share denominator. The census records employed residents, and workplace locations are unobserved. Source: 1980 and 2010 population censuses and historical village links.

Kibera.[5] Across developing countries, about 8% of urban employment is in agriculture (Gollin et al. 2016).

Agricultural employment remains substantial among residents of villages overlapping the urban core. In 2010, the mean agricultural employment share in these villages was 18.3% across 132 cities with a mapped satellite footprint. Whole villages are counted, including any territory extending across the footprint boundary.[6] This combination of farming and non-agricultural employment within an urban area is consistent with McGee (1991)'s description of Southeast Asian *desakota* regions, which combine rice cultivation and non-agricultural activity at urban densities.

[5]The Nairobi percentage measures household participation in crop cultivation. The Indonesian shares measure agricultural employment among residents.

[6]For each city, I divide agricultural workers by all employed residents in the villages overlapping its footprint, then average these shares across cities. A village is included when its polygon has positive-area overlap with the satellite footprint. The selected administrative villages include both *desa* and *kelurahan*. The calculation covers 3,413 distinct villages with observed employment counts. Separately coded forest and water areas are excluded. The 2 sample cities added through the baseline urban classification do not have a corresponding satellite footprint. These coverage limits apply to the descriptive calculation. The regression sample is unchanged.

In Palu and Kuala Tungkal, the same cities used to illustrate rural productivity exposure, agricultural workers account for 6.2% of employed residents in Palu's 44 overlapping villages and 24.1% in Kuala Tungkal's 6 overlapping villages. The village shares vary within each city, and some overlapping villages extend well beyond the urban outline (Figure 5).

Figure 5: Agricultural Employment Among Village Residents in Palu and Kuala Tungkal

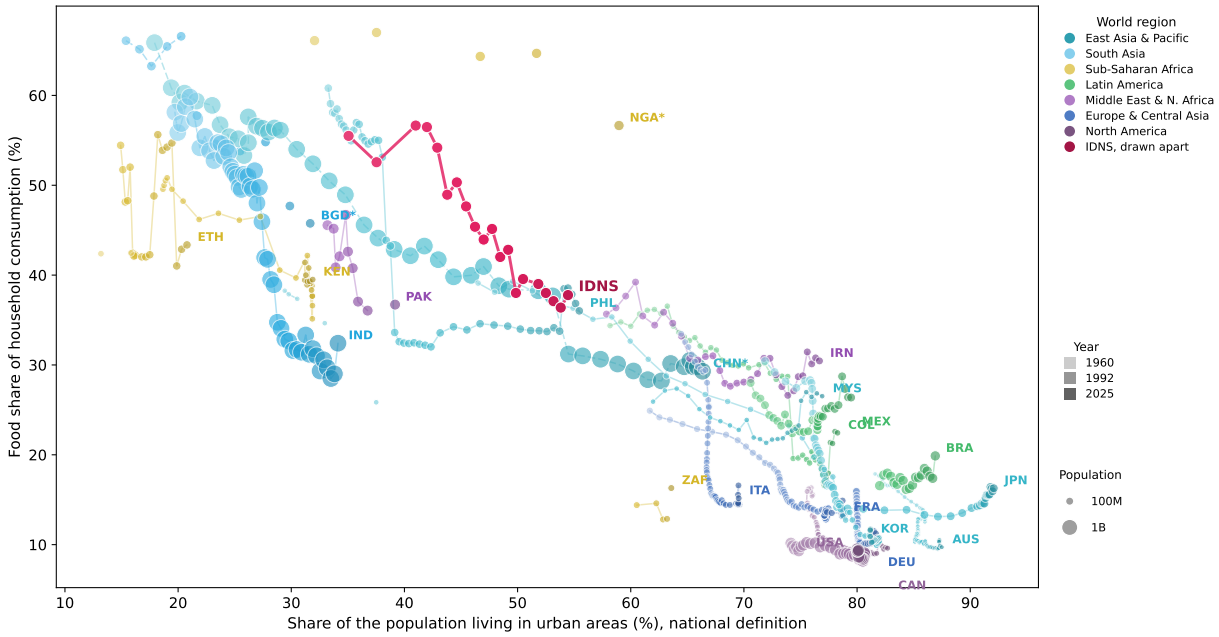



*Notes:* Agricultural workers as a share of employed residents in each mapped 2010 village, from the 2010 population census. The blue outline with a white halo is the Global Human Settlement urban-center footprint. Thin lines separate villages and gray lines separate subdistricts. Both maps use a continuous color scale capped at 25%. Shares above 25% receive the darkest shade. The reported city shares pool workers in all villages with positive-area overlap with the footprint. Whole villages are counted, including any parts extending beyond the footprint. Workplace and farm-plot locations are unobserved.

City boundaries therefore affect the measured level of urban agriculture (Appendix Table D2).

One possible reason that urban residents continue farming is the lack of a sufficient affordable food supply. If the rural hinterland is not productive, urban agriculture persists to feed the city, because otherwise prices would be too high. Indonesia's household food expenditure share fell as its urban share rose, a pattern also observed in the large countries shown in Figure 6. This aggregate association does not establish that food scarcity explains agricultural employment in the sample cities.

Figure 6: Food Share of Household Consumption and Urban Population Share, Largest Countries




*Notes:* Country paths plot the food share of household consumption against the urban population share under national definitions. Marker size indicates population. Darker shades indicate later years. Indonesia (IDNS) uses the adjusted Susenas series. Dashed paths and asterisks identify broader food definitions whose levels are not directly comparable. Series differ in coverage and food definitions, and some values are interpolated between observed years (Appendix A.3).

# 3 Data and Outcomes

To measure the urban outcomes, I use the microdata of the population census for 1980, the earliest baseline period for which I have data, and for 2010, the endline. For population, total employment and sectoral employment, both the baseline and the endline are observed, and I interpret the estimates as differences over the period. Migration measures come from the 2010 census. The additional measures, prices, food spending and consumption, come from household surveys that exist only for the later periods and capture cross-sectional variation as a function of urban exposure. The village census measures agricultural production. The spatial data define the urban sample and the travel-time measures of cost distance. I first describe the data sources and urban outcomes (Section 3.1 and Section 3.2), then construct exposure to nearby settlements (Section 3.3). I conclude with the baseline controls and measures of trade openness (Section 3.4 and Section 3.5).

## 3.1 Data Sources

I use the population censuses to measure baseline conditions, city outcomes, and village employment and settler retention. The 1980 records cover a 5% sample, expanded by census weights, and supply baseline controls.[7] The full-population records for 2010 supply city population, employment, and demographic

[7]The census sampled blocks within urban and rural strata, with institutional blocks in a separate stratum (Central Bureau of Statistics & Minnesota Population Center 2018). The city controls are estimates from this sample. Their sampling uncertainty is not separately estimated.

composition. Those for 2000 supply village agricultural-employment and settler-retention measures. Spatial maps of the administrative units exist only for the later census years (2000, 2010 and 2020), with no map for the 1980 baseline period, and the historical records can generally be linked at the subdistrict level. I aggregate these linked records to cities for the urban analysis.

Linking these census records over time requires accounting for changes in Indonesia's administrative divisions between 1980 and 2010. I construct a subdistrict correspondence from several sources. The historical BPS code links in Olken (2009), covering 1976 to 2002, connect the earlier censuses and village surveys to year-2000 subdistrict identifiers. I combine these with BPS village histories and spatial matching of later boundaries, then aggregate the linked villages to subdistricts. I validate the links using administrative codes and village names, including in areas outside the original Olken coverage, such as Aceh. The resulting correspondence permits analysis of baseline conditions and later outcomes on the same geographic units.

To measure city exposure to nearby settlements, I use the Transmigration Census records of settlement locations, establishment years, and the number of individuals initially placed in each village. I obtain these records and the village agro-climatic similarity index from the replication data of Bazzi et al. (2016) and Bazzi et al. (2019).

To measure household outcomes and local agricultural production, I use the National Socioeconomic Survey (Susenas) and the Village Potential Statistics (PODES). Susenas supplies household consumption and earnings. PODES supplies rice production. I match each Susenas wave to the common subdistrict geography and harmonize item codes, reporting units, questionnaire coverage, and survey weights to construct comparable spending and consumption measures. Table 2 summarizes the years, geographic detail, and coverage of these sources.

Satellite data complement the administrative records with spatial measures of urban settlement and population at different dates. I use the Global Human Settlement Layer (GHS) urban boundaries and the larger functional urban areas in descriptive maps, and compare the settlement classifications with census measures of urbanization (European Commission, Joint Research Centre 2024; Moreno-Monroy et al. 2021). Additional checks use the gridded population series, including observations from 1990 onward, to assess the census population patterns. These grids distribute census-based population totals across space using built-up-area information. Historical observations also depend on interpolated satellite inputs.

To measure access between the mapped cities and settlements, I construct travel times that combine overland and maritime connections. The land-travel surface of Weiss et al. (2018) combines roads, other transport access, land cover, and terrain to measure travel costs across locations. I add the shipping connections in Halpern et al. (2008), aligning the two geographic layers and expressing their costs in common units. The combined surface permits travel between Indonesian islands along shipping routes. The surface includes transport information observed after settlement. Appendix B.1 describes its construction and timing limitations.

For access to imported rice, I combine port locations with administrative customs records. Port names and coordinates come from the February 2011 *BP Shipping Marine Distance Tables* directory (AtoBviaC Plc 2011). The customs records report rice quantities in kilograms and values by port of entry and country of origin over 2000-2010. The sources are matched by port name, retaining geocoded directory ports with zero recorded rice imports. Route openness uses these port locations. Rice-import market access also uses recorded import quantities.

A historical set of maps of ports and their trade volumes gives a measure of port access dated before the program. Rutz (1987) maps the main ports of Indonesia's coastal regions with their inter-island cargo turnover in 1973, before the customs observation period. The maps were easier to obtain than the underlying statistical tables, so I digitized them, matching the ports to the nearest cities and inferring each port's trade volume from the map symbology. The 1973 volume is matched to 133 of the 134 cities and gives a pre-program test of the route measure (Appendix B.2). The maps report total cargo turnover without distinguishing agricultural goods.

### 3.2 Urban Outcomes

I aggregate the linked subdistrict records to the city areas defined above, which approximate urban labor markets. Population and employment are sums across residents, survey outcomes pool the individuals or households within these areas with their survey weights, and the cities available vary by outcome and wave (Table 2 and Appendix A.1).

**Population and employment.** The headline employment count and the sector counts use the different definitions in Table 2. Sector totals therefore need not add to headline employment. Modern employment comprises manufacturing and services. Agriculture is reported separately. These broad groups are comparable across the census years, while changes in detailed industry codes limit comparisons of smaller sectors. Together, agriculture, manufacturing, and services account for 96.3% of workers with a valid industry code on the Outer Islands in 1980 and 91.9% in 2010 (Figure 3). Construction, mining, utilities, and other industry codes remain in the sector-share denominator. The sectoral analysis also separates service activities, including trade and transportation. I report the unemployment rate and use $\ln(1+x)$ for log outcomes.

**Birthplace and recent migration.** The 2010 census records birthplace and residence five years earlier at the district level. Birthplace shares describe residents' origins. Five-year residence identifies recent moves across district or province boundaries. Both distinguish Java and Bali from other origins. These measures do not locate moves within districts or identify former transmigrants among city residents. The censuses do not follow individuals between waves. Worker-specific decompositions restrict the denominator to workers and are labeled accordingly. The 1980 census identifies birthplace at the province level, so the baseline birthplace controls distinguish origins in broader geographic groups than the district-level measures available in 2010.

Table 2: Outcome Definitions and Source Coverage

| Outcome and source | Location detail | Definition | Coverage |
|---|---|---|---|
| Population, 2010 census | Village or urban neighborhood | Resident count | Main city sample |
| Total employment, 2010 census | Village or urban neighborhood | Worked or ran a business in the preceding week | Main city sample |
| Sector employment, 2010 census | Village or urban neighborhood | Counts by reported industry, with shares divided by everyone with a valid industry code | Main city sample |
| Food outcomes, Susenas 1996, 2002, and 2011 | Household, located by subdistrict | Weighted means of spending and quantities, and price indices from item unit values | Surveyed cities, varying by wave and outcome |
| Wages and rents, Susenas 2011 | Individuals and households, located by subdistrict | Weighted median monthly earnings among positive earners, and median rent relative to the national median | Cities with the required survey data |
| Rice production, PODES 2003 | Village | Kilograms of paddy per person, using population in villages with observed output | Cities with observed rice production |
| Rice net imports, Susenas 2002 and PODES 2003 | Subdistrict | Rice consumption minus production, in tons of milled rice | Shared subdistrict coverage, aggregated to cities |

*Notes:* The main specifications apply the city sample's exposure and baseline-control requirements. Each survey or production estimate includes the cities with the observations needed to construct that outcome. Its sample size is reported with the estimate. Source geography is mapped to year-2000 subdistricts and then to cities. Household statistics use survey weights. BPS denotes Indonesia's statistical office. Susenas is the National Socioeconomic Survey and PODES the Village Potential Statistics. Appendix Table A5 reports survey coverage by wave. Sources: the 2010 population census, Susenas, and PODES.

**Household spending and consumption.** Susenas provides consumption per person, household spending, and the food share, household food spending divided by total spending. The main food-spending, calorie, staple-quantity, and agricultural-price measures use the 1996 wave, before the 2004 rice import ban. The 2002 rice quantities match the timing of the 2003 village production data used for the local rice balance. Later spending and earnings measures use 2011, with housing rents examined in the supporting analysis. These outcomes do not have comparable 1980 observations, so the regressions compare their levels across cities.

**Prices.** Agricultural price indices use survey-weighted medians of household item unit values. In 1996, these divide consumption value by quantity, including own production valued by BPS. Each index compares an item's survey-weighted median unit value over the city's households with its median over all urban-center households. It aggregates the items under the expenditure weights of all urban centers, the city's own weights, or equal weights. The rice price is the value of rice consumed over the kilograms consumed across the city's households. Later waves distinguish purchases, permitting purchase unit values. The indices can reflect both prices and the quality and composition of consumption.

**Wages and rents.** I deflate 2011 monthly earnings by local price indices to measure real wages. The housing-rent index divides the survey-weighted city median monthly rent, including actual and imputed rents, by the national weighted median. It serves as a proxy for non-tradable prices.

**Rice production and net imports.** Rice production per person uses villages with observed output and their recorded population. I calculate net imports as consumption minus production on the subdistricts covered by both Susenas 2002 and PODES 2003. Consumption per person in the 2002 survey is multiplied by the 2003 population of the villages with observed rice output, and production is expressed in milled-rice tons before subtraction. A positive difference indicates a local rice shortfall, potentially met by purchases from elsewhere in Indonesia or abroad. The consumption-production difference does not identify shipments or their origin.

### 3.3 Exposure: Agro-Climatic Similarity, Settler Counts, and Cost Distance

I use the agro-climatic similarity index of Bazzi et al. (2016) to measure cities' exposure to rural agricultural productivity.

**Village agro-climatic similarity.** Let $x_o$ and $x_v$ denote agro-climatic characteristics of origin district $o$ and the district containing destination village $v$, and let $\omega_{ov}$ be the share of settlers in $v$ born in $o$. The underlying similarity index is

$$AS_v^{\text{raw}} = -\sum_o \omega_{ov}\, D(x_o, x_v), \tag{1}$$

where $D(\cdot)$ sums absolute differences across standardized characteristics. The negative sign makes a closer match a higher value. I express similarity on a scale from 0 to 1, denoted $AS_v$, with higher values indicating a closer match between the settlers' origins and the destination village.

**Travel time and city exposure.** I combine the land-travel surface with shipping connections to calculate travel times from subdistricts and urban areas to transmigration villages. I also calculate travel costs among the urban and rural locations of the spatial model. Access to ports and shipping routes affects travel times between islands. The city-to-village travel times determine catchment membership, the inverse-travel-time weights of the settler count, and which cities have a village within 300 minutes.[8]

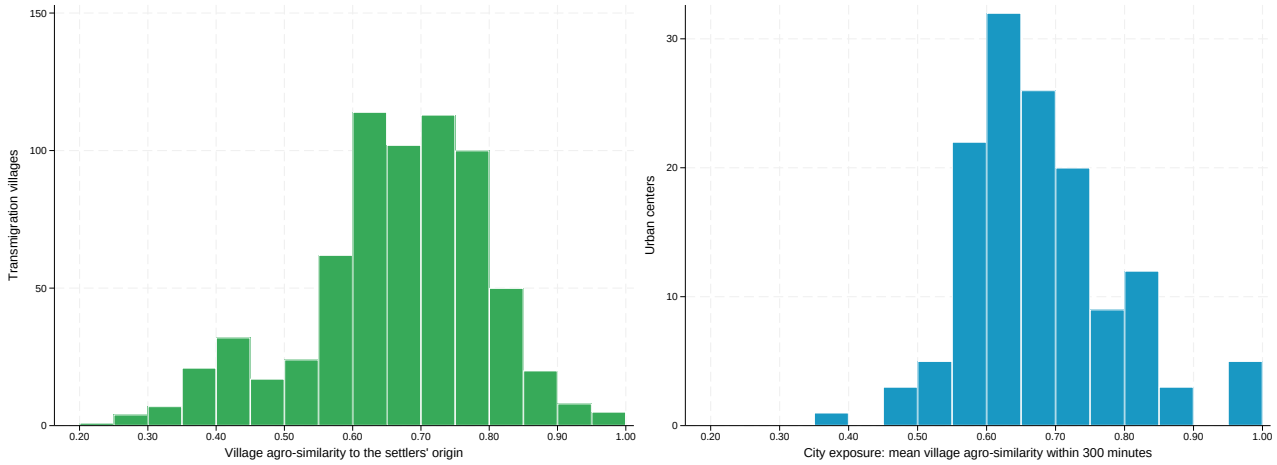
I measure average village productivity with equal village weights, accounting separately for settlement size and proximity through the settler-count control defined below. City exposure is the mean agro-climatic similarity of settlements within 300 travel minutes of city $c$:

$$AgSim_c = \frac{1}{|V_c|}\sum_{v\in V_c} AS_v, \qquad V_c = \{v : d_{cv} \le 300 \text{ minutes}\}, \tag{2}$$

[8] Appendix B.1 documents the geographic coverage, the 2,000-minute limit of the village-pair calculations, the distance adjustments, and source-date limitations.

where $d_{cv}$ is the travel time used to construct exposure and $|V_c|$ is the number of villages in the catchment.[9] The city mean remains between 0 and 1. For estimation, I subtract its mean and divide by its standard deviation across all urban centers with measured exposure, before restricting the sample to cities with complete baseline controls. One standard deviation in cities' exposure to surrounding agro-climatic similarity is 0.103 on the 0 to 1 scale of the index, 0.76 of the standard deviation across the transmigration villages (0.136), because the catchment mean averages over several villages. The estimates are reported per standard deviation in cities' exposure. Applying the 20% higher rice productivity per village standard deviation that Bazzi et al. (2016) estimate, one city standard deviation corresponds to about 15% higher rice productivity in the surrounding villages. The two sample regions in Figure 2 illustrate the comparison, and Figure 7 shows the village and city distributions.

Figure 7: Transmigration-Village Agro-Similarity and Aggregated Urban Exposure




*Notes:* Left panel: agro-similarity to the region of origin of the settlers, an index from 0 to 1, for the 680 of the 814 transmigration villages that lie within 300 travel minutes of an outer-island urban center. The remaining villages lie outside those catchments. Right panel: the exposure of each of the 138 urban centers with a village in that window, the mean agro-similarity of those villages. Averaging narrows the city distribution, which starts at 0.38 where the village index starts at 0.22. Bins are 0.05 wide on the same horizontal scale in both panels.

**Settler counts and weighting.** I also measure the scale of settlement around each city using the number of individuals initially placed in nearby villages. Let $N_v$ denote that initial count in thousands. The control sums these counts with inverse-travel-time weights over the same catchment:

$$SettlerCount_c^{IDW} = \sum_{v \in V_c} \frac{N_v}{\max\{1, d_{cv}\}}, \tag{3}$$

where the superscript $IDW$ stands for the inverse-travel-time weights and travel time is floored at 1 minute. Its unit is thousands of settlers per travel minute. It increases with the number and size of nearby settlements and gives greater weight to closer villages.

[9] For exposure construction, travel time is capped at 1 minute for villages with more than 0.1% of their polygon area inside a city polygon. Times implying speeds below 2 kilometers per hour are capped at the time implied by great-circle distance from the city centroid to the village at that speed. Sample eligibility uses travel times before these adjustments.

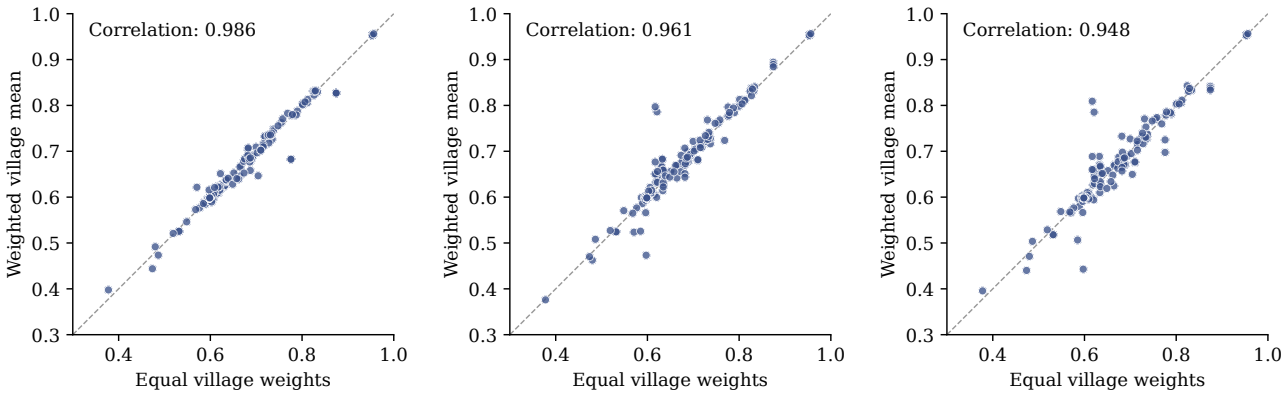
The unweighted mean summarizes agricultural productivity across surrounding settlements. The settler-count control accounts separately for their initial settler population and distance to the city. Conditional on the settler-count and destination controls, the comparison is between cities with different average agro-climatic similarity in their surrounding settlements.

Alternative definitions weight the villages by their initial settler population, inverse travel time, or both. For weights $w_{cv}$, define

$$AgSim_c^w = \frac{\sum_{v \in V_c} w_{cv} AS_v}{\sum_{v \in V_c} w_{cv}}, \qquad w_{cv} \in \left\{ N_v, \frac{1}{\max\{1, d_{cv}\}}, \frac{N_v}{\max\{1, d_{cv}\}} \right\}. \tag{4}$$

The population weight is the number of settlers initially placed in a village. Inverse-time weights give more influence to nearby villages. The weighted mean can be higher or lower depending on how similarity varies with distance. The alternative measures are closely correlated with equal-weight exposure across the same cities (Figure 8). Table 5 reports the urban-outcome estimates under settler and inverse-time weights. Appendix Figure D3 also includes their combinations.

Figure 8: Exposure Under Alternative Village Weights




*Notes:* Each point is one of the 134 main-sample cities. The horizontal axis is mean village agro-climatic similarity within 300 travel minutes with equal village weights. The vertical axis uses the alternative weights in Equation 4, on the same zero-to-one scale. Dashed lines show equality. Each panel reports the Pearson correlation across the same cities. Population weights use initial settler counts. Travel-time weights use adjusted pairwise travel time with a one-minute floor. Sources: Transmigration Census and constructed travel-time data.

**Measured exposure and the main city sample.** Applying the 300-minute cutoff to unadjusted nearest-village travel time includes 138 of the 187 urban centers, of which 134 have a 1980 baseline recorded directly in the census. The main city sample consists of the 134 cities with a nearest-village travel time within the window, measured exposure, the required baseline controls, and measured 2010 outcomes. When $V_c$ is empty, the catchment mean is undefined and exposure is left unmeasured. The regressions compare degrees of exposure among cities with measured catchments.[10]

[10]The distribution of nearest-village travel times and the full sample selection are reported in Appendix Figure A3b and Appendix Table A2.

The 250- and 400-minute windows provide neighboring checks around the 300-minute main specification (Table 3). At 200 minutes, only 105 of the 134 main-sample cities have measured exposure. This city count is too small for the robustness comparison. The 250-minute window includes 127 urban centers and 630 transmigration villages, compared with 138 centers and 680 villages at 300 minutes and 151 centers and 726 villages at 400 minutes. The nested-window comparison rebuilds exposure, village controls, and city eligibility at each window (Appendix Figure D5). A separate comparison widens exposure to 400 and 500 minutes while holding the main city sample and its 300-minute controls fixed (Table 5).

Table 3: Urban Centers and Transmigration Villages by Catchment Radius

| | Full sample | 250 min | 300 min | 400 min |
|---|---|---|---|---|
| Urban centers | 187 | 127 | 138 | 151 |
| Transmigration villages | 814 | 630 | 680 | 726 |
| Cities from the main city sample | 134 | 123 | 134 | 134 |

*Notes:* The reference column lists the full Outer Islands urban-center and program-village samples. Each radius column counts urban centers with at least one village in the catchment and distinct villages within at least one city's catchment. Villages belonging to several catchments are counted once. Catchment membership uses adjusted travel time; the nearest-village eligibility test uses measured travel time and gives the same city counts at these radii. The last row counts cities from the main sample with measured exposure at each radius. The fixed-main-sample checks retain the 300-minute controls; the separate nested-window regressions rebuild controls and city eligibility at each radius. Sources: Transmigration Census, population censuses, and the constructed travel-time data.

### 3.4 Baseline Controls

Alongside the settler-count control, the main specification includes 16 destination controls for geography, soil, and climate. These cover the same underlying characteristics used to construct agro-climatic similarity. They are drawn from the agro-climatic controls in Bazzi et al. (2016) and averaged over the same villages as exposure: for each village characteristic $X_v$, I construct $X_c = |V_c|^{-1} \sum_{v \in V_c} X_v$. Geography includes three slope measures, ruggedness, log elevation, and distances to coasts and rivers. Soil characteristics include organic carbon, exchangeable sodium percentage, pH, two texture measures, and two drainage measures. Climate includes mean rainfall and temperature over 1948-1978.[11] The tables refer to them as agro-climatic transmigration-village controls.

In addition to these catchment characteristics, I control for four city characteristics in 1980: log population, log employment, the urban population share, and log agricultural employment. Some villages were settled in 1979 and 1980, before the census that records these baselines. Section 4.3 discusses the timing. A sensitivity check adds the 1980 birthplace shares of the population (Appendix Table C1).

Among these city controls, the urban population share measures the fraction of baseline residents living in villages classified as urban by BPS. For city $c$ it is

$$UrbanShare_c = \frac{\sum_{k \in c} Pop_k^{\text{urban}}}{\sum_{k \in c} Pop_k},$$

[11] The index measures origin and destination characteristics at the district level. The controls use destination-village values averaged over the city's catchment and enter elevation in logs.

where $k$ indexes its constituent subdistricts and $Pop_k^{\text{urban}}$ is their 1980 population in villages classified as urban by BPS. The satellite boundary determines membership in the city. The census village classification determines the urban share within that geography.

### 3.5 Trade and Openness Measures

The urban response to productivity in surrounding villages may also depend on access to food from elsewhere. I measure access to imported rice using international rice routes and realized rice imports, and describe general port access using domestic travel times.[12]

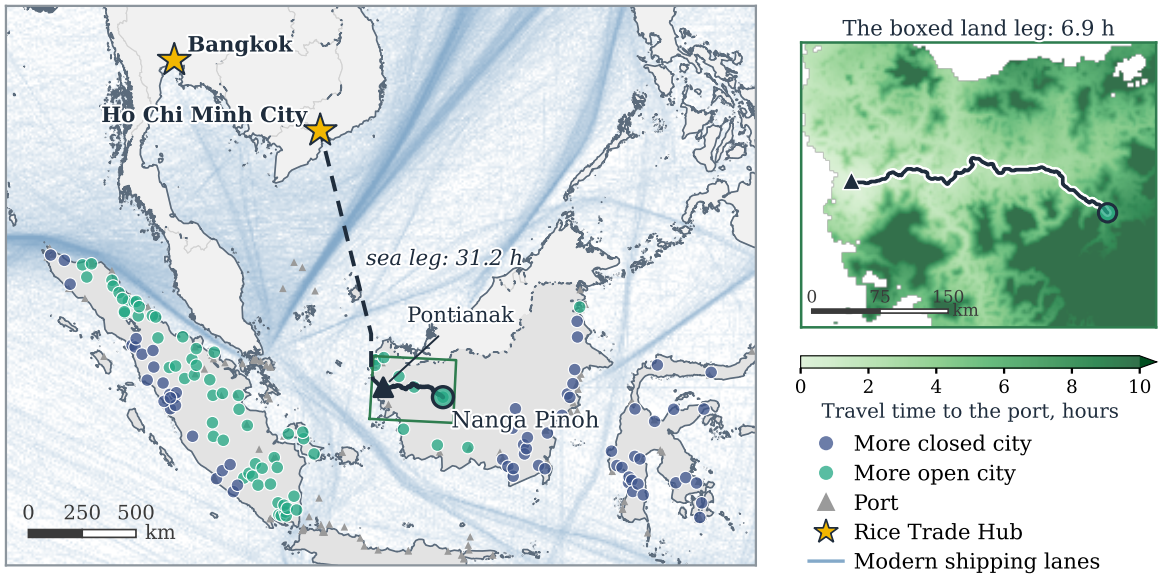
**Route openness.** The preferred measure adds least-cost travel time from the city to its nearest port to the onward sea leg to the nearer of Bangkok and Ho Chi Minh City, following the rice-route approach in Bazzi (2017). Both legs are expressed in hours. I take the domestic leg from the friction surface of Section 3.3, which includes roads, geography and the shipping routes, with a floor of half an hour, and I draw the international leg as the shortest route by sea from the port to the hub, along the coastline and through the straits, at a shipping speed of 40 kilometers per hour (the Online Supplement draws each port's route). Shorter total travel time means greater openness. The more open cities lie below the median travel time within the exposure window, and the continuous measure, route proximity, is the total with its sign reversed, standardized across all 187 urban centers so that higher values mean greater openness. Figure 9 maps the cities by this split and draws both legs for one city.

The international leg uses fixed hub locations and coastline geography. The port sample consists of 183 geocoded Indonesian ports from the shipping directory described above, including ports with zero recorded rice imports. The directory does not record port opening dates. On historical maps the ports of 98 of the 134 cities were main ports in 1973, before the program, and all but 5 appear in a 1986 harbor inventory (Table B1). The domestic leg uses road and shipping inputs observed after settlement, so the measure combines fixed international geography with domestic links that postdate the program. The 1973 cargo volumes of the historical maps give the measure a test dated before the program: the cities it ranks as more open sat in coastal regions that handled more interisland cargo in 1973, and agro-similarity does not predict that cargo (Appendix B.2).

**Realized-import access and general port access.** Gravity market access is the sum over ports of the rice imports recorded in 2000 to 2010, in kilograms, each divided by the travel time from the city in hours with a floor of half an hour. I take $\ln(1+x)$ of the sum and standardize it across the full set of cities in the source data, so that higher values mean greater access. It gives the alternative openness split in Table 6, with the more open cities above the median within the exposure window. Realized imports respond to demand and supply, so this measure is partly an equilibrium outcome. I use it to compare the estimated productivity response across measures of outside rice access. The main openness comparison uses the route measure (Appendix B.2).

[12] Rice from other Indonesian regions is another source of urban food, and the rice-import measures do not capture these domestic shipments. Appendix B.2 reports the construction of the measures and compares route openness with domestic port access to assess how much international geography adds to port proximity.

Figure 9: Route Distance to the Rice Hubs Across the Main City Sample




*Notes:* Each circle is a city of the main city sample, in two shades by its route distance to the nearer of Bangkok and Ho Chi Minh City: the more closed cities lie at or above the median over the 138 urban centers within the 300-minute window and the more open cities below it, as the legend shows. Small dots are ports in the sample and stars the two rice hubs. For Nanga Pinoh in Kalimantan the two legs of the route are drawn: the land leg to Pontianak over the friction surface of Section 3.3, enlarged in the inset with travel time to the port, and the sea leg from Pontianak to Ho Chi Minh City along the shortest route by sea. The shaded bands show the modern shipping lanes of Halpern et al. (2008). They are shown for orientation and are not used to calculate the international sea leg. The scale bar is measured on the ground.

**Instrumented market access.** To address the endogeneity of realized-import access, I instrument both rice-import market access and its interaction with agro-similarity, using route openness and its interaction with agro-similarity (Section 5.2). The main specification uses the high-openness indicators. The appendix also reports the continuous form. The interaction measures how the urban response to rural agricultural productivity varies with access to outside rice markets.

The exclusion restriction applies to both urban outcome levels and their response to agro-similarity. Conditional on agro-similarity and the main controls, route openness must affect urban outcomes only through rice-import access, and it must change the response to rural productivity only through that same access. For example, route proximity could also change how cities respond to rural productivity by improving access to markets for non-agricultural goods or facilitating migration. Such channels would violate the interaction restriction if their effects operate outside the measured rice-import access. Fixed hubs and coastline do not establish this exclusion, and the domestic transport inputs postdate settlement. I therefore interpret the IV estimates as evidence on the rice-import-access mechanism conditional on these restrictions. The first-stage and weak-instrument-robust results are reported with the estimates.

# 4 Conceptual Framework and Empirical Strategy

This section sets out how a productivity gain in a city's hinterland can reach the city (Section 4.1), the equation that estimates its effect (Section 4.2), and the identification argument behind it (Section 4.3).

## 4.1 Framework: Mechanisms Linking Rural Productivity to Urban Outcomes

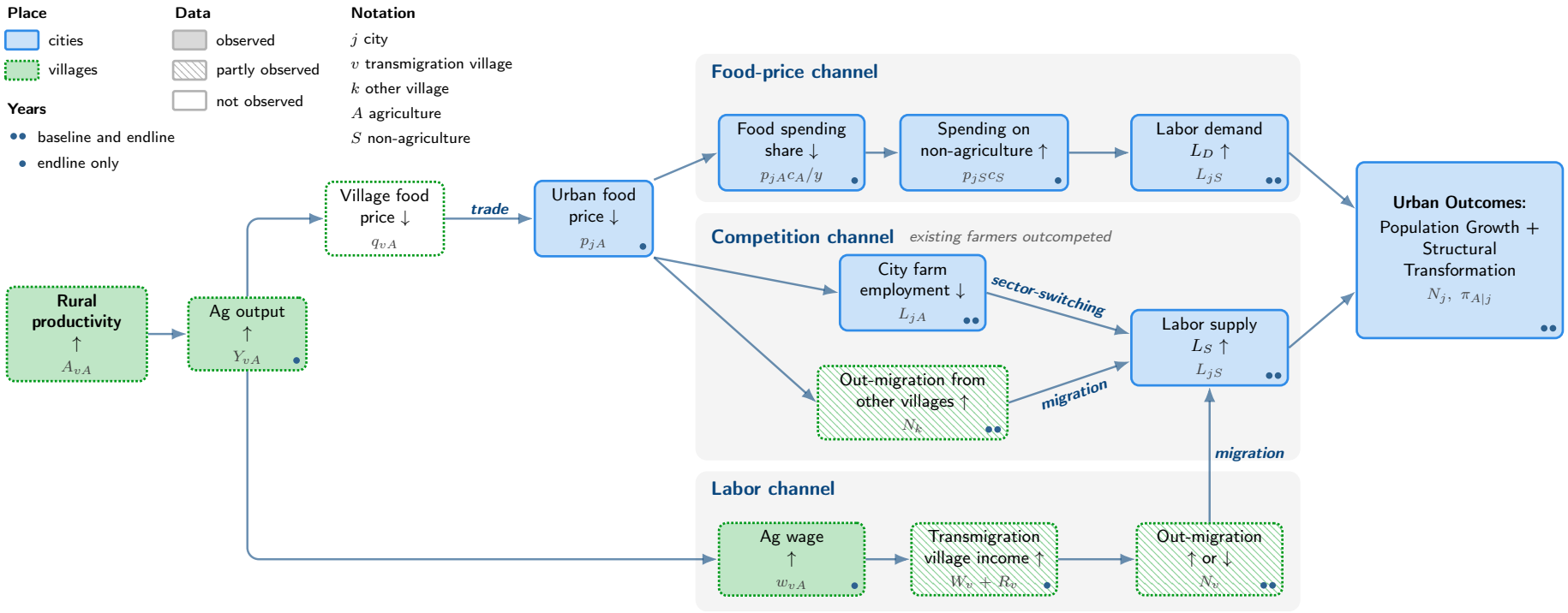
The productivity shock directly affects food production in the villages that receive it. The region is the city and its rural hinterland, which together make up one economy, as in Glaeser (2014). The city is closed to the extent that food from outside its hinterland is costly, and cities differ in that cost. The food-price channel can change urban employment composition without migration, while rural income changes can encourage or discourage migration. The relative strength of these responses is ambiguous.[13] Figure 10 summarizes the mechanisms within a city and its hinterland. The spatial equilibrium model in Section 6 also allows trade and migration between urban regions. The symbols below the shapes show the corresponding model quantities. In the map, $j$ denotes a city, $v$ a transmigration village and $k$ another village, and subscripts $A$ and $S$ denote agriculture and non-agriculture ($L_{jA}$ is agricultural employment in the city, $w_{vA}$ the agricultural wage in a transmigration village).

Food trade links rural productivity to urban employment through labor supply from urban agriculture and labor demand in the modern sector. Rural productivity raises agricultural output and therefore reduces prices. As the food price falls, urban residents with access to the expanding food supply from rural areas need to spend less on food, and as food spending falls they can spend more on non-agricultural items, raising demand for the non-agricultural sector. Overall expenditures can rise while residents spend relatively less on food. This first channel translates into potential modern-sector employment growth, since labor demand rises as residents demand more services, and the demand for the traded good can be met by producers elsewhere. At the same time, cities themselves do not need to produce as much food within their boundaries, and the remaining peri-urban agriculture can decline as workers exit urban agriculture, which is a labor supply effect. Existing farmers without a productivity shock must now compete with the other villages producing more. These supply and demand responses allow structural transformation among existing urban residents. In the model, urban employment adjusts through relative sectoral wages. Cheaper food from nearby villages matters more when alternative sources are costly. If a city can trade only with the surrounding rural areas, the effect of changing productivity is expected to be stronger.[14]

[13] The food-problem literature emphasizes the role of agricultural productivity in releasing labor (Schultz 1953; Gollin et al. 2007), and the dual-economy model sustains non-agricultural employment through cheap food (Lewis 1954). Trade opportunities and the factor bias of technical change can alter this relationship (Matsuyama 1992; Bustos et al. 2016).

[14] District price differences provide descriptive evidence of food-market frictions. Using 2011 Susenas unit values for pairs of outer-island districts, the coefficient on log travel time in a 95th-quantile regression of the absolute log rice-price gap is 0.087. The corresponding coefficient for the food basket is 0.065. Travel time explains at most 12% of the variation in pairwise price gaps. These gradients do not identify trade costs or establish that transport links are predetermined, and the quantitative exercise in Section 6 uses the trade-cost range from the literature.

Figure 10: Mechanism Map




*Notes:* The figure summarizes adjustments within a city and its rural hinterland. Signs illustrate possible responses through each channel. Signs on the food-trade branch describe the case in which additional rural supply lowers urban food prices and the cost of meeting food needs. Filled shapes indicate observed quantities, hatched shapes partly observed quantities and unfilled shapes unobserved quantities. Two dots denote a quantity measured at baseline and endline, one dot at endline only. The village food-price symbol denotes the origin supply index $q_{vA}$. The migration symbols denote village populations $N_v$ and $N_k$, whose changes represent net migration in the model. For the transmigration villages the arrows show the case in which the productivity gain raises agricultural wages and incomes. Out-migration from those villages can fall or rise, because higher income reduces the incentive to leave but also relaxes the liquidity constraints on migration. $L_S$ denotes labor available to urban non-agricultural sectors through sector switching or migration, and $L_D$ denotes labor demand through non-agricultural spending. The data measure employment $L_{jS}$, the model quantity associated with both labor supply and labor demand. The net change in urban labor supply reflects the combined sector-switching and migration responses.

The wage effect operates in the rural areas, where productivity gains raise earnings for those who experience them relative to others, which happens in the transmigration villages that are more productive, and agricultural incomes there depend on both the productivity gain and the fall in food prices. On the one hand, this could reduce out-migration by providing a sufficiently high standard of living in rural areas, reducing the push factor by offering higher income. At the same time, migration costs can be substantial, and some rural households face liquidity constraints in financing them (Bryan et al. 2014; Bazzi 2017). Higher income relaxes those constraints, so out-migration from the transmigration villages can also rise. Other rural areas face greater competition from the more productive villages and lower prices for their agricultural output, which can encourage out-migration from those areas.

Each employment response considered separately is consistent with several mechanisms, so I assess them using three sets of evidence in Section 5. First, I examine how employment composition, population, and total employment vary with exposure (Section 5.1), and whether food prices fall (Section 5.2). With food-supply conditions held fixed, an inflow of workers raises food demand and does not imply this price decline. Second, I test whether the response is larger in cities with costly access to outside food (Section 5.2). Third, I assess the contribution of net additional workers to modern-sector employment and examine migration and agricultural employment in the transmigration villages (Section 5.2 and Section 5.3). Birthplace data help assess whether residents came from the regions the migration mechanism predicts. Repeated cross-sections do not identify individual moves or occupational transitions, so the model in Section 6 compares the responses under different trade and migration costs.

### 4.2 Estimating Equation

To estimate these urban responses, I compare outcomes across cities surrounded by villages with different agro-climatic similarity, conditional on settlement scale, destination characteristics, and the city's 1980 baseline. The main population and employment outcomes are measured in the 2010 census. For urban center $c$, I estimate

$$y_c = \beta\, AgSim_c + \phi\, SettlerCount_c^{IDW} + C_{c,1980}\,\xi + X_c\,\zeta + \varepsilon_c \tag{5}$$

where $y_c$ is an urban outcome aggregated from the city's subdistricts and $AgSim_c$ is standardized city agro-climatic similarity, constructed as in Section 3.3. The coefficient $\beta$ is the conditional difference in the outcome per standard deviation of exposure within the 300-minute travel-time window, corresponding to about 15% higher rice productivity in those villages. I interpret the estimates as a reduced form in agro-similarity. Appendix Table E3 reports alternative crop and output measures, and Section 6 puts the magnitudes into economic units.

The settler count, $SettlerCount_c^{IDW}$, accounts for the scale and proximity of settlement, which the program chose. Conditional on these controls, $\beta$ is identified from differences in nearby villages'

agro-similarity. The vector $X_c$ contains the catchment's agro-climatic and geographic controls. The settler count is constructed as in Section 3.3 and the vector $X_c$ as in Section 3.4.

The baseline vector $C_{c,1980}$ contains log population, log total employment, the urban population share, and log agricultural employment. The same four baselines enter every column to improve precision and condition the endline comparison on earlier city characteristics. The outcome's own baseline is therefore included for population, total employment, and agricultural employment. Baseline agricultural and total employment also determine the agricultural share. Additional checks include the 1980 value of modern employment or its share for the corresponding outcome (Table 5 and Appendix Figure D3). The unemployment variants in the heatmap add the 1980 agricultural share and log population density. I use Conley standard errors allowing spatial correlation within 200 kilometers.

The catchments that define exposure follow travel time and can cross district or province borders. The main specification compares cities across the whole sample, conditional on settlement scale, destination characteristics, and baseline city characteristics. Island and province fixed effects restrict these comparisons geographically (Table 5).

Prices, wages, and expenditure have no 1980 measurement to condition on, so those regressions compare levels across cities within each survey year and provide suggestive evidence on the mechanisms. For the food share, the estimate is unchanged when a 1980 proxy for the missing baseline is added, population density, the agricultural employment share, or schooling, one at a time or together (Appendix Table E1). For selected 2002 and 2011 household-survey outcomes, Appendix Figure D3 varies exposure construction and controls and adds island fixed effects.

### 4.3 Identification

The key identifying assumption is that, among cities with similar observed characteristics and numbers of nearby settlers, exposure to better-matched villages is unrelated to other determinants of urban outcomes. The village assignment process motivates this assumption, but the government chose the number and locations of the settlements. I therefore control for the number of settlers and the characteristics of the destination villages. Among cities with equally large nearby settlements on similar measured land characteristics, exposure differs with the match between the places settlers came from and the places they were sent.

The program did not match migrants to destinations on agro-climatic similarity, and within a cohort it is plausible that households could have been assigned to another village that opened around the same time (Bazzi et al. 2016). Section 2.1 describes the assignment process. Families from a rice-producing region such as Bali who were sent to a destination also suited to rice are a high-productivity match. Families from the same origin sent to an area more suited to cassava are a lower-quality match. This assignment argument supports village-level variation in match quality, conditional on the destination.

The village-level validation shows a positive relationship between agro-climatic similarity and crop yields relative to each crop's national average, although the estimates are imprecise with standard errors clustered by province (Appendix Figure C4). This is evidence on the relevance of the overall index for village productivity.

A city surrounded by sites with land similar to the Java and Bali origin mix could have high exposure regardless of which settlers arrived. Following Borusyak & Hull (2023), I hold the settlement sites fixed and reassign settlers' origins to distinguish exposure associated with the sites' land characteristics from exposure associated with the origins of the settlers assigned there. The main specification uses both sources of variation, conditional on the controls. I examine their separate contributions in Section 5.1, with the detailed reassignment and validation tests in Appendix C.1.

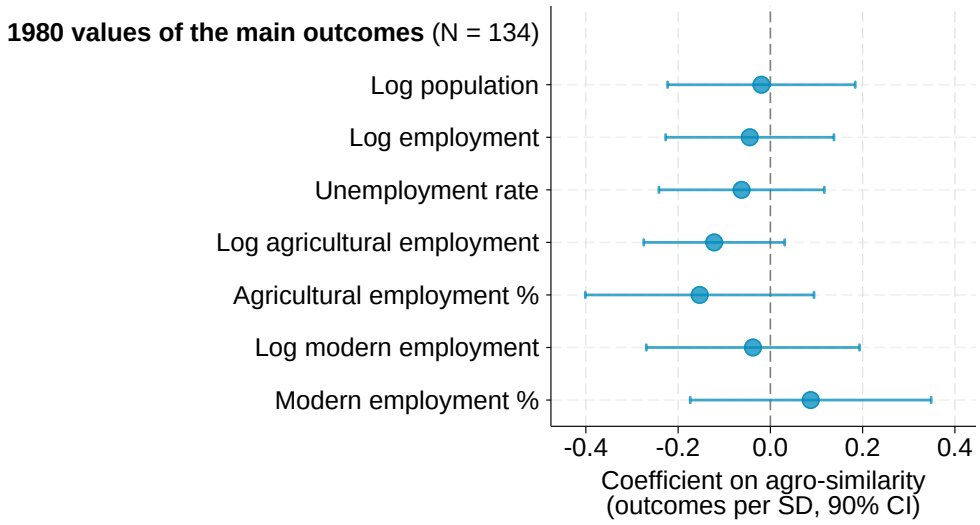
Transport geography and city boundaries introduce additional identification concerns. Travel times determine which villages contribute to a city's exposure (Appendix B.1)[15], and the transport surface uses road data observed after settlement, so an endogenous transport network is a concern. The road checks hold the 1950 road density fixed, condition the road-condition measure on its 1990 level, and give imprecise city estimates (Appendix E.5). More exposed cities lie farther from the Thai and Vietnamese rice hubs by sea. Controlling for route proximity preserves the headline results, with one exception reported in Appendix B.2.

Reliable satellite-based urban boundaries from before the program are unavailable for much of the study region. Historical satellite products include earlier dates, but their coverage is incomplete, as described in Section 2.2.1. I therefore combine post-program urban boundaries with the census-based reclassification that adds areas already urban in 1980 but omitted by the satellite measure, and hold this definition fixed across census years. Section 5.1 reports estimates that exclude these additional areas and that vary the urban cutoffs. The urban-center area and the land area of a city's member subdistricts do not respond detectably to exposure, while the number of member subdistricts rises (Online Supplement). The estimates describe outcomes in the sampled urban centers and do not identify the effect on the formation of new cities.

The productivity exposure of cities is uncorrelated with the baseline levels of the seven main outcomes. Conditional on the settler count and agro-climatic controls, population, employment and sectoral employment in 1980 do not differ significantly between cities later surrounded by more and by less productive villages (Figure 11). Some villages were established in 1979 and 1980, before these baseline data were collected. Reconstructing exposure and village-level controls using only villages established in 1981 or later preserves balance on all seven outcomes, holding the city sample fixed

[15] I vary the exposure window in two ways: rebuilding the city sample and village controls at the neighboring 250- and 400-minute windows, and widening exposure on the main city sample to 400 and 500 minutes while holding the 300-minute controls fixed (Table 5 and Appendix Figure D5). Appendix Table D1 defines the specifications, with further checks in Appendix D.2 and Appendix Figure D3. Section 5.1 reports the results.

Figure 11: Baseline Balance: Main Urban Outcomes in 1980




*Notes:* Each row regresses the standardized 1980 value of a main urban outcome on standardized agro-climatic similarity. Each outcome is standardized within the main city sample. Agro-climatic similarity is standardized over cities with a measured exposure. Controls are the inverse-travel-time-weighted settler count and the unweighted means of the 16 agro-climatic characteristics of villages within 300 travel minutes. All rows use the main sample of 134 cities. Bars are 90% confidence intervals based on Conley standard errors allowing spatial correlation within 200 kilometers. Modern employment combines manufacturing and services. The outcomes use 1980 census data. The village exposure uses the transmigration data of Bazzi et al. (2016). Appendix Figure C2 repeats the comparison using only villages established in 1981 or later to construct exposure and village-level controls.

(Appendix Figure C2). Rebuilding exposure and the village-level controls from these later villages also keeps the size, sign and statistical significance of the four sectoral estimates (Appendix Table A1).[16]

## 5 Reduced-Form Results

Cities surrounded by more productive villages, compared with cities surrounded by less productive ones, differ substantially in employment structure, with more employment in manufacturing and services and imprecisely estimated differences in population and total employment (Section 5.1). Rural productivity can affect cities through food trade and migration. I examine consumption, food prices and employment by trade openness, then migration and the contribution of additional workers to modern-sector employment (Section 5.2). I then compare rural and urban adjustment and assess public provision, capital and rural demand as additional links from the villages to the cities (Section 5.3). The spatial model examines trade and migration as mechanisms of these responses (Section 6).

[16] Adding the four baseline amenity levels as controls attenuates the sectoral estimates, which retain their signs and statistical significance on the same cities (Appendix Table A1). Appendix Table E12 reports the amenity comparisons.

### 5.1 Urban Structural Transformation

Across the 134 cities in the main sample, exposure to more productive villages shifts employment from urban agriculture to manufacturing and services.[17] Each column of Table 4 compares 2010 outcomes across cities whose exposure differs by one standard deviation, conditional on their 1980 characteristics and the other controls.[18] The agricultural employment share is lower by 5.1 percentage points, and modern-sector employment, combining manufacturing and services, is higher by about 17%. Agricultural employment also falls, and the modern employment share rises.[19] These are conditional endline differences. The regressions do not follow individual workers as they change jobs.

[17]The sectoral pattern also appears under alternative city definitions (Appendix Table A1). The sectoral responses keep their directions across exposure windows. Estimates at the neighboring 250- and 400-minute windows are imprecise when the city sample and village controls are rebuilt. Widening exposure on the main city sample while holding the 300-minute controls fixed gives significant modern-employment and population responses at 400 and 500 minutes, and the agricultural-share response remains negative and less precise (Table 5 and Appendix Figure D5).

[18]The compositional results are similar with baseline birthplace controls, alternative settler-count controls and the settlers' share of the pre-existing rural population (Appendix Table C1 and Table A1). Settler weights give similar estimates, while inverse-time weights reduce the modern-employment response and its significance. Island fixed effects preserve the agricultural-share and modern-employment results. Province fixed effects reduce their size and leave them statistically insignificant, with standard errors of similar size (Table 5).

[19]The agricultural-employment estimate is marginally significant. Some uniform-kernel spatial variances are unstable at wider distances. Bartlett estimates retain significant sectoral responses. Under the wild bootstrap over catchment components, log modern employment remains significant, the modern employment share is marginally significant, and the agricultural employment share is insignificant (Appendix Table D4).

Table 4: Urban Growth and Structural Transformation: From Agriculture to the Modern Sector

| | Population + Employment | | | Agric Empl | | Modern Empl | |
|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| | ln pop | ln emp | unemp rate | ln L agric | agric % | ln L modern | modern % |
| Agro-similarity | 0.051 | 0.024 | 0.008 | -0.166* | -0.051*** | 0.158** | 0.066*** |
| | (0.059) | (0.057) | (0.007) | (0.093) | (0.018) | (0.062) | (0.015) |
| Settler count | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Baseline Pop 1980 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Urban share 1980 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Total Empl 1980 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Agric Empl 1980 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Agro-climatic controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Outcome mean | 11.842 | 10.933 | 0.122 | 9.822 | 0.374 | 10.232 | 0.516 |
| Outcome SD | 0.854 | 0.840 | 0.040 | 0.896 | 0.183 | 0.975 | 0.156 |
| Observations | 134 | 134 | 134 | 134 | 134 | 134 | 134 |

*Notes:* Each column is one regression of a 2010 population census outcome on standardized agro-similarity on the main city sample, with the main specification controls marked by the check marks: the settler count and 16 agro-climatic covariates within 300 minutes, and four 1980 baselines. Outcomes are log population, log total employment and the unemployment rate (columns 1 to 3), log agricultural employment and the agricultural employment share (columns 4 and 5), and log modern-sector (manufacturing and services) employment and the modern-sector employment share (columns 6 and 7); shares and the unemployment rate are fractions, and the four sector columns are shaded. In the labels: ln (the natural log), L (a count of workers), emp and Empl (employment), unemp rate (the unemployment rate), and agric % and modern % (the two sector shares); Baseline Pop 1980, Total Empl 1980 and Agric Empl 1980 are the logs of 1980 population, total employment and agricultural employment, and Urban share 1980 is the urban share of the 1980 population; SD is the standard deviation. Conley SE within 200 km in parentheses, with stars from the same standard errors: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Appendix C.1 reports randomization inference for the design-based estimates, the Online Supplement every randomization-inference p-value, and Appendix Table D4 other spatial-inference radii; variants with polynomial terms in the settler count are in the Online Supplement (Section S3.7).

The sectoral differences are large relative to the structural transformation these cities experienced. The contrast between cities at the 90th and 10th percentiles of exposure amounts to about 63% of the average decline in the agricultural employment share between 1980 and 2010.[20] Services account for most of the increase in the modern employment share (Appendix Table E2). Population and total employment also have positive estimates, but their responses and the unemployment response are imprecisely estimated. Modern-sector employment thus expands relative to total employment.[21]

Modern employment shares rise with both sources of exposure distinguished in Section 4.3: the settlement sites' land characteristics and the origins of the settlers assigned there. The difference between their coefficients is imprecisely estimated. The agricultural employment share falls with the site-land component, while its response to the origin component is imprecise, with an interval that

[20] This comparison describes the exposure gradient within the observed sample. Zero exposure lies far outside its observed range.

[21] Modern employment responds more in cities with larger baseline populations, while its response does not differ detectably by initial agricultural intensity (Appendix Table D3).

Table 5: Robustness of the Main Estimates to Specification Variations

| | (1) Log population | (2) Agricultural employment share | (3) Log modern employment | Observations |
|---|---|---|---|---|
| Main specification | 0.051 | -0.051*** | 0.158** | 134 |
| | (0.059) | (0.018) | (0.062) | |
| *Controls* | | | | |
| Unweighted settler count | 0.047 | -0.053*** | 0.150** | 134 |
| | (0.071) | (0.018) | (0.074) | |
| Weighted covariates | 0.100* | -0.045*** | 0.187*** | 134 |
| | (0.060) | (0.014) | (0.060) | |
| Own-outcome baseline | 0.051 | -0.051*** | 0.177*** | 134 |
| | (0.059) | (0.011) | (0.063) | |
| *Exposure weights* | | | | |
| Population-weighted | 0.057 | -0.044*** | 0.155** | 134 |
| | (0.057) | (0.015) | (0.061) | |
| 1/d | 0.002 | -0.052*** | 0.110* | 134 |
| | (0.045) | (0.018) | (0.062) | |
| $1/d^2$ | -0.023 | -0.047*** | 0.075 | 134 |
| | (0.036) | (0.017) | (0.058) | |
| *Wider catchment, main sample* | | | | |
| 400 min | 0.118** | -0.023 | 0.188** | 134 |
| | (0.059) | (0.032) | (0.094) | |
| 500 min | 0.137*** | -0.044* | 0.268*** | 134 |
| | (0.048) | (0.026) | (0.052) | |
| *FE and inference* | | | | |
| Province FE | 0.060 | -0.018 | 0.084 | 134 |
| | (0.074) | (0.017) | (0.056) | |
| Island FE | 0.043 | -0.049** | 0.152** | 134 |
| | (0.059) | (0.020) | (0.060) | |
| Cluster: province | 0.051 | -0.051** | 0.158* | 134 |
| | (0.077) | (0.022) | (0.084) | |
| Island FE, cluster province | 0.043 | -0.049* | 0.152* | 134 |
| | (0.078) | (0.025) | (0.084) | |
| Outcome mean | 11.842 | 0.374 | 10.232 | |
| Outcome SD | 0.854 | 0.183 | 0.975 | |

*Notes:* Each cell is the coefficient on standardized agro-similarity from one city-level regression of the 2010 census outcome in the column heading, where the modern sector is manufacturing plus services, with its standard error in parentheses. The first row is the main specification of Table 4, on the main city sample, with the full set of controls and Conley SE within 200 km. Each row below it changes the design choice its label names, as Appendix Table D1 defines; the exposure-weight rows replace the equal weights of the villages within 300 minutes, with $d$ the travel time from a city to a transmigration village, the wider-catchment rows average over the villages within 400 or 500 minutes and keep the main city sample and its 300-minute controls, and the fixed-effect rows add an indicator for each province or for each island. Standard errors are clustered by province (17 provinces) in the two rows labeled as clustered, whose stars use the $t$ distribution. Shaded cells clear the 10% level, and the shade deepens at the 5% and 1% levels. The Observations column gives the number of cities in each row. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

includes both zero and the main estimate.[22] Controlling for expected exposure under a reshuffle of village scores preserves significant estimates for both employment shares (Appendix Table A1).

To explore whether the results are driven by the agricultural peripheries of cities, I repeat the analysis for the urban core, using the baseline non-agricultural employment thresholds described in Appendix D.3. Keeping subdistricts that were at least one-quarter non-agricultural in 1980 gives a slightly larger decline in the agricultural employment share than in the main sample, while modern employment still rises. Raising the threshold to one-half makes the estimated agricultural-share decline more substantial, although the modern-employment response becomes less precise (Appendix Table A1).

[22]The modern-share test of equal coefficients gives $p = 0.809$. Randomization tests on settler origins give $p = 0.090$ within province and settlement wave and 0.116 within province (Panel C of Appendix Table C2). The village-productivity and baseline-outcome tests in Appendix C.1 qualify the causal interpretation of the origin component.

The decline in the agricultural employment share therefore extends to established urban areas, with a larger estimate when the most agricultural parts are excluded.

### 5.2 Food Trade and Migration

In the food-market mechanism, cheaper food lets residents meet their basic needs and spend more on non-agricultural goods, raising the demand for services and manufactured products and, with it, the demand for labor in those sectors. A city that can import food cheaply has less to gain from a more productive hinterland. Its food supply depends less on nearby harvests, so the food-market mechanism predicts a smaller employment response.

Food-price responses are consistent with cheaper food in cities surrounded by more productive villages, particularly those less connected to international rice markets. The pooled decline is imprecisely estimated in the smaller observed-price sample. In more closed cities, the decline is marginally significant and differs significantly from the positive, imprecise response in more open cities (Figure 12a).[23] Raw-food price estimates are also negative in rural subdistricts with settlements, although their pooled and group estimates and the difference between groups are imprecise. These subdistricts combine settled and preexisting villages, so their price response does not isolate prices in the settlements.

Unit values combine price and quality choices. Rice unit values provide an imprecise city comparison, and the matched-location variety check does not establish that shifts toward higher-quality rice mask a price decline (Appendix Figure E4a).[24]

In more exposed cities the food expenditure share falls, and households consume more food. Total and food monthly spending per person rise in the 1996 survey, before the 2004 rice import ban. Calories and staple quantities per person also rise, with significant estimates under the main spatial inference (Figure 12b). Food spending increases even as its share of the household budget falls. Spending measures nominal consumption value, including own production valued by the survey, so greater quantities, household resources and product composition can offset a price decline. The consumption comparison by openness does not establish gains concentrated in more closed cities (Appendix Figure E4b).

Within cities, the food share falls as spending per person rises. Applying that slope to the response of log spending per person predicts 77% of the food-share decline in 1996 and 55% in 2011. This descriptive comparison does not isolate an income mechanism, because spending can respond to both prices and earnings. The remaining change is negative and statistically insignificant in both waves, so the food-share evidence does not establish a decline beyond the observed Engel relationship (Appendix Figure E5).

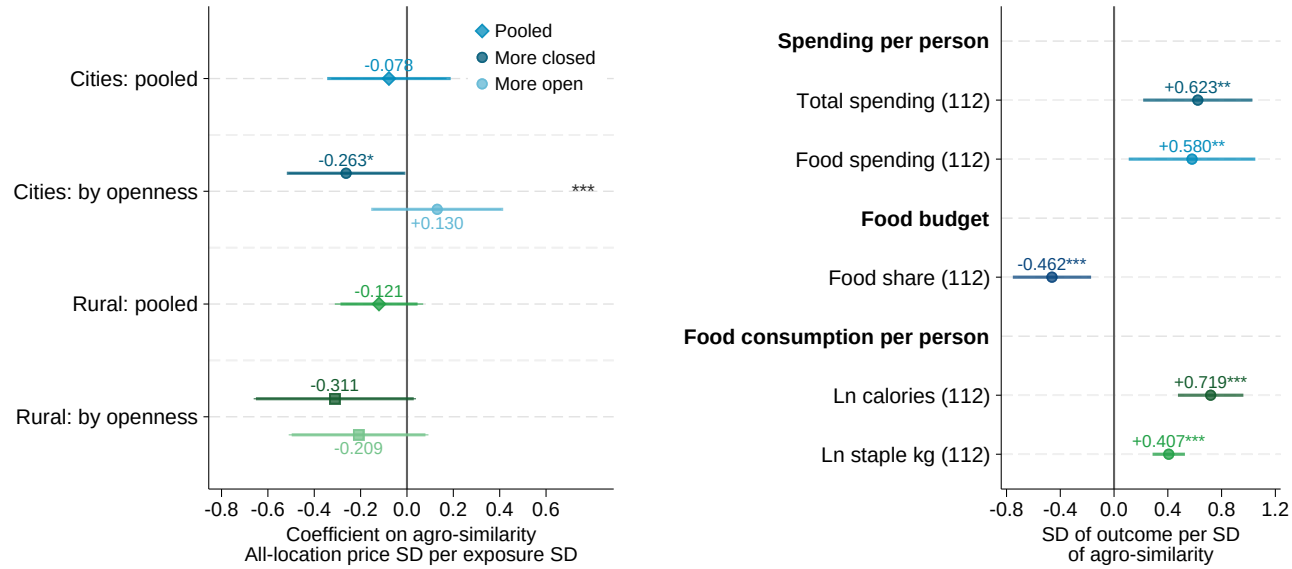
[23] The closed-city decline is marginally significant under both uniform and Bartlett spatial weights, and the difference between groups is significant under both. With province clustering, the contrast is marginally significant and the closed-city decline is insignificant.

[24] The other pooled price measures also have negative, insignificant estimates (Appendix Figure D4). Across thirds of route distance, the city agricultural-price response differs marginally between the most closed and most open third, while the other price responses do not differ detectably (Appendix Table B2).

Figure 12: Raw-Food Prices, City Spending and Consumption, 1996

(a) Raw-Food Price Responses by Trade Openness

(b) Pooled City Spending and Consumption




*Notes:* Susenas 1996. Panel (a) uses 112 main-sample cities and 117 rural subdistricts with a settlement outside cities and within 300 travel minutes of a transmigration village. Diamonds are pooled slopes from separate regressions. The paired slopes come from one regression per location type with agro-similarity interacted with the openness groups and the openness indicator. More open locations are strictly below the location-specific median route hours to Bangkok or Ho Chi Minh City, fixed before restricting to observed prices. Ties enter the more closed group. Regressions use the full main-specification controls, with the settler count in overlapping villages for settlement subdistricts. The raw-food index uses consumption unit values, including own production valued by the survey, and can reflect quality. Prices use the SD across all measured cities and rural subdistricts in the source population. Exposure uses separate city and rural SDs. Thick and thin bars are 90% intervals using uniform and Bartlett spatial weights within 200 km. Stars at right test more open minus more closed under uniform spatial inference. Panel (b) uses 112 cities in the main sample with observed consumption outcomes. Each row is a pooled city regression in the main specification with the full set of controls and Conley SE within 200 km. Spending is monthly nominal consumption per person in levels, including own production valued by the survey. Food share is the person-weighted mean household food-budget share. Calories are per person per day and staples are kilograms per person per year, both transformed as ln(1+x). Each outcome uses its SD across measured main-sample cities. Exposure uses the existing city SD. Bars are 90% confidence intervals. In both panels, household aggregates retain survey weights, location regressions have equal weights, and coefficient stars test zero under uniform spatial inference: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: Susenas 1996 and the route-access measure.

Rice production within cities rises with exposure, while the estimates for output per person and consumption less local production are imprecise (Appendix Table E3 and Figure E3). Consumption less production combines 2002 consumption with 2003 production in milled-rice weight. It measures an implied net requirement and does not identify where the rice comes from, since I do not observe trade flows between a city and its hinterland.[25]

The expenditure pattern persists in the later survey. In 2011 the non-food expenditure share is higher by 1.6 percentage points, statistically significant, and staples per person do not differ detectably with exposure. Household income per worker in 1996 is higher by 26.1% per standard deviation of

[25] Home-grown food, produced by the household itself or received as a gift, is a small share of the raw-food consumption value of city households in the 2002 survey, and exposure does not lower it. Its response is positive and insignificant for both all households of the member subdistricts and households the survey classifies as urban.

exposure, statistically significant, while the 2011 non-agricultural wage, overall wage and real wage estimates are statistically insignificant (Appendix Table E4, Panels A and B; Appendix Table E3).

I test the prediction of a smaller employment response in more open cities (Table 6, Panel A) using route proximity to the Thai and Vietnamese hubs that supply most of Indonesia's rice imports, a measure adapted from Bazzi (2017). This comparison supports the mechanism if route proximity does not also capture other determinants of the response to rural productivity, including urban labor demand. The transport-source and timing qualifications in Section 3.5 apply to this measure.

The median split estimates a separate exposure response in the more closed and more open cities:

$$\begin{aligned} y_c = \beta_1 \, AgSim_c + \beta_2 \, AgSim_c \times HighOpen_c + \chi \, HighOpen_c \\ + \phi \, SettlerCount_c^{IDW} + C_{c,1980} \, \xi + X_c \, \zeta + \varepsilon_c \end{aligned} \tag{6}$$

Here $HighOpen_c$ indicates the 68 cities with above-median openness (the other 66 are the more closed cities), and the controls are those of Equation 5. The exposure response is $\beta_1$ in the more closed cities and $\beta_1 + \beta_2$ in the more open cities. The interaction $\beta_2$ tests whether the responses differ.

Modern-sector employment responds more in the cities less connected to international rice markets. Both openness groups have a lower agricultural employment share and higher modern-sector employment with greater exposure, with larger responses in the more closed cities. The differences in modern employment and its share are statistically significant, and the agricultural-share difference is marginally significant (Table 6, Panel A).[26] Population and employment estimates are also positive and larger in the more closed cities, although both differences are imprecisely estimated. The larger shift of employment into the modern sector in the more closed cities is consistent with Glaeser's closed economy, in which a better harvest raises the share of workers in the urban sector. In his model the urban sector produces traded manufacturing goods. Here the added modern-sector workers are mostly in services, so cities can also grow in services (Gollin et al. 2016; Jedwab & Vollrath 2015) as the cost of meeting food needs falls.

The 1996 and 2002 price waves cover fewer cities than the census, and the pooled employment responses are weaker on those observed-price samples. The larger modern-employment response in the more closed cities is significant on the 1996 sample and marginally significant on the 2002 sample (Appendix Table E5, Panels A and B). The price contrast can also reflect retail and transport costs, household income responses, and changes in purchased quality.

Realized rice imports give a stronger agricultural-share contrast, but can respond to the mechanism being tested. Panel B splits cities by gravity market access, constructed from realized import volumes. A city whose productive hinterland replaces imported rice can appear more closed because of

[26] The closed-city modern-employment and agricultural-share responses are significant. The corresponding open-city estimates are imprecise, while the modern employment share rises significantly in the more open cities.

that substitution. The larger agricultural-share difference in this panel is consistent with this source of endogeneity. It does not measure the bias in the route-based estimate. Modern-sector results have similar signs across the two measures (Appendix B.2).

In Panel C, the gravity split of Panel B and its interaction with exposure are instrumented by the route split of Panel A and its interaction with exposure. Identification requires the route split and its interaction with agro-similarity to affect urban outcome levels and their response to rural productivity only through rice-import access, conditional on agro-similarity and the main controls (Section 3.5). The continuous form's estimating and first-stage equations and its first-stage statistic are in Appendix B.2.

The first stage is weak, with a Kleibergen-Paap statistic of 4.37 against a Stock-Yogo critical value of 7.03. I therefore report the Anderson-Rubin test of the joint hypothesis that both instrumented coefficients are zero. It rejects this hypothesis in four of the seven columns.[27] This joint test does not establish that the exposure responses differ by openness. The continuous form of the instrument, with a first-stage statistic of 2.53, is in Appendix Table B3.

The geographic overlap between trade access and other regional characteristics limits this comparison. The modern-employment difference stays significant when island indicators are interacted with exposure. I do not have baseline data on prices and food availability, so I cannot test directly whether food supply was less of a binding constraint in the more open cities, which can import food from anywhere. The openness comparison supports the food-market mechanism if the other mechanisms respond similarly in the two groups of cities after controls. Migration, amenities, and rural demand could also vary with trade access. Migration is examined next. Rural demand and public provision are considered in Section 5.3.

Rural productivity can also affect migration through labor supply in the settlements and labor demand in the cities. A labor-saving productivity gain would release settlers to the cities, as in Bustos et al. (2016). A pull from the cities, as in Harris & Todaro (1970), would show up as migration into the more exposed cities. Food trade can raise urban labor demand and induce migration from origins outside the settlements. The variation here is in the settlers' farming skills at their destination, conditional on the number of settlers placed, so it is a productivity shock to the people the program moved and holds their number fixed. For the settlers, a better match raises what farming pays where they are, so it can retain labor in the settlements. Food trade may increase competition for surrounding agricultural producers while raising labor demand in the cities.

The recent-migration results are consistent with population reallocation in response to the productivity shock. More exposed cities have a higher share of residents who lived in another district five years earlier. On the main city sample, a one-standard-deviation increase in exposure raises this share of 2010 residents by +1.92 percentage points. The share who lived in another province also rises, with

[27] The rejection count uses $p < 0.10$.

both estimates significant under the main spatial inference (Panel A of Appendix Table E8).[28] District moves include moves within a province as well as across provinces. These measures describe recent arrivals among current residents and do not identify departures, earlier moves, or the transmigration villages from which migrants came.

Recent migration rises more in the more closed cities. The district-move difference between the openness groups is statistically significant and the province-move difference is imprecise (Panel B of Appendix Table E8). Recent district moves are also more common among workers in more exposed cities. This comparison uses workers with an observed district of residence five years earlier and includes return migrants. Its openness difference is imprecise under both spatial inference methods. These estimates leave migration as a possible contributor to the larger employment response in the more closed cities.

The recent interprovincial share also rises significantly in the groups with larger initial populations or higher initial agricultural employment shares (Appendix Table D3, column 4). The estimates for smaller and less agricultural cities are positive and imprecise, and the differences between groups are imprecisely estimated in both splits.

In the comparison of residents born from 1955 to 1990, the share born in another district of the same province is significantly higher in more exposed cities (Figure 13). This pattern is consistent with people moving into urban centers from nearby districts, possibly from other rural areas. The birthplace data do not establish whether migrants' previous residences were nearby or rural.

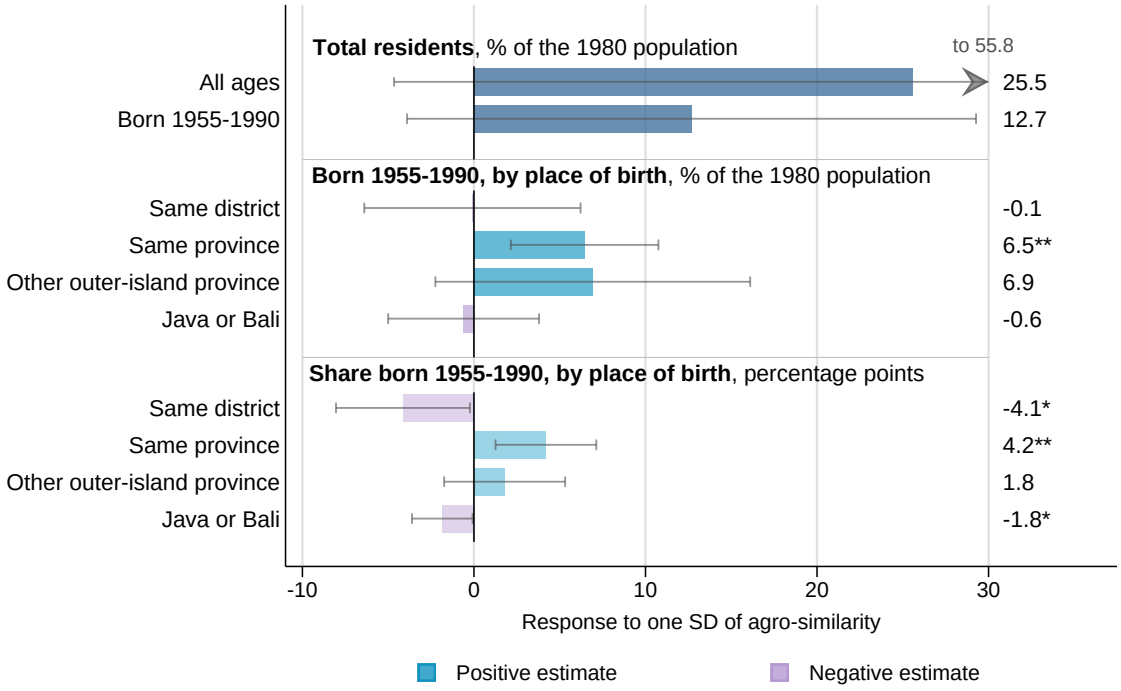
Within this cohort, I use Java or Bali birthplace as the best available proxy for first-generation transmigrants. It excludes their children born on the Outer Islands and includes migrants from Java or Bali who moved outside the program. The share of these potential first-generation transmigrants is, if anything, smaller in more exposed cities.

The city and village evidence provides little support for first-generation transmigrants leaving more productive transmigration villages as the source of this reallocation. The linear gradient of the measured settler stay-share is statistically insignificant (Online Supplement). Figure 14 plots the local-linear fit of a broader definition. The definitions in the figure count Java- or Bali-born residents relative to the original settlers, excluding locally born children. They do not follow the original settlers individually or locate the origins of urban in-migrants.

Rural subdistricts without a settlement shift employment out of agriculture while their total employment does not rise (Figure 17). This spatial pattern is consistent with adjustment in areas whose output competes with that of the productive settlements. The censuses do not locate the migrants' origin villages or trace flows between these places.

[28] With Bartlett distance weights, the province estimate is marginally significant and the district estimate remains significant.

Figure 13: 2010 Urban Residents by Birthplace




*Notes:* Each bar is one regression of the row's outcome on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200 km. Whiskers are 90% confidence intervals from the same standard errors. The whole-population interval runs past the axis and ends in an arrowhead, with its upper bound written beside it. The first block is the city's whole population and its residents born from 1955 to 1990 in the 2010 census, in % of its 1980 census population. The whole-population response is 10% of that outcome's mean of 244%, against a log population response of 5% in Table 4, since the two outcomes weight the cities differently and a regression in levels gives more weight to the cities that grew most. The second block splits that cohort, in the same units, into four birthplace groups that add up to it: born in the same district as the city, elsewhere in the same province, in another outer-island province, and on Java or Bali. The third block gives each group's share of the cohort in percentage points, with standard errors in Appendix Table C1. The hue of a bar marks the sign of its estimate, and its shade marks the block. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

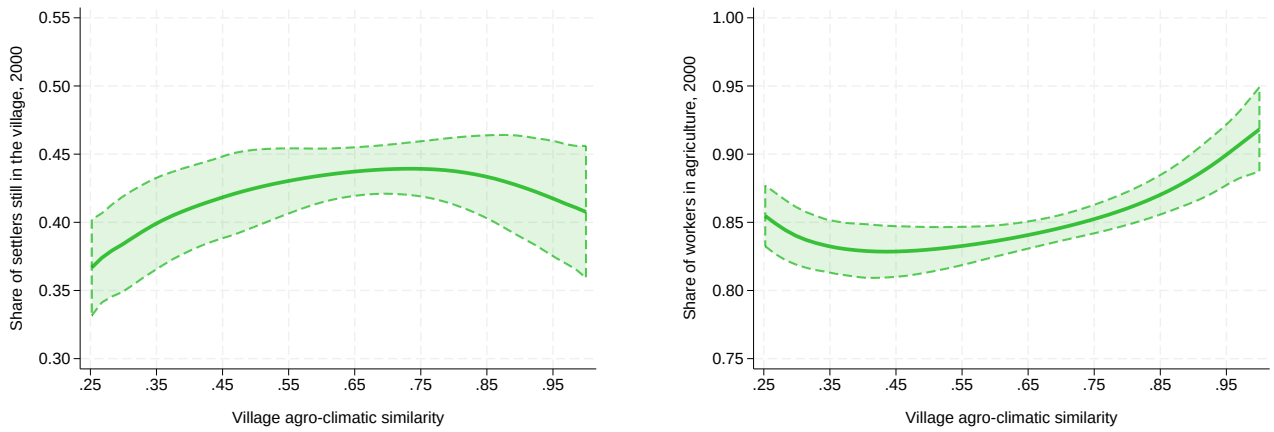
At the mean city, the modern-sector employment gain exceeds the estimated increase in the city's workforce (Figure 15). To assess how much of the gain additional workers could fill, I hold the incumbent workforce fixed, assume departures and labor-force entry and exit do not offset arrivals, and assign every additional worker to a modern-sector job. Under these conditions, the workers added could fill 24% of the modern-job gain, with a one-sided 95% upper bound of 82%. The estimates are imprecise, so the calculation gives a bound. In the main specification, the additional workers could fill only part of the modern-sector expansion.

Net employment growth does not measure gross migration when arrivals replace departing workers or coincide with changes in labor-force participation. At the measured modern-employment rate of in-migrants, the workers added could fill 16% of the gain, with an upper bound of 54%. Counts of residents aged 15 and over are less informative because they include nonworkers, and they do not reject that the residents added could fill the entire modern-job gain if every additional resident worked in

Figure 14: Settler Stay-Share and the Agricultural Employment Share Against Agro-Climatic Similarity

(a) Settler Stay-Share

(b) Agricultural Employment Share, 2000




*Notes:* Semiparametric local linear fit (Epanechnikov kernel, bandwidth 0.25) of the village settler stay-share in panel (a), and of the share of a village's workers in agriculture in the 2000 population census in panel (b), on the village's agro-similarity (0 to 1 scale), with their pointwise 90% confidence bands, partialing out the village's ruggedness, slope classes, log elevation, distance to the coast and 1948 to 1978 mean temperature, estimated on the 808 transmigration villages with a measured 2000-census agricultural employment share and a measured stay-share below 2 and plotted over the 0.25 to 1 similarity range. The bands treat the villages as independent. Panel (a) of Appendix Figure E1 reports the linear gradient of the agricultural share with its island-clustered, wild-bootstrap and Conley inference. Appendix Figure D1 draws both panels at a bandwidth of 0.30 and with each village weighted by its 2000 population. The plotted stay-share is a village's 2000-census residents born on Java or Bali over the settlers the program placed there. The stay-share named in the text also removes residents who lived in Java or Bali five years before the census and residents born after the village was settled.

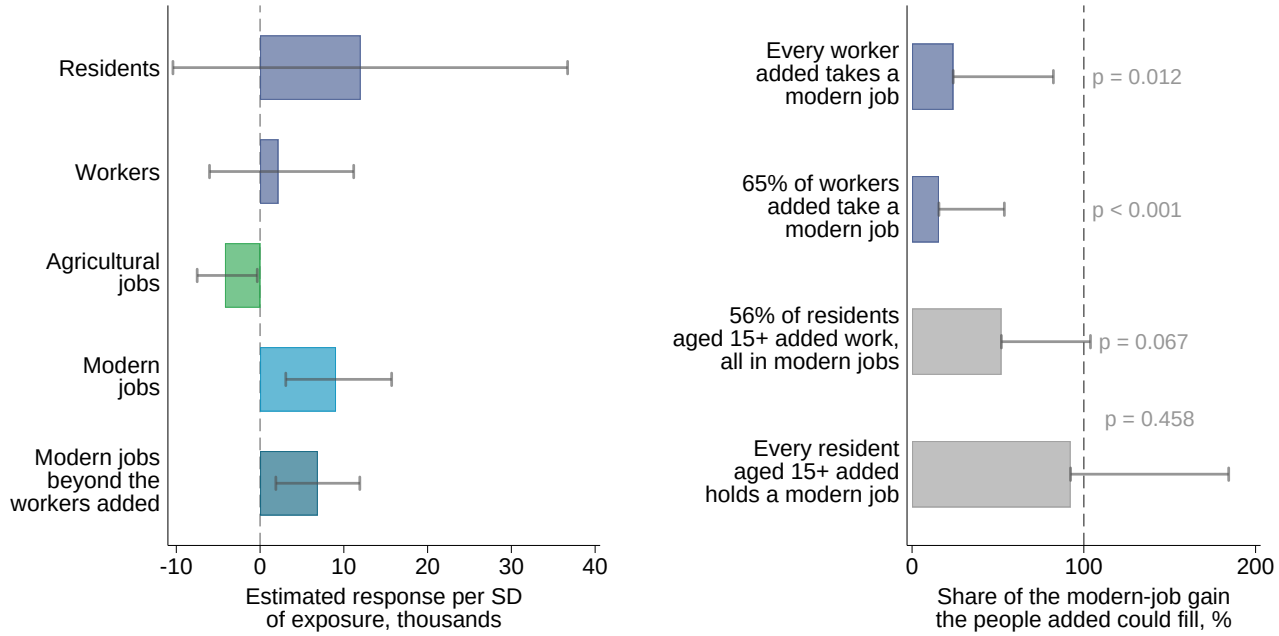
that sector. With the expected-exposure control used above, the worker-based upper bound includes the whole gain when every additional worker takes a modern-sector job (Appendix E.3; Panel B of Appendix Table E9). Employment among workers born in the city's district and children and youths per native-born adult do not rise detectably with exposure. These proxies cannot exclude labor-force or fertility responses (Appendix Table E10).

Most of the agricultural-share response occurs within birthplace groups. Decomposing the full 2010 workforce by birthplace and industry attributes 84% of the fall to differences within groups, with smaller contributions from workforce composition and its interaction with within-group changes (Appendix Figure E7). A separate partition by birthplace and prior residence also attributes most of the fall to differences within groups: workers born in the current district, lifetime migrants living there five years before the census, and lifetime migrants living elsewhere five years earlier (Online Supplement). This partition distinguishes lifetime migrants by residence five years earlier, while the recent district-migration measure also includes return migrants. Both decompositions use the full workforce, including birth cohorts outside the 1955 to 1990 range of Figure 13. Repeated cross sections combine individual sectoral switching, cohort replacement, and selection within groups. They cannot identify the share due

Figure 15: Workers Added and Modern Jobs Gained at the Mean City

(a) Estimated Responses of People and Jobs per SD

(b) Share of New Modern Jobs People Added Could Fill




*Notes:* The main city sample, in the main specification with the full set of controls and Conley SE within 200 km. Panel (a) turns columns 1, 2, 4 and 6 of Table 4 into the estimated response of each outcome's level at the mean city per SD of exposure (the 2010 mean times $e^b - 1$), with 90% intervals. The last bar is the modern jobs gained less the workers added, with its interval from the joint covariance of the two coefficients. Employment outside agriculture and the modern sector, estimated directly in the same specification, changes by −0.047 log units (SE 0.084), about 520 fewer workers at the mean city. The bars need not sum to the employment bar because each comes from a separate regression in logs. Panel (b) gives the share of the modern-job gain the people added could fill under a fixed incumbent workforce, no offsetting departures or labor-force entry and exit, and the sector mix stated by each row, its one-sided 95% upper bound (whisker) and the one-sided $p$-value of a share of 100% or more (dashed line). The first two bars count workers and the last two residents aged 15 and over. Appendix E.3 gives the construction. Source: 1980 and 2010 population censuses.

to each. This distinction matters in Indonesia, where Budí-Ors (2026) attributes much of the aggregate decline in the agricultural share to generational replacement.[29]

The price contrast and the employment response by openness support a food-trade mechanism, with precision that varies across uncertainty calculations. Pooled consumption describes the broader response and does not identify the contribution of food prices. Some migration into the cities does occur. More exposed cities have a higher share of residents born in another district of the same province, and recent district moves are more common among their workers, consistent with a response to urban labor demand. Newcomers from other districts take up some of the modern-sector job growth. The productive transmigration villages, if anything, keep their settlers. The conditional workforce calculation and within-group decompositions suggest additional adjustment in employment composition.

[29] Hamory et al. (2021) use individual histories in the Indonesian Family Life Survey (IFLS) to study sectoral switching. Its coverage of the outer-island cities studied here is insufficient for the city-level exposure design.

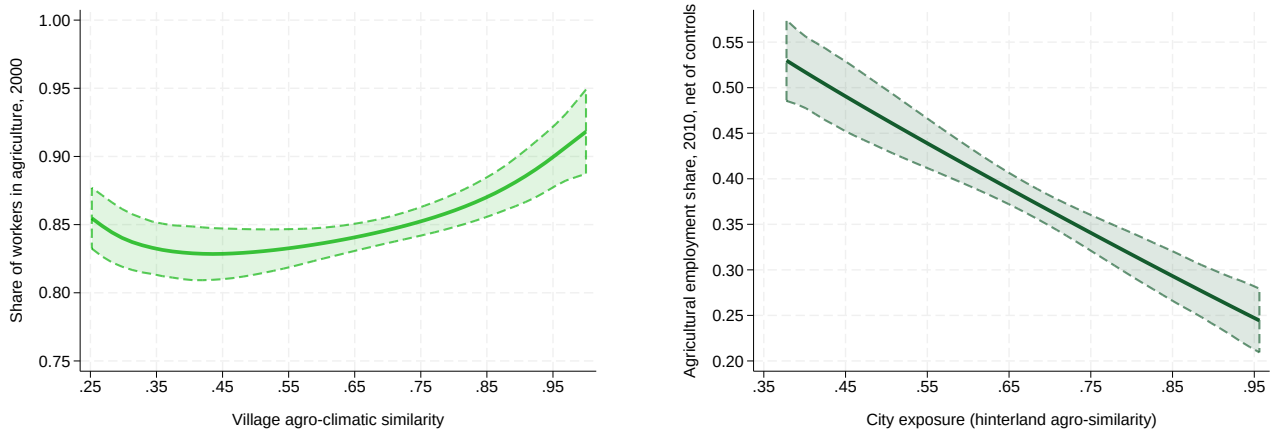
### 5.3 Rural Adjustment and Alternative Channels

The village agricultural employment share has a positive, statistically insignificant gradient with agro-similarity, while the city agricultural share declines (Figure 16, and Appendix Figure E1 gives the linear gradients). In Matsuyama's world, the villages are the open economy that receives the shock: they can sell their food to the cities and, if anything, specialize further in agriculture, their comparative advantage. Nearby cities respond as a city does in the closed economy of Glaeser (2014), moving employment out of agriculture and into the modern sector, and a test confined to the shocked villages misses this pairing.

Figure 16: The Agricultural Employment Share Against Agro-Similarity in the Transmigration Villages and in the Nearby Cities

(a) Transmigration Villages, 2000

(b) Nearby Cities, 2010




*Notes:* Semiparametric local linear fits (Epanechnikov kernel) of the agricultural employment share on agro-similarity, with pointwise 90% confidence bands, after partialing out each panel's controls. **(a)** The share of a village's workers in agriculture in the 2000 population census against the village's own agro-similarity (0 to 1 scale), on the 808 transmigration villages with a measured agricultural share and a measured settler stay-share below 2, partialing out the village's ruggedness, slope classes, log elevation, distance to the coast and 1948 to 1978 mean temperature, at a bandwidth of 0.25, as in panel (b) of Figure 14. **(b)** The agricultural employment share in the 2010 population census against city exposure, the mean agro-similarity of the transmigration villages within 300 travel minutes, on the main city sample, partialing out the controls of the main specification, at a bandwidth of 0.30. The bands treat the villages and the cities as independent, so they are narrower than intervals from Conley SE. At a bandwidth of 0.30 the kernel weights every city at each point of panel (b), so its curve is close to linear by construction. Appendix Figure E1 gives each linear gradient with its clustered, wild-bootstrap and Conley inference.

The fitted curve of the village agricultural share is flat over the lower half of the similarity range and rises over the upper half, most steeply among the best-matched villages (panel (b) of Figure 14). Its statistically insignificant linear gradient limits the evidence for stronger agricultural specialization. The city estimates provide clearer evidence of structural transformation.

Better-matched transmigration villages are more likely to grow rice, the staple consumed in the cities, under both the village controls used here and those of Bazzi et al. (2016). That study finds less similar villages allocating more crop revenue to cash crops.[30]

[30]The rice-growing probability rises by 0.079 (standard error 0.017) with the village geography controls, by 0.058 (0.026) with province-by-placement-year cells, and by 0.127 (0.022) under the Table 7 controls of Bazzi et al. (2016). Under the

Productivity gains in the settlements can affect surrounding rural areas as well as cities. The broader comparison includes entire rural subdistricts with and without a settlement, alongside the cities (Figure 17). Both rural groups lie outside every city and within 300 travel minutes of a transmigration village. Outcomes and exposure are standardized separately among cities and among the pooled rural subdistricts. The coefficients describe changes relative to each population's variation, so their magnitudes are not effects in common physical units. Regressions use the same controls, with the settler count measured within 300 travel minutes. For rural subdistricts with a settlement, it is the number of settlers placed in the transmigration villages that overlap the subdistrict.

The cities show evidence of structural transformation, and modern-sector employment rises while the population estimate is positive and imprecise. The rural subdistricts with no transmigration village match the cities to an extent, in a similar sectoral reallocation of employment. Their agricultural employment share falls and modern share rises, with both responses precisely estimated. The modern-sector employment response is positive there, but the point estimate on overall employment is negative, so the pattern is not one of overall growth in those areas. The areas that experienced the positive productivity shock show little change. The point estimates on population and on overall employment are positive and indistinguishable from zero, and the reallocation out of agriculture into the modern sector is small and statistically insignificant, with composition estimates close to zero and smaller standard errors than in the rural group without settlements. The pattern is consistent with the village economies being open economies that trade with the cities. They continue in agriculture.[31]

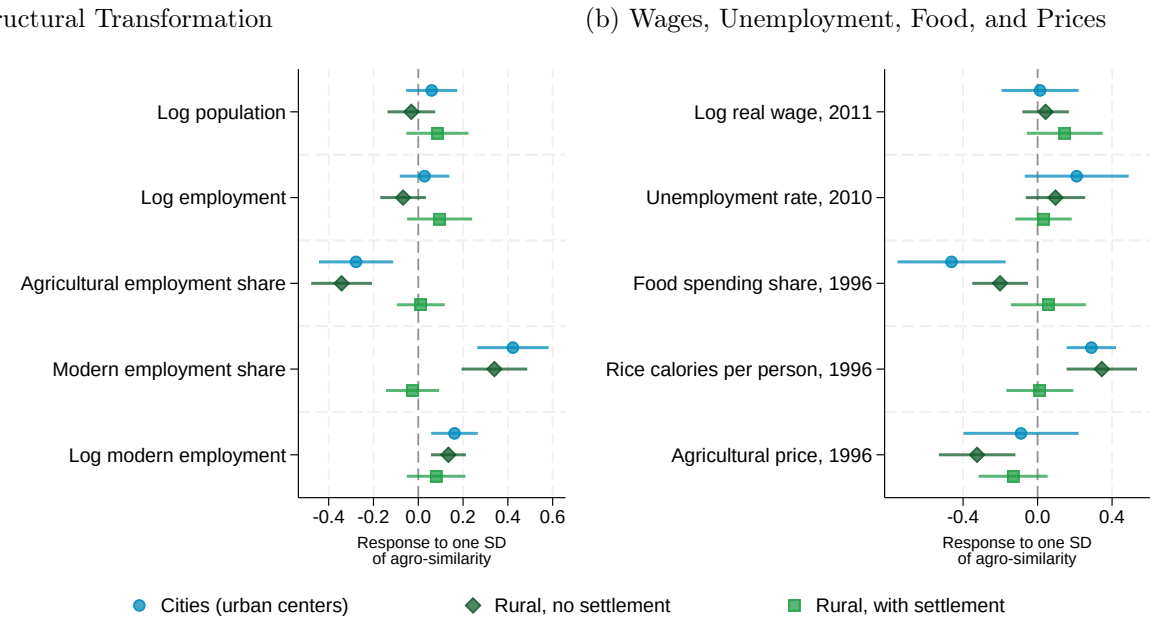
Wages do not respond detectably in any location type, so the sectoral reallocation comes without a clear gain in what a worker earns. The 2011 nominal and real wage estimates in rural subdistricts with a settlement are also insignificant. Raw-food price estimates in 1996 are negative in rural subdistricts with and without settlements, with a significant decline in the latter group (Appendix Table E6).

The food expenditure share falls where employment moves out of agriculture. In 1996 it falls in cities and rural subdistricts without settlements, with a significant estimate in the latter group. The estimate in subdistricts with settlements is imprecise. Calories per person rise in cities and do not move detectably in either rural group. Rice calories (Figure 17, panel (b)) rise in cities and rural subdistricts without settlements (Appendix Table E6). Unemployment responses are imprecise throughout. These outcomes do not establish a welfare ranking across places.

village controls, the estate-crop estimates are not statistically significant, and the cash-crop revenue weight is close to zero.

[31] In one regression that gives each location type its own slope and controls and a common Conley variance, the agricultural-share response of the cities differs from that of the subdistricts with a settlement by −0.289 (standard error 0.115), and their modern-share responses differ by 0.448 (0.117), in the standardized units of Figure 17. Between the subdistricts with and without a settlement, the two responses differ by 0.354 (0.105) and −0.365 (0.117). All four gaps are statistically significant. The composition estimates of the subdistricts with a settlement stay close to zero and statistically insignificant when their settler count is the inverse-travel-time-weighted count within 300 minutes, as in the city regressions, and when their exposure is the settler-weighted agro-climatic similarity of their own transmigration villages.

Figure 17: The Response by Location Type: Cities and Rural Subdistricts With and Without a Settlement



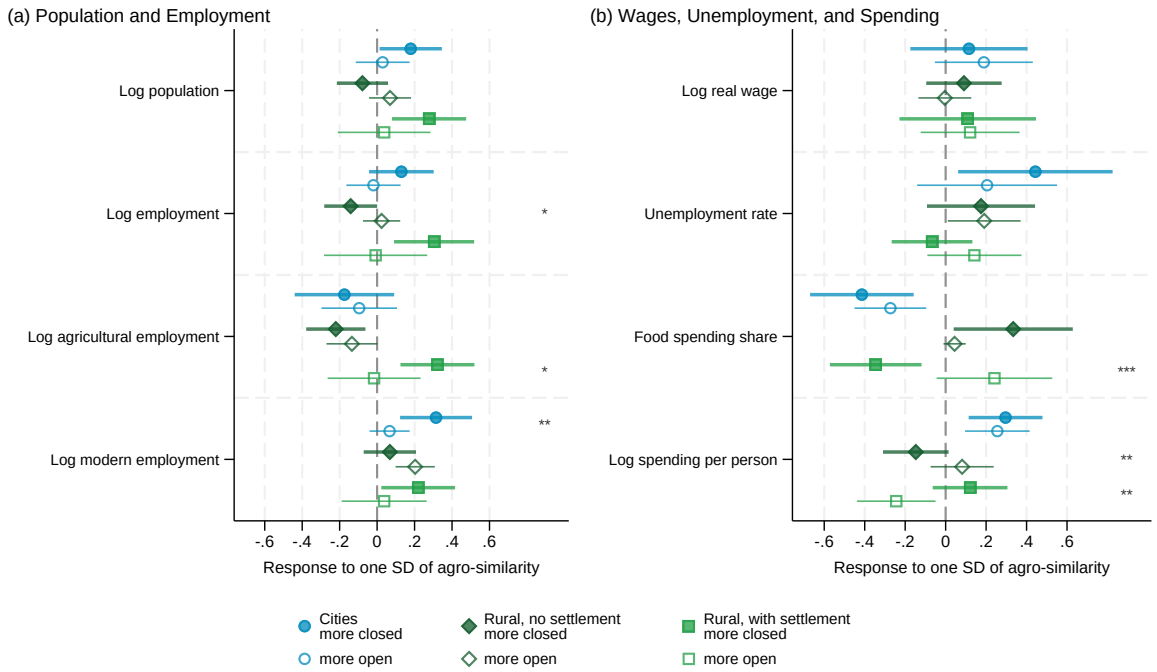

*Notes:* Each marker is the response to a one-SD increase in agro-similarity, the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, and its bar is the 90% confidence interval from Conley SE within 200 km. Cities are the main city sample. The rural subdistricts with and without a settlement lie outside every city and within 300 travel minutes of a transmigration village. Outcomes and agro-similarity are standardized once among the cities and once among the pooled rural subdistricts, at exposure SDs of 0.103 (138 cities with a measured exposure) and 0.103 (794 subdistricts), so city and rural coefficients use different standardization scales. The two rural groups share a scale. The city regressions are in the main specification with the full set of controls. The subdistrict regressions have the same controls, except that the settler count of a subdistrict with a settlement is the number of settlers placed in the transmigration villages that overlap it. **(a)** Every row is a 2010 census outcome on the main city sample, 191 rural subdistricts with a settlement and 603 without. **(b)** The unemployment rate is from the 2010 census. The real wage is the median earner's wage over the local price index, from the 2011 household-survey (Susenas) wave used here, among observed positive earners in retained subdistricts. The food share of spending, rice calories per person (rice kilograms at a fixed calorie content) and the agricultural price (the raw-food index) are from the 1996 wave, before the 2004 rice import ban. Each row uses the units its wave covers: 112 to 134 cities, 117 to 191 subdistricts with a settlement and 288 to 603 without. The rural wage source retains survey subdistricts with at least five observed workers. Every outcome is in Appendix Table E6 with its SE.

Rural responses also differ with access to outside food markets. Among rural subdistricts with a settlement, the more closed subdistricts have larger agricultural-employment responses, a marginally significant difference (Figure 18). The total-employment and population differences have the same sign and are imprecise. Spending per person rises in the more closed subdistricts and falls in the more open ones, and the food-expenditure-share response is lower in the more closed subdistricts. The expenditure differences remain statistically significant with alternative spatial standard errors and province clustering, while the real-wage responses do not differ detectably. Each location type has its own median route-distance split, so these tests compare responses within location types at potentially different levels of access. Employment is measured in 2010, and wages and spending in 2011. The larger agricultural-employment response in more closed settlement areas is consistent with continued farming where outside food access is weaker.

Food-crop villages account for a larger population share in more exposed settlement subdistricts in both openness groups, and their responses do not differ detectably. The population share in plantation

villages falls in the more open subdistricts and does not change detectably in the more closed ones, with a statistically significant difference (Appendix Table E7). The estimates do not indicate a shift toward plantations in either group.

Figure 18: Trade Openness and the Response by Location Type: Population, Employment, Wages, and Spending




*Notes:* Each marker is the response of its row's outcome to a one-SD increase in agro-similarity, in the specification and samples of Figure 17: the main city sample and the rural subdistricts with and without a settlement, which lie outside every city, with each outcome standardized over the cities or over the pooled rural subdistricts. Each location type is split at the median route distance to the Thai and Vietnamese rice hubs among its units within 300 travel minutes of a transmigration village. The coefficients come from one regression per outcome and location type of the outcome on agro-similarity interacted with indicators for the more closed and the more open group. Filled markers on thick intervals denote the more closed group and hollow markers on thin intervals the more open group. Intervals are 90%, from Conley SE within 200 km. Stars indicate that the coefficients of the two groups differ: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Panel (a): the 2010 census outcomes. Panel (b): the unemployment rate (2010 census) and three household-survey outcomes. The real wage and the two spending rows are from the 2011 household-survey (Susenas) wave, which covers fewer subdistricts than the census. Wages summarize observed positive earners. The rural wage source retains survey subdistricts with at least five observed workers. Regressions are in the main specification with the full set of controls, except that the settler count of a subdistrict with a settlement is the number of settlers placed in the transmigration villages that overlap it.

Public investment, capital through banks, and rural demand could contribute to this adjustment. I assess each by its predicted response in the settlements or the rural subdistricts that overlap them.[32]

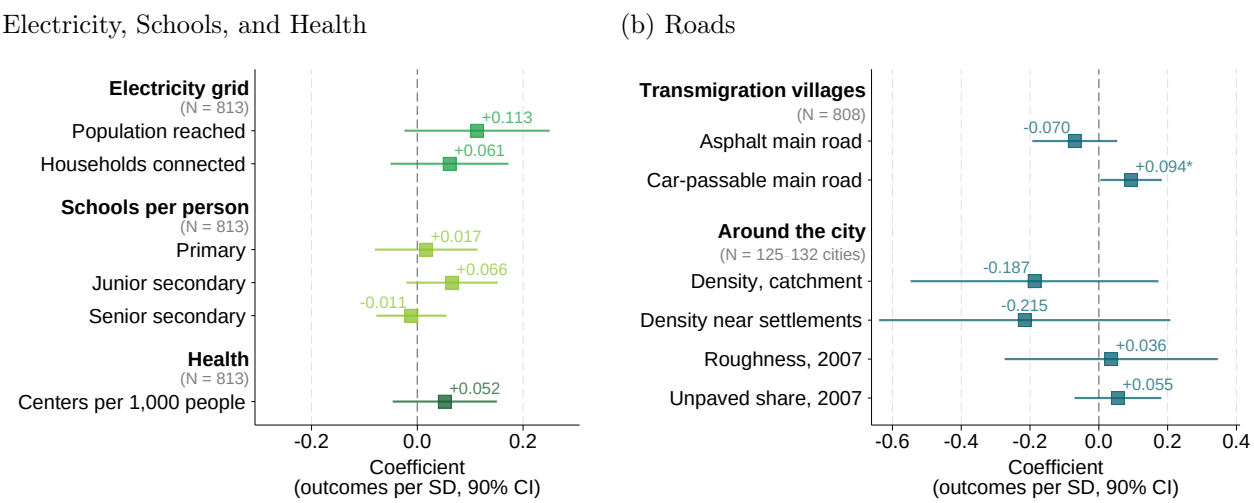
If the state invests more in productive settlements, their roads, electricity, schools and health centers would improve more. Most provision estimates are statistically insignificant, in the transmigra-

[32]The own-city sectoral results are broadly similar after controlling for neighboring exposure, while some specifications show negative city-size responses to neighboring exposure. The slopes do not differ detectably in the conflict provinces, although excluding them reduces the sectoral coefficients and their precision. The conflict comparison covers Aceh, West and Central Kalimantan, and Central Sulawesi (Barter & Côté 2015; van Klinken 2007). Exposure does not detectably predict becoming a new district seat after 1999. The modern-employment result survives excluding education, health, social and government services and cities with new district seats (Appendix Table E14 and Appendix E.5). These checks do not cover processing of farm output, commodity booms or every government program.

tion villages, in the rural subdistricts with a settlement and in the cities, and the pre-1980 INPRES (Presidential Instruction) school allocation does not favor more exposed cities (Figure 19; Appendix Table E13 and Table E15; Appendix E.5). The village comparisons do not have comparable pre-settlement provision baselines.

The countryside around more productive settlements did gain grid access and car-passable roads after the baseline (Appendix Table E14). These responses can reflect both public supply and household demand for connections. Public investment remains a possible contributor to rural food production. The tests do not quantify its contribution to city employment.

Figure 19: Public Provision in Transmigration Villages and Roads Around the City

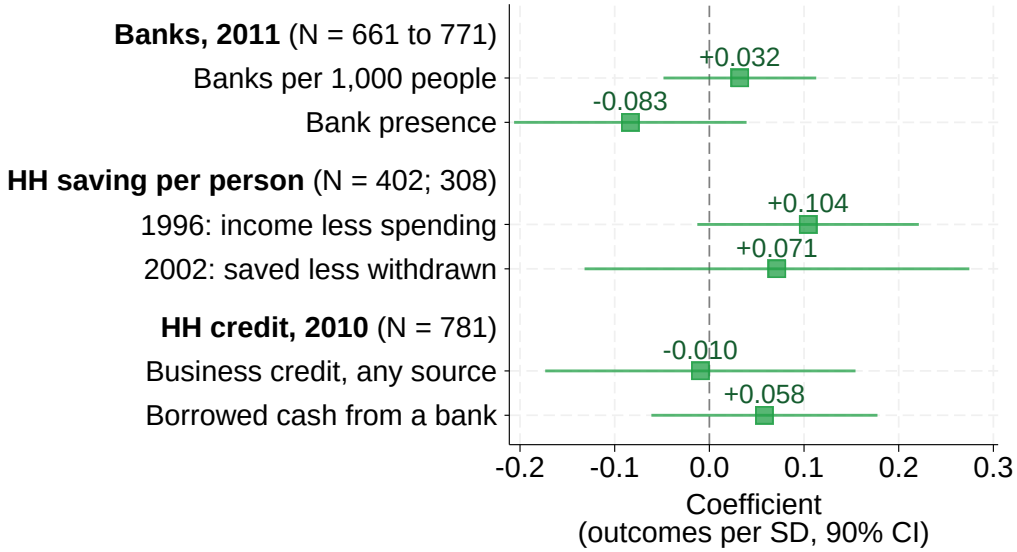



*Notes:* Markers give responses per standard deviation of agro-similarity, with outcomes standardized over each regression sample. Bars are 90% confidence intervals from Conley standard errors within 200 km. The labels print the signed coefficients to three decimals. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The transmigration-village rows relate 2011 provision to each village's own agro-similarity, standardized over the transmigration villages, with village geography controls. These comparisons do not include pre-settlement provision baselines. The household grid-connection share divides connected households by all measured households in the village. Grid reach and roads are population shares. Schools and health centers (puskesmas and puskesmas pembantu) are measured per 1,000 people. Each transmigration village combines its 2011 descendants. Their sample labels count villages. City-road rows use the main city sample, its exposure and full main controls: road density holds the 1950 density fixed, and road condition is the 2007 survey, controlling for its 1990 level (the regression's outcome is the 1990 to 2007 change). Their labels count cities. The car-passable-road response becomes imprecise with Bartlett weights or province clustering (Appendix E.5). The baseline-controlled subdistrict evidence remains in Appendix Table E15, where grid connections are conditional on villages reached by the grid. These outcomes do not measure all public investment, earlier provision changes, or service quality. Sources: village census (PODES), 2011; historical and current road maps; Indonesian Road Management System surveys, 1990 and 2007.

A capital-supply channel begins with greater saving in productive rural areas, which banks lend to firms elsewhere (Bustos et al. 2020). In rural subdistricts with a settlement, household saving and the changes in bank density and presence are imprecisely estimated (Figure 20). The bank-presence interval runs from −3.4 to +0.7 percentage points of the population per standard deviation, against a mean change of 2.0 points since 1980 across all rural subdistricts in that regression, so it excludes an expansion of bank presence larger than a third of the average change. Saving is measured as income less spending in 1996 and as money saved less withdrawn in 2002. In 2010, business credit from any source and cash borrowing from a bank are imprecisely estimated. The saving intervals allow increases, and

these household measures do not cover all deposits available to banks. The village censuses record bank presence but do not trace deposits or lending from rural savers to city firms. The evidence therefore leaves capital transmission possible.

Figure 20: The Capital Channel in Rural Subdistricts With a Settlement




*Notes:* Each marker is the coefficient on standardized agro-similarity for rural subdistricts with a settlement, from one regression per row over the rural subdistricts of the main sample: the outcome, standardized to mean 0 and SD 1 over the row's subdistricts, on agro-similarity times an indicator for a settlement in the subdistrict and times its complement, the indicator, the settler count (the settlers placed in the subdistrict's own transmigration villages where it has one and, where it has none, the settlers of the villages within 300 travel minutes weighted by inverse travel time), the four 1980 census baselines of the subdistrict and the 16 agro-climatic covariates. The 90% intervals are from Conley SE within 200 km. Banks: the 2011 village census, controlling for the outcome's 1980 level (the regression's outcome is the 1980 to 2011 change, standardized over the row's subdistricts). Bank presence is the population share living in a village with a bank, and it leaves out West Kalimantan and the West Sumatra subdistricts recorded as nagari in 2011. HH saving: monthly household (HH) income less expenditure in 1996, and net saving (money saved less savings withdrawn) in 2002, each in Rupiah per person per month as its inverse hyperbolic sine (Susenas 1996 and 2002). HH credit: the shares of survey households with business credit from any source and that borrowed cash from a bank (Susenas 2010). The sample sizes in the headings count all rural subdistricts in each regression, including those without a settlement. The plotted coefficients describe the subdistricts with a settlement. Sources: village censuses (PODES) of 1980 and 2011; Susenas 1996, 2002 and 2010.

A rural-demand mechanism predicts higher purchases of city goods by households in productive settlements. Total spending and its non-food share provide indirect measures of this demand. Their 2011 exposure responses are statistically insignificant for both the countryside with settlements and the whole hinterland, each aggregated by city (Appendix E.5). These late spending measurements leave earlier rural-demand responses unresolved.

The evidence supports a food-market explanation for urban adjustment, with recent migration and rural infrastructure as possible contributors. The evidence does not quantify the contribution of each mechanism. Section 6 examines trade and migration within a spatial model.

Table 6: Urban Responses by Access to Rice-Import Routes and Markets

| | (1) ln pop | (2) ln emp | (3) unemp rate | (4) **ln L agric** | (5) **agric %** | (6) **ln L modern** | (7) **modern %** |
|---|---|---|---|---|---|---|---|
| *Panel A: High openness = below-median route distance to the Thai and Vietnamese rice hubs* | | | | | | | |
| **Agro-similarity, more closed cities:** $\beta_1$ | **0.153*** | **0.109** | **0.018*** | **-0.156** | **-0.071***** | **0.307***** | **0.098***** |
| | (0.086) | (0.088) | (0.009) | (0.145) | (0.022) | (0.114) | (0.018) |
| Agro-similarity, more open cities: $\beta_1+\beta_2$ | 0.025 | -0.016 | 0.008 | -0.085 | -0.024 | 0.065 | 0.035** |
| | (0.075) | (0.074) | (0.008) | (0.110) | (0.019) | (0.063) | (0.015) |
| Difference, open − closed: $\beta_2$ | -0.128 | -0.125 | -0.009 | 0.071 | 0.047* | -0.242** | -0.062*** |
| | (0.088) | (0.097) | (0.011) | (0.152) | (0.026) | (0.109) | (0.020) |
| High openness (=1) | 0.099 | 0.032 | 0.015** | 0.230 | 0.039 | -0.003 | -0.029 |
| | (0.106) | (0.101) | (0.006) | (0.148) | (0.032) | (0.109) | (0.029) |
| *Panel B: High openness = above-median market access over realized rice-import volumes* | | | | | | | |
| **Agro-similarity, more closed cities:** $\beta_1$ | **0.109** | **0.073** | **0.014**** | **-0.223**** | **-0.078***** | **0.231**** | **0.083***** |
| | (0.085) | (0.085) | (0.007) | (0.105) | (0.018) | (0.092) | (0.016) |
| Agro-similarity, more open cities: $\beta_1+\beta_2$ | -0.002 | -0.022 | 0.003 | -0.105 | -0.025 | 0.086 | 0.049*** |
| | (0.068) | (0.057) | (0.007) | (0.084) | (0.020) | (0.077) | (0.018) |
| Difference, open − closed: $\beta_2$ | -0.111 | -0.096 | -0.011* | 0.118** | 0.053** | -0.145 | -0.034* |
| | (0.091) | (0.082) | (0.007) | (0.054) | (0.021) | (0.121) | (0.020) |
| High openness (=1) | 0.025 | 0.010 | -0.005 | 0.084 | 0.008 | -0.036 | -0.008 |
| | (0.074) | (0.067) | (0.011) | (0.082) | (0.026) | (0.096) | (0.023) |
| *Panel C: Two-stage least squares, the Panel B split instrumented by the Panel A split* | | | | | | | |
| **Agro-similarity, more closed cities:** $\beta_1$ | **0.143** | **0.129** | **0.013** | **-0.288*** | **-0.103***** | **0.379***** | **0.130***** |
| | (0.107) | (0.101) | (0.014) | (0.174) | (0.031) | (0.131) | (0.031) |
| Agro-similarity, more open cities: $\beta_1+\beta_2$ | -0.019 | -0.070 | 0.006 | -0.012 | 0.005 | -0.050 | 0.001 |
| | (0.120) | (0.125) | (0.013) | (0.150) | (0.026) | (0.129) | (0.025) |
| Difference, open − closed: $\beta_2$ | -0.161 | -0.200 | -0.007 | 0.276 | 0.108** | -0.429* | -0.129*** |
| | (0.175) | (0.187) | (0.020) | (0.252) | (0.049) | (0.221) | (0.050) |
| High openness (=1) | 0.224 | 0.083 | 0.032*** | 0.480 | 0.078 | 0.021 | -0.055 |
| | (0.257) | (0.233) | (0.012) | (0.374) | (0.069) | (0.251) | (0.066) |
| First-stage $F$ (Kleibergen-Paap) | 4.37 | 4.37 | 4.37 | 4.37 | 4.37 | 4.37 | 4.37 |
| *Panel D: High 1973 cargo = above-median 1973 total turnover of the city's coastal region (Rutz 1987, Map 2)* | | | | | | | |
| **Agro-similarity, more closed cities:** $\beta_1$ | **0.103** | **0.096** | **0.002** | **-0.099** | **-0.043**** | **0.226***** | **0.073***** |
| | (0.086) | (0.077) | (0.009) | (0.128) | (0.019) | (0.075) | (0.018) |
| Agro-similarity, more open cities: $\beta_1+\beta_2$ | -0.010 | -0.055* | 0.008 | -0.137* | -0.035* | 0.053 | 0.044*** |
| | (0.026) | (0.028) | (0.007) | (0.081) | (0.019) | (0.040) | (0.013) |
| Difference, open − closed: $\beta_2$ | -0.113* | -0.151** | 0.006 | -0.038 | 0.008 | -0.173*** | -0.028** |
| | (0.063) | (0.065) | (0.008) | (0.104) | (0.020) | (0.065) | (0.014) |
| High 1973 cargo (=1) | 0.050 | 0.070 | -0.009 | 0.123 | 0.042* | -0.035 | -0.052*** |
| | (0.103) | (0.097) | (0.012) | (0.131) | (0.025) | (0.083) | (0.019) |
| Observations, cities with a 1973 total | 133 | 133 | 133 | 133 | 133 | 133 | 133 |
| Outcome mean | 11.842 | 10.933 | 0.122 | 9.822 | 0.374 | 10.232 | 0.516 |
| Outcome SD | 0.854 | 0.840 | 0.040 | 0.896 | 0.183 | 0.975 | 0.156 |
| Observations | 134 | 134 | 134 | 134 | 134 | 134 | 134 |

*Notes:* Each column of each panel is one regression of a 2010 outcome of Table 4 on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200 km. Panels A and B interact agro-similarity with an indicator for high openness, as in Equation 6, split at the median over all 138 urban centers within the 300-minute window: route distance (Section 3.5) is the travel time from a city through its nearest port to the nearer of Bangkok and Ho Chi Minh City, and market access (Appendix B.2) sums the rice imports of every port of the list, discounted by the travel time from the city. Realized imports can respond to local food substitution, so Panel B is a sensitivity comparison. The openness result relies on Panel A, subject to the transport-source and timing qualifications in Section 3.5. Panel C is two-stage least squares on the two median splits: the high-openness indicator of market access and its interaction with agro-similarity are instrumented by the high-openness indicator of route proximity and its interaction, so its rows are those of Panels A and B. The Kleibergen-Paap first-stage $F$ under the Conley variance is 4.37, below the Stock-Yogo critical value of 7.03. The Anderson-Rubin test, which stays valid with weak instruments, rejects the hypothesis that both instrumented coefficients are zero at $p < 0.10$ in four of the seven columns. This joint test does not identify which coefficient differs from zero. Stars in Panel C refer to conventional coefficient tests and do not provide weak-instrument-robust inference for individual coefficients. Panel D splits the cities at the median of the 1973 interisland cargo turnover of their coastal region on the maps of Rutz (1987), a measure dated before the program that agro-similarity does not predict (0.011, standard error 0.149). The cities of one region share a value, so the two groups differ in size, 78 more closed and 55 more open cities with the region at the median in the more closed group, and the panel uses the 133 cities whose region has a 1973 turnover. SE in parentheses. The four sector columns are shaded. The beta terms are the coefficients of Equation 6. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

# 6 A Spatial Equilibrium Model With Trade and Migration

The empirical results show a substantial shift from agriculture to modern employment in cities exposed to more productive rural hinterlands, alongside a modest and imprecisely estimated population response. I use a spatial model to examine how agricultural productivity gains in rural areas can change employment within nearby cities. Households choose both location and sector, allowing the model to distinguish changes in a city's employment composition from changes in its population.

Cities need to feed their residents. They can obtain food from their rural hinterlands, buy from more distant suppliers, or farm within the urban boundary. Access to these sources changes how agricultural productivity affects structural transformation. In Matsuyama (1992), openness changes the relationship between agricultural productivity and industrialization. Glaeser (2014) shows why rural prosperity supports urbanization when outside food is unavailable. Here, the productivity gain occurs in rural locations supplying the city. The need for agriculture in urban areas is driven by a lack of sufficiently affordable food supply from the hinterland. Once the hinterland becomes more productive, the regions surrounded by more productive hinterlands can benefit from the greater availability of food and expand their modern sector, since the subsistence constraint is then satisfied. Affordable supplies from outside the hinterland reduce the city's dependence on its rural neighbors.

Matsuyama (1992), and before him Lewis (1954) and Harris & Todaro (1970), take a simplified view in which trade is either frictionless in the background or, as an exogenous condition, shapes the structural transformation response. In a large economy, the country-level averages of agricultural contraction and modern-sector expansion conceal considerable within-country variation in how the gains to agricultural productivity growth accrue to the regions that experience it, to the regions that benefit from it indirectly through trade, and to the workers who respond to it as their incomes adjust. To capture this spatial variation, I design a model in which households choose a location and a sector, every location has both agricultural and non-agricultural production, and the locations, with different sectoral compositions, trade with each other. The model allows me to decompose how an agricultural productivity change affects the region where it occurs and other regions, distinctly through trade and migration. I hold urban productivity fundamentals fixed to study this transmission through food markets. Food and traded non-agricultural output move across locations at a cost, while local services are produced where they are consumed.[33] In the counterfactuals I vary the levels of trade and migration: the economy can have high levels of trade, through lower trade costs, or higher or lower levels of migration, depending on the barriers, which correspond to the frictions in trade and in migration.

I use differences in rural agro-climatic similarity to study how agricultural productivity affects employment in nearby cities. That focus connects the city-hinterland relationship in Nagy (2023), who studies city formation and growth, to the spatial structural change analyzed by Eckert & Peters (2025), where technology diffusion shapes regional adjustment. The distinction between goods-market and

[33] The trade and choice structure draws on Allen & Arkolakis (2014); Redding (2016); Khanna et al. (2025), and food demand is non-homothetic (Kongsamut et al. 2001; Gollin et al. 2007). Local services connect household spending to urban employment, while traded non-agricultural production responds to demand across locations.

labor-market adjustment also follows Tombe & Zhu (2019), who quantify the aggregate productivity effects of trade and migration frictions.

### 6.1 Setup and Equilibrium

The model focuses on the reallocation of workers between agriculture and non-agriculture in an economy with $J$ locations. On the consumption and production side, non-agricultural output is split between non-traded local services, $S$, and traded goods and services, $M$, alongside food, $A$. There are therefore three goods and two sectors for worker choice. Food costs affect how much households can spend on local services, while traded production connects urban labor demand to markets elsewhere.[34]

A fixed mass $\bar{L}$ of workers chooses where to live and whether to work in agriculture or non-agriculture. Within each location, workers move freely between local and traded non-agricultural activities, which pay the same wage and form a common labor market. At location $j$, agricultural employment is $L_{jA}$ and total non-agricultural employment is $L_{jS}$, so $N_j = L_{jA} + L_{jS}$ and $\sum_j N_j = \bar{L}$. The model abstracts from unemployment and natural population growth. Changes in a location's population therefore represent net migration.

In the data, modern employment combines manufacturing and services, and this total corresponds to $L_{jS}$. The model's distinction between local and traded output concerns where goods and services are consumed and does not coincide with the census distinction between manufacturing and services. The employment comparison is between agriculture and the modern sector. The two non-agricultural goods allow local spending and demand from other locations to affect labor demand differently within that common labor market.

**Production and land.** Agriculture and local services use labor and land. Traded non-agricultural production uses labor with constant returns:

$$\begin{aligned} Y_{jA} &= (A_{jA}T_{jA})^{1-\alpha}L_{jA}^{\alpha}, \\ Y_{jS} &= \bar{A}_{jS}N_j^{\eta}T_{jH}^{1-\gamma_S}(L_{jS}^{\text{local}})^{\gamma_S}, \\ Y_{jM} &= A_{jM}L_{jM}, \qquad L_{jS} = L_{jS}^{\text{local}} + L_{jM}. \end{aligned} \tag{7}$$

The parameter $A_{jA}$ multiplies agricultural land in production, while $\bar{A}_{jS}$ and $A_{jM}$ govern local and traded non-agricultural productivity. The main specification sets the agglomeration elasticity $\eta$ to zero. Local and traded non-agricultural activities pay the common wage $w_{jS}$, and labor is allocated between them according to their demands. The fixed land endowment $\bar{T}_j$ is divided between agriculture and local services, $T_{jA} + T_{jH} = \bar{T}_j$. Land rents are equalized across active uses and distributed equally among local residents as a per-person rebate $R_j$. Land use and employment composition can therefore adjust with population held fixed. Appendix F.1 gives the factor-price and land-market conditions.

[34] The traded category $M$ includes manufacturing and services sold across locations. Consumption travel provides one reason for treating services as tradable across locations. Blanchard et al. (2025) show how rural residents purchase urban services through consumption trips to cities.

**Preferences and expenditure.** Households consume food $c_A$, local services $c_S$, and traded non-agricultural output $c_M$. Stone-Geary preferences introduce a subsistence food requirement:

$$U(c_A, c_S, c_M) = (c_A - \bar{a})^{\psi_A} c_S^{\psi_S} c_M^{\psi_M}, \quad \psi_A + \psi_S + \psi_M = 1, \quad \bar{a} > 0. \tag{8}$$

The subsistence requirement makes the food expenditure share fall with income.[35] Let $y_j = W_j + R_j$ denote expenditure resources, where $W_j$ is the sector-inclusive wage index defined below. For $y_j > p_{jA}\bar{a}$, households pay the subsistence bill and allocate the remainder across the goods in fixed shares. The food expenditure share is

$$\frac{p_{jA}c_A}{y_j} = \psi_A + (1 - \psi_A)\frac{p_{jA}\bar{a}}{y_j}. \tag{9}$$

Holding expenditure resources fixed, cheaper food reduces the subsistence bill and raises spending on both non-agricultural goods. Demand for local services increases within the city. Demand for the traded good can be met by producers elsewhere. Appendix F.2 derives the expenditure equations.

**Sector and location choice.** Workers differ in their returns to sectors and locations. Fréchet heterogeneity gives the agricultural employment share

$$\pi_{A|j} = \frac{w_{jA}^{\theta_S}}{w_{jA}^{\theta_S} + w_{jS}^{\theta_S}}, \qquad L_{jA} = N_j \pi_{A|j}, \qquad L_{jS} = N_j(1 - \pi_{A|j}). \tag{10}$$

The sector-choice elasticity $\theta_S$ governs the response of employment composition to relative wages. A larger $\theta_S$ makes the ratio of agricultural to non-agricultural employment more responsive to relative wages. The wage index is $W_j = (w_{jA}^{\theta_S} + w_{jS}^{\theta_S})^{1/\theta_S}$. Location choice also reflects amenities $\Phi_L^j$, the subsistence food bill, and the consumption price index $P_j = p_{jA}^{\psi_A} p_{jS}^{\psi_S} p_{jM}^{\psi_M}$:

$$V_j = \Phi_L^j \frac{W_j + R_j - p_{jA}\bar{a}}{P_j}, \qquad \pi_j = \frac{V_j^{\theta_L}}{\sum_k V_k^{\theta_L}}, \qquad N_j = \bar{L}\pi_j. \tag{11}$$

The location value $V_j$ combines amenities with the purchasing power of resources remaining after subsistence food needs are met. The location-choice elasticity $\theta_L$ governs population adjustment to differences in these values: a larger $\theta_L$ makes location shares more responsive to a given change in relative attractiveness. Moving costs modify these choices in the counterfactuals. The model abstracts from the liquidity constraints on migration discussed in Section 4.1.

**Trade and equilibrium.** Food and traded non-agricultural varieties are differentiated by origin, with substitution elasticity $\sigma > 1$. Trade costs increase with the travel-time measure introduced in Section 3.3. Own-location trade costs equal one. The food supply-price index is inverse agricultural marginal labor productivity, $q_{kA} = 1/(\partial Y_{kA}/\partial L_{kA})$. The traded non-agricultural good has unit labor

[35] See Matsuyama (1992) and Kongsamut et al. (2001).

cost $q_{kM} = w_{kS}/A_{kM}$. Destination prices are

$$p_{jg} = \left[\sum_k (\tau_{kjg} q_{kg})^{1-\sigma}\right]^{1/(1-\sigma)}, \qquad g \in \{A, M\}. \tag{12}$$

Local-service prices equate local expenditure with the value of local output. Together with sector wages, land rents and worker choices, these prices determine the allocation across locations and activities. The model does not impose goods-market clearing by location. Appendix F.3 states the complete system.

**Transmission to urban employment.** A rural productivity gain lowers the origin's food-supply index at given inputs. Its effect on the city's food price is larger when the city buys more of its food from that origin. Let $\lambda^A_{vj}$ denote the share of city $j$'s food expenditure on origin $v$. Holding other origins' supply indices fixed,

$$\frac{\partial \ln p_{jA}}{\partial \ln A_{vA}} = \lambda^A_{vj} \frac{\partial \ln q_{vA}}{\partial \ln A_{vA}} < 0. \tag{13}$$

Cheaper food raises households' purchasing power and reduces urban farmers' returns at given inputs. Local-service demand, traded sales, land use and wages then adjust. The sector-choice equation implies

$$\frac{d\pi_{A|j}}{d \ln A_{vA}} = \theta_S \pi_{A|j}(1 - \pi_{A|j}) \frac{d \ln(w_{jA}/w_{jS})}{d \ln A_{vA}}. \tag{14}$$

With population fixed, agricultural employment falls when the agricultural wage falls relative to the non-agricultural wage. Population also responds to changes in the attractiveness of cities and rural locations. The counterfactuals quantify these joint responses.

## 6.2 Taking the Model to the Data

**Geography and inputs.** The model economy consists of 342 locations: the 134 cities in the main empirical sample, 42 additional cities outside the settlement exposure window, and 166 rural district remainders. Rural locations include both areas with transmigration villages and areas without them.[36] The additional cities participate in trade and migration but do not enter the exposure regressions. All reported city responses use the main empirical sample. The displayed food-price comparison uses its 112 cities with an observed 1996 price.

Census population and employment, Susenas consumption and wages, village-census crop area and infrastructure, and satellite land measures determine the location inputs. Crop area and built-up land determine land endowments. Agricultural employment relative to crop area informs agricultural productivity. Non-agricultural employment and land inform local-service productivity. Traded productivity matches census employment valued at reference wages, scaled to aggregate traded-good expendi-

[36] A rural district remainder combines the subdistricts outside the model's cities within that district. Of the 166 rural locations, 77 contain a transmigration village and 89 do not. Rural aggregation limits the size of the bilateral trade system and the burden of repeated equilibrium calculations. An earlier implementation with 1,465 rural subdistricts did not produce a reliable calibration.

ture. Amenities reflect density, built-up land and infrastructure. Appendix F.4 gives the constructions and parameter sources.

Table 7 lists the parameters set outside the model, with their sources (Panel A), and the two choice elasticities estimated in it, with the slopes they are fit to (Panel B).

**Calibration and model fit.** I calibrate the location-choice and sector-choice elasticities to match the estimated effects of rural agro-similarity on city population and the agricultural employment share. The population response is informative about movement across locations, and the employment-share response is informative about reallocation within them. Wages and prices connect the two choices, so I select the elasticities jointly. For each calibration, I apply the same rural productivity increase and estimate the empirical specification on the model's city responses, using the same controls and standardized units. The comparison asks how simulated population and agricultural-share changes vary with exposure, conditional on the observed controls. The remaining outcomes assess implications that were not used to choose the elasticities.

The model fits these population and agricultural-share responses most closely at $(\theta_L, \theta_S) = (1.0, 3.8)$.[37] Several calibrations are consistent with the empirical confidence intervals, leaving substantial uncertainty about the elasticities. I report counterfactuals at this pair and at $(0.45, 1.6)$ to show their sensitivity to location and sectoral responsiveness.

At the best-fitting calibration, the population coefficient is $+0.147$ compared with $+0.060$ in the data, and the agricultural-share coefficient is $-0.246$ compared with $-0.278$. Both lie inside the empirical confidence intervals. All 4 untargeted responses also lie inside their intervals, although the food-price estimate is imprecise. These comparisons assess the pattern of responses across cities. Standardized fit does not establish the size of the response in original outcome units.

### 6.3 Counterfactuals

I cannot randomize cities into some trading more and some receiving more migrants. The 30-year changes in the data are a general equilibrium adjustment, with prices and wages, population and trade volumes all adjusting in the long run. I compare migration with costly trade to trade with costly migration. At each cost setting, I first solve for wages, prices, population and employment before the productivity increase. I then raise rural productivity, keep those trade and moving costs fixed, and solve for the new allocation. I calculate each city's response from its change between these two allocations. Each comparison therefore allows the initial allocation to adjust to its trade and moving costs.

[37] Fit is measured by the squared differences between the model and empirical coefficients, weighted by the inverse of their estimated variances. This pair provides the best fit among the 17 completed calibrations satisfying $\theta_L < \theta_S$. Several nearby parameter combinations could not be solved within the calculation limit. The available comparisons therefore do not establish a global optimum. I use the objective as a descriptive fit distance. The objective and the parameter comparison are reported in Appendix F.4.

Table 7: Calibrated Parameters and Their Sources

*Panel A. Set outside the model*

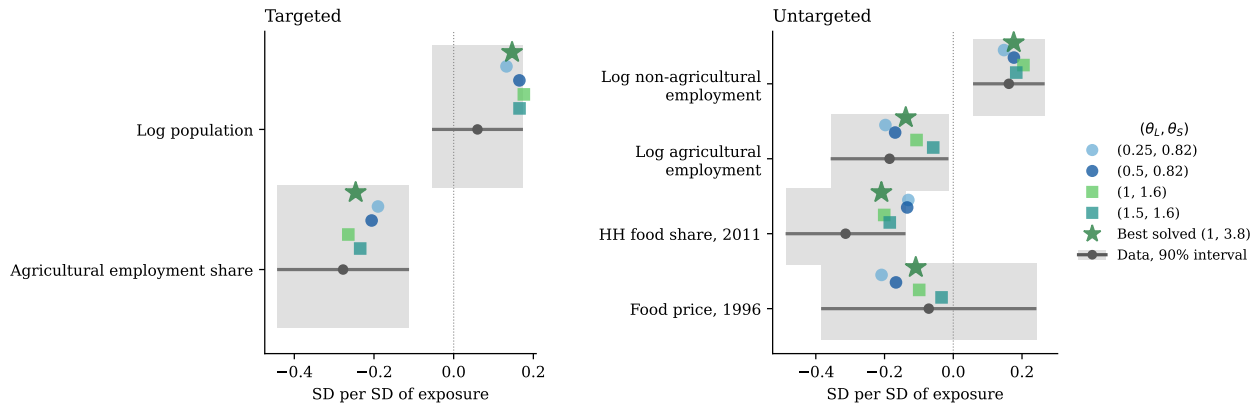
| Symbol | Parameter | Value | Origin | Source or target |
|---|---|---|---|---|
| $\sigma$ | Substitution across origins | 4 | Lit. | Broda & Weinstein (2006): median 2.9, 5th to 95th percentile 2.1 to 5.2 |
| $\delta_A$ | Food trade-cost elasticity | 0.5 | Lit. | Tombe & Zhu (2015): 0.51 on log distance |
| $\delta_M$ | Traded-output trade-cost elasticity | 1.0 | **Data**† | IFLS service-trip gravity: 1.091 (0.033); outer islands 0.948, 90% interval 0.863 to 1.034; published band 0.33 to 1.0 |
| $-(\sigma-1)\delta_A$, $-(\sigma-1)\delta_M$ | Distance elasticity of trade flows: food, traded output | $-1.5$, $-3$ | Implied | Head & Mayer (2014): median $-0.89$ over 1,835 estimates, on km |
| $\theta_L\nu$ | Moving-cost elasticity | None | None | No moving cost at the baseline |
| $\alpha$ | Agricultural labor share | 0.5 | Lit. | Fuglie (2010): 0.46 |
| $\gamma_S$ | Local-services labor share | 0.7 | Set | Between the US labor shares of value added, 0.66 in services and 0.71 in manufacturing |
| $\psi_A$ | Food preference weight | 0.550 | **Data** | Susenas 2011 purchased-food share |
| $\psi_M$ | Traded-output preference weight | 0.082 | **Data** | Susenas 2011 purchased-services items |
| $\psi_S$ | Local-services weight | 0.368 | **Data** | $1-\psi_A-\psi_M$ |
| $\bar{a}$ | Subsistence food | 0.108 | **Data** | BPS food poverty line, 2000, over the reference wage |
| | Location fundamentals: $A_{A,j}$, $A_{M,j}$, $A_{S,j}$, $T_j$, $\Phi_{L,j}$, $d_{kj}$ | | **Data** | 2010 census, 2003 crop area, built-up area, 1980 amenity index, travel times; 342 locations |
| $\kappa$ | Amenity scale | 0.5 | Set | Held fixed |
| $\eta$ | Agglomeration | 0 | Set | Switched off |
| $\Delta \ln A_A$ | Agricultural productivity gain at hosts | 0.05 | Norm. | The program's own gain is not measured |

*Panel B. Estimated in the model*

| Symbol | Parameter | Value | Origin | Source or target |
|---|---|---|---|---|
| $\theta_L$ | Location choice elasticity | 1.0 | Fit | Log population slope: data +0.060 (0.069), model +0.147; published 1.0 to 2.5 |
| $\theta_S$ | Sector choice elasticity | 3.8 | Fit | Agricultural employment share slope: data $-0.278$ (0.101), model $-0.246$; published 1.6 to 3.1 |
| | Distance $J$ on the two slopes | 1.66 | | Lowest of the 17 completed pairs satisfying $\theta_L < \theta_S$; descriptive fit distance |

*Notes: Panel A.* The model has 342 locations, the cities and the rural remainders of their districts. Origin: Lit., published estimate; Data, measured input (shaded rows, label in bold); Set, chosen parameter; Norm., normalization; Implied, calculated from other parameters; None, moving costs absent at baseline. † A data value within the published range shown. IFLS outpatient trips proxy for traded-output costs. Poisson pseudo-maximum likelihood with origin and destination fixed effects gives $\delta_M = -\beta/(\sigma-1)$; the chosen value is within the reported outer-island interval. The distance elasticity of trade flows, $-(\sigma-1)$ times the decay, is the slope of log flows on log distance; distance in the model is travel time, and Head and Mayer's estimates are on kilometers. Agricultural productivity uses 2010 employment per unit of crop land; non-agricultural productivities use 2010 employment. Land combines 2003 crop area and 1980 built-up area. Amenities use 1980 satellite measures and the 1983 village census, scaled by $\kappa$. The food weight uses purchased spending; the model's total food share adds subsistence spending. Purchased-service items inform the traded-output weight; its employment basket is broader. *Panel B.* The elasticities jointly minimize $J$, the sum of squared model-data coefficient differences in empirical standard-error units. Coefficients use standardized outcomes and exposure, the main controls, 134 cities and Conley standard errors within 200 km. Among 17 completed pairs with $\theta_L < \theta_S$, $J$ ranges from 1.66 to 2.83; the best pair is at the boundary of completed calculations. $J$ is descriptive and does not give a model test or parameter confidence set. Published ranges: Khanna et al. (2025); Eckert & Peters (2025); Zárate (2022); Takeda (2022). The values tried in the sensitivity runs are in Table F2.

Figure 21: Model Fit: Targeted and Untargeted Responses



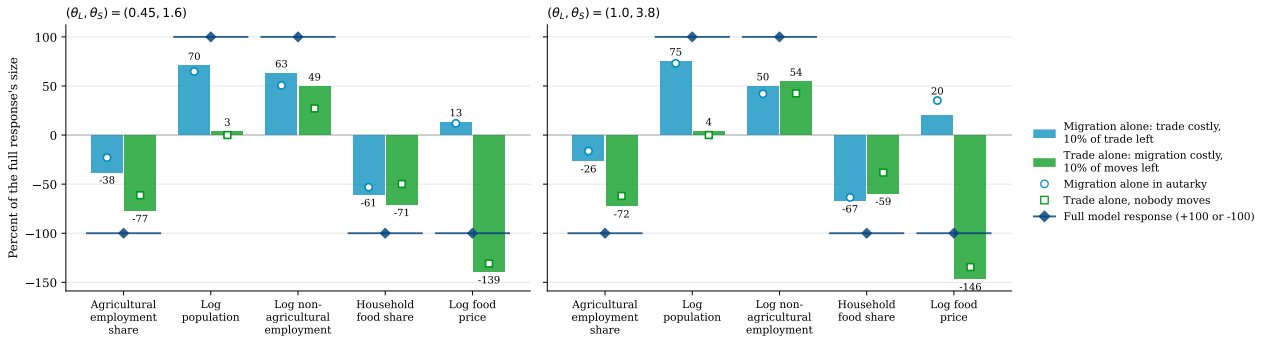

*Notes:* Points compare empirical exposure coefficients with model responses to a 0.05 increase in log agricultural productivity at rural settlement hosts. Outcomes and exposure are standardized. Regressions use the main empirical controls. The shaded bands and whiskers show empirical 90% confidence intervals with spatial standard errors at a 200-kilometer cutoff. Log population and the agricultural employment share are targeted. Log non-agricultural employment, log agricultural employment, the 2011 food expenditure share, and the 1996 raw-food price index are untargeted. Quantity and expenditure-share responses use 134 cities. The price comparison uses the 112 cities with a 1996 price. The empirical price outcome is $\ln(\text{price} + 1)$, while its model equivalent is the standardized log food-price change. Circles and squares are four other completed calibrations. The star indicates the best-fitting completed calibration. The model points do not have estimated confidence intervals.

The experiment raises $A_{jA}$, which augments land in agricultural production, by 0.05 log units in rural district remainders containing transmigration villages. The increase applies to the whole host remainder, holding urban productivity and other fundamentals fixed. I report the responses of the 134 cities in the main empirical sample. For each urban outcome, I estimate the exposure coefficient using the empirical specification's controls and exposure scale. The full model's urban coefficient, with both channels open, is the reference for each restricted comparison. The figure divides each restricted urban coefficient by the absolute size of the full urban coefficient, retaining the direction of the response. Its notes report the full coefficients and their units.

Consider first a world where trade is costly, so that migration remains available while trade is restricted. I multiply all between-location food and traded-output costs until the economy-wide share of food expenditure on purchases from other locations falls to 10% of its reference value. Prices remain free to adjust. In the trade comparison, moving costs rise until shock-induced population reallocation is 10% of its reference value. This reallocation is half the sum of absolute population changes across all model locations. Dividing by two counts each net population transfer once, since a loss at an origin is matched by a gain at a destination. The measure summarizes changes in where people live and does not count offsetting migration flows. The common remaining fraction compares food purchases from other locations with net population redistribution. It does not equate the economic costs of restricting trade and migration. Both channels remain open. Autarky and fixed population are reference cases.

At $(0.45, 1.6)$, migration with costly trade reproduces 38% of the full urban agricultural-share exposure response. At $(1.0, 3.8)$, it reproduces 26%. With costly migration, trade reproduces 77% and 72%, respectively. Trade accounts for more of the agricultural-share exposure response at these

Figure 22: Urban Responses With Trade or Migration Made Costly




*Notes:* Responses of the 134 reporting cities to a 0.05 increase in the log land-augmenting agricultural productivity parameter at rural settlement hosts. Each bar is an urban exposure coefficient from the main specification, divided by the absolute value of the full-model urban coefficient and multiplied by 100. Migration alone raises all between-location trade costs until the economy-wide share of food expenditure on purchases from other locations is 10% of its reference value. Trade alone raises moving costs until half the sum of absolute shock-induced population changes across model locations is 10% of its reference value. The unshocked economy is solved separately at each cost. Open markers show autarky and fixed population. Outcome units and exposure standardization are common across comparisons. The elasticity pairs are $(0.45, 1.6)$ and $(1.0, 3.8)$. At the latter pair, the full agricultural-share coefficient is $-0.0058$ percentage points per standard deviation of exposure. The population and non-agricultural employment coefficients are $+0.0047$ and $+0.0168$, in 100 times log changes per standard deviation. These conditional comparisons do not report sampling uncertainty.

costs and elasticities. For non-agricultural employment the restricted responses are closer: migration reproduces 63% and 50%, while trade reproduces 49% and 54%. The classical dual-sector models of Lewis (1954) and Harris & Todaro (1970) take trade between sectors as frictionless in the background, with adjustment through labor movement. In this setting the two channels act together, and without either of them the sectoral reallocation within cities would have been smaller.

In a separate comparison, food and traded-output prices and traded-output employment are fixed at their unshocked levels while population adjusts. The corresponding trade comparison fixes population while allowing prices to adjust. With the different parameter estimates in closely fit models, migration and trade separately produce similar magnitudes, each substantially smaller than the full model response. Under these restrictions, migration produces a larger agricultural-share decline at 8 of the 17 elasticity pairs (Online Supplement) and in 2 of the 11 cost settings of Appendix F.6. The tighter the migration barrier, the more of the response trade reproduces. The IFLS moving-cost settings lower the migration-only share and raise the trade-only share of the urban agricultural-share response relative to the reference cost setting. The cost of trade is plausibly lower than the cost of migration. The percentages compare restricted economies and can overlap because changing trade also changes the wages and prices that govern migration, while migration changes local demand and production. They need not sum to 100%.

The exposure coefficient describes differences across cities. Over this period the cities grow overall, so the decline is relative to a new, larger urban baseline across cities. Cities surrounded by less productive villages have relatively more farmers. A productivity gain can therefore reduce population in every city and still produce a positive exposure slope if more exposed cities lose fewer residents, and the pooled urban change, which also includes responses common to cities and changes in their employment

weights, can rank the channels differently. The historical benchmark below uses observed urban change and is a separate empirical calculation.

**Historical magnitudes.** Between 1980 and 2010, the aggregate agricultural employment share fell from about 48% to 30% in the 134 main-sample cities, while agricultural employment rose by about 1.4 million workers.[38]

Regressing the change in the agricultural share on exposure with the main specification's controls gives about −5.1 percentage points per standard deviation of exposure, the same size as the regression on the 2010 level of the share in the main results table, and the 90% intervals of the two estimates overlap (Table 4). A lower-match comparison illustrates its scale. Suppose each city's exposure above the full main sample's tenth percentile is reduced to that reference, with lower exposures held fixed. Applying the estimated slope and holding observed 2010 employment weights fixed gives a difference of about −5.5 percentage points between the observed and lower-match profiles, about 30% of the observed aggregate decline, and its 90% interval excludes zero (Table 8). This reference is an observed exposure level; it has not been established as an absence of match-related productivity gains. Shared village catchments also mean this city-level comparison need not correspond to a feasible reassignment of settlers.

Historical attribution requires the match-related productivity component at both endpoints and its effect after prices, wages, and employment weights adjust. A persistent matching advantage can have different urban effects as other economic conditions change. Cross-city exposure differences do not identify an effect common to all cities. The magnitude comparison therefore does not identify the share of the historical decline caused by agricultural similarity. A model calculation of that share requires productivity mappings and the corresponding counterfactual at both endpoints, with settlement placements and initial settler counts held fixed.

Table 8 compares the observed historical decline with the lower-match exposure benchmark.

[38] The denominator includes all workers with a reported industry, and the aggregate share divides total agricultural employment by total employment. Counts follow the main analysis's fixed city membership, historical crosswalk, and allocation weights. Some later-created member codes lack a separate historical observation because their predecessor population is assigned to another member. A sensitivity requiring every member code to be observed retains 121 cities and gives a change of −18.68 percentage points; label completeness does not establish more accurate historical boundaries.

Table 8: Observed Structural Change and the Lower-Match Benchmark

| | Observed Change | Lower-Match Benchmark |
|---|---|---|
| Comparison | 2010 minus 1980 | Observed minus lower-match exposure |
| Share change or difference in changes (percentage points) | −18.35 | −5.54 |
| 90% coefficient-based interval | — | [−7.44, −3.64] |
| Ratio to the observed decline | Reference denominator | 30.20% |
| Employment weights | Observed at each date | Observed 2010 weights held fixed |

*Notes:* Both columns describe the 134 main-sample cities. The observed share divides total agricultural employment by total industry-reported employment. The benchmark lowers exposure above the full main sample's tenth percentile to that reference and preserves exposure below it. It is the predicted difference in share changes, observed exposure minus the lower-match profile. The ratio divides this difference by the signed observed decline. The displayed interval for the difference in share changes propagates the change regression's Conley uncertainty conditional on the reference, exposure profile, employment weights, and observed denominator. The dash indicates a descriptive historical change without a reported coefficient-based interval. This magnitude comparison does not identify the historical contribution of agricultural similarity or a feasible reassignment of settlers. Source: the saved census and exposure calculations underlying this subsection.

# 7 Conclusion

Cities surrounded by more productive villages shift employment from agriculture to manufacturing and services, with a positive but imprecisely estimated population response. I use Indonesia's Transmigration Program to study how productivity differences across transmigration villages affect nearby cities. A causal interpretation requires city exposure to be conditionally exogenous. The policy targeted only villages. The cities around them restructured their employment.

Cheaper food can enable urban households to satisfy the subsistence constraint. They have more resources available to spend on non-agricultural items. That can create demand in that sector and create jobs in cities. Cheaper food can also reduce the return to farming within the city. I do not observe trade flows specifically, so the evidence on trade is suggestive. Urban households in more exposed cities consume more food while devoting a smaller share of their budgets to it. The modern-sector response is larger in cities weakly connected to outside food markets, where raw-food price estimates suggest a decline. Recent migration also rises with exposure, and the data do not separate its contribution from other changes in urban employment composition. In the model, the role of these channels varies with trade and migration costs.

I show that within the country, differential rural productivity can drive different urban outcomes. As governments are evaluating different policies that often target rural areas, they might underestimate the effects that these policies can have on urban areas, intentional or unintentional.

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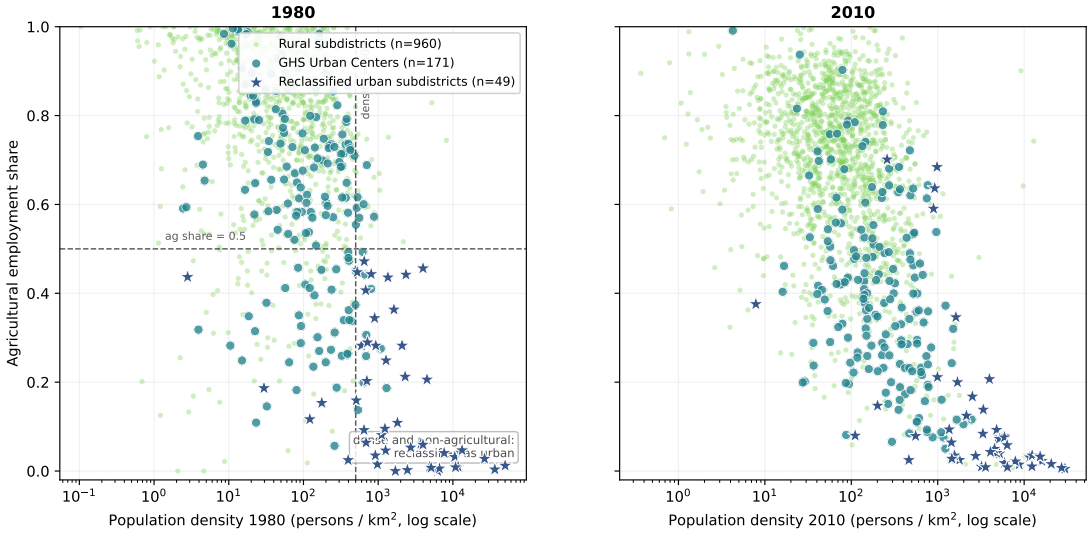
# A Data and Sample

This part documents the sources behind every number in the paper and the conditions for inclusion in the main city sample. The Online Supplement adds further detail on data sources, variable construction and validation beyond this appendix.

## A.1 Urban Sample Construction

The decline in the agricultural employment share was widespread across the city sample (Figure A2). Most cities lie below the diagonal, indicating a lower agricultural employment share in 2010 than in 1980.

Figure A1: Reclassification Diagnostics: Density and Agricultural Employment Share




*Notes:* Each point is a place, positioned by population density (horizontal axis, log scale) and agricultural employment share (vertical axis) in 1980 (left) and 2010 (right), the two characteristics used to classify subdistricts as urban. Small circles are rural subdistricts, larger circles the satellite-defined urban centers, and stars the reclassified urban subdistricts. The legend reports the places with valid density and agricultural-share measurements in 1980. Coverage differs in 2010. The satellite-center count excludes centers without both 1980 measurements, and the reclassified-subdistrict count includes the 42 reattached edges and the subdistricts forming standalone additions. The dashed lines on the left mark the rule's two criteria in 1980, density above 500 persons per square kilometer and an agricultural employment share below 0.5, and the stars cluster in the dense, non-agricultural quadrant.

**Urban-sample definition.** The urban sample combines Urban Center overlap with the pre-treatment administrative reclassification described below. A subdistrict enters the urban sample if it overlaps a GHS Urban Center polygon; overlapping subdistricts are aggregated into one analysis unit per urban center; and the administrative reclassification described in Section 2.2.1 reattaches edge subdistricts and adds urban centers, giving the 187-center sample, every center of which has a travel time to a transmigration village. The Urban Center layer itself follows one global rule: on a one-kilometer population and built-up grid, contiguous high-density cells (at least 1,500 inhabitants per square kilometer) forming clusters of at least 50,000 people are classified as urban centers, and the polygon layer collects these clusters (Dijkstra et al. 2021), so a single satellite-based rule identifies cities consistently across the world.

**Aggregation to the urban center.** To identify what an urban center is, I combine data at three levels. The outcome data are available at the village and urban-neighborhood level (administrative level 4), the finest spatial units available; these make up the subdistricts (administrative level 3) that the 1980 baseline data cover. Because the specification controls for 1980 levels, the baseline and endline units have to match, so the finest level the data afford is

the level-3 subdistrict. The household surveys are at most available at the subdistrict level as well. Multiple contiguous subdistricts overlap with the same urban area, and because the analysis studies the city as one labor market, with exposure based on travel time around the city, there is little meaningful variation across subdistricts within a city. I therefore merge contiguous subdistricts into one urban-center area, the unit that corresponds to the meaningful urban labor market. Five subdistricts each contain two urban centers. I split each between the two in proportion to the employment in the villages that each center's built-up area covers, assigning a village inside neither to the nearer center, so every worker is counted once (Appendix E.4).

**The 2010 totals each city is built from.** A city's 2010 population and employment are the sums of its member year-2000 subdistricts, and those subdistrict totals rely on a village-level link. Under the boundary overlay, 203 outer-island subdistricts recorded a 2010 population under half or over two and a half times their year-2000 population, against a median ratio of 1.15; under the administrative link that falls to 54, of which 38 are in Aceh, in northern Sumatra, where the year-2000 census itself undercounts.[39]

**From the satellite layer to the city sample.** The GHS layer has 182 Urban Center polygons on the Outer Islands, and one of them leaves before the analysis, a census twin: two polygons fall on a single subdistrict in the census geography, so they are treated as one unit throughout. That leaves 181 unique satellite-defined centers. The administrative reclassification adds six standalone units through the two rules described below, giving 187 urban centers. It also merges 42 edge subdistricts, drawn from 25 districts, into 22 existing urban centers. Figure A1 plots agricultural employment shares against population density and shows the 1980 classification cutoffs. The sample consists of all 187 centers.

**Administrative additions and reattachments.** The first rule requires 1980 population density above 500 persons per square kilometer and an agricultural employment share below one half. It adds three dense administrative centers that fall below the satellite population threshold and reattaches dense edge subdistricts to existing centers. The second rule applies to the remaining subdistricts of each district, after excluding satellite-center members and subdistricts already included through the first rule. A remainder qualifies when its combined 1980 urban population share exceeds one half. Under this rule, three enter as their own units (Palangka Raya on Kalimantan, Sawahlunto on Sumatra, and Batam off Sumatra), two reattach to their city's existing urban center (Palu on Sulawesi and Samarinda on Kalimantan), and one is dropped because its kota is fully urban, so the additional urban center would duplicate the city itself (Pangkalpinang).

**Eligibility for the main city sample.** Two restrictions then define the main city sample. The 300-minute travel-time window leaves 138 centers, of which 134 have a 1980 baseline the census recorded directly, which is the main city sample (Table A2). A center enters only if all three required 1980 baselines are observed, log population, log employment, and the urban-population share, and a partial baseline does not qualify it.

**Baseline controls.** The main specification conditions on a fourth 1980 baseline, log agricultural employment, as a control; for two in-sample centers I compute it from the measured 1980 agricultural employment level. The same four baselines enter every column. The own-outcome checks add the 1980 value of modern employment or its share for the corresponding outcome (Table 5 and Figure D3). For unemployment, the heatmap adds the 1980 agricultural share and log population density.

**Centers excluded for missing baselines.** Four in-window centers are excluded, and for two distinct reasons. Three, in Bengkulu on Sumatra, West Kalimantan and Central Kalimantan, have no 1980 census record at all, and their

[39] The 2000 census enumerated only 40.3% of Aceh's villages, so a 2000-to-2010 comparison for Aceh cannot be repaired by any subdistrict link. The paper uses the year-2000 census as Bazzi et al. (2016) do, for the transmigration villages: the village agricultural employment share of Figure 16 and the settler stay-share of Figure 14. Its subdistrict map and its village list are the geography every city total is built on, and its counts serve here only as a check on that link. Every city-level quantity taken from the population census comes from its 1980 and 2010 rounds.

Table A1: Main Estimates Under Alternative Samples and Specifications

| | Population + Employment | | | Agric Empl | | Modern Empl | |
|---|---|---|---|---|---|---|---|
| | (1) ln pop | (2) ln emp | (3) unemp rate | (4) ln L agric | (5) agric % | (6) ln L modern | (7) modern % |
| *Main specification, main city sample* | | | | | | | |
| Agro-similarity | 0.051 | 0.024 | 0.008 | -0.166* | -0.051*** | 0.158** | 0.066*** |
| | (0.059) | (0.057) | (0.007) | (0.093) | (0.018) | (0.062) | (0.015) |
| Observations | 134 | 134 | 134 | 134 | 134 | 134 | 134 |
| **Part I: Samples** | | | | | | | |
| *Panel A: Excluding every area the reclassification created or enlarged* | | | | | | | |
| Agro-similarity | 0.068 | 0.036 | 0.012 | -0.152 | -0.059*** | 0.179** | 0.067*** |
| | (0.071) | (0.067) | (0.008) | (0.093) | (0.018) | (0.076) | (0.018) |
| Observations | 114 | 114 | 114 | 114 | 114 | 114 | 114 |
| *Panel B: Panel A further restricted to at least 25 residents per km$^2$ in 1980* | | | | | | | |
| Agro-similarity | -0.001 | -0.054 | 0.017** | -0.332*** | -0.080*** | 0.122* | 0.087*** |
| | (0.047) | (0.049) | (0.007) | (0.090) | (0.021) | (0.063) | (0.020) |
| Observations | 90 | 90 | 90 | 90 | 90 | 90 | 90 |
| *Panel C: Panel A further restricted to at most 80% agricultural employment in 1980* | | | | | | | |
| Agro-similarity | 0.059 | 0.040 | 0.008 | -0.139 | -0.055*** | 0.159** | 0.057*** |
| | (0.079) | (0.075) | (0.007) | (0.098) | (0.017) | (0.073) | (0.017) |
| Observations | 100 | 100 | 100 | 100 | 100 | 100 | 100 |
| *Panel D: Panel A further restricted by both cutoffs of Panels B and C* | | | | | | | |
| Agro-similarity | -0.013 | -0.052 | 0.012 | -0.304*** | -0.067*** | 0.092* | 0.074*** |
| | (0.046) | (0.048) | (0.008) | (0.099) | (0.023) | (0.049) | (0.020) |
| Observations | 81 | 81 | 81 | 81 | 81 | 81 | 81 |
| *Panel E: Cities rebuilt from their subdistricts at least 25% non-agricultural in 1980* | | | | | | | |
| Agro-similarity | 0.042 | 0.015 | 0.008 | -0.241** | -0.062*** | 0.138*** | 0.066*** |
| | (0.060) | (0.059) | (0.007) | (0.110) | (0.018) | (0.048) | (0.015) |
| Observations | 115 | 115 | 115 | 115 | 115 | 115 | 115 |
| *Panel F: Cities rebuilt from their subdistricts at least 50% non-agricultural in 1980* | | | | | | | |
| Agro-similarity | -0.039 | -0.040 | 0.009 | -0.440*** | -0.071*** | 0.080 | 0.077*** |
| | (0.044) | (0.045) | (0.007) | (0.109) | (0.017) | (0.060) | (0.015) |
| Observations | 68 | 68 | 68 | 68 | 68 | 68 | 68 |
| *Panel G: The city and its rural subdistricts within 50 km* | | | | | | | |
| Agro-similarity | 0.041 | 0.010 | 0.006 | -0.134* | -0.054*** | 0.129** | 0.055*** |
| | (0.056) | (0.048) | (0.006) | (0.069) | (0.020) | (0.056) | (0.014) |
| Observations | 134 | 134 | 134 | 134 | 134 | 134 | 134 |

*Notes:* Each cell is the coefficient on standardized agro-similarity from one regression of the 2010 census outcome in its column, in the main specification with the full set of controls and Conley SE within 200 km. Standard errors are in parentheses, and the stars use the same uniform-kernel standard errors. Agro-similarity is standardized once over every urban center with a measured exposure, so the panels share one scale. The top row is the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline), as in Table 4. Panel A drops every center the reclassification created or enlarged, and Panels B, C and D keep the Panel A centers with at least 25 residents per square kilometer, with at most 80% of employment in agriculture, or with both, in 1980. Panels E and F rebuild each city from its member subdistricts whose own 1980 employment was at least one quarter (Panel E) or one half (Panel F) outside agriculture, summing the 2010 outcomes and the four 1980 baselines over them. Agro-similarity, the settler count and the covariates remain those of the whole city, a subdistrict two cities share is split between them by its measured weights, and the 21 member subdistricts with no 1980 employment count stay in every panel. A city with no subdistrict above the threshold is absent from the panel, 19 cities in Panel E and 64 in Panel F, and Panel F also loses 2 cities whose kept subdistricts have no 1980 employment count, so a 1980 baseline cannot be formed. Panel G adds to each city the rural subdistricts, those outside every city, whose nearest city it is within 50 km in a straight line, measured to the nearest member subdistrict of each city on the same island, and sums the 2010 outcomes and the four 1980 baselines over that region: 709 rural subdistricts join, 6 cities do not gain a rural subdistrict, and the 57 rural subdistricts within 50 km that the 1980 census does not record are left out. Shares and the unemployment rate are fractions. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

baselines cannot be recovered from the earlier village censuses, which count households by activity and do not count workers. The fourth, an urban center added by the reclassification in West Sumatra, is represented only by 1980 census

Table A1: Main Estimates Under Alternative Samples and Specifications, Continued

| | Population + Employment | | | Agric Empl | | Modern Empl | |
|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| | ln pop | ln emp | unemp rate | ln L agric | agric % | ln L modern | modern % |
| **Part I: Samples, continued** | | | | | | | |
| *Panel H: Without the conflict provinces* | | | | | | | |
| Agro-similarity | 0.059 | 0.040 | -0.003 | -0.029 | -0.024 | 0.081 | 0.037 |
| | (0.098) | (0.096) | (0.009) | (0.153) | (0.022) | (0.092) | (0.026) |
| Observations | 108 | 108 | 108 | 108 | 108 | 108 | 108 |
| *Panel I: Without the cities hosting a new district seat in a member subdistrict* | | | | | | | |
| Agro-similarity | 0.065 | | | | -0.053*** | 0.172*** | |
| | (0.066) | | | | (0.017) | (0.064) | |
| Observations | 101 | | | | 101 | 101 | |
| *Panel J: Panel I without the cities whose town is the capital of a new district* | | | | | | | |
| Agro-similarity | 0.056 | | | | -0.045*** | 0.137*** | |
| | (0.066) | | | | (0.015) | (0.052) | |
| Observations | 98 | | | | 98 | 98 | |
| **Part II: Exposure and controls** | | | | | | | |
| *Panel K: Exposure from the 687 villages established after the 1980 census* | | | | | | | |
| Agro-similarity | 0.066 | 0.042 | 0.007 | -0.159* | -0.050*** | 0.164*** | 0.063*** |
| | (0.056) | (0.057) | (0.006) | (0.092) | (0.019) | (0.062) | (0.017) |
| Observations | 134 | 134 | 134 | 134 | 134 | 134 | 134 |
| *Panel L: Controlling for the expected exposure, double-robust form* | | | | | | | |
| Agro-similarity | 0.034 | 0.024 | -0.006 | -0.112 | -0.040** | 0.112* | 0.052*** |
| | (0.077) | (0.074) | (0.006) | (0.121) | (0.020) | (0.062) | (0.017) |
| Observations | 134 | 134 | 134 | 134 | 134 | 134 | 134 |
| *Panel M: Controlling for the settler share of the hinterland* | | | | | | | |
| Agro-similarity | 0.066 | 0.038 | 0.008 | -0.154* | -0.051*** | 0.169*** | 0.065*** |
| | (0.055) | (0.055) | (0.007) | (0.089) | (0.018) | (0.064) | (0.015) |
| Observations | 134 | 134 | 134 | 134 | 134 | 134 | 134 |

*Notes, continued:* Panel H leaves out the 26 cities of the provinces where communal or separatist violence displaced people around 2000: Aceh, on Sumatra (8 cities, the separatist conflict), West Kalimantan (8) and Central Kalimantan (5) (the Dayak-Madurese violence), and Central Sulawesi (5, the Poso conflict), so 108 cities in 13 provinces remain and every control coefficient is estimated again on them. Maluku and North Maluku, the provinces of the Maluku conflict, hold no city of the main city sample. Panels I and J report log population, the agricultural employment share and log modern-sector employment without the 33 cities hosting the seat of a district created 1999 to 2010 in a member subdistrict (Panel I), and also without the 3 cities whose town is the capital of such a district although the seat subdistrict is not a member (Panel J). A city hosts a seat when one of its member subdistricts holds the regent's or mayor's office, located in the village censuses of 2011 and 2014. Panel K rebuilds agro-similarity, the inverse-travel-time-weighted settler count and the 16 agro-climatic covariates from the 687 transmigration villages established after the 1980 census that lie within 300 travel minutes of the city, and standardizes the rebuilt agro-similarity over every urban center with a measured value. The four 1980 baselines are unchanged. Panel L also controls for the expected exposure under two random reassignments of village agro-climatic similarity, one within province and one within province and five-year plan (1979 to 1983 or 1984 to 1988), 2,000 draws each and both standardized. This is the double-robust specification of Borusyak & Hull (2023), consistent if either the controls or the reassignment model is correct. Panel M adds settlers placed in rural subdistricts within 100 kilometers of the city as a share of their 1980 census population. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

records with a known district and a missing subdistrict assignment, so its baseline does not qualify; its 2010 outcomes are also missing. The Aceh centers are not excluded on this ground: the extension to the 1980 crosswalk described in Section 3.1 recovered them.

**Census and travel-time exceptions.** Bireuen, an Aceh city on Sumatra whose count in the 2000 population census (SP2000), 357, is an enumeration shortfall against 78,876 people in 1980 and 128,959 in 2010, is in the satellite-defined count above, because the size, inclusion, and outlier cutoffs do not use year-2000 quantities. One center, in the Sangihe island group off North Sulawesi, has no transmigration village within reach of its island group. Its travel time to the nearest transmigration village, 1,120 minutes, is recovered from the all-subdistrict travel-time matrix and places it far outside any window used here.

Table A1: Main Estimates Under Alternative Samples and Specifications, Continued

| | Population + Employment | | | Agric Empl | | Modern Empl | |
|---|---|---|---|---|---|---|---|
| | (1) ln pop | (2) ln emp | (3) unemp rate | (4) ln L agric | (5) agric % | (6) ln L modern | (7) modern % |
| **Part II: Exposure and controls, continued** | | | | | | | |
| *Panel N: Four 1980 amenity levels* | | | | | | | |
| Amenity controls added | | | | | -0.045** | 0.154** | 0.052*** |
| | | | | | (0.020) | (0.061) | (0.019) |
| Main specification, same cities | | | | | -0.053*** | 0.165*** | 0.067*** |
| | | | | | (0.018) | (0.063) | (0.015) |
| Observations | | | | | 133 | 133 | 133 |
| *Panel O: Four levels and 1950 road density* | | | | | | | |
| Amenity controls added | | | | | -0.043* | 0.185*** | 0.058*** |
| | | | | | (0.023) | (0.071) | (0.021) |
| Main specification, same cities | | | | | -0.045*** | 0.149** | 0.063*** |
| | | | | | (0.016) | (0.058) | (0.014) |
| Observations | | | | | 130 | 130 | 130 |
| *Panel P: Four levels, 1950 roads, 1983 access* | | | | | | | |
| Amenity controls added | | | | | -0.036 | 0.130 | 0.046* |
| | | | | | (0.030) | (0.110) | (0.025) |
| Main specification, same cities | | | | | -0.050** | 0.099 | 0.059*** |
| | | | | | (0.022) | (0.081) | (0.018) |
| Observations | | | | | 103 | 103 | 103 |
| *Panel Q: Nine levels and 1950 road density* | | | | | | | |
| Amenity controls added | | | | | -0.048** | 0.154*** | 0.061*** |
| | | | | | (0.022) | (0.052) | (0.019) |
| Main specification, same cities | | | | | -0.045*** | 0.149** | 0.063*** |
| | | | | | (0.016) | (0.058) | (0.014) |
| Observations | | | | | 130 | 130 | 130 |
| *Panel R: Agro-similarity interacted with the conflict provinces* | | | | | | | |
| Agro-similarity | 0.056 | 0.045 | 0.007 | -0.122 | -0.042** | 0.170** | 0.063*** |
| | (0.084) | (0.079) | (0.007) | (0.119) | (0.021) | (0.081) | (0.020) |
| × conflict province | -0.021 | -0.066 | 0.009 | -0.126 | -0.027 | -0.015 | 0.015 |
| | (0.098) | (0.089) | (0.008) | (0.167) | (0.032) | (0.084) | (0.035) |
| Observations | 134 | 134 | 134 | 134 | 134 | 134 | 134 |
| *Panel S: Agro-similarity of every other measured city, weighted by inverse distance* | | | | | | | |
| Agro-similarity | 0.129* | 0.101 | 0.004 | -0.100 | -0.052*** | 0.236*** | 0.068*** |
| | (0.068) | (0.068) | (0.006) | (0.107) | (0.019) | (0.074) | (0.019) |
| Agro-similarity of neighbors | -0.607** | -0.601** | 0.035** | -0.516 | 0.006 | -0.613** | -0.013 |
| | (0.281) | (0.272) | (0.015) | (0.384) | (0.075) | (0.307) | (0.075) |
| Observations | 134 | 134 | 134 | 134 | 134 | 134 | 134 |

*Notes, continued:* In each of Panels N to Q the first row adds the controls the panel title names and the second row omits them on the same cities, the cities of the main city sample that hold every control of the panel, and the two sector shares and log modern employment are estimated. The four amenity levels are those Appendix Table E12 finds imbalanced in the 1980 village census: senior-secondary schools and polyclinics per 1,000 people, the population share in villages on the state grid, and the share of households connected. The nine add primary and junior-secondary schools and health centers per 1,000 people and the population shares with a paved road and with a road a four-wheel vehicle can pass. Over the subdistricts that host the city's transmigration villages within 300 minutes, the 1950 road density is the log of one plus road kilometers per 100 square kilometers on the 1950 road atlas, and the 1983 access is the population share on a road a four-wheel vehicle can pass in the 1983 village census. Panel R adds agro-similarity times an indicator of the four conflict provinces, and the indicator itself. Its first row is the slope outside the conflict provinces and its second the difference inside them, with every control coefficient common to both groups. Panels S and T add the agro-similarity of neighboring cities beside the city's own: in Panel S the mean of the standardized agro-similarity of every other city with a measured exposure (137 cities for each city), each weighted by 1 over its great-circle distance in km, defined for all 134 cities of the main city sample, and in Panel T the plain mean over the other measured cities within 100 km, which leaves out the 9 cities with no measured city that close, so 125 remain. City centers are the coordinates the Conley SE use. Both neighbor measures are means of standardized agro-similarity, so each coefficient is the change in the outcome per SD of agro-similarity, and on the main city sample the Panel S measure has an SD of 0.29 and the Panel T measure an SD of 0.83. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A1: Main Estimates Under Alternative Samples and Specifications, Continued

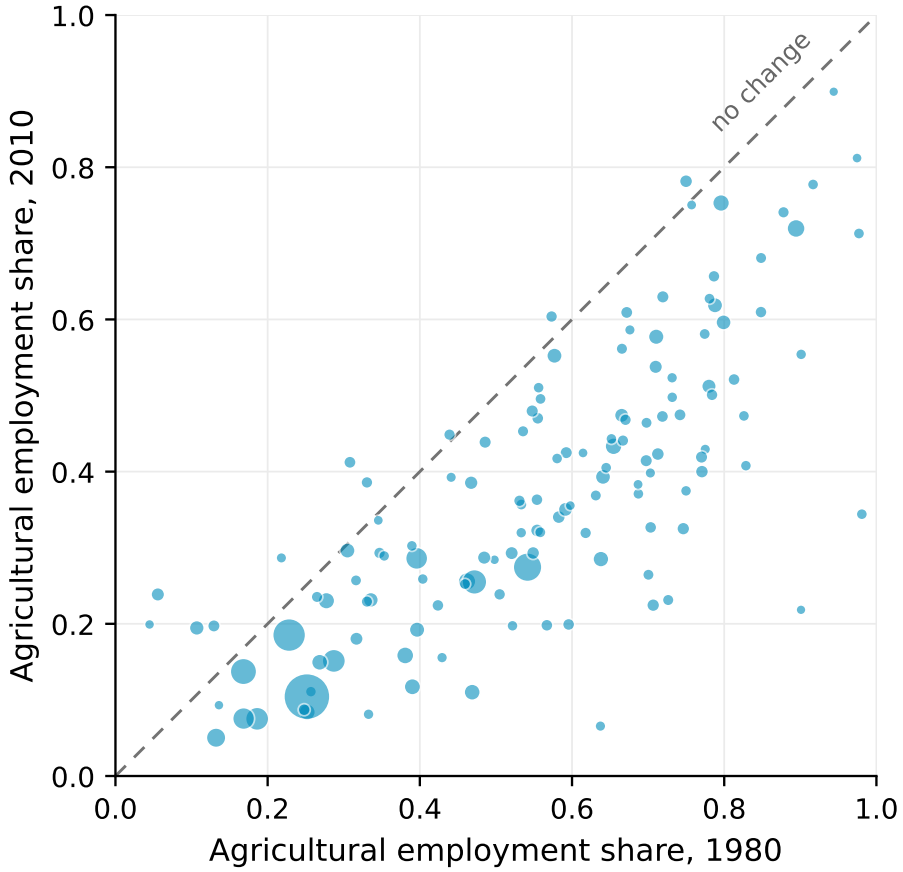
| | Population + Employment | | | Agric Empl | | Modern Empl | |
|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| | ln pop | ln emp | unemp rate | ln L agric | agric % | ln L modern | modern % |
| **Part II: Exposure and controls, continued** | | | | | | | |
| *Panel T: Agro-similarity of the measured cities within 100 km, plain mean* | | | | | | | |
| Agro-similarity | 0.100 | 0.058 | 0.014* | -0.164 | -0.059** | 0.210** | 0.072*** |
| | (0.085) | (0.085) | (0.008) | (0.121) | (0.024) | (0.100) | (0.021) |
| Agro-similarity of neighbors | -0.133 | -0.107 | -0.006 | -0.110 | -0.014 | -0.063 | 0.012 |
| | (0.085) | (0.084) | (0.006) | (0.132) | (0.027) | (0.074) | (0.024) |
| Observations | 125 | 125 | 125 | 125 | 125 | 125 | 125 |
| Settler count | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Baseline Pop 1980 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Urban share 1980 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Total Empl 1980 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Agric Empl 1980 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Agro-climatic controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

*Notes, continued:* Sources: population censuses of 1980 and 2010, Transmigration Census, village censuses of 1980, 1983, 2011 and 2014, the 1950 road atlas, and the World Bank district proliferation crosswalk. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

**Distance to the nearest transmigration village, and the 300-minute cutoff.** The 300-minute cutoff falls at the 73.8th percentile of travel time to the nearest transmigration village across the 187 centers, keeping the closer three-quarters of the sample. The cutoff is not a natural break in this distribution; it is chosen as a meaningful, achievable travel time. The median center is 145 minutes from its nearest transmigration village (Figure A3b).

**Robustness to the sample construction.** Panels A to D of Table A1 re-estimate the main effects on four subsamples: first dropping every center the reclassification touched, then imposing a 1980 population-density floor, a 1980 agricultural-share ceiling, and finally both. In every subsample the agricultural-share estimate stays negative and the modern-share estimate positive, both statistically significant, and each estimate of log population, log modern employment and the two employment shares falls inside the headline's 90% confidence interval. The agricultural-share screen is the demanding one, since it conditions on the 1980 level of the agricultural share itself and so works against finding a response in that share.

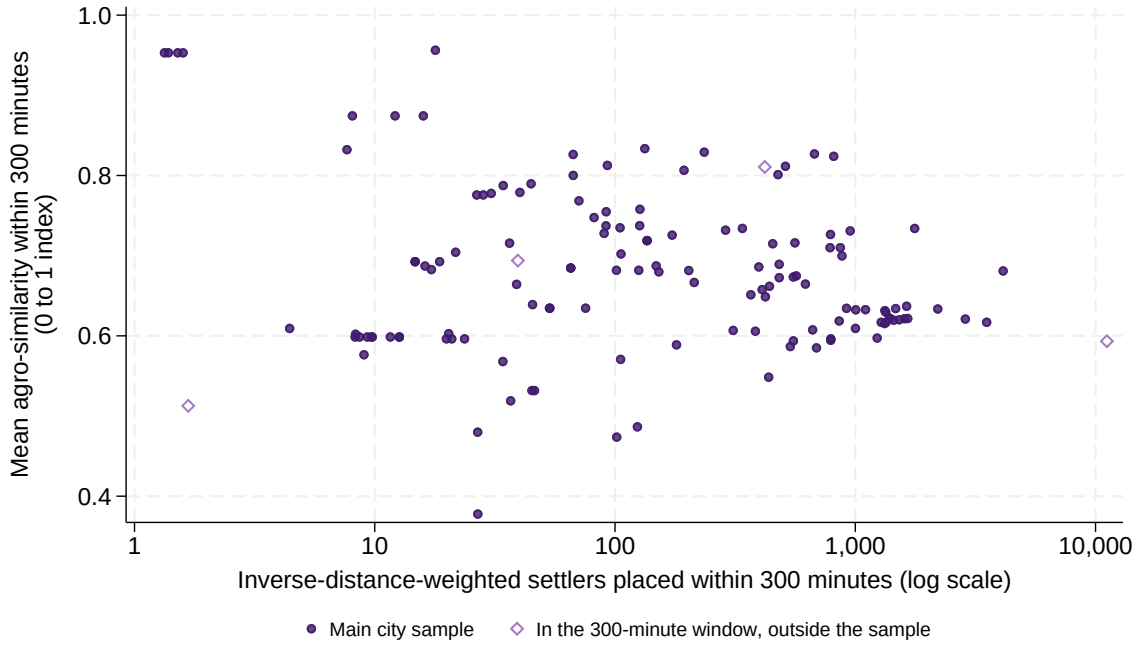
Figure A2: Agricultural Employment Shares Across Cities, 1980 and 2010



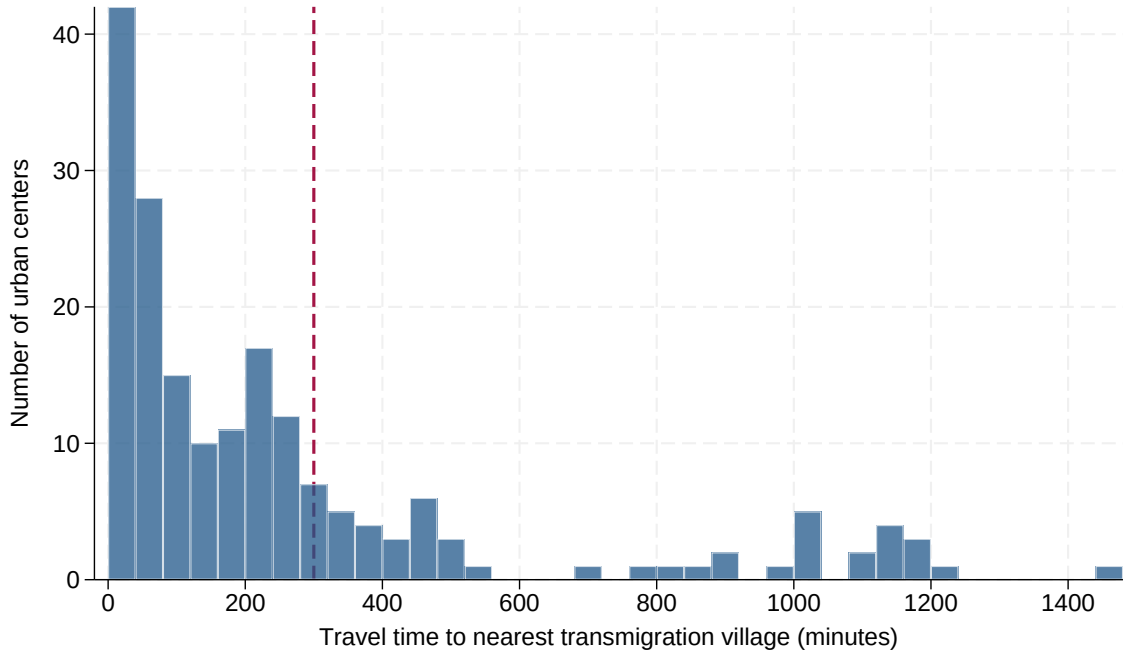

*Notes:* Each point is a city in the main city sample, with its 1980 agricultural employment share on the horizontal axis and its 2010 share on the vertical axis. Marker area is proportional to 2010 census population. The dashed line is the 45-degree line. City membership is held constant across years. Source: 1980 and 2010 population censuses.

Figure A3: Settler Placement and the Travel Time From Each City to the Nearest Transmigration Village

(a) Agro-Similarity Against the Settler Count, Cities Within 300 Minutes




(b) Urban Centers by Travel Time to the Nearest Transmigration Village




*Notes:* Panel (a): the 138 urban centers within 300 travel minutes of a transmigration village, with the main city sample (those with a 1980 census baseline) drawn as solid circles and the four in-window centers it excludes as hollow diamonds. The vertical axis is agro-similarity, the mean agro-climatic similarity of the transmigration villages within 300 minutes on its 0 to 1 index scale, and the horizontal axis is the number of settlers placed in those villages, weighted by inverse travel time, on a log scale. Panel (b): the 187-center sample by travel time in minutes to the nearest transmigration village, in 40-minute bins. The dashed line marks the 300-minute cutoff, which keeps 138 of the 187 centers, the 73.8th percentile of this distribution; the median center is 145 minutes away.

Table A2: The Urban Sample, From the Satellite Layer to the Main City Sample

| | Urban centers |
|---|---|
| Urban Center polygons on the Outer Islands (GHS layer) | 182 |
| Less a census twin, two polygons on one subdistrict | −1 |
| Satellite-defined urban centers | 181 |
| Plus centers the reclassification adds as standalone units | +6 |
| City sample | 187 |
| Less centers more than 300 travel minutes from a transmigration village | −49 |
| Within the 300-minute window | 138 |
| Less centers without a 1980 baseline the census recorded directly | −4 |
| Main city sample | 134 |

*Notes:* Counts of outer-island urban centers at each step of the sample construction. The Urban Center polygons are the GHS Urban Center Database layer; the census twin is two polygons that fall on a single subdistrict in the census geography and are treated as one unit. The reclassification adds two kinds of standalone center: dense administrative centers that fall below the satellite population threshold, and district remainders whose combined 1980 urban population share exceeds one half. Remainders exclude subdistricts already included through satellite overlap or the density rule. It also reattaches 42 edge subdistricts to 22 existing centers, which leaves the count unchanged. The window keeps the centers at most 300 travel minutes from a transmigration village. The main city sample keeps the in-window centers whose three 1980 baselines, log population, log employment and the urban-population share, the census records directly; the text of this subsection gives the reason each of the other four is left out.

## A.2 Transmigration-Village and Sample-Group Descriptives

The transmigration villages whose agro-climatic similarity to the settlers' origin regions generates the city-level agro-similarity exposure are small agricultural settlements (mean 1,885 settlers placed at founding, 69% of employment in agriculture) established between 1979 and 1988. The program-wide total of more than two million resettled people cited in Section 2 is the official aggregate, which includes placements outside this village census. The village sample underlying the exposure measure records approximately 1.5 million placed individuals. The summary statistics cover all 814 transmigration villages, with the 2010 village-population row available for the 716 villages with digitized polygon footprints (Table A3).

Table A3: Summary Statistics: The Transmigration Villages

**Panel A: Levels (settlers placed, village population)**

| | Mean | SD | Median | Min | Max | N |
|---|---|---|---|---|---|---|
| Settlers placed (count) | 1,885 | 1,011 | 1,792 | 136 | 8,852 | 814 |
| Village population 2010 | 2,235 | 5,228 | 817 | 0 | 85,296 | 716 |

**Panel B: Economic structure, similarity, and agro-climatic covariates**

| | Mean | SD | Median | Min | Max | N |
|---|---|---|---|---|---|---|
| Agro-climatic similarity to migrant origins | 0.667 | 0.136 | 0.684 | 0.000 | 1.000 | 814 |
| Agricultural employment share | 0.689 | 0.241 | 0.758 | 0.009 | 0.993 | 814 |
| Urban village (share) | 0.029 | 0.169 | 0.000 | 0.000 | 1.000 | 814 |
| Years of schooling | 3.998 | 0.899 | 3.978 | 0.713 | 7.653 | 814 |
| Agricultural productivity (normalized) | 0.996 | 2.647 | 0.435 | 0.001 | 27.316 | 770 |
| Ethnolinguistic fractionalization | 0.428 | 0.202 | 0.440 | 0.021 | 0.889 | 814 |
| Ruggedness | 0.311 | 0.129 | 0.310 | 0.006 | 0.691 | 814 |
| Distance to coast (index) | 10.553 | 1.138 | 10.844 | 5.795 | 11.925 | 814 |
| Soil organic carbon | 4.765 | 6.214 | 1.685 | 0.584 | 25.065 | 814 |
| Soil pH | 5.069 | 0.435 | 4.950 | 4.256 | 7.790 | 814 |
| Rainfall, 1948 to 1978 | 224.998 | 35.188 | 227.616 | 90.999 | 331.816 | 814 |
| Temperature, 1948 to 1978 | 26.253 | 1.709 | 26.748 | 17.225 | 28.174 | 814 |

*Notes:* Unit of observation is the transmigration village: the 814 villages from which the agro-similarity exposure of the urban centers is built, settled between 1979 and 1988 (mean 1983). Columns are the mean, standard deviation, median, minimum, maximum and number of villages with a value. Village population in 2010 is estimated from the Global Human Settlement population grid for the 716 villages with digitized settlement footprints, and normalized agricultural productivity is available for 770 villages. The agricultural employment share comes from the 2000 population census in Bazzi et al. (2016). Agricultural productivity is the index of Bazzi et al. (2016): the revenue-weighted geometric mean across crops of the village's yield per hectare relative to the crop's national average yield in the 2003 village census, so a value of one means every crop yields the national average. Similarity to the origin regions of the settlers is an index from 0 to 1, soil organic carbon is a percentage and temperature is in degrees Celsius. Settler counts, placement years, agro-climatic similarity and the covariates come from the transmigration program records and the construction of Bazzi et al. (2016). N is the number of villages with a value; the rainfall and temperature rows are averages over 1948 to 1978, and the urban village row is the share of villages classified as urban.

**Sample-group descriptives.** Cities are larger and far less agricultural than rural subdistricts, while agro-similarity exposure is balanced across the three sample groups (Table A5). This descriptive balance does not establish that city exposure is exogenous conditional on the controls; the identifying comparison is discussed in Section 4.3.

**The main city sample and the other outer-island urban centers.** The comparison with urban centers outside the main city sample uses the same 1980 baselines and 2010 outcomes as the city summary statistics (Table A4 and Table 1).

Table A4: The Main City Sample Against the Other Outer-Island Urban Centers

| | Main city sample | Other centers | Difference | SE | N sample | N other |
|---|---|---|---|---|---|---|
| *1980 baselines* | | | | | | |
| Log population, 1980 | 11.093 | 11.139 | -0.046 | (0.205) | 134 | 50 |
| Log employment, 1980 | 9.846 | 9.861 | -0.015 | (0.200) | 134 | 50 |
| Log agricultural employment, 1980 | 9.249 | 9.280 | -0.031 | (0.204) | 134 | 50 |
| Urban population share, 1980 | 0.260 | 0.237 | 0.023 | (0.032) | 134 | 50 |
| Agricultural employment share, 1980 | 0.560 | 0.548 | 0.012 | (0.036) | 134 | 50 |
| *2010 outcomes* | | | | | | |
| Log population, 2010 | 11.842 | 11.717 | 0.125 | (0.168) | 134 | 52 |
| Log employment, 2010 | 10.933 | 10.779 | 0.154 | (0.177) | 134 | 52 |
| Log agricultural employment, 2010 | 9.822 | 9.753 | 0.069 | (0.223) | 134 | 52 |
| Log modern employment, 2010 | 10.232 | 10.069 | 0.163 | (0.171) | 134 | 52 |
| Agricultural employment share, 2010 | 0.374 | 0.406 | -0.032 | (0.035) | 134 | 52 |
| Modern employment share, 2010 | 0.516 | 0.510 | 0.007 | (0.030) | 134 | 52 |
| Unemployment rate, 2010 | 0.122 | 0.133 | -0.012 | (0.013) | 134 | 52 |

*Notes:* Each row compares the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline) with the 53 other outer-island urban centers: 49 lie farther than 300 travel minutes from every transmigration village, and 4 lie inside that window and lack a directly recorded 1980 census baseline. The urban center added by the reclassification in West Sumatra also lacks 2010 outcomes, so the 2010 comparison has one fewer center. The first two columns are group means; the third is the coefficient from a regression of the row variable on an indicator for the main city sample, with its Conley SE within 200 km (uniform kernel) in parentheses; the last two count the centers in each group with a value for the row. Population and employment counts are in natural logs, and shares and the unemployment rate are fractions; the modern sector is manufacturing plus services. A Wald test that the five 1980 rows jointly predict membership in the main city sample has $p = 0.735$ under the uniform kernel, which weights every pair of centers inside the cutoff equally, and $p = 0.727$ under the Bartlett kernel, which weights a pair by one minus its distance over the cutoff. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

**Coverage of the household survey by wave.** The 2011 wave of the household survey (Susenas) includes the whole main city sample but follows Indonesia's 2004 rice import ban, which moved food prices nationwide for reasons unrelated to local agro-climatic conditions. The agricultural price in the main results therefore comes from a wave before the ban, and the 2002 wave also matches the timing of the 2003 village census (PODES) production data. Of the two waves before the ban, the 1996 wave covers more cities and more rural subdistricts, with and without a settlement, than the 2002 wave (Table A5).

**Price measures from household unit values.** The price measures of Figure D4 use the 1996 household survey (Susenas). A household's unit value for a food item divides its consumption value, including own production, by quantity. I winsorize each item's unit values at the 5th and 99th percentiles within province and urban status, then take the survey-weighted median over the city's households as the city's item price.

Each index divides the city's item price by the item's median over households in all surveyed urban centers, the national price. The raw-agricultural-food indices aggregate these price relatives using the urban centers' pooled expenditure weights (Laspeyres), the city's own expenditure weights (Paasche), or equal weights (a geometric mean). The crops index takes the geometric mean over the rice, corn and cassava items and their processed forms.

The rice unit value divides the value of rice consumed by the kilograms consumed, each summed over the city's households with survey weights. It includes local and higher-quality varieties, with own production valued by BPS. Every unit-value measure reflects price and quality together.

Table A5: Descriptive Statistics and Household-Survey Coverage by Location Type

| | Cities | | Rural, no settlement | | Rural, with settlement | | All | |
|---|---|---|---|---|---|---|---|---|
| *Panel A: Descriptive statistics* | | | | | | | | |
| | mean | SD | mean | SD | mean | SD | mean | SD |
| Population 1980 | 118,652 | (220,180) | 25,661 | (18,569) | 17,949 | (13,326) | 37,501 | (91,311) |
| Population 2010 | 229,366 | (404,467) | 36,971 | (24,057) | 52,180 | (35,899) | 67,882 | (168,977) |
| Employed 2010 | 90,259 | (152,082) | 16,121 | (10,851) | 22,864 | (15,625) | 28,214 | (64,050) |
| Agricultural employment share 2010 | 0.374 | (0.183) | 0.654 | (0.169) | 0.694 | (0.130) | 0.622 | (0.193) |
| Manuf employment share 2010 | 0.047 | (0.042) | 0.032 | (0.038) | 0.025 | (0.025) | 0.033 | (0.037) |
| Services employment share 2010 | 0.470 | (0.145) | 0.254 | (0.122) | 0.218 | (0.085) | 0.278 | (0.143) |
| Urban population share 1980 | 0.260 | (0.228) | 0.027 | (0.102) | 0.011 | (0.052) | 0.058 | (0.148) |
| Agro-similarity (raw) | 0.680 | (0.103) | 0.680 | (0.102) | 0.675 | (0.107) | 0.679 | (0.103) |
| Travel time to nearest transmigration village (min) | 109 | (86) | 163 | (79) | 1 | (0) | 122 | (96) |
| Observations | 134 | | 603 | | 191 | | 928 | |
| *Panel B: Units covered by the household survey* | | | | | | | | |
| Universe | 134 | | 603 | | 191 | | | |
| Sample, Susenas 1996 | 112 | | 288 | | 117 | | | |
| Sample, Susenas 2002 | 109 | | 206 | | 104 | | | |
| Sample, Susenas 2011 | 134 | | 582 | | 188 | | | |

*Notes:* Columns are the samples of the structural-transformation regressions by location type: cities (the main city sample: outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline), rural subdistricts without a settlement, rural subdistricts with a settlement (one of the 814 transmigration villages), and all three pooled. The rural subdistricts are in 2000 boundaries, lie outside every city and within 300 travel minutes of a transmigration village, and have a measured 1980 baseline and 2010 outcomes. Panel A: cells give the group mean with the standard deviation in parentheses. The Observations row gives the number of observations, and every row is measured on all of them. Employment counts those who worked or ran a business in the previous week, and employment shares divide each sector count by everyone with an industry recorded. Agro-similarity is the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, before standardization. Travel time is in minutes to the nearest transmigration village, set to one in a subdistrict with a settlement. The population and employment rows come from the 1980 and 2010 population censuses. Panel B: counts of units. Universe: the units of each sample that hold agro-similarity and every control of the main specification. Sample: the units of the universe that a wave's surveyed households live in, pooled to the city from its own households and to the year-2000 subdistrict, on the most widely observed consumption, spending or food-price outcome in the wave. Coverage can be lower for the other outcomes in these groups. Employment shares and income are outside this coverage comparison. The 2011 wave follows the 2004 ban on rice imports.

## A.3 Cross-Country Food Consumption and Urbanization

The cross-country comparison provides context for Indonesia's changes in food spending and urbanization (Figure 6). Differences in national urban definitions and food-expenditure measurement limit comparisons of levels across countries. The series plot national food expenditure shares against urban population shares over the years available for each country, between 1960 and 2025. The main food-share source is the United Nations National Accounts household final consumption expenditure by purpose (Table 3.2). I divide expenditure on food and non-alcoholic beverages by total household consumption expenditure. The World Bank's World Development Indicators supply population and the urban population share, using each country's administrative definition of urban. Marker size represents population and darker shades indicate later years.

**Country coverage.** The country set starts with the 20 most populous countries in the United Nations panel, ranked by the latest available population observation. Poland and Iraq are omitted because their urban shares change little over the period. China, the United States, Pakistan, Bangladesh, and South Africa are added from national statistical sources,

giving 23 countries. Russia, which the United Nations source also lacks, is omitted because of price-measurement problems in its household data. Egypt, the Democratic Republic of the Congo, Viet Nam, Türkiye, and Thailand are absent because the United Nations source lacks their household-consumption breakdown.

**National survey sources.** Published household-survey rounds supply the United States (1984 to 2024), Pakistan (2001 to 2024), South Africa (2006 to 2023), Nigeria (1986 to 2019), and Bangladesh (1981 to 2022). The United States series uses the Bureau of Labor Statistics Consumer Expenditure Surveys: food at home is divided by the consumption subtotal, which excludes personal insurance, pensions, and cash contributions. Pakistan's Household Integrated Economic Survey series removes tobacco and ready-made food from earlier rounds using the expenditure components in the published reports; later rounds report the narrower food category directly. South Africa's published Income and Expenditure Survey and Living Conditions Survey rounds report food and non-alcoholic beverages relative to household consumption expenditure. Nigeria and Bangladesh retain their broader published food shares, which include meals away from home and tobacco. Nigeria's change from diary collection to a shorter recall period further limits comparisons over time.

**China.** China's series is the Engel coefficient published by its National Bureau of Statistics, covering 1978 to 2025. Before the national series begins in 2013, I combine the published urban and rural coefficients using urban and rural population weights. The urban and rural expenditure definitions differ, and the 2013 survey change introduces a break. China's food category includes tobacco, liquor, and meals away from home. Dashed paths and asterisks identify China, Nigeria, and Bangladesh because these broader definitions prevent direct comparison of their levels with the narrower food shares.

**Indonesia.** Indonesia (IDNS) uses the national household-survey series from Susenas for 1994 to 2017. I remove catering, alcohol, and tobacco using item-level consumption data available in 7 of the 20 observed years. The adjustment is the share of reported food spending retained after these exclusions: it is measured in 1996, 2002, 2011, and 2013 to 2016, interpolated linearly between those years, and held at the nearest measured value outside that interval. I multiply each year's broad food share by this retained share. I then add the mean difference between the United Nations and adjusted Susenas shares over their overlapping years, +6.3 percentage points, to the entire Susenas series. This constant adjustment aligns its level with the national-accounts source while preserving its changes in percentage points.

# B Measures

## B.1 Land-and-Sea Travel Times and Pairwise Distances

Cost distance measures the minimum accumulated travel time between two locations, allowing the fastest route to differ from the shortest geographic path. I construct these measures for the paper's mapped urban areas, subdistricts, and transmigration villages before applying the sample restrictions. Unadjusted travel time determines nearest-village eligibility for the city sample. Adjusted travel time determines village catchment membership and inverse-time weights, while the main exposure remains the unweighted mean of village agro-similarity within the catchment.

**Inherited land surface.** I construct the pairwise travel measures used in this paper from the Malaria Atlas Project (MAP) 2015 land-travel friction surface, which assigns the time required to cross each location. The inherited surface combines information on transport access, land cover, and terrain from global accessibility mapping (Weiss et al. 2018). The shipping speeds follow the construction described by Nelson (2008). My spatial construction starts with the land surface: I add the missing maritime connections, align the geographic layers, locate the paper's origins and destinations, and compute their pairwise cost distances. The published distance to the nearest city is a separate output of the global accessibility studies; the distances used here are calculated for this paper's locations.[40]

[40] The land input agrees with the MAP 2015 surface; the VMap-based input of the earlier construction is not available. The earlier construction described by Nelson (2008) uses VMap0 roads, released in 1997 as an update to the Digital

**Maritime connections and geographic alignment.** The land surface does not supply the shipping connections needed to travel between Indonesian islands. I add the commercial-shipping layer from Halpern et al. (2008), based on vessel observations during the 12 months beginning October 2004. I project the land and shipping layers into the same coordinate system for Southeast Asia, clip them to a common geographic extent, and align their grid cells. I express the land friction values in minutes per kilometer. For shipping, I set travel costs at 3 minutes per kilometer in cells with positive recorded intensity and 1.5 minutes per kilometer in major shipping corridors, the cells where the layer records a value of 3 or more, corresponding to assumed speeds of 20 and 40 kilometers per hour. I add the land and water friction values where both are present, including coastal cells, and set costs at 30 minutes per kilometer in cells without an observed component. This last value is an assumed cost of travel outside the mapped connections. The shipping observations describe vessel activity after settlement, and their coverage is incomplete.

**Origins, destinations, and least-cost paths.** I link the census subdistrict polygons, GHS urban-center boundaries, and transmigration-village locations to their administrative identifiers and project their coordinates onto the combined surface. I calculate distances to transmigration villages separately for subdistricts and urban locations. For the urban calculations, I intersect each constituent subdistrict with the union of mapped urban footprints and use the centroid of that intersection when it lies inside the footprint, otherwise an interior point. Where the intersection is empty, the calculation retains the subdistrict centroid. For the surrounding-subdistrict calculations, origins are the subdistrict centroids; destinations are the transmigration-village locations matched to the replication data of Bazzi et al. (2016).

For each origin, the least-cost calculation accumulates travel costs across adjacent cells and selects the path with the smallest sum to each destination. The calculations use an approximately 1-kilometer grid and average the friction values of adjacent cells, with diagonal steps weighted by $\sqrt{2}$. They treat a horizontal or vertical step as one nominal kilometer.[41] I match the resulting pair records by subdistrict and village identifiers. For the revised urban endpoints, I calculate the change in least-cost travel time between the original and revised locations on the same surface and grid, calibrate it to the scale of the original calculation, and add that change to the pairwise times. This preserves the original travel-cost convention while changing the location represented by the urban endpoint. Pairs newly included by the endpoint change use the calibrated revised calculation, and I round the resulting values to whole minutes with a minimum of one minute. Travel time from a city to a village is the minimum across the city's constituent subdistrict locations, with membership updated to the city definition used in the analysis. For members without urban-origin pair records, I use the corresponding subdistrict-origin calculations.

**Pairwise times and exposure.** The pairwise calculations cover a broader set of locations than the main city sample. The urban and subdistrict pair records cover journeys up to 2,000 travel minutes, a limit separate from the 300-minute catchment used to construct the main exposure measure. I separately calculate great-circle distances from geographic coordinates for the straight-line comparisons. I retain unadjusted travel time alongside adjusted travel time used in constructing exposure: the adjustments set travel time to one minute when more than 0.1% of a village's polygon area overlaps the urban footprint and bound excessive travel times using the city-centroid-to-village great-circle distance at 2 kilometers per hour. Weighted exposure variants are described in the main text.

**Other geographic pairs.** I also construct great-circle distances between urban-member subdistrict centroids and other subdistrict centroids in the same island group. For the spatial model, I construct pairwise land-and-sea travel costs among its urban and rural locations. The rural model locations aggregate subdistricts into the rural remainders of districts. These calculations have distinct geographic scopes from the urban-area-to-village and subdistrict-to-village travel measures described above.

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Chart of the World developed in 1992. The documentation of the land input identifies the MAP 2015 surface associated with Weiss et al. (2018), which uses later road sources. The available records therefore do not establish that the road component of the travel times is restricted to VMap0.

[41] The projected cells are approximately 1.09 kilometers wide. The calculation does not separately multiply its costs by that cell width. The reported travel times use this grid convention.

**Road timing and identification.** The later road sources and shipping observations prevent interpreting the combined surface as a transport network observed before the program. If transport responses affect catchment membership or nearest-village eligibility, holding those catchments fixed in the expected-exposure exercises does not remove the resulting selection.

**Travel time and straight-line distance.** Geographic connections matter for access around Tana Paser. Two transmigration villages inside the same 50-kilometer radius of the functional urban area have sharply different travel times (Figure B1a). The northern village is labeled at about an hour, while the southwestern village is labeled at over five hours. This illustration uses travel time to the nearest functional urban area; the empirical distances between cities and villages use the urban-center definitions and pairwise construction described above. Figure B1b traces the least-cost route from three cities to their nearest transmigration village.

## B.2 Trade-Openness Measure: Rice-Import Market Access

**Source data.** To measure how exposed each urban center is to imported food, I use administrative records of *rice imports by port of entry*: volumes (kg) and values disaggregated by importing port and origin for 2000-2010, matched by port name to geocoded Indonesian ports in the February 2011 shipping directory (AtoBviaC Plc 2011). The port sample retains directory ports with zero recorded rice imports. Because the traded commodity is rice, the resulting index is specifically a measure of *food*-trade openness, the form of openness most directly relevant to a local agricultural-productivity shock. Summing import throughput to the port level gives, for each port of entry $p$, a total volume $\text{kg}_p$ and a location. The compilation stores port coordinates in degrees and minutes; taken as decimal degrees, each port falls 20 to 40 km from its berth, so all port-distance and market-access calculations use coordinates expressed as decimal degrees, which place the ports on the coast (median 2.8 km from the coastline against 13.0 km) and put the named terminals on their real berths.

**Construction.** I distinguish port proximity from access to the volume of rice imported through reachable ports. Proximity is the great-circle distance from a city to its nearest port, with a separate indicator for a port within 50 km. Rice-import market access uses the domestic land-and-sea travel times:

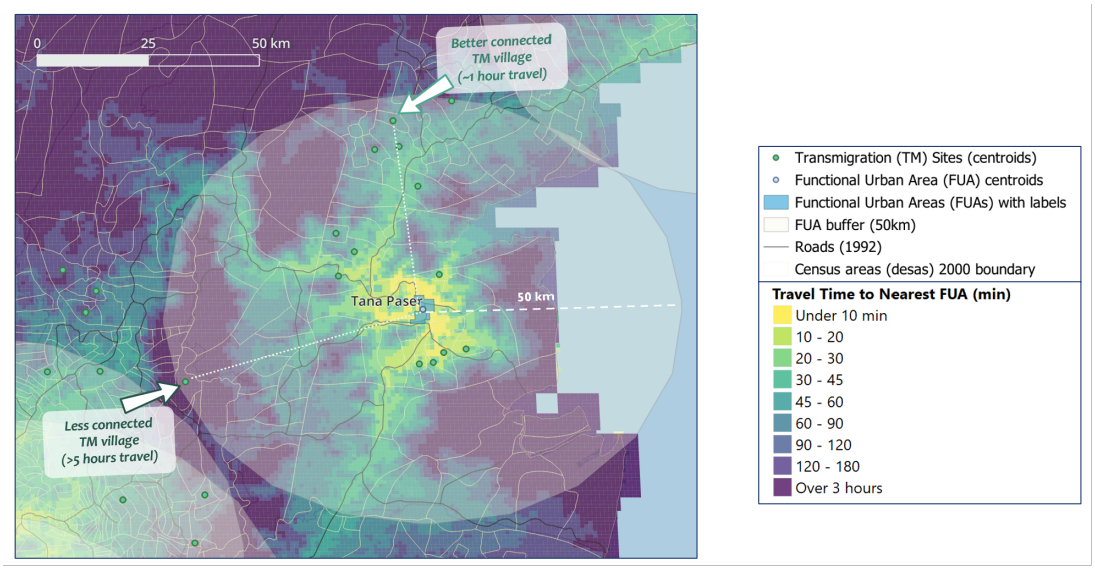
$$MA_i = \sum_p \frac{\text{kg}_p}{\max\{0.5, t_{ip}\}},$$

where $\text{kg}_p$ is the port's total recorded rice imports over 2000 to 2010 and $t_{ip}$ is travel time from city $i$ to port $p$ in hours. The half-hour floor bounds access for cities at or beside a port. I take $\ln(1 + MA_i)$ and standardize it across the full set of cities in the source data. The high-access indicator splits cities at the median within the exposure window. Each port's contribution increases with its recorded import volume and decreases with travel time. Because realized imports respond to demand and supply, this measure is partly an equilibrium outcome. Panel B of Table 6 compares it with route openness; differences between the two measures do not identify the direction of endogeneity bias.

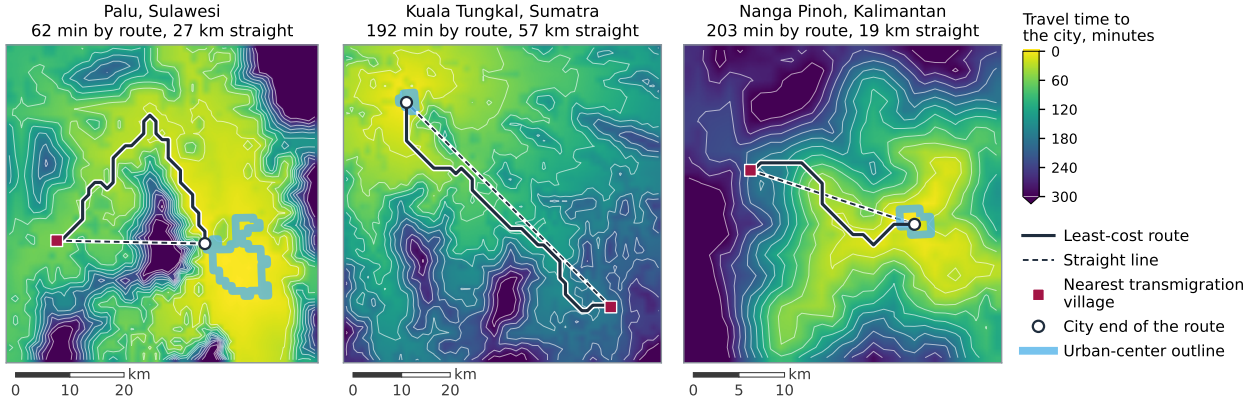
**Gravity market access and port distance.** The gravity index uses realized import volumes. Administrative records of rice imports by port of entry over 2000 to 2010 give each port a total import volume, and the market-access index sums those volumes over every port, discounting by travel time. Realized import volumes respond to the substitution this paper studies: a city whose hinterland becomes more productive buys more of its food there, its ports handle less imported rice, and the gravity measure classifies it as more closed. This substitution can make measured import access respond to hinterland productivity, creating an endogeneity concern for the gravity comparison. On the agricultural employment share the openness offset is 0.053 (standard error 0.021) on the gravity split against 0.047 (standard error 0.026) on the route split. On the modern-sector outcomes the two splits agree in sign, and the gravity offsets are smaller: $-0.034$ against $-0.062$ on the modern employment share, the first marginally significant and the second statistically significant, and $-0.145$ against $-0.242$ on log modern-sector employment, where only the route estimate is statistically significant. The main openness comparison uses the route split. The gravity offsets are slightly larger on the agricultural share and closer to zero on the two modern-sector outcomes. These differences describe sensitivity to the openness measure; they

Figure B1: Travel Time and Straight-Line Distance

(a) Straight-Line Distance and Travel-Time Connectivity Around Tana Paser, Kalimantan




(b) Least-Cost Routes From Three Cities to Their Nearest Transmigration Village




*Notes:* Panel (a): the background shows calculated travel time to the nearest functional urban area (FUA), Tana Paser in this region, in bins from under 10 minutes to over 3 hours. Filled dots identify transmigration-village locations, and the small ring identifies the urban-area centroid. The circle shows a 50-kilometer radius, and dotted lines connect the two highlighted villages to the urban area. The 1992 roads are displayed for orientation and do not establish the road source underlying the travel-time surface. Village boundaries follow the 2000 census, and urban-area boundaries come from the Global Human Settlement Layer. This is a descriptive map, with assumed travel costs and no statistical inference. Panel (b): each map shows one city of the main city sample (Palu on Sulawesi, Kuala Tungkal on Sumatra, Nanga Pinoh on Kalimantan) and the transmigration village with the shortest travel time to it. The background is the travel time to the city in minutes, on the grid of the travel-time calculation, with white lines every 30 minutes. The solid line is the least-cost path between the village and the city's origin point, traced on the travel-speed friction surface, and the dashed line is the straight line between the same two points. The map titles give the travel time of the pair used in the paper and the great-circle length of the straight line. Source: my geographic construction and travel-time calculations; transmigration-village locations from Bazzi et al. (2016); urban-center boundaries from the Global Human Settlement Layer.

do not isolate the bias from endogenous trade volumes. It is the alternative openness split in Panel B of Table 6, whose Panel A is the route split, and it correlates at 0.45 with route openness. Rutz (1987) records a 1973 cargo turnover at the ports through which 93.5% of the 2000 to 2010 rice tonnage moved, six years before the main placement wave of 1979 to 1984. The historical maps establish earlier activity for part of the port sample; the domestic travel-time input uses later

roads and shipping observations. The simplest measure is the great-circle distance to the nearest port. Bare distance is nearly uninformative about reachable trade, because ports in the sample are common (the median city sits 45 km from one) and nearest-port distance does not incorporate trade volumes. By great-circle distance the nearest port is a 1973 main port for 84 of the 134 cities, and a harbor the Directorate General of Sea Communications already listed in 1986 for 121 of them; Table B1 gives these counts beside those for the nearest port by travel time.

The differences of Table 6 that this section cites keep their significance with the Bartlett kernel: the route split's open-minus-closed differences have Bartlett $p$-values of 0.033 on log modern employment, 0.004 on the modern share and 0.075 on the agricultural share, and the gravity split's are 0.011, 0.170 and 0.077 on the agricultural share, log modern employment and the modern share.

Because the more open cities lie farther west, I also interact exposure with standardized longitude, and separately with distance to Jakarta. The open-minus-closed differences keep their signs in both cases. With longitude, the log modern-employment difference is −0.470 (standard error 0.157) and stays statistically significant, while the differences on the agricultural share, 0.090 (0.055), and on the modern share, −0.067 (0.045), lose precision. With distance to Jakarta, the differences are 0.044 (0.031) on the agricultural share, −0.350 (0.130) on log modern employment and −0.070 (0.027) on the modern share, the two modern-sector differences statistically significant.

Table B1: Vintage of the Ports Behind the Two Openness Measures

| | Route measure (nearest port by travel time) cities | Great-circle port distance (nearest port by great circle) cities | Rice imports (2000 to 2010) share of tons |
|---|---|---|---|
| 1973 main port (Map 2) | 98 (73%) | 84 (63%) | 93.5% |
| Other 1986 harbor (Map 1) | 31 (23%) | 37 (28%) | 5.6% |
| On neither map | 5 (4%) | 13 (10%) | 0.9% |
| In the 1986 inventory | 129 (96%) | 121 (90%) | 99.1% |
| Cities | 134 | 134 | |
| *Route openness using restricted port samples* | | | |
| 1973 main ports only | same side of the median: 130 of 134 | | correlation 0.973 |
| 1986 inventory only | same side of the median: 132 of 134 | | correlation 0.997 |

*Notes:* The sample is the main city sample. Each city's port is its nearest of the 183 on the port list, by travel time over land and sea for the route measure and by great-circle distance for the simplest measure, bare distance to the nearest port; the first two columns count cities by the category of that port, with the share of cities in parentheses, and the third gives the share of 2000 to 2010 rice-import tonnage through the ports of each category. A 1973 main port is the main port of one of 49 coastal regions at which Map 2 of Rutz (1987) draws a 1973 cargo circle; Map 1 prints the 237 harbors in their 1986 state. The main placement wave of the program ran 1979 to 1984. The lower panel recalculates route openness using the restricted port samples and compares its below-median high-openness split with the full-list split. Each correlation is between the route measure rebuilt on the restricted ports and the full-list route measure across the cities.

**Proximity versus the gravity composite.** Gravity market access is negatively related to the great-circle distance to the nearest port of the list, in the expected direction, but only weakly ($\rho = -0.11$ over the 138 cities within the exposure window): at a given distance there is wide variation in access, because the composite also loads the size of the reachable ports and the contribution of more distant ones, while bare proximity is nearly uninformative when every port of the list is included (the median city sits 45 km from one). Measuring distance to the nearest port that recorded rice imports in 2000-2010 tightens the correlation to $\rho = -0.51$, a comparison that conditions on the trade outcomes themselves. The gravity index incorporates import volumes in addition to port proximity.

**Route proximity to the Thai and Vietnamese rice hubs.** The gravity composite uses realized import volumes, which are an equilibrium outcome. The preferred openness measure in Section 5.2 uses route proximity, following

the "distance to Ho Chi Minh City/Bangkok (port)" construction in Bazzi (2017) (his original uses great-circle distance to the nearest port, a published shipping distance from that port, and the average of the two hubs; I use least-cost travel time to the port, the shortest route by sea from it, and the nearer hub): for each urban center, the route-aware least-cost travel time to its nearest port (over the same land-and-sea friction surface used for every cost distance in the paper) plus the length of the shortest route by sea from that port to the nearer of Bangkok and Ho Chi Minh City, the hubs from which the bulk of Indonesian rice imports ship, expressed in hours at the friction surface's shipping-lane speed so the two legs share one unit. Smaller totals mean closer alignment with the rice-import routes, hence higher measured openness; the high-openness indicator in Table 6 is a below-median split inside the estimation window. The international geography is fixed. The domestic leg uses the land-and-sea surface described in Section B.1, whose road and shipping inputs postdate settlement. The shipping directory does not record port opening dates. On the Rutz maps, the ports of 98 of the 134 cities were main ports with a recorded cargo turnover in 1973 and so predate the program; 129 appear in the 1986 harbor inventory or earlier, a date inside the program period of 1979 to 1988; and 5 are absent from both maps and may postdate it (Table B1). Three properties matter in practice. First, the international leg dominates the composite (median 51 hours against 1.1 domestic), which adds the west-to-east proximity-to-the-routes gradient that nearest-port distance discards: route proximity correlates 0.45 with the gravity composite but only 0.15 with domestic-only nearest-port proximity, and the measure built on the port that minimizes each city's total travel time correlates at 0.73 with the one built on its nearest port. Second, the sea leg follows a route a ship can sail, the shortest path over a 1 kilometer grid of the coastline (the Online Supplement draws each port's route), so a leg from a western port rounds the northern tip of Sumatra or passes the Sunda Strait. A great circle from the port to the hub crosses more than 200 kilometers of land for 105 of the 134 cities and ranks them differently: the rank correlation between the two versions of the measure is 0.60, and 33 cities change side of the median, 30 of them in Sumatra. Third, the port sample includes ports with and without recorded rice imports. Whether a given port appears in the 2000-2010 rice-import records is itself an equilibrium outcome, and one the mechanism of this paper could move, since a city that substitutes hinterland food for outside supply leaves its port less rice-import business. The sample therefore includes every geocoded directory port, so inclusion does not depend on recorded rice-import volume. Restricting it to the 58 ports that did record rice volume changes little. The two sets place 130 of the 138 in-window cities on the same side of the openness median, and the eight they disagree on split four in each direction. Restricting the port set instead to the 1973 main ports leaves 130 of the 134 on the same side of that median, at a correlation of 0.973 with the measure using the full port sample. The high-openness offset in Table 6 goes from +0.047 to +0.025 on the agricultural share and from −0.062 to −0.041 on the modern share (standard errors 0.024 and 0.023 on the restricted set): both keep their sign at roughly half to two thirds of their size; on the restricted set the modern-share offset is marginally significant and the agricultural-share offset is insignificant. The reason the two agree at all is the first property above, since the international leg accounts for 98% of the composite's cross-city variance and no port's rice-import status can move it. With the candidate ports restricted to the 44 harbors of the 1973 map, the complete set of main ports before the program, 130 of the 134 cities stay on their side of the median, and the route split returns the same differences: −0.178 (standard error 0.086) on log modern employment and −0.054 (0.014) on the modern share, against −0.242 (0.109) and −0.062 (0.020) on the full port list, all four statistically significant. The historical-port comparison assesses sensitivity to port selection; it does not establish the timing of the domestic transport network.

**Baseline levels by openness group.** The more open cities were larger in 1980 than the more closed ones (151,000 against 85,000 residents, a statistically significant difference in levels and an insignificant one in logs) and had a lower modern employment share (34.3% against 42.4%, statistically significant) with a higher agricultural share (58.9% against 53.0%, insignificant); the urban population shares do not differ (25.9% and 26.2%). The openness regressions control for 1980 log population, log total employment, log agricultural employment and the urban population share (Panel D of Table B4).

**Continuous market access.** The continuous form replaces the high-openness indicator in Equation 6 with standardized rice-import market access (Table B3):

Table B2: 1996 Food Prices by Third of Route Distance to the Rice Hubs

| | Cities | | | Rural, with settlement | | | Rural, without settlement | | |
|---|---|---|---|---|---|---|---|---|---|
| | Agric. price | Staple crops | Rice u.v. | Agric. price | Staple crops | Rice u.v. | Agric. price | Staple crops | Rice u.v. |
| Agro-similarity, most open third | 0.043 | 0.302 | 0.156 | -0.221 | -0.711*** | -0.543** | -0.371*** | -0.124 | -0.143 |
| | (0.163) | (0.203) | (0.218) | (0.199) | (0.251) | (0.229) | (0.117) | (0.172) | (0.153) |
| Agro-similarity, middle third | 0.141 | 0.174 | -0.006 | 0.010 | 0.001 | -0.123 | -0.151 | -0.079 | -0.122 |
| | (0.175) | (0.181) | (0.209) | (0.268) | (0.353) | (0.317) | (0.215) | (0.185) | (0.160) |
| Agro-similarity, most closed third | -0.189 | -0.008 | -0.131 | -0.554** | -0.322 | -0.439 | -0.190 | -0.151 | -0.083 |
| | (0.175) | (0.195) | (0.227) | (0.241) | (0.271) | (0.268) | (0.229) | (0.169) | (0.150) |
| p-value, closed minus open | 0.070 | 0.122 | 0.262 | 0.351 | 0.324 | 0.773 | 0.413 | 0.916 | 0.799 |
| Observations | 112 | 112 | 112 | 117 | 117 | 116 | 288 | 288 | 286 |

*Notes:* Each column is one regression of the 1996 price, standardized over the cities or over the pooled rural subdistricts, on agro-similarity interacted with indicators for the thirds of route distance to the Thai and Vietnamese rice hubs within the location type, the most open third being the nearest, in the main specification with the full set of controls and Conley SE within 200 km in parentheses; for rural subdistricts with a settlement, the settler count is the number of settlers placed in the transmigration villages that overlap the subdistrict. Coefficients are in SD of the price per SD of agro-similarity. The $p$-value tests that the most closed and the most open third respond alike. The agricultural price is the raw-food index, the staple-crops price covers rice, corn and cassava, and the rice unit value is consumption value over consumption kilograms, all from the 1996 household survey (Susenas), before the 2004 rice import ban. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B3: Agro-Similarity by Trade Openness: Market Access Instrumented by Route Proximity

| | (1) ln pop | (2) ln emp | (3) unemp rate | (4) **ln L agric** | (5) **agric %** | (6) **ln L modern** | (7) **modern %** |
|---|---|---|---|---|---|---|---|
| **Agro-similarity, mean closed city** | **0.095** | **0.054** | **0.018**** | **-0.241** | **-0.072***** | **0.264**** | **0.112***** |
| | (0.129) | (0.120) | (0.008) | (0.169) | (0.022) | (0.123) | (0.021) |
| Agro-similarity, mean open city | -0.048 | -0.058 | 0.001 | -0.206 | -0.045 | 0.037 | 0.039 |
| | (0.157) | (0.145) | (0.009) | (0.236) | (0.038) | (0.113) | (0.034) |
| Market access (z) | 0.467 | 0.419 | 0.002 | 0.720 | 0.079 | 0.272 | -0.067 |
| | (0.376) | (0.342) | (0.014) | (0.465) | (0.082) | (0.303) | (0.067) |
| First-stage $F$ (Kleibergen-Paap) | 2.53 | 2.53 | 2.53 | 2.53 | 2.53 | 2.53 | 2.53 |
| Outcome mean | 11.842 | 10.933 | 0.122 | 9.822 | 0.374 | 10.232 | 0.516 |
| Outcome SD | 0.854 | 0.840 | 0.040 | 0.896 | 0.183 | 0.975 | 0.156 |
| Observations | 134 | 134 | 134 | 134 | 134 | 134 | 134 |

*Notes:* Two-stage least squares on the main city sample, in the main specification with the full set of controls and Conley SE within 200 km. Standardized rice-import market access, discounted by travel time, and its interaction with standardized agro-similarity are instrumented by standardized route proximity, the travel time to the Thai and Vietnamese rice hubs with its sign reversed, and its interaction with agro-similarity. Rows give the agro-similarity effect at the mean market access of the cities below and above the median of market access ($-0.83$ and 0.85 SD) and the coefficient on market access; the outcomes are measured in 2010, with the four sector columns shaded. Standard errors are in parentheses. The Kleibergen-Paap $F$ statistic of the first stage, under the same Conley variance, is below the Stock-Yogo 10% maximal-size critical value of 7.03 for two endogenous regressors and two instruments. Agro-similarity is standardized, and the beta terms are the coefficients of (15). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

$$y_c = \beta_1 \, AgSim_c + \beta_2 \, AgSim_c \times MA_c + \chi \, MA_c + \phi \, SettlerCount_c^{IDW} + C_{c,1980} \, \xi + X_c \, \zeta + \varepsilon_c \tag{15}$$

For this continuous specification, I restandardize log rice-import access within the exposure window. Here $MA_c$ is market access in those standard deviations, so $\beta_1$ is the exposure effect at mean market access and $\beta_2$ is its change

per standard deviation of market access. Market access and its interaction with exposure are instrumented by route proximity, standardized across all 187 urban centers as in Section 3.5, and its interaction with exposure:

$$\begin{aligned} MA_c &= \pi_1 \, Route_c + \pi_2 \, AgSim_c \times Route_c + \pi_3 \, AgSim_c + W_c \, \kappa + u_c \\ AgSim_c \times MA_c &= \rho_1 \, Route_c + \rho_2 \, AgSim_c \times Route_c + \rho_3 \, AgSim_c + W_c \, \lambda + v_c \end{aligned} \tag{16}$$

$Route_c$ rises with proximity to the rice hubs, and $W_c$ includes the settler count, the 1980 baselines, and the agro-climatic controls of Equation 5. The excluded instruments are $Route_c$ and $AgSim_c \times Route_c$, while the endogenous regressors are $MA_c$ and $AgSim_c \times MA_c$. Their validity requires both instruments to be uncorrelated with the outcome-equation error after conditioning on agro-similarity and $W_c$. Economically, route proximity must affect outcome levels and the response to agro-similarity only through rice-import access. A route-related effect on the productivity response through non-agricultural trade or migration would violate this restriction if it operates outside measured rice-import access (Section 3.5). Table B3 evaluates $\beta_1 + \beta_2 MA_c$ at the mean market access of the more closed and more open groups. Panel C of Table 6 runs the same two-stage least squares on the two high-openness indicators, the gravity split instrumented by the route split, so its rows are those of Panels A and B.

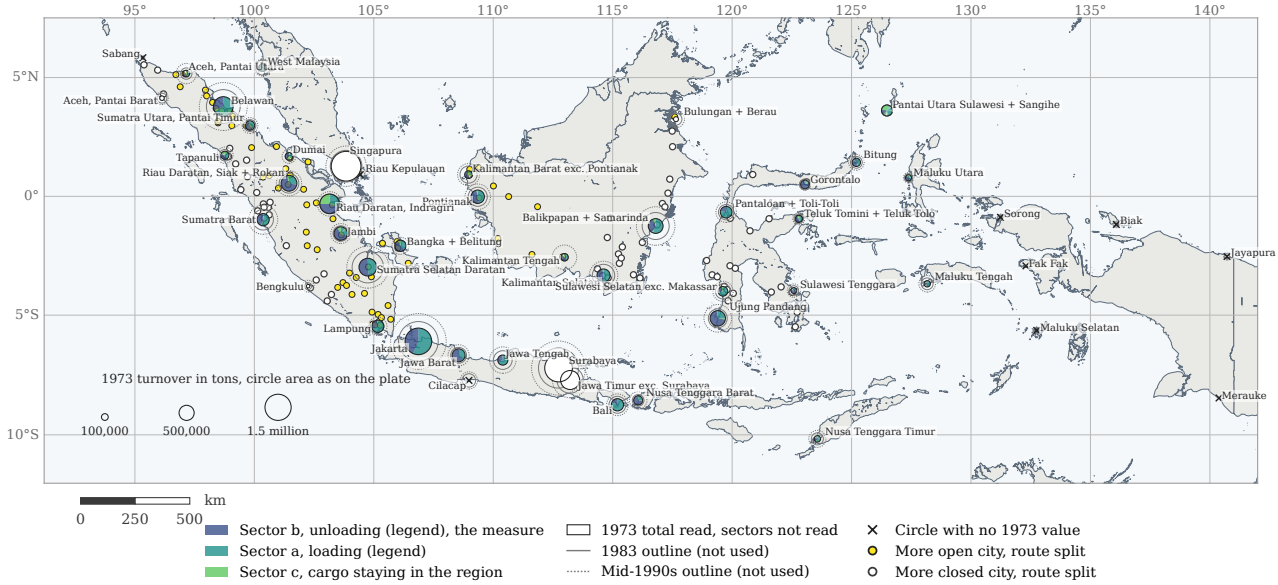
**The continuous instrumented form.** This form replaces the two indicators of Panel C with continuous market access and continuous route proximity and evaluates the exposure response at the mean market access of the more closed and the more open cities (Table B3). The modern-employment interaction is −0.135 log units per standard deviation of market access (standard error 0.108) and the modern-share interaction −0.043 (standard error 0.027), both statistically insignificant. The first stage of this form is the weaker of the two. Its Kleibergen-Paap $F$ statistic is 2.53, below the critical value of 7.03, so the estimates lean toward the ordinary least squares estimates and their standard errors understate the uncertainty. The Anderson-Rubin test stays valid with weak instruments, and it rejects the hypothesis that both instrumented coefficients are zero at the 10% level in three of the seven columns.

**The 1973 cargo maps.** The historical maps were easier to obtain than the underlying statistical tables of the sea-transport directorate, so I digitized them. Map 2 of Rutz (1987) draws the main port of each coastal region with three circles whose area is the region's interisland cargo turnover in 1973, 1983 and the mid-1990s. I measured the 1973 circle of each port on a scan of the printed plate, turned its radius into tons with the map's own nomogram, matched each port to the cities whose nearest port it is, and assigned each city the 1973 turnover of its port's region, which covers 133 of the 134 cities. The map's legend and the article text disagree on which sector of each circle is unloading, so the total turnover is the measure. Figure B2 shows the digitized circles beside the route split.

**The 2004 rice-import ban and three sensitivity checks.** Indonesia's 2004 rice-import ban is the event the route measure was designed around: Bazzi (2017) documents faster post-ban rice-price growth in the places closer to the import routes, with the pre-ban gradient reversed. Across the cities within 300 travel minutes of a settler village, post-ban (2002 to 2008) rice-price growth rises by 0.001 per standard deviation of route proximity (standard error 0.013), and by 0.016 (0.013) with agro-similarity and the main specification's controls, both statistically insignificant. The cities the route measure ranks as more open lie in coastal regions whose harbors handled more interisland cargo in 1973, before the program. Across the 133 cities whose region has a 1973 cargo turnover on the Rutz (1987) maps, route proximity correlates 0.37 with that turnover (0.29 with both demeaned by island), and across the 124 cities with a 1973 unloading value it correlates 0.52 with unloading under the map legend's sector labels and 0.25 under the article text's. Per 1980 resident of the region, one SD of route proximity predicts 0.628 SD of the 1973 total turnover (0.715 within islands), both statistically significant, with the settler count and the agro-climatic controls and no 1980 baseline, since the outcome precedes them. Agro-similarity does not predict the region's 1973 total turnover (0.011, standard error 0.149), where it does predict route proximity (−0.387, standard error 0.105), so the 1973 cargo is a measure of sea trade before the program that is balanced on exposure. Panel D of Table 6 splits the cities at the median of this measure, and Figure B2 shows the digitized circles. Three sensitivity checks assess the route comparison (Table B4). More exposed cities lie farther from the rice hubs by sea (−0.39 SD per SD, standard error 0.10), so Panel A adds route proximity to the main

specification: six of the seven headline estimates keep their signs and significance, the three sectoral coefficients at 78% to 111% of the main ones with standard errors close to the main specification's, and log agricultural employment falls from −0.166 to −0.079 (standard error 0.095) at an unchanged standard error, so its marginal significance goes with the smaller coefficient; route proximity itself enters positively on agricultural employment. With the square of agro-similarity added to the route split, the open-minus-closed differences in modern employment and the modern share keep their sign and stay statistically significant. Panel C of the same table takes the split on the region's 1973 cargo unloading, whose sector the map's legend and the article text label differently: on log modern employment the open-minus-closed difference is −0.116 (standard error 0.063) under the legend's labels and 0.026 (0.067) under the article text's.

Figure B2: The 1973 Cargo Turnover of the Coastal Regions, as Digitized




*Notes:* One circle per coastal region at its main port, on the printed plate's projection and symbol scale, with the circle's area the region's 1973 total turnover and its wedges the sectors under the map legend's labels. The cities of the main city sample (134) are drawn by the route split of Table 6, Panel A, filled dots the more open cities and open circles the more closed. Source: Rutz (1987), Map 2, digitized; land from GADM 4.1.

## B.3 Standardized ($z$-on-$z$) Reduced-Form Estimates: Model Calibration Targets

The model's response slopes are read in standardized ($z$-on-$z$) units, with both the treatment and each outcome standardized, so that the slopes, which are unit-free, are compared with the data's in sign and relative size. Table B5 reports the reduced-form estimates in those units on the main city sample, with the Susenas- and PODES-based rows on the sample each source covers, reported per row. The elasticities are chosen to fit two of these slopes, for log population and the agricultural employment share ($m_1$ and $m_2$); at the best-fitting completed calibration, both model coefficients have the same signs as the empirical estimates and lie within their 90% confidence intervals on the main city sample (Section F.4). Modern employment ($m_3$) and the remaining rows are untargeted.

Table B4: Checks on the Route Measure

| | (1) ln pop | (2) ln emp | (3) unemp rate | (4) **ln L agric** | (5) **agric %** | (6) **ln L modern** | (7) **modern %** |
|---|---|---|---|---|---|---|---|
| *Panel A: The main specification with route proximity as a control* | | | | | | | |
| Agro-similarity, main specification | 0.051 | 0.024 | 0.008 | -0.166* | -0.051*** | 0.158** | 0.066*** |
| | (0.059) | (0.057) | (0.007) | (0.093) | (0.018) | (0.062) | (0.015) |
| Agro-similarity, with route proximity | 0.097 | 0.066 | 0.007 | -0.079 | -0.040** | 0.176** | 0.053*** |
| | (0.072) | (0.070) | (0.007) | (0.095) | (0.016) | (0.073) | (0.013) |
| Route proximity | 0.120* | 0.110* | -0.002 | 0.227*** | 0.029 | 0.046 | -0.033* |
| | (0.072) | (0.065) | (0.004) | (0.078) | (0.019) | (0.071) | (0.017) |
| *Panel B: The route split without and with agro-similarity squared* | | | | | | | |
| Open minus closed | -0.128 | -0.125 | -0.009 | 0.071 | 0.047* | -0.242** | -0.062*** |
| | (0.088) | (0.097) | (0.011) | (0.152) | (0.026) | (0.109) | (0.020) |
| Open minus closed, with agro-similarity squared | -0.178* | -0.187* | -0.004 | 0.023 | 0.048* | -0.286** | -0.055*** |
| | (0.102) | (0.110) | (0.011) | (0.163) | (0.027) | (0.119) | (0.018) |
| Observations | 134 | 134 | 134 | 134 | 134 | 134 | 134 |

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B4: Checks on the Route Measure, Continued

| | (1) ln pop | (2) ln emp | (3) unemp rate | (4) **ln L agric** | (5) **agric %** | (6) **ln L modern** | (7) **modern %** |
|---|---|---|---|---|---|---|---|
| *Panel C: The split on the region's 1973 cargo unloading, under each set of sector labels* | | | | | | | |
| Open minus closed, legend's sector labels | -0.113* | -0.125* | 0.005 | 0.008 | 0.013 | -0.116* | -0.017 |
| | (0.064) | (0.064) | (0.007) | (0.104) | (0.021) | (0.063) | (0.017) |
| Open minus closed, article text's sector labels | 0.053 | 0.051 | -0.003 | 0.201** | 0.029 | 0.026 | -0.024 |
| | (0.070) | (0.077) | (0.007) | (0.099) | (0.018) | (0.067) | (0.021) |
| Observations | 124 | 124 | 124 | 124 | 124 | 124 | 124 |

*Panel D: 1980 city characteristics by trade-openness group*

| | Mean | | Open minus | |
|---|---|---|---|---|
| 1980 characteristic | Closed | Open | closed | $p$ |
| Population, thousands | 85.15 | 151.16 | 66.01** | 0.049 |
| | | | ( 33.50) | |
| Log population | 10.95 | 11.23 | 0.28 | 0.214 |
| | | | ( 0.23) | |
| Urban population share, % | 26.2 | 25.9 | -0.3 | 0.940 |
| | | | ( 3.7) | |
| Log employment | 9.66 | 10.03 | 0.37 | 0.107 |
| | | | ( 0.23) | |
| Log agricultural employment | 9.01 | 9.48 | 0.47* | 0.091 |
| | | | ( 0.28) | |
| Agricultural employment share, % | 53.0 | 58.9 | 5.9 | 0.156 |
| | | | ( 4.1) | |
| Modern employment share, % | 42.4 | 34.3 | -8.0** | 0.020 |
| | | | ( 3.5) | |
| Cities | 66 | 68 | | |

*Notes:* Panels A to C use the main city sample, in the main specification with the full set of controls and Conley SE within 200 km. Their seven columns are the outcomes of Table 4, measured in 2010, with the sector columns shaded as there. Panel A estimates the main specification and then adds route proximity, the travel time to the nearer of the Thai and Vietnamese rice hubs through the nearest port with its sign reversed and standardized, as one more control; its last row is that control's own coefficient. Panel B estimates the route split of Panel A of Table 6, whose difference row is the agro-similarity effect in the more open cities less that in the more closed ones, and then adds the square of standardized agro-similarity. Panel C estimates the same difference on the split by the region's 1973 cargo unloading on Rutz (1987) Map 2, above the median over the window cities, under the map legend's sector labels and under the article text's; the cities of one region share a value, and the panel uses the 124 cities whose region has a 1973 unloading value. Panel D compares the main city sample's more closed and more open cities, split at the median route distance to the Thai and Vietnamese rice hubs over the cities within the sample window (52.3 hours), 66 and 68 cities. Mean: the group average of the 1980 census characteristic, shares in percent. Open minus closed: the coefficient on the more-open indicator in a regression of the characteristic on it, with Conley SE within 200 km in parentheses and its two-sided $p$-value. Logs are the natural log of one plus the count. The modern employment share is manufacturing and services over all employed. Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B5: Reduced-Form Responses to Agro-Similarity in Standardized Units

| | **Main specification** | | $z$**-on-**$z$ | |
|---|---|---|---|---|
| Outcome | Coef. | (SE) | response | $N$ |
| *Outcomes the calibration targets* | | | | |
| $m_1$ Log population | 0.051 | (0.059) | +0.060 | 134 |
| $m_2$ Agricultural employment share | $-0.051^{***}$ | (0.018) | −0.278 | 134 |
| *Untargeted outcomes* | | | | |
| $m_3$ Log modern employment | $0.158^{**}$ | (0.062) | +0.162 | 134 |
| Log agricultural employment | $-0.166^{*}$ | (0.093) | −0.185 | 134 |
| Modern employment share | $0.066^{***}$ | (0.015) | +0.420 | 134 |
| Housing rent | −0.011 | (0.108) | −0.013 | 134 |
| Agricultural (raw-food) price index | 0.016 | (0.022) | +0.132 | 134 |
| Log wage | 0.019 | (0.032) | +0.062 | 134 |
| Log real wage | 0.003 | (0.029) | +0.014 | 134 |
| Household food expenditure share | $-0.016^{***}$ | (0.006) | −0.313 | 134 |
| Log rice production | $0.736^{***}$ | (0.251) | +0.314 | 132 |

*Notes:* Each coefficient is from the main specification of Table 4, with the full set of controls and Conley SE within 200 km, on the main city sample; the treatment is standardized agro-similarity. The $z$-on-$z$ response divides each coefficient by the standard deviation of the outcome over its estimation rows, so it is the response in standard deviations of the outcome per standard deviation of agro-similarity; the model's response slopes are computed in the same form and are compared with these in sign and relative size. The structural estimate in Section 6.2 is fitted to the log-population and agricultural-employment-share slopes ($m_1$ and $m_2$) in this table. At the best-fitting completed calibration, both model coefficients have the same signs as the empirical estimates and lie within their 90% confidence intervals (Section F.4). Modern employment ($m_3$) and the remaining rows are untargeted. The housing rent, price, wage and food-share rows are from Susenas 2011 and the rice-production row from the 2003 village census, each on the urban centers the source covers; $N$ is the number of urban centers in each regression.
$^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

# C Identification Support

This part reports the balance, placebo, and recentering tests for the exposure design. The tests by source of variation also identify a 1980 birthplace-share association; as discussed below, settlement timing prevents that comparison from separating prior sorting from early program effects.

## C.1 Design-Based Recentering and the Placebo Tests

I use two reassignment exercises. The whole-match reshuffle exchanges village agro-climatic similarity scores among sites within province and settlement wave. The origin redraw exchanges settler-origin compositions within the same groups while holding each site's destination agro-climate fixed. The second exercise separates expected exposure associated with site land from the deviation associated with arriving settlers. The component regressions describe these sources, while the reassignment tests evaluate them under the specified assignment process. Their interpretation remains conditional on that process.

**Exposure by source.** The origin redraw holds each settlement site's land fixed and exchanges settler-origin compositions among sites in the same province and five-year plan. Its expected exposure is the land component; realized exposure less this expectation is the component associated with the settlers who arrived. After the main specification controls, the land component predicts 85% of the variation in exposure used by the headline regression. Panel C of Table C2 gives each component its own coefficient, in units of one SD of headline agro-climatic similarity. The modern employment share rises by +0.065 with site land and +0.070 with settler origins; the randomization tests for the origin component give $p = 0.090$ within province and wave and 0.116 within province. The agricultural employment share falls with site land. Its origin coefficient is −0.016, with a 90% interval from −0.086 to 0.055 that includes the headline estimate. The differences between the two components' coefficients are imprecisely estimated for the four sectoral outcomes: log agricultural employment, the agricultural employment share, log modern employment and the modern employment share.

**Productivity and 1980 outcomes by source.** In the rice-productivity specification of Bazzi et al. (2016), the land coefficient is positive (+0.346), whereas the origin coefficient is negative (−0.169; Panel A of Table C2). The randomization test on the origin component gives $p = 0.149$ within province and wave and 0.218 within province. This exercise does not establish a positive productivity effect of the origin component. That component has 0.406 times the standard deviation of the land component on the rice sample, which limits precision.

The origin component is associated with a lower 1980 share of residents born in another province (−0.129 per SD; Panel B of Table C2). Its randomization $p$-value is 0.015 within province and wave and 0.072 within province. The remaining 1980 outcomes do not reject under either origin reassignment. Because 127 transmigration villages were established by 1980, birthplace composition can already reflect the program. These tests do not distinguish prior sorting from early program effects and do not establish balance of the origin component.

**Design-based recentering.** The exposure measure combines the match between arriving settler cohorts and settlement sites with the sites' location and agro-climate. A city's expected exposure reflects the similarity of those sites to the origins from which settlers were drawn. Borusyak & Hull (2023) show how controlling for expected exposure or subtracting it can address this nonrandom component under a valid model of assignment.

For the whole-match reshuffle, the sites, their locations, when each was settled, how many settlers it received, and which sites lie within each city's travel-time catchment all stay where the program put them. I reshuffle the village-level agro-climatic similarity across the sites of the same province and five-year-plan (Repelita) settlement wave, rebuild each

Table C1: Headline Estimates Controlling for the 1980 Birthplace Composition

| | Main specification, as reported (1) | Main specification, same cities as (3) (2) | Adding the 1980 birthplace shares (3) |
|---|---|---|---|
| *Panel A: Urban growth and structural transformation* | | | |
| Log population | 0.051 | 0.051 | 0.053 |
| | (0.059) | (0.059) | (0.049) |
| *p-value* | 0.389 | 0.389 | 0.274 |
| Log employment | 0.024 | 0.024 | 0.027 |
| | (0.057) | (0.057) | (0.048) |
| *p-value* | 0.674 | 0.674 | 0.583 |
| Unemployment rate | 0.008 | 0.008 | 0.009 |
| | (0.007) | (0.007) | (0.006) |
| *p-value* | 0.216 | 0.216 | 0.158 |
| Log agricultural employment | -0.166* | -0.166* | -0.161** |
| | (0.093) | (0.093) | (0.078) |
| *p-value* | 0.075 | 0.075 | 0.039 |
| Agricultural employment share | -0.051*** | -0.051*** | -0.051*** |
| | (0.018) | (0.018) | (0.018) |
| *p-value* | 0.006 | 0.006 | 0.005 |
| Log modern employment | 0.158** | 0.158** | 0.161*** |
| | (0.062) | (0.062) | (0.060) |
| *p-value* | 0.011 | 0.011 | 0.007 |
| Modern employment share | 0.066*** | 0.066*** | 0.066*** |
| | (0.015) | (0.015) | (0.014) |
| *p-value* | 0.000 | 0.000 | 0.000 |
| Observations | 134 | 134 | 134 |
| *Panel B: 2010 residents by birthplace* | | | |
| Born in the same district | -0.041* | -0.041* | -0.045*** |
| | (0.024) | (0.024) | (0.015) |
| *p-value* | 0.081 | 0.081 | 0.004 |
| Born elsewhere in the same province | 0.042** | 0.042** | 0.043** |
| | (0.018) | (0.018) | (0.018) |
| *p-value* | 0.019 | 0.019 | 0.016 |
| Born in another outer-island province | 0.018 | 0.018 | 0.015 |
| | (0.021) | (0.021) | (0.013) |
| *p-value* | 0.404 | 0.404 | 0.224 |
| Born on Java or Bali | -0.018* | -0.018* | -0.014 |
| | (0.011) | (0.011) | (0.009) |
| *p-value* | 0.085 | 0.085 | 0.129 |
| Observations | 134 | 134 | 134 |

*Notes:* Each cell is one regression on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200 km. Beneath each coefficient are its standard error in parentheses and the two-sided Conley $p$-value, shaded by significance level. Column (3) adds two of the three 1980 census birthplace shares of the city's population (born in another outer-island province, born on Java or Bali; born in this province is the reference), and column (2) repeats the main specification on column (3)'s cities, those with the 1980 shares measured. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C1: Headline Estimates Controlling for the 1980 Birthplace Composition, Continued

*Notes, continued:* Panel A gives the seven 2010 census outcomes of Table 4 and Panel B the four birthplace shares of Figure 13, whose 2010 shares sum to one over each city's cohort born 1955 to 1990; shares and the unemployment rate are fractions.

Table C2: Village Rice Yield and City Outcomes by Source of Agro-Climatic Similarity

| | Component | | | Randomization $p$, origins | |
|---|---|---|---|---|---|
| | Site land | Settler origins | $p$-value, equal effects | Province and wave | Province |
| *Panel A: Village log rice yield* | | | | | |
| Bartlett 150 km | 0.346*** | -0.169* | | | |
| | (0.112) | (0.102) | | | |
| HC1 | 0.346*** | -0.169 | | 0.149 | 0.218 |
| | (0.078) | (0.120) | | | |
| District clusters | 0.346*** | -0.169 | | | |
| | (0.123) | (0.103) | | | |
| Villages | 600 | | | | |
| *Panel B: 1980 city outcomes* | | | | | |
| Log population | -0.229 | 0.755* | | 0.192 | 0.175 |
| | (0.179) | (0.436) | | | |
| Log employment | -0.246 | 0.704* | | 0.200 | 0.243 |
| | (0.168) | (0.419) | | | |
| Unemployment rate | -0.000 | -0.003 | | 0.536 | 0.273 |
| | (0.003) | (0.006) | | | |
| Log agricultural employment | -0.274 | 0.405 | | 0.509 | 0.484 |
| | (0.194) | (0.409) | | | |
| Agricultural employment share | -0.015 | -0.102 | | 0.275 | 0.247 |
| | (0.050) | (0.090) | | | |
| Log modern employment | -0.295 | 0.854 | | 0.161 | 0.195 |
| | (0.249) | (0.574) | | | |
| Modern employment share | 0.005 | 0.061 | | 0.517 | 0.607 |
| | (0.043) | (0.083) | | | |
| Log born in same province | -0.257 | 0.939** | | 0.117 | 0.134 |
| | (0.176) | (0.437) | | | |
| Log born in other province | -0.241 | 0.241 | | 0.829 | 0.619 |
| | (0.350) | (0.684) | | | |
| Share born in other province | 0.017 | -0.129*** | | 0.015 | 0.072 |
| | (0.024) | (0.045) | | | |
| Cities | 134 | | | | |

*Notes:* Each coefficient row is one regression with both components of agro-climatic similarity, its standard errors in parentheses beneath. The site-land component is the expected similarity when the origin composition of each settlement site's settlers is re-drawn 2,000 times among the sites settled in the same province and five-year plan, using the origin-destination similarity grid of Bazzi et al. (2016), with each site keeping its location and its land. The settler-origin component is realized similarity minus that expectation, so the two components sum to the similarity measure. Panel A: the outcome is log rice yield in the 600 villages of the original rice-productivity sample in Bazzi et al. (2016), Table 3, column 1. Both components enter with the original site controls and island fixed effects, in units of one SD of village agro-climatic similarity on that sample, and the three rows give Bartlett spatial standard errors within 150 km, heteroskedasticity-robust HC1 standard errors and standard errors clustered by district. The replication classifies the other 214 transmigration villages as having no rice production and does not report their rice yield. Panel B: 1980 census outcomes of the 134 cities of the main city sample, with the settler count and the 16 agro-climatic controls and without the four 1980 baseline controls, because the outcomes are dated 1980, in units of one SD of headline agro-similarity, with HC1 standard errors. The randomization $p$-values are for the settler-origin coefficient: the share of 2,000 re-draws of settler origins, within province and settlement wave or within province, whose heteroskedasticity-robust $t$ lies at least as far from the re-draws' mean as the observed one. In Panel A they stand on the HC1 row, and the re-draws reassign origins across all linked transmigration villages before restricting to the rice sample. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C2: Village Rice Yield and City Outcomes by Source of Agro-Climatic Similarity, Continued

| | Component | | | Randomization $p$, origins | |
|---|---|---|---|---|---|
| | Site land | Settler origins | $p$-value, equal effects | Province and wave | Province |
| *Panel C: 2010 city outcomes* | | | | | |
| Log population | 0.136 | -0.263 | 0.094 | 0.249 | 0.575 |
| | (0.092) | (0.168) | | | |
| Log employment | 0.095 | -0.238 | 0.185 | 0.379 | 0.407 |
| | (0.096) | (0.171) | | | |
| Unemployment rate | 0.017** | -0.022* | 0.007 | 0.389 | 0.730 |
| | (0.007) | (0.013) | | | |
| Log agricultural employment | -0.130 | -0.298 | 0.522 | 0.198 | 0.269 |
| | (0.122) | (0.196) | | | |
| Agricultural employment share | -0.060*** | -0.016 | 0.259 | 0.728 | 0.651 |
| | (0.021) | (0.036) | | | |
| Log modern employment | 0.228** | -0.100 | 0.188 | 0.742 | 0.794 |
| | (0.093) | (0.187) | | | |
| Modern employment share | 0.065*** | 0.070*** | 0.809 | 0.090 | 0.116 |
| | (0.017) | (0.020) | | | |
| Cities | 134 | | | | |

*Notes, continued:* Panel C: 2010 population census outcomes of the main city sample, in the main specification with the full set of controls and Conley SE within 200 km, in units of one SD of headline agro-similarity. Where the two effects are equal each coefficient equals the headline coefficient, and the equal-effects $p$-value tests that they are equal. The randomization $p$-values are computed as in Panel B. Modern-sector employment is manufacturing and services, and shares and the unemployment rate are fractions. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

city's unweighted 300-minute catchment mean under each of $S = 2{,}000$ reshuffles, and average them:

$$z_c = \frac{1}{|V_c|} \sum_{v \in V_c} AS_v, \qquad \mu_c = \frac{1}{S} \sum_{s=1}^{S} z_c^{(s)}, \tag{17}$$

where $V_c$ is the set of transmigration villages within 300 minutes of city $c$, $AS_v$ is the agro-climatic similarity of village $v$, and $z_c^{(s)}$ is the city's exposure under draw $s$. I then either control for the expected exposure or subtract it from the treatment:

$$\begin{aligned} y_c &= \beta\, z_c + \gamma_P\, \mu_c^P + \gamma_{PW}\, \mu_c^{PW} + X_c'\zeta + \varepsilon_c, \\ y_c &= \tilde{\beta}\left(z_c - \mu_c^{PW}\right) + X_c'\zeta + \varepsilon_c, \end{aligned} \tag{18}$$

where $X_c$ holds the main specification controls, $\mu_c^{PW}$ and $\mu_c^P$ are the expected exposures under whole-match reshuffles within province-by-wave cells and within provinces, and exposure is standardized. I report the control form with both expectations as the double-robust specification. It estimates the coefficients on realized exposure and both expectations separately. Subtracting the province-by-wave expectation constrains its coefficient to equal the negative of the realized-exposure coefficient and omits the province-only expectation before standardization. The regressions on separately standardized recentered exposure also use a different exposure unit. In Figure C1, I instrument headline exposure with its recentered value so that every series uses one SD of headline exposure. These regressions do not establish the validity of the reassignment schemes or of the identification assumptions in Section 4.3.

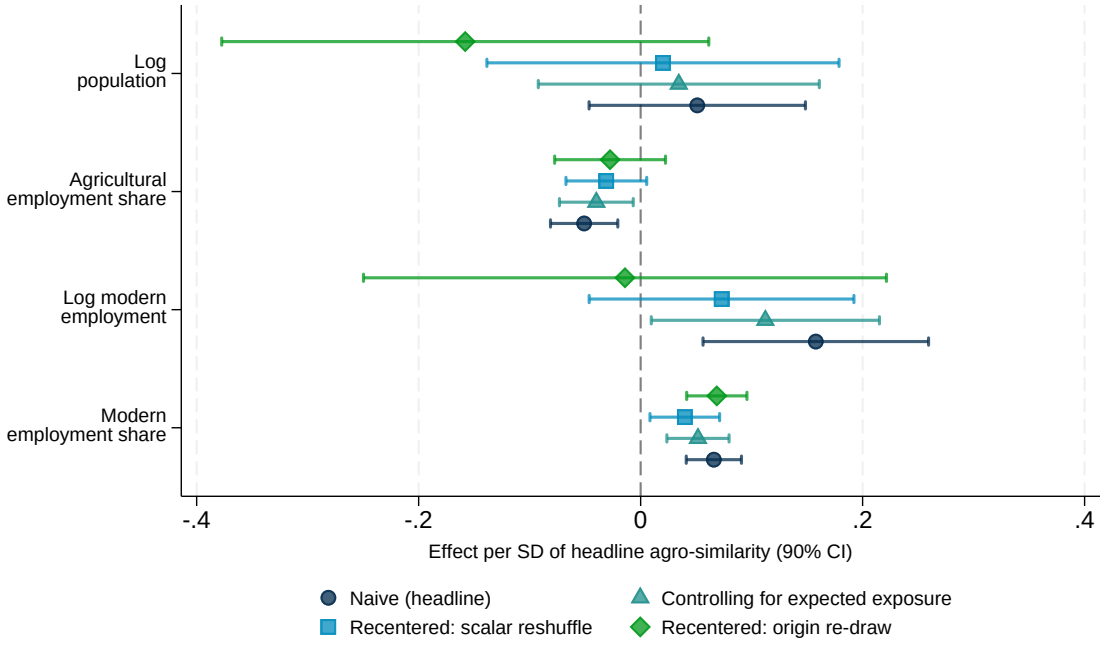
For the common-unit figure, let $Z_c$ denote standardized headline exposure and $R_c^j$ the standardized realized-minus-expected exposure under reassignment $j$, either the whole-match reshuffle or the origin redraw. Each difference is formed on the scale of the corresponding assignment calculation, then standardized across cities with observed values before restricting to the main sample. The first stage and the instrumental-variables estimator are

$$Z_c = \alpha_j + \pi_j R_c^j + X_c'\delta_j + u_c^j, \qquad \widehat{\beta}_j^{IV} = \frac{\sum_c \ddot{R}_c^j \ddot{y}_c}{\sum_c \ddot{R}_c^j \ddot{Z}_c}, \tag{19}$$

where dots denote each variable less its fitted value from a regression on $X_c$ and a constant. The controls are the inverse-travel-time-weighted settler count, four 1980 city baselines, and 16 agro-climatic means. On the same 134 cities as the figure, the first-stage coefficient is +0.518 (Conley SE 0.067) for the whole-match reshuffle and +0.397 (SE 0.067) for the origin redraw. Their partial $R^2$ values are 0.525 and 0.290. These measure association after conditioning on the controls. The Online Supplement reports the conditional standard deviations and alternative standard errors; these diagnostics do not establish instrument validity or supply a weak-instrument-robust confidence set.

The structural-transformation results hold under the double-robust form (Table A1). The agricultural employment share falls by 0.040 and the modern employment share rises by 0.052, both statistically significant, and log modern-sector employment rises by 0.112, marginally significant; the two share estimates are about four fifths of the headline estimates. Subtracting the expected exposure instead removes most of the variation in the treatment, because a city's realized and expected exposure move closely together: they correlate at 0.75 under the whole-match reshuffle and 0.96 under the origin redraw. In the headline exposure units, the modern employment-share estimate is positive under both recenterings, with wider intervals for the other outcomes (Figure C1).

Figure C1: Headline Estimates Under Design-Based Recentering

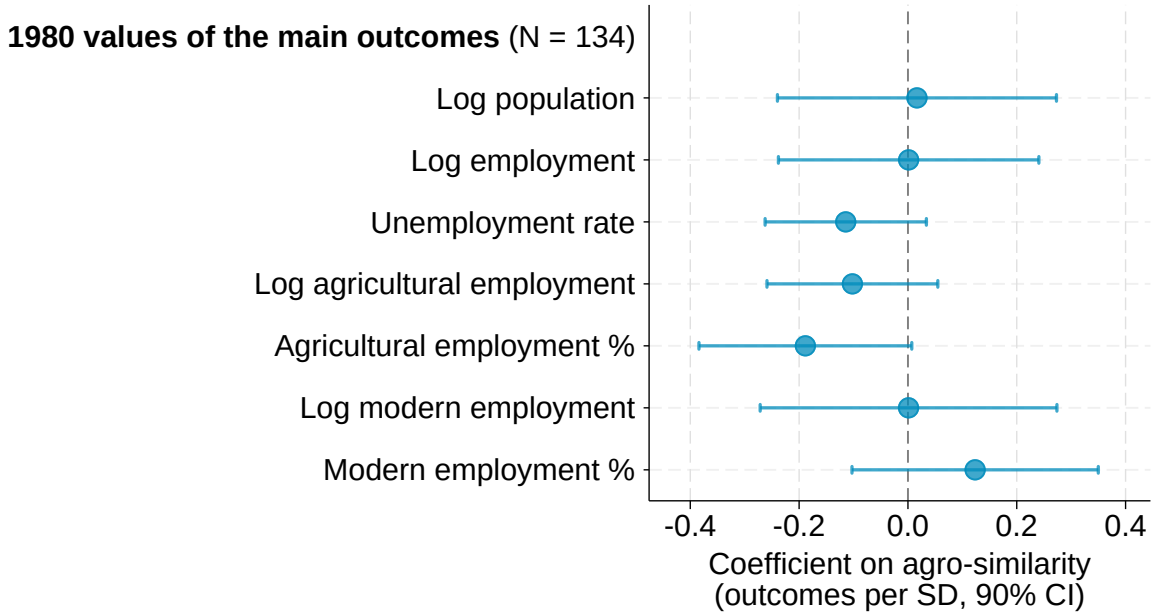



*Notes:* Rows are 2010 census outcomes; the modern sector is manufacturing plus services. Each marker is an effect per SD of headline agro-climatic similarity, from one specification on the main city sample, in the main specification with the full set of controls and Conley SE within 200 km; bars are 90% confidence intervals from the same standard errors. "Naive" is the headline specification. "Controlling for expected exposure" adds the expected exposure under the province and the province-by-settlement-wave reshuffles as controls (Table A1). "Scalar reshuffle" recenters the exposure against a permutation of the village match within province-by-settlement-wave cells. "Origin redraw" recenters against a permutation that holds each site's destination agro-climate fixed and redraws only the settler-origin composition, using the origin-destination similarity grid of Bazzi et al. (2016). Each recentered series instruments headline exposure with exposure less its average over 2,000 permutations, following Borusyak & Hull (2023). The Online Supplement reports the first-stage diagnostics and regressions on separately standardized recentered exposures.

The same 2,000 whole-match reshuffles provide a simulation-based assignment test for the double-robust specification: I reestimate the regression under each reshuffle and ask how often a placebo $t$-statistic lies at least as far from the placebo average as the observed one. The test rejects the hypothesis of no effect on the modern employment share ($p = 0.050$ with reshuffles within province and wave, and 0.053 within province) and does not reject it for the agricultural share ($p = 0.196$) or for log modern-sector employment ($p = 0.181$). The test compares $t$-statistics because, once its

expectation and the controls are taken out, a reshuffled exposure has about half the standard deviation of the realized exposure remaining after those controls, so a coefficient estimated on it is spread more widely than the observed coefficient; dividing each by its own standard error puts the observed and the placebo estimates on one scale. The Online Supplement reports every randomization-inference $p$-value, the coefficient form included.

Figure C2: Baseline Balance: Exposure From Villages Established in 1981 or Later




*Notes:* Each row regresses a standardized 1980 urban outcome on standardized agro-climatic similarity. Each outcome is standardized within the main city sample; agro-climatic similarity is standardized over cities with a measured exposure. Exposure, the inverse-travel-time-weighted settler count, and the unweighted means of all 16 agro-climatic controls are reconstructed using only villages established in 1981 or later within 300 travel minutes. All rows retain the main sample of 134 cities used in Figure 11. Bars are 90% confidence intervals based on Conley standard errors allowing spatial correlation within 200 kilometers. Modern employment combines manufacturing and services. The outcomes use 1980 census data; village characteristics and establishment years come from the transmigration data of Bazzi et al. (2016).

**Baseline placebo test, in levels.** Most estimated associations between exposure and the cities' 1980 outcomes are imprecise. I regress ten baseline outcomes, urban center-level aggregates of individual 1980 census records, on village agro-climatic similarity. The two exposure constructions use the unweighted mean and population×inverse-travel-time weighting within 300 minutes, each over all 814 transmigration villages and again over the 687 settled after the census. The right-hand block of Table C3 also rebuilds every village-derived control over those later villages. One of the forty estimates is marginally significant. These comparisons assess observed baseline differences; they do not establish the absence of prior sorting.

**Main estimates from the later villages.** Exposure built from the 687 villages established after the 1980 census correlates at 0.99 with the main exposure across the main city sample. With that exposure, and with the settler count and the agro-climatic covariates rebuilt from the same villages, the coefficient is −0.050 (standard error 0.019) on the agricultural employment share, 0.164 (0.062) on log modern-sector employment and 0.063 (0.017) on the modern employment share, each statistically significant and close to its value in the main specification (Table A1).

**Controlling for the 1980 birthplace composition directly.** The compositional headline does not depend on the 1980 birthplace composition of the population. With the 1980 birthplace shares added to the main specification, the coefficients on the agricultural and modern employment shares are nearly unchanged and remain statistically

Table C3: Placebo Baseline Test: 1980 Outcomes Under Two Exposure Constructions and Two Village Sets

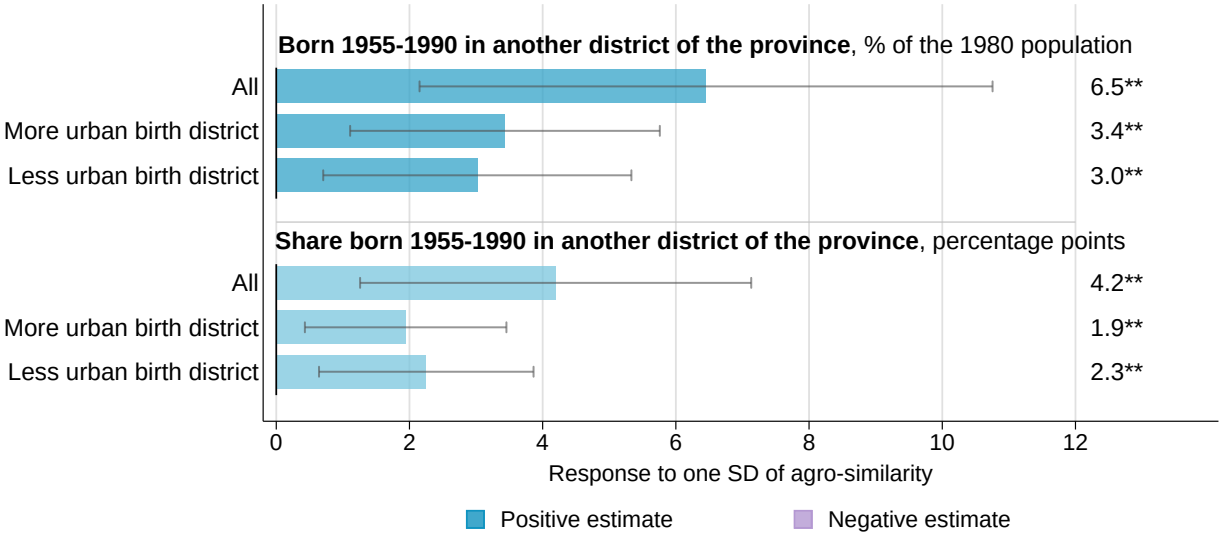
| | **Outcome, main city sample** | | | **All transmigration villages** | | **Villages settled after the 1980 census** | |
|---|---|---|---|---|---|---|---|
| 1980 baseline outcome | Mean | SD | *N* | Simple mean | Settler weighted | Simple mean | Settler weighted |
| Log population | 11.093 | 0.99 | 134 | -0.019 | -0.022 | +0.016 | -0.109 |
| | | | | (0.122) | (0.166) | (0.154) | (0.165) |
| Log employment | 9.846 | 0.96 | 134 | -0.043 | -0.042 | +0.001 | -0.118 |
| | | | | (0.106) | (0.151) | (0.140) | (0.154) |
| Unemployment rate | 0.017 | 0.02 | 134 | -0.001 | -0.002 | -0.002 | -0.003* |
| | | | | (0.002) | (0.002) | (0.002) | (0.002) |
| Log agricultural employment | 9.249 | 1.06 | 134 | -0.129 | -0.092 | -0.108 | -0.171 |
| | | | | (0.099) | (0.107) | (0.101) | (0.109) |
| Agricultural employment share | 0.560 | 0.22 | 134 | -0.033 | -0.021 | -0.041 | -0.020 |
| | | | | (0.033) | (0.032) | (0.026) | (0.028) |
| Log modern employment | 8.801 | 1.31 | 134 | -0.050 | -0.052 | +0.001 | -0.149 |
| | | | | (0.185) | (0.237) | (0.218) | (0.241) |
| Modern employment share | 0.383 | 0.20 | 134 | +0.017 | +0.007 | +0.024 | +0.005 |
| | | | | (0.031) | (0.031) | (0.027) | (0.029) |
| Log born same province | 10.934 | 1.00 | 134 | -0.002 | -0.008 | +0.033 | -0.084 |
| | | | | (0.132) | (0.175) | (0.164) | (0.172) |
| Log born other province | 8.342 | 1.88 | 134 | -0.139 | -0.125 | -0.103 | -0.267 |
| | | | | (0.211) | (0.235) | (0.225) | (0.264) |
| Born other province, share | 0.134 | 0.14 | 134 | -0.014 | -0.011 | -0.014 | -0.020 |
| | | | | (0.021) | (0.015) | (0.019) | (0.017) |
| *# significant / 10* | | | | *0* | *0* | *0* | *1* |

*Notes:* Each cell is the coefficient from a separate placebo regression of a 1980 census outcome on one standardized construction of agro-similarity, with the settler count and 16 agro-climatic covariates within 300 minutes as controls, all built over the village set of the block; Conley SE within 200 km in parentheses. Each column estimates on the main city sample where that construction and that outcome are measured, 134 urban centers. The ten outcomes are the seven headline outcomes at baseline (modern employment is manufacturing plus services) and three rows on province of birth, the finest level the 1980 census records. Mean, standard deviation and number of urban centers describe each outcome on the sample of the first coefficient column. In each village set, the first column is the exposure of the main specification, the simple mean of village agro-similarity within 300 travel minutes, and the second weights each village by its settler count times its inverse travel time. The left block builds every construction and control over all 814 transmigration villages; the right block rebuilds them over the 687 villages settled after the 1980 census (the other 127, placed by 1980, held 19.5% of all settlers). The last row counts the coefficients with at least one star in each column. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

significant, and the 2010 birthplace pattern holds, with the locally born share still falling and the same-province share still rising (Table C1).

**Rural and urban birth districts.** The census records the district of birth, so the only way the data could separate moves from rural areas from moves between urban areas is the urban share of that district. I split the residents born in another district of the same province by whether the 1980 census urban share of their birth district is above or below that of the median district of their province (Figure C3). Both groups rise significantly with exposure, by similar amounts. Both being significant does not mean that rural and urban origins respond alike: a district's urban share follows the census definition and describes the whole district, and migrants may come from any part of it. The split therefore does not distinguish rural-to-urban from urban-to-urban migration.

Figure C3: Same-Province Migrants by the Urban Share of Their Birth District

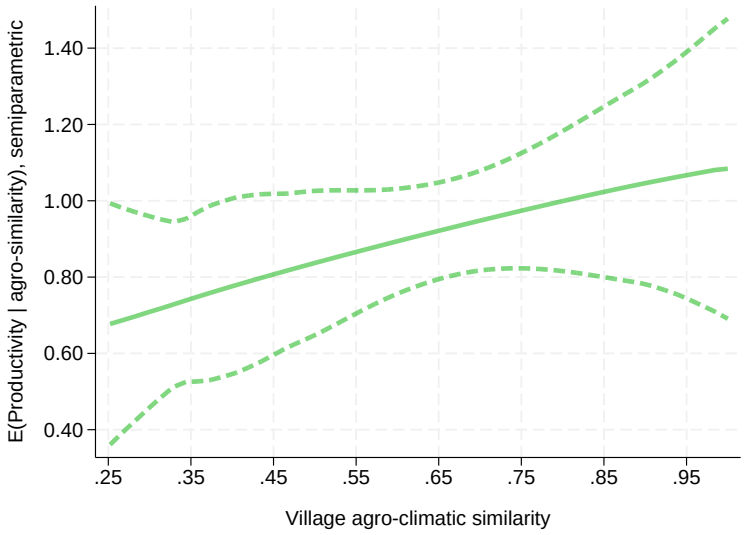



*Notes:* Each bar is one regression of the row's outcome on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200km. Whiskers are 90% confidence intervals. Rows: the city's residents born 1955 to 1990 in another district of the same province, all and split by whether the 1980 census urban share of their birth district is above the median district of their province. First block: head counts in % of the city's 1980 population. Second block: shares of the city's residents born 1955 to 1990, in percentage points. In each block the two groups add up to *All*, which equals the same-province bars of Figure 13. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## C.2 Village-Level Analysis

Across the transmigration villages, agricultural productivity is positively associated with agro-climatic similarity, but the slopes are imprecisely estimated with province clustering; the raw full-range estimate is marginally significant. This association supports the relevance of the overall index and does not establish the validity of city-level exposure or the productivity effect of its origin component. The productivity measure is the agricultural productivity index of Bazzi et al. (2016): the yield per hectare of each crop the village grows is divided by that crop's average yield across all Indonesian villages that grow it, and the index is the geometric mean of these ratios across crops, weighted by each crop's share of the village's crop revenue at national prices. A village whose crops all yield the national average scores one, and the median transmigration village scores 0.43. The index comes from the 2003 village census (PODES), which records the 2001-2002 growing season, and it is observed for 770 of the 814 transmigration villages, while agro-similarity is defined for all 814. The notes of Figure C4 report the linear estimates with raw and winsorized productivity, over the plotted and full ranges.

Figure C4: Village-Level Relevance: Agricultural Productivity Against Agro-Climatic Similarity




*Notes:* Local linear fit (Epanechnikov kernel, bandwidth 0.30) of village agricultural productivity on agro-climatic similarity, an index from 0 to 1, across the 768 transmigration villages with observed values of the productivity index and a similarity between 0.25 and 1; the solid line is the fit and the dashed lines its pointwise 90% confidence band, which treats villages as independent. Productivity is the agricultural productivity index of Bazzi et al. (2016) from the 2003 village census (PODES), the revenue-weighted geometric mean across crops of the village's yield per hectare relative to the crop's national average yield, so a value of one means every crop yields the national average; it is winsorized at the 1st and 99th percentiles. The linear slope of productivity on the similarity index, its only regressor, is 0.525 (standard error 0.395) on these villages and 0.767 (0.447) with raw productivity, and over the full similarity range of 770 villages it is 0.535 (0.366) winsorized and 0.772 (0.419) raw. Standard errors are clustered by province, with the number of province groups minus one as the degrees of freedom, and only the raw full-range slope is marginally significant.

# D Robustness

This part reestimates the headline results under alternative constructions of the exposure measure and alternative specifications.

The two village curves of Figure 14 are drawn again unweighted at a bandwidth of 0.30, and with each village weighted by its 2000 population at bandwidths of 0.25 and 0.30 (Figure D1). In a regression of the village agricultural share on agro-similarity and its square, with the same controls on the same 808 villages, the squared term is +0.767 (Conley SE 0.189 within 200 km), significant under those standard errors and insignificant under a wild cluster bootstrap over the four islands ($p = 0.142$).

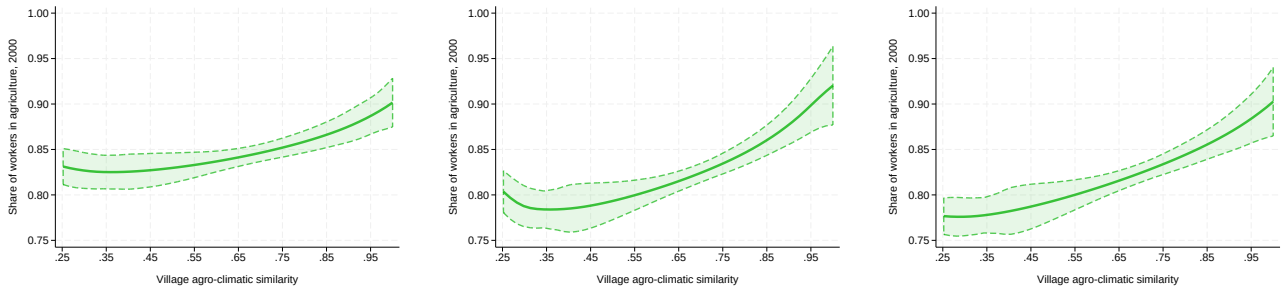
## D.1 Settlement Size Relative to the Pre-Existing Rural Population

Settlers placed in rural subdistricts within 100 km of a city amount to about 7% of those subdistricts' 1980 population on average and 4% at the median. All subdistricts belonging to an urban center are excluded from this rural area. Distance is the shortest great-circle distance from the rural subdistrict's centroid to the centroids of the city's member subdistricts, providing a fixed geographic measure of the local scale of resettlement. Among the 134 cities, 22 have settler shares of at least 15%. I add this share to the main specification to assess whether the sectoral results reflect differences in the scale of resettlement relative to the existing rural population (Table A1).

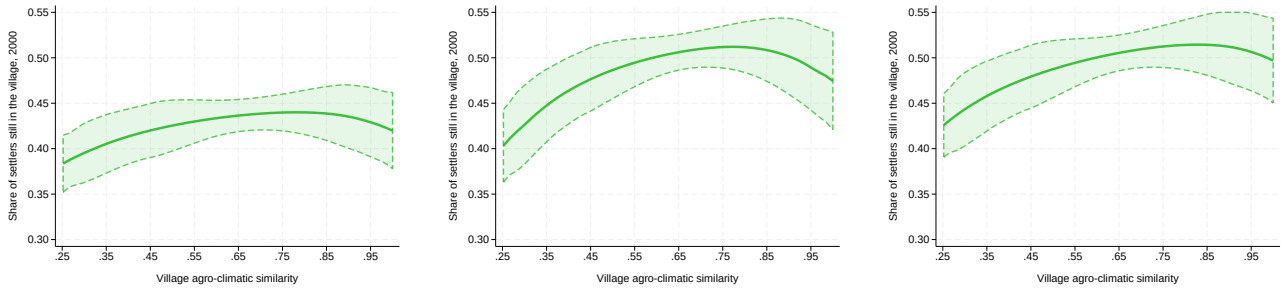
With this additional control, the agricultural employment-share response remains negative and statistically significant, while the modern-sector employment level and share responses remain positive and significant. The agricultural em-

Figure D1: The Village Agricultural Share and Settler Stay-Share at Bandwidth 0.30 and With Population Weights

(a) Unweighted, 0.30 (b) Population Weights, 0.25 (c) Population Weights, 0.30




(d) Unweighted, 0.30 (e) Population Weights, 0.25 (f) Population Weights, 0.30



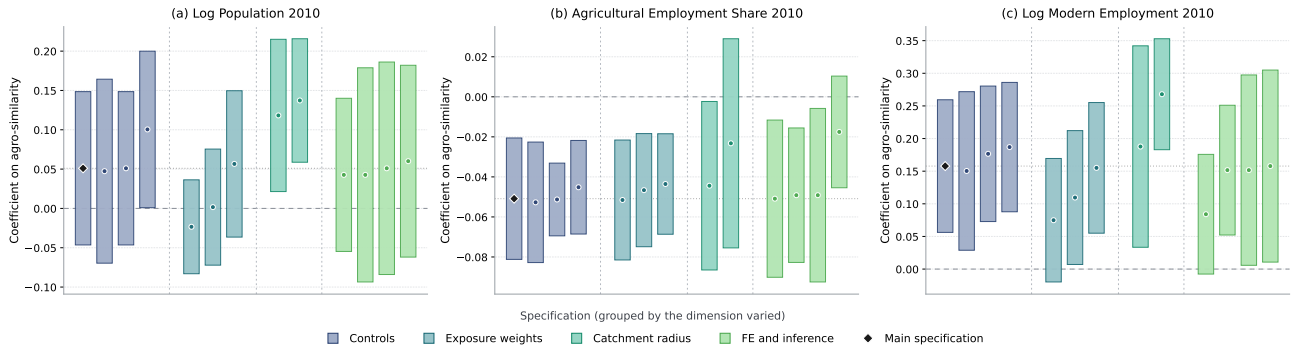

*Notes:* Semiparametric local linear fits (Epanechnikov kernel) of the share of a village's workers in agriculture in the 2000 Population Census (top row) and of the settler stay-share (bottom row) on the village's agro-similarity, partialing out the village controls of Figure 14 on the same 808 transmigration villages: unweighted at a bandwidth of 0.30 (left), and with each village weighted by its 2000 population at bandwidths of 0.25 (middle) and 0.30 (right), where the weighted fits also remove the controls with the Epanechnikov kernel. Each band is a pointwise 90% confidence band that treats the villages as independent. The axes are those of Figure 14; a band that leaves its axis is cut at the axis limit, here in the population-weighted stay-share at 0.30 (band up to 0.552).

ployment coefficient remains negative and marginally significant; the population, total employment, and unemployment-rate coefficients remain insignificant. The main specification already controls for the number of settlers placed near each city using the travel-time window, while this additional control measures settlement relative to rural population within the fixed geographic radius. The denominator includes some early arrivals because some villages were established before the 1980 census. This check addresses settlement scale; it does not establish exogeneity of village placement.

## D.2 The Specification Curve and Price-Measure Robustness

**Specification variants on the main city sample.** The main-sample table and grouped curve vary controls, village weights, wider exposure windows, and geographic fixed effects (Table 5 and Figure D2). The 400- and 500-minute rows retain the main cities and their 300-minute controls. The separate nested-window figure rebuilds the sample, exposure, and village controls over windows from 250 to 500 minutes (Figure D5). At the neighboring 250- and 400-minute windows, the agricultural-share and modern-employment directions agree with the main estimates, but their coefficients are imprecise. The sensitivity summary refers to the displayed specifications. The weighting checks compare equal village weights within 300 minutes with settler weights and inverse travel time; the heatmap also includes their combinations (Figure D3). The agricultural-share estimates remain negative. Settler weights give a similar modern-employment estimate, while inverse-time weights reduce its size and significance. Province fixed effects also reduce both coefficients and leave them statistically insignificant, whereas island fixed effects preserve the sectoral results (Table 5).

Figure D2: Specification Curve by the Dimension Varied




*Notes:* Each panel is one 2010 population census outcome. Each marker is the coefficient on standardized agro-similarity under one of 13 specifications on the main city sample, grouped by controls, exposure weights, catchment radius, and fixed effects and inference (Table D1). The diamond marks the main specification with the full set of controls and Conley SE within 200 km; the 400- and 500-minute catchments widen the exposure and keep the controls at 300 minutes. Within a group, markers are ordered by the estimate, with the main specification first in the controls group; shaded bars are 90% confidence intervals. Conley intervals use normal critical values; the two province-clustered variants use Student $t$ critical values with the number of provinces minus one as the degrees of freedom. The dotted horizontal line marks the main estimate. For log population the own-outcome baseline coincides with the main specification, whose controls already include the 1980 population.

The heatmap reports 31 specifications for 17 outcomes (Figure D3). The agricultural employment share is negative and the modern-sector share positive across nearly all displayed specifications; population and price estimates are less precise. Rows vary exposure weights, controls, travel-time windows, and island fixed effects. The heatmap excludes province fixed effects and province-clustered inference, which are reported in the main-sample robustness table. Province fixed effects reduce the sectoral coefficients and leave them insignificant. With province-clustered standard errors, the sectoral estimates remain significant or marginally significant.

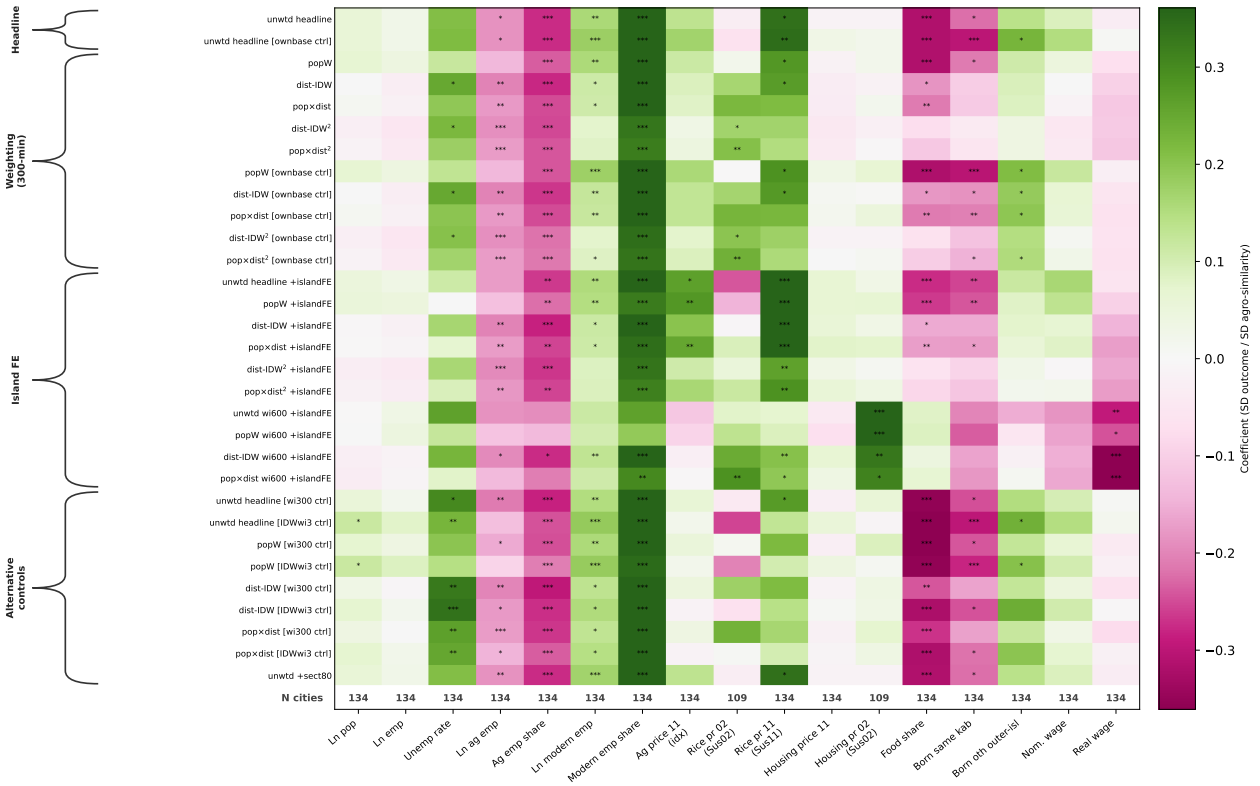
**Robustness of the urban price result across measures.** The urban food-price response is small, negative and imprecise under every measure of the local price. Figure D4 reestimates the headline main specification reduced form with the dependent variable swapped across 5 1996 price measures: the broad agricultural-goods index under national (Laspeyres), city (Paasche), and equal-weight aggregation, the equal-weight crops index, and a rice unit value. The agro-similarity coefficient on the log price runs from $-0.028$ to $-0.004$ across the 5 measures, and none of them is statistically significant, in logs or in index levels. The four indices are price relatives with a median near one and enter as the log of one plus the index, so their coefficients read as about half of a proportional change; the rice unit value is a price in rupiah per kilogram, and its coefficient reads as a proportional change.

The price indices and rice unit value are correlated across cities (pairwise log correlations of 0.76 to 0.94 among the national and city agricultural indices and the rice unit value, and as low as 0.42 for a pair that includes an equal-weight index). Traded food prices are spatially compressed (coefficient of variation $\approx 0.11$) relative to the non-traded housing benchmark (0.78).

The 1996 survey also splits rice into a local and a high-quality variety. Per standard deviation of agro-similarity, the high-quality share of rice spending is 19.3 percentage points higher (standard error 6.1), from a mean of 26.8%, and the share of rice buyers who buy any of it is 18.2 points higher (standard error 6.3), from 26.2%, on 112 cities (Table S21 of the Online Supplement).

Market rice prices (the World Bank's district-level market-price panel, matched to urban centers by market coordinates) do not move with agro-similarity in 2011 levels; the estimate is statistically insignificant. In the 2011 household survey, households near productive transmigration villages buy 4.2% fewer kilograms of rice per person, a

Figure D3: Specification-Curve Heatmap: Constructions and Controls




*Notes:* Each cell is a coefficient in outcome SD per SD of agro-similarity. Each exposure construction is standardized across all cities where it is measured. Outcome SDs use measured outcomes among cities within 300 travel minutes, before restricting to complete baseline data. Each column's regression sample is fixed across rows within the main city sample; its size is below the column. Row labels name the village weights of agro-similarity, unwtd for equal weights (the main specification), popW for settler weights, dist-IDW and dist-IDW$^2$ for inverse travel time and its square, and pop×dist and pop×dist$^2$ for settlers times those weights; wi600 builds exposure within 600 minutes and +islandFE adds island fixed effects; [ownbase ctrl] adds the 1980 agricultural employment share, log modern employment, or modern employment share for the corresponding outcome. For unemployment, birthplace, and household-survey outcomes it adds the 1980 agricultural share and log population density. Population, total employment, and agricultural employment already have their own baselines; [wi300 ctrl] uses the unweighted settler count within 300 minutes and [IDWwi3 ctrl] the 16 covariates weighted by inverse travel time; +sect80 adds the 1980 agricultural or modern-sector share to the matching sectoral outcomes. In the column labels, 02 and 11 are the 2002 and 2011 Susenas waves, idx an index, kab the district and outer-isl another outer-island province. All rows use Conley SE within 200 km. Four island fixed-effects rows use exposure within 600 minutes. Outcomes come from the 2010 census and Susenas 2002 and 2011; birthplace shares describe residents born from 1955 to 1990. The two 2002 columns use 109 cities. Shading deepens with magnitude, one hue for positive and another for negative cells; stars use the same standard errors: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

marginally significant fall, while their spending on rice does not move, so they pay more per kilogram for less rice. Over 2002 to 2008 the rice price in the city's consumer price index does not grow faster than the index as a whole: both growth responses are statistically significant, and their difference is not.

## D.3 City Boundary Definitions and the Matched Cutoff

Three checks bound what the city definition can do to the results. Table D2 measures the cities under four definitions, each on the cities it covers, from the whole member subdistrict to the built-up core intersected with the census's own urban classification: the level of urban farming falls as the city is drawn tighter, and its 2010 mean is still about 14% under the tightest cut. Most urban centers are a combination of a few subdistricts, so I also rebuild every city from its member subdistricts whose own 1980 employment was at least one quarter, or at least one half, outside agriculture, summing the

Table D1: Specification Variants in the Grouped Specification Curve

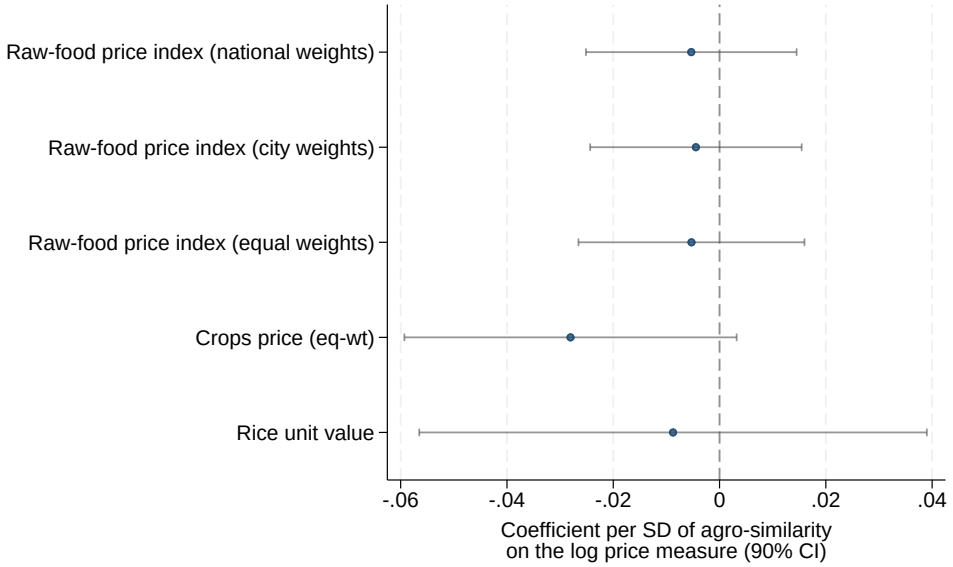
| Variant | What it changes relative to the main specification | $N$ |
|---|---|---|
| *Controls* | | |
| Main specification | The settler count as an inverse-travel-time-weighted sum over the transmigrant villages inside the 300-minute catchment, the four 1980 baselines (log population, log employment, the urban-population share and log agricultural employment), and the 16 agro-climatic covariates as unweighted means over the same villages. | 134 |
| Unweighted settler count | The settler count entered as an unweighted sum over the same villages. | 134 |
| Weighted covariates | The 16 agro-climatic covariates entered as means over the same villages weighted by inverse travel time. | 134 |
| Own-outcome baseline | Adds the outcome's 1980 value. For log population this coincides with the main specification, which already includes its 1980 value. | 134 |
| *Exposure weights* | | |
| Population-weighted | Villages enter weighted by their settler count. | 134 |
| 1/d | Villages enter weighted by the inverse of travel time, so nearer villages count for more. | 134 |
| $1/d^2$ | The same, with the inverse of squared travel time, a steeper decay. | 134 |
| *Catchment radius* | | |
| Within 400 min | The unweighted exposure mean uses villages within 400 travel minutes. The main city sample and its 300-minute controls are held fixed. | 134 |
| Within 500 min | The unweighted exposure mean uses villages within 500 travel minutes. The main city sample and its 300-minute controls are held fixed. | 134 |
| *FE and inference* | | |
| Province FE | Adds province fixed effects, so the comparison uses differences between cities within the same province. | 134 |
| Island FE | Adds island fixed effects. | 134 |
| Cluster: province | The same point estimate, with standard errors clustered at the province level instead of Conley. | 134 |
| Island FE, cluster province | Island fixed effects together with province-clustered standard errors. | 134 |

*Notes:* Each row defines one bar in each panel of Figure D2, the coefficient on standardized agro-similarity under that specification for 2010 log population, the agricultural employment share and log modern-sector (manufacturing and services) employment, from the population census. The first row is the main specification with the full set of controls and Conley SE within 200 km. The last row combines two of these changes, island fixed effects and province-clustered standard errors. $N$ is the number of urban centers, and every row estimates on the main city sample. Unclustered rows use the main specification's Conley variance estimator. The two province-clustered rows use Student $t$ intervals and tests with the number of clusters minus one degrees of freedom. In the Variant column, d is the travel time from a city to a transmigration village and FE marks fixed effects.

2010 outcomes and the four 1980 baselines over those subdistricts (Table A1). The agricultural employment share then falls by 6.2 and 7.1 percentage points per standard deviation of agro-similarity, against 5.1 for the whole city, and farm employment falls by more. Modern-sector employment rises at the quarter threshold, and its estimate loses precision at the half. The half threshold also removes old town centers, since many town subdistricts were still mostly agricultural in 1980, and 40 of the 102 cities with a 1980 urban population under the village-based allocation lose the subdistrict that held most of it. This assignment uses urban shares from the 1980 village census to distribute each parent subdistrict's urban population among its successors, preserving the parent total. The BPS-urban row of Table D2 applies the parent subdistrict's population allocation weights to its urban census workers.

Panel G of Table A1 adds to each city the rural subdistricts within 50 km, and in that region the agricultural employment share falls by 5.4 percentage points per standard deviation of agro-similarity, against 5.1 for the whole city,

Figure D4: The Effect of Agro-Similarity on Food Prices Across Price Measures, 1996




*Notes:* Each row is one regression of the log of one plus a 1996 price measure on standardized agro-similarity on the main city sample, 112 centers with available 1996 household-survey (Susenas) data for the dependent variable, in the main specification with the full set of controls and Conley SE within 200 km. Bars are 90% confidence intervals from the same standard errors. Four measures are price indices built from the city's households: the survey-weighted median unit value of each raw agricultural food item over those households, relative to its median over the households of all urban centers, aggregated under national, city and equal weights, and over the rice, corn and cassava items under equal weights. These indices have a median near one, so a coefficient on their log is about half of a proportional change. The fifth measure is the city's rice unit value, the value of the rice consumed over the kilograms consumed, each summed over its households with survey weights, and its coefficient reads as a proportional change.

while modern-sector employment and the modern share rise by less and population and employment stay imprecisely estimated. Table D3 splits the sample at the median 1980 agricultural share, 57.9% of 1980 employment, and runs the split as an interaction, so both slopes come from one regression on all 134 centers and their coefficient difference has its own standard error. The differences by initial agricultural intensity are imprecisely estimated. The agricultural-share coefficient is $-0.046$ in the less agricultural cities and $-0.056$ in the more agricultural ones, the modern-employment coefficient is 0.155 and 0.158, and no difference between them is distinguishable from zero, the smallest of the four $p$-values being 0.423. The same design on 1980 population does separate the sample. Log modern employment responds 0.152 more in the larger cities than in the smaller ones (standard error 0.055) and log population 0.150 more (standard error 0.041), both statistically significant, and the agricultural-share response does not differ detectably across that population split.

## D.4 Inference: The Conley Cutoff and Catchment Overlap

**The Conley cutoff.** The headline uses a 200-kilometer Conley cutoff. This distance enters the variance estimator and leaves the point estimates unchanged; it is distinct from the exposure window measured in travel minutes. A shorter 150-kilometer cutoff gives well-defined standard errors for all seven outcomes, but it does not uniformly increase uncertainty. The agricultural-share and modern-share standard errors are slightly larger at 150 kilometers than at 200 (0.020 against 0.018 and 0.016 against 0.015, respectively). Table D4 compares these estimates with wider distances and catchment-based clustering.

**Catchment overlap and inference at wider radii.** The relevant direction to vary the cutoff is wider, and the reason is in the construction of the treatment. Exposure is an unweighted mean over the transmigration villages

Table D2: Urban Agricultural Employment Share Under Four Definitions of the City

| Year | Definition | City mean | Pooled | Cities |
|---|---|---|---|---|
| 1980 | K whole subdistrict (as reported) | 0.560 | 0.483 | 134 |
| 1980 | U BPS-urban villages | 0.142 | 0.091 | 105 |
| 2010 | K whole subdistrict | 0.374 | 0.299 | 134 |
| 2010 | C built-up core | 0.181 | 0.113 | 130 |
| 2010 | U BPS-urban villages | 0.159 | 0.100 | 132 |
| 2010 | CU core and BPS-urban | 0.136 | 0.085 | 128 |

*Notes:* Agricultural employment share of the cities of the main city sample under four definitions of the city, from loosest to tightest: K sums every member subdistrict; C keeps the villages whose polygon meets the satellite footprint of the city; U keeps the villages the Indonesian census classifies as urban, a classification made independently of the satellite footprint; CU requires both. City mean averages the city shares, Pooled divides total agricultural employment by total employment, and Cities counts the cities measured; the data are the population censuses of 1980 (5% sample) and 2010.

inside a city's 300-minute catchment, and neighboring catchments intersect, so two cities can be built partly from the same village draws and their exposure errors are correlated at whatever distance that sharing reaches. Measured directly on the travel-time matrix the exposure itself is built from, the average city in the main city sample draws on 30 in-window villages, 80% of those villages enter more than one city's catchment, and one village enters as many as 19. Of the 8,911 city pairs in the sample, 765 share at least one village, at a median separation of 143 kilometers and a maximum of 459; 27% of those sharing pairs sit farther apart than the 200-kilometer cutoff, which assumes independence beyond it. Grouping cities into the connected components of that sharing graph, the finest partition across which no village is shared at all, gives 14 components over the main city sample, and the largest holds 75 cities: roughly half the sample is linked into a single block by shared catchments.

The uniform-kernel variance estimates require caution. At 200 kilometers, the variance matrix has five or six negative eigenvalues out of 23, depending on the outcome, so it is not positive semidefinite. Across the ladder, four of the 28 uniform-kernel coefficient variances are negative and therefore do not have real-valued standard errors. All six available standard errors at 500 kilometers are smaller than at 200. Those wider uniform-kernel rows cannot establish reliable precision. Every Bartlett cell has a standard error, and the Bartlett variance matrix at 200 kilometers is positive semidefinite; from 200 to 500 kilometers the Bartlett standard error falls by less than the uniform one in five of the six columns with both standard errors at 500 kilometers, and for the unemployment rate it does not fall.

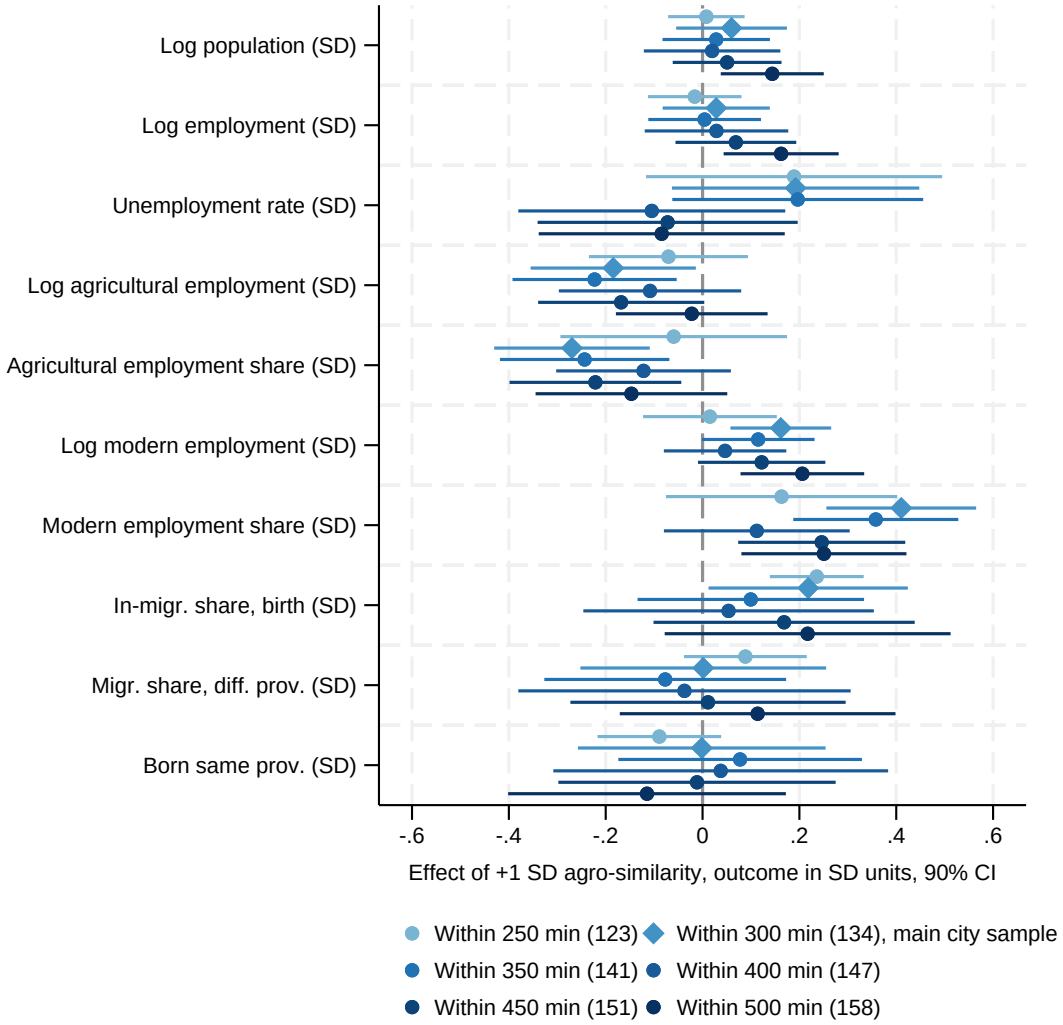
I compare inference over distances from 150 to 500 kilometers and under clustering by the catchment components (Table D4). The agricultural share, log modern employment, and the modern share are statistically significant under the Bartlett and component-clustered estimates. The wild cluster bootstrap over the components gives weaker inference: log modern employment remains significant, the modern share is marginally significant, and the agricultural share is insignificant. Population and total employment remain imprecisely estimated under these alternatives. The small number of catchment components limits the precision of this comparison.

Table D3: Exposure Effects by the City's 1980 Agricultural Share and by Its 1980 Population

| | (1) agric % | (2) ln L modern | (3) ln pop | (4) In-migrant share |
|---|---|---|---|---|
| *Panel A: Split at the median 1980 agricultural share, 57.9%* | | | | |
| Below median (less agricultural) | -0.046*** | 0.155*** | 0.055 | 0.004 |
| | (0.017) | (0.060) | (0.063) | (0.004) |
| Above median (more agricultural) | -0.056*** | 0.158** | 0.044 | 0.007*** |
| | (0.018) | (0.080) | (0.060) | (0.002) |
| Above-median indicator (=1) | 0.089*** | -0.125** | -0.008 | -0.002 |
| | (0.022) | (0.061) | (0.056) | (0.003) |
| Difference (above minus below) | -0.010 | 0.003 | -0.011 | 0.002 |
| | (0.014) | (0.046) | (0.039) | (0.003) |
| $p$-value of the difference | 0.462 | 0.940 | 0.780 | 0.423 |
| $p$-value, Bartlett kernel | 0.500 | 0.946 | 0.797 | 0.367 |
| Observations | 134 | 134 | 134 | 134 |
| *Panel B: Split at the median 1980 population, 67,979 residents, the same design* | | | | |
| Below median (smaller) | -0.047** | 0.089 | -0.017 | 0.004 |
| | (0.022) | (0.066) | (0.081) | (0.004) |
| Above median (larger) | -0.057*** | 0.241*** | 0.133** | 0.007*** |
| | (0.018) | (0.080) | (0.065) | (0.002) |
| Above-median indicator (=1) | -0.018 | 0.163* | 0.151 | 0.001 |
| | (0.018) | (0.098) | (0.098) | (0.003) |
| Difference (above minus below) | -0.010 | 0.152*** | 0.150*** | 0.003 |
| | (0.018) | (0.055) | (0.041) | (0.002) |
| $p$-value of the difference | 0.580 | 0.006 | 0.000 | 0.153 |
| $p$-value, Bartlett kernel | 0.567 | 0.013 | 0.013 | 0.201 |
| Observations | 134 | 134 | 134 | 134 |

*Notes:* Each column of each panel is one regression of a 2010 census outcome on the main city sample, in the main specification with the full set of controls and Conley SE within 200 km. Standardized agro-similarity enters interacted with an indicator for each side of the split, so the below-median and above-median rows are the two slopes and the indicator row is the shift in the outcome on the above-median side; the row labeled Difference is the above-median slope minus the below-median slope, from the same regression. Each split is cut at the sample median: 57.9% of 1980 employment in agriculture in Panel A, 67,979 residents in 1980 in Panel B. Standard errors (uniform kernel) are in parentheses; the Bartlett-kernel $p$-value tests the same difference, weighting a pair of cities by one minus its distance over the cutoff, and its cells are shaded by significance level. Modern employment is manufacturing plus services; the agricultural share is a fraction of 2010 employment, and the in-migrant share is the fraction of the 2010 population that lived in a different province five years earlier. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure D5: The Agro-Similarity Effect Across Nested Travel-Time Windows, 250 to 500 Minutes




*Notes:* Each row is one 2010 population census outcome, standardized over all urban centers so its SD stays fixed as the sample widens: the seven outcomes of Table 4 (modern-sector employment is manufacturing plus services) and the shares of residents born outside the current district, born in a different province, and born in the same province. The six points in a row are coefficients from separate regressions on standardized agro-similarity over nested catchments: the urban centers whose nearest transmigration village lies within 250 to 500 travel minutes, in 50-minute steps, with a 1980 census baseline and measured 2010 outcomes (123 to 158 urban centers, as the legend reports). At each radius the exposure, the settler count and the 16 agro-climatic covariates are rebuilt over the villages within it, and the four 1980 baselines enter as in the main specification. The 300-minute point, marked with a diamond, is the main specification on the main city sample. Bars are 90% confidence intervals from Conley SE within 200 km.

Table D4: The Headline Estimates Under a Ladder of Spatial-Inference Radii

| | Population + Employment | | | Agric Empl | | Modern Empl | |
|---|---|---|---|---|---|---|---|
| | ln pop | ln emp | unemp rate | ln L agric | agric % | ln L modern | modern % |
| Agro-similarity | 0.051 | 0.024 | 0.008 | -0.166* | -0.051*** | 0.158** | 0.066*** |
| Conley 200 km (main), SE | (0.059) | (0.057) | (0.007) | (0.093) | (0.018) | (0.062) | (0.015) |
| Conley 200 km (main), $p$ | 0.389 | 0.674 | 0.216 | 0.075 | 0.006 | 0.011 | 0.000 |
| Conley 150 km, SE | (0.059) | (0.054) | (0.006) | (0.098) | (0.020) | (0.065) | (0.016) |
| Conley 150 km, $p$ | 0.389 | 0.658 | 0.188 | 0.091 | 0.010 | 0.015 | 0.000 |
| Conley 450 km, SE | (.) | (.) | (0.007) | (0.059) | (0.014) | (.) | (0.011) |
| Conley 450 km, $p$ | . | . | 0.209 | 0.005 | 0.000 | . | 0.000 |
| Conley 500 km, SE | (0.027) | (0.013) | (0.006) | (0.054) | (0.014) | (.) | (0.011) |
| Conley 500 km, $p$ | 0.055 | 0.070 | 0.185 | 0.002 | 0.000 | . | 0.000 |
| Bartlett 200 km, SE | (0.055) | (0.052) | (0.006) | (0.094) | (0.019) | (0.059) | (0.016) |
| Bartlett 200 km, $p$ | 0.352 | 0.649 | 0.152 | 0.078 | 0.006 | 0.008 | 0.000 |
| Bartlett 150 km, SE | (0.054) | (0.052) | (0.006) | (0.093) | (0.018) | (0.058) | (0.015) |
| Bartlett 150 km, $p$ | 0.345 | 0.647 | 0.133 | 0.074 | 0.005 | 0.006 | 0.000 |
| Bartlett 450 km, SE | (0.049) | (0.045) | (0.006) | (0.081) | (0.017) | (0.052) | (0.014) |
| Bartlett 450 km, $p$ | 0.293 | 0.599 | 0.193 | 0.042 | 0.003 | 0.002 | 0.000 |
| Bartlett 500 km, SE | (0.046) | (0.043) | (0.006) | (0.079) | (0.017) | (0.048) | (0.013) |
| Bartlett 500 km, $p$ | 0.267 | 0.576 | 0.194 | 0.036 | 0.002 | 0.001 | 0.000 |
| Components, SE | (0.042) | (0.040) | (0.007) | (0.061) | (0.018) | (0.058) | (0.013) |
| Components, $p$ | 0.245 | 0.558 | 0.255 | 0.017 | 0.014 | 0.017 | 0.000 |
| Components, wild bootstrap $p$ | 0.177 | 0.520 | 0.292 | 0.150 | 0.140 | 0.044 | 0.058 |
| Observations | 134 | 134 | 134 | 134 | 134 | 134 | 134 |

*Notes:* Every column repeats the main specification of Table 4, a regression of a 2010 census outcome on standardized agro-similarity on the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline), with the full set of controls; only the variance estimator changes across rows, so the coefficients are identical, and their stars come from the Conley $p$-value at 200 kilometers. The Conley rows use a uniform kernel, which gives every pair of cities inside the cutoff the same weight, and the Bartlett rows use the Bartlett kernel, which weights a pair by one minus its distance over the cutoff. Each $p$-value cell takes one shade at 0.10 or above and deepening shades below 0.10, 0.05 and 0.01. The radii bracket the distance over which cities share transmigration villages: 80% of the villages within the window enter more than one city's catchment, and two cities sharing a village sit up to 459 kilometers apart. The last three rows cluster on the connected components of the village-sharing graph, the finest partition that keeps every shared village inside one group: 14 components over the 134 cities, the largest holding 75, so the clustered standard error uses few effective groups and the wild cluster bootstrap over them, with Webb weights, is the version to read; it has little power at this count. Under the uniform kernel four of the 28 cells have no standard error, because the estimated variance of the exposure coefficient is negative there. In each row SE is the standard error of the coefficient in the first row and $p$ its two-sided $p$-value, and (main) labels the radius of the main specification. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

# E Additional Results

This part reports additional food-market and rural outcomes, the descriptive Engel comparison, and the migration calculations. The workforce bound and within-group decompositions distinguish the restrictions needed to attribute employment changes to additional workers. Public provision and the other alternative-channel tests follow, with their coverage and inference qualifications.

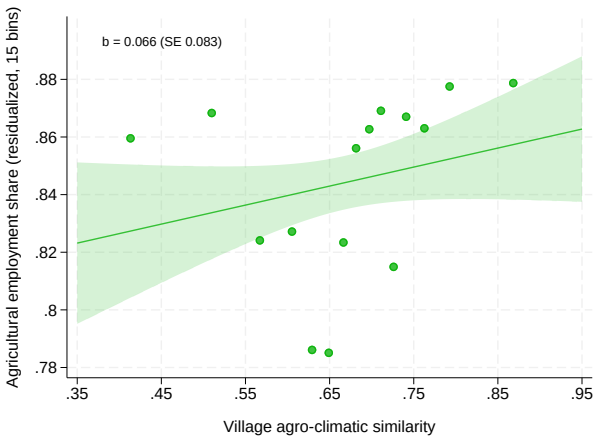
## E.1 Additional Reduced-Form Outcome Tables

**The response by location type, full set of outcomes.** Every outcome behind Figure 17a and Figure 17b is estimated for the three location types, each with its standard error. The size and composition outcomes, which all three areas measure on the same units, form Panel A, and the prices, wages and consumption outcomes, which come from the household survey and cover a different number of units in each row, form Panel B. The agricultural and housing prices are the 2002 Susenas wave, which precedes the 2004 rice import ban; the real wage is the 2011 wave. Panel C holds the eight 1996 rows; Figure 17b plots three of them, the food share, rice calories per person and the raw-food price, beside the 2011 real wage and the 2010 unemployment rate.

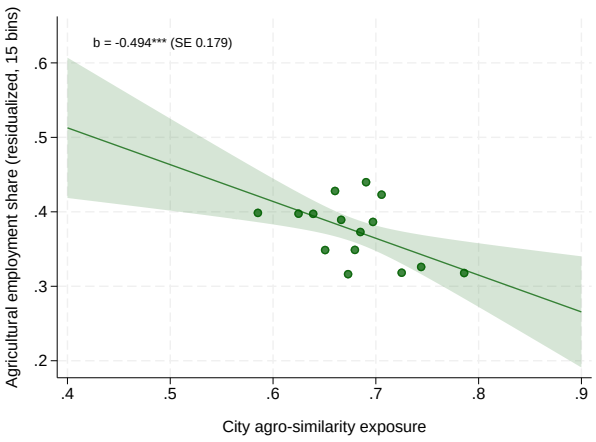
**Linear gradients of the agricultural employment share.** Figure E1 draws the binned scatter plots and linear fits behind the semiparametric curves of Figure 16, with the slope of each and its inference.

Figure E1: Linear Gradients of the Agricultural Employment Share in the Transmigration Villages and the Nearby Cities

(a) Transmigration Villages, 2000




(b) Nearby Cities, 2010

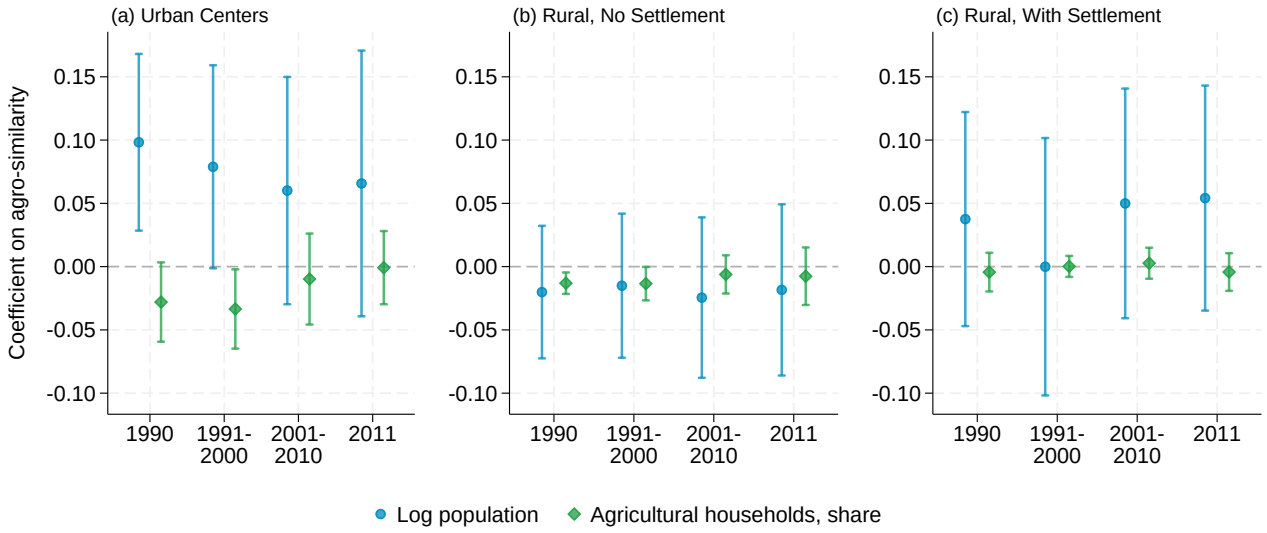



*Notes:* Binned scatter plots: dots are 15 quantile-bin means of the outcome and of agro-similarity on its 0 to 1 index scale, both residualized on the panel's controls with means added back. The line and its 90% band come from ordinary least squares on every residualized unit of the sample, and the in-plot coefficient is its slope with the panel's standard error and stars under that standard error (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). **(a)** The share of a village's workers in agriculture in the 2000 population census against the village's own agro-similarity, on the 808 transmigration villages with a measured agricultural share and a measured settler stay-share below 2, with village terrain, geographic and climate controls; the standard error is clustered by island (wild cluster bootstrap over the four island clusters, Webb weights, $p = 0.83$; Conley SE within 200 km, $p = 0.32$). **(b)** The agricultural employment share in the 2010 population census against city exposure, the mean agro-similarity of the transmigration villages within 300 travel minutes, on the main city sample, in the main specification with the full set of controls and Conley SE within 200 km; column (5) of Table 4 gives the same relationship in standardized units.

**Village-census dynamics by location type.** The village censuses show how population and the agricultural-household share respond over time (Figure E2).[42] In cities, both the population response and the decline in the share of agricultural households occur early and dissipate over time. The early population estimates are positive and the agricultural-household-share estimates negative, each statistically significant in one of the first two periods. In the later periods both effects are closer to zero and the population estimates are less precisely estimated. Rural subdistricts without settlements also have lower agricultural-household shares early in the period, without detectable population growth. These household classifications differ from the individual employment sectors measured in the population census, so their later attenuation does not establish that the worker-sector response also diminished.

This pattern is consistent with production in the transmigration villages replacing some of the farming by pre-existing households in these areas. The population responses in subdistricts with transmigration villages are imprecise, while their agricultural-household-share estimates are close to zero with tighter confidence intervals than in cities. The figure reports separate regressions and does not test differences between location groups.

Figure E2: Population and the Agricultural Household Share, 1990 to 2011, by Location Type




*Notes:* Each point is the coefficient on standardized agro-similarity from one regression, in the main specification with the full set of controls, on one cell of village census (PODES) waves, with 90% confidence intervals from Conley SE within 200 km. The cells are the 1990 wave, the 1993 and 1996 waves, the 2003, 2005 and 2008 waves, and the 2011 wave. Each panel keeps the units that report both outcomes in every cell: in panel (a) 127 urban centers of the main city sample, and in panels (b) and (c) 594 rural subdistricts without a settlement and 191 with one, lying outside every city and within 300 minutes of a transmigration village. The settler count of a subdistrict with a settlement is the number of settlers placed in the transmigration villages that overlap it. The other sectors are reported inconsistently across waves, so the agricultural share is the sectoral measure available over time. The series start in 1990, because earlier waves cover the agricultural share for less than two-thirds of the sample, and end in 2011, the last wave that releases population and household counts.

The goods-market outcomes behind the mechanism discussion follow. Table E3 reports production by crop category, consumption per person, the production-minus-consumption surplus by category, and food prices. The non-food side (non-food spending, housing rents, and wages) is reported in Table E4, and the manufacturing/services composition of the modern sector in Table E2 below.

The main results treat the modern sector as a single object; its manufacturing and services components are reported separately here, in levels and shares, in the modern-sector layout (Appendix Table E2). Columns 5 to 7 report

[42] These PODES village censuses record households reported by village heads, while the population census records individual workers by primary industry.

Table E1: Baseline Proxy Controls and the Cross-Sectional Food-Share Estimate

| | (1) Population density | (2) Agricultural employment share | (3) Schooling | (4) All three |
|---|---|---|---|---|
| Exposure, without the proxy | -0.0164*** | -0.0164*** | -0.0164*** | -0.0164*** |
| | (0.0055) | (0.0055) | (0.0055) | (0.0055) |
| Exposure, with the proxy | -0.0167*** | -0.0164*** | -0.0159*** | -0.0160*** |
| | (0.0057) | (0.0053) | (0.0056) | (0.0055) |
| Movement, in standard errors | -0.06 | -0.01 | 0.09 | 0.06 |
| Urban centers | 134 | 134 | 134 | 134 |

*Notes:* The outcome is the 2011 household food share, the mean share of household spending on food in the national socioeconomic household survey (Susenas). Each column adds to the main specification one 1980 census proxy for the unobserved baseline of the outcome: population per square kilometer, the agricultural employment share, the number of residents who had completed secondary school, and all three together. Each regression is of the food share on standardized agro-similarity, in the main specification with the full set of controls and Conley SE within 200 km. The two rows of each column use the same urban centers, those with a transmigration village within 300 travel minutes and a 1980 census baseline that have the food share and the proxy measured, counted in the bottom row. Movement is the with-proxy estimate minus the without-proxy estimate, over the without-proxy standard error. Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

the shares of workers outside both sectors: construction, the remaining industry codes, and the two together. The agricultural share of Table 4, the two modern-sector shares and the share outside both sectors add up to one.

Finally, the agricultural production and food consumption/surplus responses are estimated across the crop-category hierarchy, in raw and standardized form, and Table B5 reports the standardized rice-production response among the outcomes the calibration does not target. The raw estimates are in Panels A to C of Table E3.

The earliest wave, household Susenas 1996 outcomes aggregated to the same urban centers, shows higher per-capita expenditure one to two decades after placement, while the 1996 rice price and the two food-price indices show little response (Table E4).

The cross-sectional outcomes have no 1980 baseline, so they are identified off balance in levels. I add a 1980 variable that proxies for the unobserved baseline of the outcome and reestimate on the same urban centers (Table E1). The estimate is essentially unchanged under every proxy and remains statistically significant in every column.

**Employment on the observed-price samples.** The earlier price surveys cover fewer main-sample cities. On those samples, the pooled agricultural-share and log modern-employment responses are smaller and imprecisely estimated, while the modern-employment difference by openness is statistically significant on the 1996 sample and marginally significant on the 2002 sample (Table E5, Panels A and B). The openness groups and exposure scale are fixed before the coverage restriction. The 2011 raw-food price sample equals the main city sample.

**The crop mix by openness.** In every location type, higher agro-similarity moves the 2011 crop mix toward food crops and away from plantations by more in the more open group than in the more closed one; the groups differ significantly in the plantation share of the settlement subdistricts and in the food-crop share of the rural subdistricts without a settlement, with no other significant difference. The 2003 plantation share of harvested area falls with agro-similarity in the more open cities, a marginally significant difference from the more closed ones (Table E7).

Table E2: The Modern Sector Split and Employment Outside Agriculture and the Modern Sector

| | Log employment | | Modern-sector shares | | Outside both sectors | | |
|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| | ln L manuf | ln L serv | manuf empl % | serv empl % | constr % | other % | outside % |
| Agro-similarity | 0.249** | 0.144** | 0.010** | 0.056*** | -0.002 | -0.013* | -0.015* |
| | (0.120) | (0.067) | (0.005) | (0.018) | (0.004) | (0.008) | (0.009) |
| Settler count | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Baseline Pop 1980 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Urban share 1980 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Total Empl 1980 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Agric Empl 1980 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Agro-climatic controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Outcome mean | 7.556 | 10.136 | 0.047 | 0.470 | 0.060 | 0.049 | 0.109 |
| 1980 sector share | 0.063 | 0.319 | 0.063 | 0.319 | 0.035 | 0.022 | 0.057 |
| Outcome SD | 1.356 | 0.966 | 0.042 | 0.145 | 0.027 | 0.065 | 0.078 |
| Observations | 134 | 134 | 134 | 134 | 134 | 134 | 134 |

*Notes:* Each column is one regression of a 2010 population census outcome on standardized agro-similarity on the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline), in the main specification with the full set of controls and Conley SE within 200 km. Standard errors are in parentheses. Columns 1 and 2 are the logs of manufacturing and of services employment, the two parts of the modern sector, and columns 3 and 4 their shares of the employment of workers with an industry code (fractions). Columns 5 to 7 are the shares of construction (census industry code 10), of the other industry codes (mining and quarrying, electricity and gas, and the census code others: codes 7, 9 and 19), and of the two together, the employment outside agriculture and the modern sector; every share divides by the workers who report an industry, so the agricultural share of Table 4 and columns 3 to 6 sum to one, and column 7 is the sum of columns 5 and 6. The 1980 sector share row is the mean 1980 share of the same sector across the sample. In the column labels, ln is the natural log, L and empl are employment, manuf and serv are manufacturing and services, constr is construction, other the remaining codes and outside the two together, and % is a share of the employment of workers with an industry code, a fraction; the settler count weights the settlers of each transmigration village within 300 travel minutes by the inverse of its travel time, in thousands; Baseline Pop 1980, Total Empl 1980 and Agric Empl 1980 are the logs of 1980 population, employment and agricultural employment; and SD is the standard deviation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E3: Urban Agricultural Goods: Production, Consumption, Surplus, and Prices

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| *Panel A: Production by crop category, 2003 village census* | | | | |
| | ln staples | ln rice | ln legumes | ln coffee+tea+spices |
| Agro-similarity | 0.608*** | 0.736*** | 0.424 | 0.103 |
| | (0.197) | (0.251) | (0.360) | (0.501) |
| Outcome mean | 8.953 | 8.455 | 4.250 | 4.341 |
| Outcome SD | 2.007 | 2.343 | 2.220 | 2.655 |
| Observations | 133 | 132 | 117 | 103 |
| *Panel B: Log consumption per person (kg), household survey* | | | | |
| | Staples, 2002 | Staples, 2011 | Rice, 2002 | |
| Agro-similarity | 0.129*** | -0.028 | 0.112*** | |
| | (0.037) | (0.027) | (0.037) | |
| Outcome mean | 4.765 | 4.579 | 4.649 | |
| Outcome SD | 0.203 | 0.162 | 0.208 | |
| Observations | 108 | 132 | 108 | |
| *Panel C: Surplus, output minus consumption* | | | | |
| | $\sinh^{-1}$ staples | $\sinh^{-1}$ rice | Rice, tons | |
| Agro-similarity | -0.154 | -0.141 | -345 | |
| | (1.543) | (1.208) | (1,805) | |
| Outcome mean | -3.403 | -4.597 | -5,522 | |
| Outcome SD | 9.319 | 8.282 | 21,675 | |
| Observations | 108 | 108 | 108 | |
| *Panel D: Prices, 2011 household survey* | | | | |
| | rice price index | agric price index | | |
| Agro-similarity | 0.043 | 0.016 | | |
| | (0.027) | (0.022) | | |
| Outcome mean | 0.999 | 1.062 | | |
| Outcome SD | 0.143 | 0.121 | | |
| Observations | 132 | 134 | | |
| Settler count | ✓ | ✓ | ✓ | ✓ |
| Baseline Pop 1980 | ✓ | ✓ | ✓ | ✓ |
| Urban share 1980 | ✓ | ✓ | ✓ | ✓ |
| Total Empl 1980 | ✓ | ✓ | ✓ | ✓ |
| Agric Empl 1980 | ✓ | ✓ | ✓ | ✓ |
| Agro-climatic controls | ✓ | ✓ | ✓ | ✓ |

*Notes:* Each coefficient is from one regression of the outcome above it on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200 km. Standard errors are in parentheses. Panel A outcomes are the log of one plus 2003 village-census (PODES) production in tons, by crop group: staples (rice, corn, cassava, sweet potato, taro, other tubers and sorghum), paddy rice, legumes (peanut, soybean and mungbean), and coffee, tea, pepper, nutmeg and clove. Panel B is the log of one plus annual consumption per person in kilograms from the household survey (Susenas): staples (rice, corn, cassava and tubers) in 2002 and 2011, and rice in 2002. In Panel C, columns 1 and 2 are the inverse hyperbolic sine of the staples and rice surplus in tons, and column 3 is the rice surplus in tons: 2003 village-census production, with rice output in milled weight (0.6402 tons of milled rice per ton of paddy, from the 2018 conversion survey of Statistics Indonesia), minus 2002 Susenas consumption per person over the city's surveyed subdistricts, times the city's 2003 population (the whole city for staples, the villages with a rice output figure for rice). Output and population include all member subdistricts with the relevant production data. Consumption per person is measured on surveyed subdistricts. The rice-import figure restricts all terms to surveyed subdistricts with reported rice output. Panel D is the rice price index of each city and its agricultural price index, an index of raw-food prices, both from Susenas 2011. Coefficients in Panels A and B and in columns 1 and 2 of Panel C are approximate semi-elasticities. Survey outcomes are observed on the centers each source covers, reported in the Observations row. In the labels, $\sinh^{-1}$ is the inverse hyperbolic sine, used because a surplus can be negative. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E4: Household Survey Outcomes in 1996 and 2011

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) |
|---|---|---|---|---|---|---|---|---|---|---|
| *Panel A: 1996 household survey* | | | | | | | | | | |
| | ln rice | Raw-food price index | All-food | House idx | ln rent | ln pcExp | Food share | agric % | modern % | ln wage |
| Agro-similarity | -0.009 | -0.012 | 0.007 | 0.335 | 0.215 | 0.182** | -0.033*** | 0.048 | -0.032 | 0.232*** |
| | (0.029) | (0.025) | (0.025) | (0.237) | (0.140) | (0.087) | (0.013) | (0.040) | (0.040) | (0.072) |
| Outcome mean | 6.84 | 1.06 | 1.07 | 1.56 | 9.84 | 11.14 | 0.65 | 0.44 | 0.47 | 12.43 |
| Observations | 112 | 112 | 112 | 112 | 112 | 112 | 112 | 119 | 119 | 105 |
| *Panel B: 2011 household survey* | | | | | | | | | | |
| | non-food % | housing price | ln wage nonag | ln wage | ln real wage | | | | | |
| Agro-similarity | 0.016*** | -0.011 | 0.027 | 0.019 | 0.003 | | | | | |
| | (0.006) | (0.108) | (0.030) | (0.032) | (0.029) | | | | | |
| Outcome mean | 0.420 | 1.586 | 14.062 | 13.968 | 13.749 | | | | | |
| Outcome SD | 0.052 | 0.910 | 0.273 | 0.314 | 0.234 | | | | | |
| Observations | 134 | 134 | 134 | 134 | 134 | | | | | |
| Settler count | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Baseline Pop 1980 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Urban share 1980 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Total Empl 1980 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Agric Empl 1980 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Agro-climatic controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

*Notes:* Each column of a panel is one regression of an outcome from the national socioeconomic household survey (Susenas) on standardized agro-similarity, on the urban centers of the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline) that the survey covers, counted in the Observations row, in the main specification with the full set of controls and Conley SE within 200 km. Standard errors are in parentheses. Panel A outcomes, from the 1996 survey, are, in order, the log rice unit value (rupiah per kilogram), price indices of raw agricultural foods and of all foods (city prices relative to the median over the households of all urban centers), the median monthly housing rent relative to the national median and the log of that median rent, log mean monthly spending per person, the mean household food spending share, the shares of workers in agriculture and in manufacturing and services, and log median household income per employed worker, which stands in for the wage. Panel B outcomes are from the 2011 survey. Non-food % is the non-food share of household expenditure, a fraction. Housing price is the weighted median rent of the urban center relative to the national figure. The wage columns are the logs of the median monthly wage of non-agricultural workers, of all workers, and of that all-worker median deflated by a local food and housing price index. In the labels, ln is the natural log and nonag is non-agricultural workers. The settler count weights the settlers of each transmigration village within 300 travel minutes by the inverse of its travel time, in thousands. Baseline Pop 1980, Total Empl 1980 and Agric Empl 1980 are the logs of 1980 population, employment and agricultural employment, and SD is the standard deviation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E5: Employment Responses on the Observed-Price Samples, Pooled and by Openness

| | Main city sample and 2011 prices | | | Cities with observed prices in 1996 | | | Cities with observed prices in 2002 | | |
|---|---|---|---|---|---|---|---|---|---|
| | agric % | ln L modern | modern % | agric % | ln L modern | modern % | agric % | ln L modern | modern % |
| *Panel A: Pooled* | | | | | | | | | |
| Agro-similarity | -5.086*** | 0.158** | 6.595*** | -3.854 | 0.071 | 5.726** | -2.157 | 0.097 | 3.865 |
| | (1.845) | (0.062) | (1.513) | (2.847) | (0.079) | (2.228) | (2.774) | (0.070) | (2.445) |
| | [1.861] | [0.059] | [1.556] | [2.667] | [0.084] | [2.159] | [2.566] | [0.080] | [2.230] |
| Cities | 134 | 134 | 134 | 112 | 112 | 112 | 109 | 109 | 109 |
| *Panel B: By openness* | | | | | | | | | |
| More closed | -7.069*** | 0.307*** | 9.788*** | -7.323*** | 0.233** | 9.575*** | -4.198* | 0.197** | 7.545*** |
| | (2.172) | (0.114) | (1.844) | (2.249) | (0.103) | (2.139) | (2.477) | (0.089) | (2.158) |
| | [2.569] | [0.110] | [2.068] | [2.568] | [0.097] | [2.210] | [2.584] | [0.090] | [2.044] |
| More open | -2.387 | 0.065 | 3.529** | -0.815 | -0.091 | 2.253 | -0.362 | 0.014 | 0.818 |
| | (1.892) | (0.063) | (1.507) | (4.080) | (0.088) | (2.980) | (3.277) | (0.067) | (2.479) |
| | [1.914] | [0.069] | [1.697] | [3.543] | [0.100] | [2.801] | [2.956] | [0.094] | [2.395] |
| More open minus more closed | 4.683* | -0.242** | -6.259*** | 6.508* | -0.324** | -7.321** | 3.836 | -0.183* | -6.727*** |
| | (2.613) | (0.109) | (2.005) | (3.544) | (0.143) | (2.854) | (2.972) | (0.109) | (2.202) |
| | [2.621] | [0.114] | [2.163] | [3.361] | [0.134] | [2.756] | [2.972] | [0.109] | [2.218] |
| Cities | 134 | 134 | 134 | 112 | 112 | 112 | 109 | 109 | 109 |
| More-closed | 66 | 66 | 66 | 57 | 57 | 57 | 55 | 55 | 55 |
| More-open | 68 | 68 | 68 | 55 | 55 | 55 | 54 | 54 | 54 |

*Notes:* The first three columns use the main city sample, which is also the observed-price sample in 2011. The other columns retain main-sample cities with the raw-food price index observed in the indicated wave. Exposure and outcome scales remain fixed. These are coverage sensitivities. Differences across samples do not identify an effect of the import restrictions. Panel A pools the cities of each sample. In Panel B, the more open cities have route hours below the median among all cities within 300 travel minutes of a settlement, and this division is fixed before restricting to measured prices. Both group slopes are estimated in one regression with a group indicator and common controls. The difference is more open minus more closed, using their covariance. Agro-similarity is standardized over the full city file before any sample restriction. All regressions include the settler count within 300 travel minutes, the 1980 population, employment, urban population share and agricultural employment, and 16 agro-climatic controls. Agricultural and modern employment shares are in percentage points, and modern employment is ln(1 + count). Modern employment combines manufacturing and services. Sources: population censuses of 1980 and 2010, and the household survey (Susenas) of 1996, 2002 and 2011. Conley SE within 200 km in parentheses (uniform kernel, which the stars use) and Conley SE with Bartlett distance weights within 200 km in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E6: All Outcomes by Location Type, Main Specification

| | Cities | Rural, with settlement | Rural, no settlement |
|---|---|---|---|
| *Panel A: Size and composition* | | | |
| Log population | +0.060 | +0.085 | −0.031 |
| | (0.069) | (0.084) | (0.065) |
| Log employment | +0.028 | +0.095 | −0.068 |
| | (0.067) | (0.088) | (0.062) |
| Log agricultural employment | −0.185* | +0.093 | −0.206** |
| | (0.104) | (0.080) | (0.081) |
| Agricultural employment share | −0.278*** | +0.012 | −0.342*** |
| | (0.101) | (0.065) | (0.082) |
| Log modern employment | +0.162** | +0.080 | +0.135*** |
| | (0.063) | (0.080) | (0.047) |
| Modern employment share | +0.423*** | −0.025 | +0.340*** |
| | (0.097) | (0.072) | (0.089) |
| Observations | 134 | 191 | 603 |
| *Panel B: Prices, wages, and consumption* | | | |
| Unemployment rate | +0.210 | +0.032 | +0.096 |
| | (0.170) | (0.092) | (0.097) |
| | [134] | [191] | [603] |
| Agricultural price, 2002 | +0.126 | −0.239** | −0.047 |
| | (0.262) | (0.121) | (0.075) |
| | [109] | [104] | [206] |
| Housing price, 2002 | −0.017 | −0.004 | +0.067 |
| | (0.141) | (0.087) | (0.077) |
| | [109] | [98] | [197] |
| Log nominal wage, 2011 | +0.062 | −0.031 | −0.020 |
| | (0.101) | (0.124) | (0.057) |
| | [134] | [190] | [590] |
| Log real wage, 2011 | +0.014 | +0.146 | +0.043 |
| | (0.126) | (0.123) | (0.076) |
| | [134] | [188] | [580] |
| Food spending share, 2011, restricted sample | −0.313*** | +0.006 | +0.061 |
| | (0.106) | (0.130) | (0.053) |
| | [134] | [188] | [581] |
| Log food spending, 2011 | +0.161 | −0.082 | +0.032 |
| | (0.115) | (0.067) | (0.094) |
| | [134] | [188] | [582] |
| Food spending share, 2011, pooled households | −0.313*** | −0.063 | +0.107* |
| | (0.106) | (0.135) | (0.062) |
| | [134] | [188] | [582] |
| Rice calories per person | +0.470*** | +0.075 | +0.015 |
| | (0.171) | (0.147) | (0.112) |
| | [108] | [104] | [206] |

Table E6: All Outcomes by Location Type, Main Specification, Continued

| | Cities | Rural, with settlement | Rural, no settlement |
|---|---|---|---|
| *Panel C: Household survey, 1996* | | | |
| Log total household spending, 1996 | $+0.511^{*}$ | $+0.119$ | $-0.149$ |
| | (0.263) | (0.128) | (0.096) |
| | [112] | [117] | [288] |
| Log food spending per person, 1996 | $+0.460$ | $+0.119$ | $-0.089$ |
| | (0.287) | (0.118) | (0.108) |
| | [112] | [117] | [288] |
| Food spending share, 1996 | $-0.462^{***}$ | $+0.058$ | $-0.201^{**}$ |
| | (0.177) | (0.122) | (0.091) |
| | [112] | [117] | [288] |
| Agricultural price (raw food), 1996 | $-0.089$ | $-0.131$ | $-0.325^{***}$ |
| | (0.188) | (0.113) | (0.125) |
| | [112] | [117] | [288] |
| Rice unit value, 1996 | $-0.048$ | $-0.206$ | $-0.147$ |
| | (0.210) | (0.186) | (0.126) |
| | [112] | [116] | [286] |
| Log calories per person, 1996 | $+0.718^{***}$ | $+0.088$ | $-0.000$ |
| | (0.148) | (0.184) | (0.147) |
| | [112] | [117] | [288] |
| Rice calories per person, 1996 | $+0.289^{***}$ | $+0.012$ | $+0.345^{***}$ |
| | (0.081) | (0.109) | (0.115) |
| | [112] | [117] | [288] |
| Log staple kg per person, 1996 | $+0.407^{***}$ | $-0.232$ | $+0.131$ |
| | (0.073) | (0.157) | (0.105) |
| | [112] | [117] | [288] |

*Notes:* Each coefficient is the response, in SD of the outcome, to a one-SD increase in agro-similarity, with Conley SE within 200 km in parentheses. Cities are the main city sample (outer-island urban centers with a transmigration village within 300 travel minutes and a 1980 census baseline); the rural subdistricts with and without a settlement lie outside every city and meet the same travel-time and baseline rule. The city column is in the main specification with the full set of controls; the two rural columns have the same controls, except that the settler count of a subdistrict with a settlement is the number of settlers placed in the transmigration villages that overlap it. Outcomes and agro-similarity are standardized once among the cities and once among both rural groups together, at exposure SDs of 0.103 over the 138 cities with a measured exposure and 0.103 over the 794 sample subdistricts, so one SD is about the same amount of exposure on either side, with outcomes on separate scales. Panel A and the unemployment rate are from the 2010 population census, the prices and rice calories from the 2002 household survey (Susenas), and the wage and food spending rows from its 2011 wave, with food spending and wages in logs. Panel C is the 1996 wave, pooled over the surveyed households of each city and subdistrict, with the agricultural price on the raw-food index and logs taken as $\ln(1 + x)$. Panel A's observation count applies to every row of that panel; each Panel B and C row gives its own count in brackets. The restricted-sample 2011 food spending share is available for subdistricts with at least one observed wage earner. The pooled food share averages all surveyed households of each subdistrict, as food spending does; for the cities the two food shares are identical. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

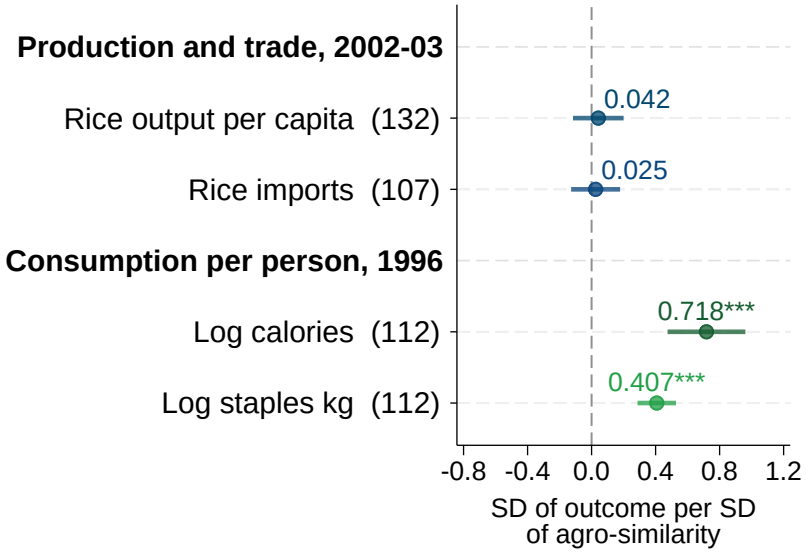
Table E7: The Crop Mix by Trade Openness

| | Mean, % | | Response, pp per SD | | Open minus | |
|---|---|---|---|---|---|---|
| Location | Closed | Open | Closed | Open | closed | Obs. |
| *Plantation area share, 2003* | | | | | | |
| Cities | 27.1 | 47.0 | 0.03 | -11.67*** | -11.70* | 127 |
| | | | (6.30) | (3.77) | (6.92) | |
| Rural, with settlement | 49.5 | 59.1 | -3.21 | -4.02 | -0.81 | 179 |
| | | | (2.97) | (5.14) | (4.93) | |
| Rural, without settlement | 28.6 | 42.7 | -14.33*** | -6.92*** | 7.41* | 541 |
| | | | (3.41) | (2.50) | (3.99) | |
| *Farming as main income, 2011* | | | | | | |
| Cities | 55.6 | 60.7 | -1.93 | 3.32 | 5.25 | 134 |
| | | | (4.14) | (2.76) | (3.92) | |
| Rural, with settlement | 92.4 | 95.1 | 1.96 | -1.49 | -3.44 | 191 |
| | | | (1.94) | (0.91) | (2.12) | |
| Rural, without settlement | 93.0 | 93.6 | 1.30 | -1.16 | -2.45 | 603 |
| | | | (1.37) | (0.86) | (1.59) | |
| *Food crops as main commodity, 2011* | | | | | | |
| Cities | 34.5 | 23.0 | 7.49 | 10.12*** | 2.62 | 134 |
| | | | (4.92) | (2.47) | (4.28) | |
| Rural, with settlement | 38.9 | 28.3 | 4.84 | 10.68** | 5.84 | 191 |
| | | | (6.06) | (5.28) | (7.51) | |
| Rural, without settlement | 58.0 | 45.3 | 6.84 | 10.91*** | 4.07 | 603 |
| | | | (5.03) | (3.97) | (4.85) | |
| *Plantations as main commodity, 2011* | | | | | | |
| Cities | 14.5 | 29.9 | -4.75 | -5.65** | -0.91 | 134 |
| | | | (5.64) | (2.52) | (4.38) | |
| Rural, with settlement | 44.5 | 63.2 | -0.99 | -16.87*** | -15.88** | 191 |
| | | | (5.21) | (4.64) | (6.58) | |
| Rural, without settlement | 25.4 | 40.8 | -9.55** | -11.81*** | -2.26 | 603 |
| | | | (4.05) | (3.91) | (4.41) | |
| *Plantations among farming villages, 2011* | | | | | | |
| Cities | 21.7 | 49.5 | -6.51 | -13.23*** | -6.72 | 134 |
| | | | (8.11) | (3.56) | (6.43) | |
| Rural, with settlement | 47.6 | 66.8 | -1.25 | -16.63*** | -15.37** | 191 |
| | | | (5.39) | (5.11) | (6.94) | |
| Rural, without settlement | 27.2 | 43.6 | -10.22** | -11.91*** | -1.68 | 603 |
| | | | (3.99) | (3.84) | (4.21) | |
| *Plantations as main commodity, villages, 2011* | | | | | | |
| Cities | 19.1 | 39.3 | -4.43 | -8.66*** | -4.24 | 134 |
| | | | (6.46) | (2.66) | (5.32) | |
| Rural, with settlement | 45.6 | 64.4 | -0.19 | -14.41*** | -14.23** | 191 |
| | | | (5.35) | (4.88) | (6.76) | |
| Rural, without settlement | 26.5 | 41.5 | -10.29*** | -12.79*** | -2.50 | 603 |
| | | | (3.88) | (3.82) | (4.31) | |

*Notes:* Means are percentages; responses are percentage points per SD of agro-similarity, with Conley standard errors within 200 km in parentheses. Each outcome/location-type regression interacts agro-similarity with the more closed and more open indicators, split at the location-type median route distance to the Thai and Vietnamese rice hubs. Regressions use the main controls; settlement subdistricts use the settlers placed in their overlapping transmigration villages. The final response column is the open response less the closed response; stars test equality: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The 2003 plantation-area share is rubber, oil palm, coffee, tea and cocoa over all harvested crop area, missing where a unit records no crop area. The 2011 rows measure population shares in villages with agriculture as the main income source and food crops (paddy or secondary food crops) or plantations as the main agricultural commodity. Plantations among farming villages uses the farming-village population as denominator; the last block uses village counts. Samples include 127 to 134 main-sample cities, 179 to 191 rural subdistricts with a settlement and 541 to 603 without one. Source: village censuses (PODES), 2003 and 2011.

**Rice production, implied imports, and physical consumption.** The production and consumption data provide an implied net rice requirement, without identifying the origins of the rice supplied to cities. The production and implied-import estimates are imprecise. Food-crop output per person from the 2003 village census, which the figure does not show, rises by 0.156 standard deviations of the outcome per standard deviation of exposure (standard error 0.095), a marginally significant estimate, while the staples and legumes surpluses per person are statistically insignificant. The calorie and staple-quantity rows report the same pooled physical-consumption responses as the main consumption display (Figure E3; Figure 12b).

Figure E3: Rice Production, Implied Imports, and Food Consumption



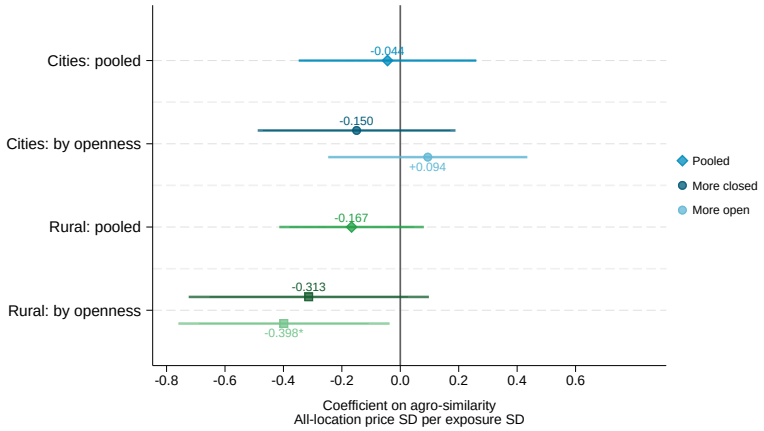

*Notes:* Each row is one regression of the standardized outcome on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200 km. Bars are 90% confidence intervals; parentheses give the number of cities. Rice output per capita is 2003 village-census (PODES) rice production in kilograms of unhusked paddy divided by the population recorded for villages with a rice output figure. Rice imports are 2002 Susenas rice consumption per person times the population of those villages, minus production in milled-rice weight at the official rate of 0.6402 tons per ton of paddy, over subdistricts with surveyed households and a rice output figure. This implied net requirement combines survey years and does not measure bilateral trade. The appendix surplus applies consumption per person to all rice-reporting villages in the city, including those in subdistricts without surveyed households. Calories over the whole diet and staple kilograms per person are from Susenas 1996, include own production, and use ln(1+x). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: PODES 2003 and Susenas 2002 and 1996.

**Rice unit values and quality.** Rice unit values combine prices with the varieties and qualities households consume. The pooled city estimate is negative and imprecise, and the more closed and more open city responses do not differ detectably. In settlement subdistricts, the decline in the more open group is marginally significant under uniform spatial weights, significant under Bartlett weights, and insignificant under province clustering; its difference from the more closed group is imprecise (Figure E4a).

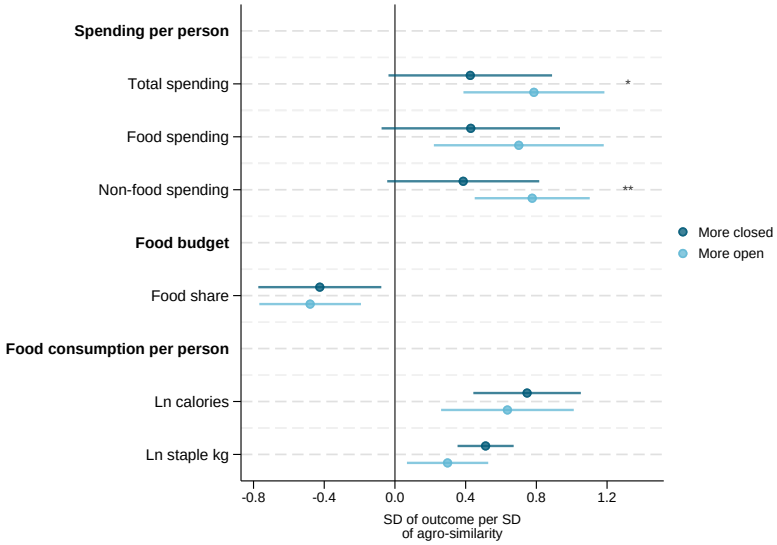
In cities, the high-quality expenditure share has positive estimates in both openness groups, with an imprecise difference. Comparing the combined rice unit value with the local-rice unit value on the same locations does not reveal a detectable difference in either location type. Both matched-sample city point estimates are positive. The observed shift toward a category labeled high-quality therefore does not establish that quality changes mask a rice-price decline. Holding locations fixed also does not hold household selection or within-category quality fixed.

Figure E4: Rice Unit Values and City Consumption by Trade Openness, 1996

(a) Rice Unit Values (Price Proxy)




(b) City Consumption




*Notes:* Panel (a): Susenas 1996, 112 main-sample cities and 116 rural subdistricts with a settlement outside cities and within 300 travel minutes of a transmigration village. Rice unit values divide survey-weighted consumption value by kilograms for local and high-quality rice, including own production valued by the survey. Diamonds are pooled slopes from separate regressions, and paired slopes come from one regression per location type with agro-similarity interacted with the openness groups and the openness indicator. More open locations lie strictly below the location-specific route-hour median, fixed before restricting to observed unit values. Regressions use the full main controls, with settlers in overlapping villages for settlement subdistricts, and equal location weights. Unit values use the SD across all measured cities and rural subdistricts, and exposure uses separate city and rural SDs. Thick and thin bars are 90% intervals using uniform and Bartlett spatial weights within 200 km. Coefficient stars test zero. Panel (b): Susenas 1996, 112 cities in the main sample with observed consumption outcomes (57 more closed and 55 more open). Each pair of slopes comes from one regression with agro-similarity interacted with the openness groups, the openness indicator, and the full set of main-specification controls, with Conley SE within 200 km. More open cities are below the current city median of route hours to the Thai and Vietnamese rice hubs, and the split is fixed before restricting to observed consumption. Spending is nominal monthly consumption per person in levels, including own production valued by the survey. Food share is the person-weighted mean household food-budget share. Calories are per person per day and staples per person per year, both transformed as ln(1+x). Each outcome uses its SD across measured main-sample cities, and exposure uses the existing city SD. Household aggregates retain survey weights, and city regressions have equal weights. Bars are 90% confidence intervals. In both panels, stars at right test the direct more-open minus more-closed difference under uniform spatial inference: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: Susenas 1996, its household aggregates, and the route-access measure.

**City consumption by trade openness.** Figure E4b compares the consumption responses of more closed and more open cities. Total, food, and non-food spending per person have positive point estimates in both groups, with larger responses in more open cities. The non-food spending difference is statistically significant under spatial inference, the total-spending difference is marginally significant, and the food-spending difference is imprecise. Province clustering weakens the evidence for the monetary differences. Food shares decline by similar amounts in both groups. Calorie and staple-quantity point estimates are larger in more closed cities, but their direct group differences are imprecise.

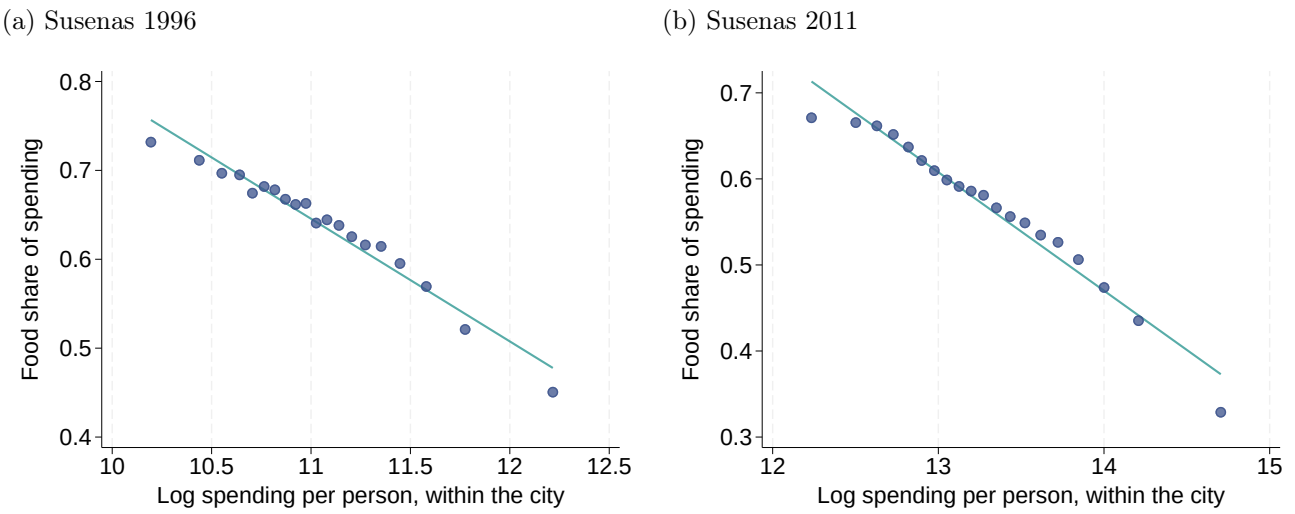
Lower food prices do not require lower nominal food spending. Greater quantities can offset a price decline, and household resources and product composition can change. The ordering of the spending and quantity point estimates is compatible with cheaper food in more closed cities, but the imprecise quantity contrasts do not establish that explanation. The food-spending aggregate, calorie measure, staple basket, and raw-food price index differ in coverage and aggregation, so ratios of their regression coefficients would not recover a price response. These results do not establish consumption gains concentrated in more closed cities or identify the contribution of food prices to the consumption response.

## E.2 Food Expenditure Shares and the Engel Relationship

Household regressions relate the food share to log spending per person, with city fixed effects and survey weights. Multiplying the slope by the city mean of log spending per person gives the predicted food-share component. Figure E5 shows the within-city relationship in each wave, and its notes report the exposure responses of spending, the observed food share, the predicted component and the remaining change.

This comparison applies a within-city household relationship to differences in city-level spending. It is descriptive: spending can respond to both prices and earnings, so the predicted component does not isolate an income mechanism. The remaining food-share change is negative and imprecisely estimated in both waves.

Figure E5: The Food Share and Spending per Person Within Cities




*Notes:* Households of the main city sample. Each point is the weighted mean of one of 20 bins of equal survey weight, after the city's weighted mean is subtracted from the food share and from log spending per person and the wave's mean is added back. The line shows the survey-weighted within-city slope, −0.138 in 1996 and −0.138 in 2011, from a household regression with city fixed effects and standard errors clustered by city. In city regressions on the main city sample (112 cities in 1996 and 134 in 2011), in the main specification with the full set of controls and Conley SE within 200 km, one SD of agro-similarity raises log spending per person by 0.183 (SE 0.082) in 1996 and 0.066 (0.022) in 2011 and changes the food share by −0.033 (0.013) and −0.016 (0.006). The predicted component, the slope times the city mean of log spending per person, changes by −0.025 (0.011) and −0.009 (0.003), 77% and 55% of the measured change. The part not predicted, −0.008 (0.013) and −0.007 (0.006), is not statistically significant in either wave, and adding the slope's sampling error leaves its standard errors unchanged at three decimals. The other three responses are statistically significant in both waves. Spending per person is monthly, in rupiah of the survey year. Logs are ln(1 + x). Source: Susenas 1996 household module and 2011 food and non-food modules.

## E.3 The In-Migration Required for the Agricultural-Share Fall

**Direct recent migration.** The five-year residence question measures recent arrivals among the residents observed in 2010. The province and district shares increase with exposure on the main sample (Panel A of Table E8). They provide direct evidence of recent moves, while the employment-based calculations below ask how much adjustment additional workers could explain under restrictions on the incumbent workforce.

**Recent migration by openness.** Province and district moves have larger exposure responses in the more closed cities. The district-move difference between the openness groups is statistically significant and the province-move difference is imprecise (Panel B of Table E8). The worker comparison uses all workers with a recorded prior district and includes return migrants. It uses the same current fractional subdistrict allocation as the main employment outcomes, and its employment totals agree city by city. The lifetime-migrant recency partition used in the decomposition excludes return migrants from its recent-arrival category. The resident and worker shares therefore answer different questions. They do not identify individual sector changes.

**The identity.** A city has $L$ workers in 2010, a share $s$ of them in agriculture. Suppose it takes in $M$ extra workers, a share $a$ of whom also work in agriculture, while the incumbent workforce and its sector composition stay fixed. Its agricultural share becomes $(sL + aM)/(L + M)$. Writing the gross inflow rate as $g = M/L$, the fall in the share is $g(s - a)/(1 + g)$, so producing a fall of $\Delta$ requires

$$g(a) \;=\; \frac{\Delta}{s - a - \Delta}, \tag{20}$$

Table E8: Recent Migration, All Cities and by Trade Openness

| | Residents | | Workers |
|---|---|---|---|
| | Province | District | District |
| *Panel A: All cities* | | | |
| Agro-similarity (per SD) | +0.52** | +1.92*** | |
| | (0.26) | (0.49) | |
| Cities | 134 | 134 | |
| *Panel B: By trade openness* | | | |
| More closed | 0.59* | 2.15*** | 2.00*** |
| | (0.34) | (0.58) | (0.61) |
| | [0.36] | [0.82] | [0.88] |
| More open | 0.31 | 1.11** | 1.17* |
| | (0.38) | (0.48) | (0.63) |
| | [0.35] | [0.57] | [0.66] |
| More open minus more closed | -0.27 | -1.04*** | -0.83 |
| | (0.40) | (0.38) | (0.51) |
| | [0.38] | [0.55] | [0.67] |
| Cities | 134 | 134 | 134 |
| More closed | 66 | 66 | 66 |
| More open | 68 | 68 | 68 |

*Notes:* Each cell regresses the share of 2010 residents who lived in another province or district five years earlier (the first two columns), or the share of workers who lived in another district (the third column), on standardized agro-similarity, using the main city sample and the full main controls: settler count, 1980 population, employment, urban population share and agricultural employment, and the 16 agro-climatic covariates. Panel A pools the cities. Panel B interacts agro-similarity with the fixed openness groups and adds their group indicator: both group slopes are estimated in one regression with common controls, and the difference is more open minus more closed, using their covariance. The more open cities have route hours below the median among all cities within 300 travel minutes of a settlement, and this division is fixed before restricting to measured prices. Coefficients and standard errors are in percentage points per SD of exposure. Conley standard errors within 200 km with uniform weights, which the stars use, are in parentheses, and in Panel B those with Bartlett weights are in brackets on the row beneath. With Bartlett distance weights, the province response of Panel A is marginally significant and its district response remains significant. The third column uses workers with an observed district of residence five years earlier as its denominator and includes return migrants. Prior district is observed for at least 94.90% of workers in each city, and 1,901 workers have an unrecorded prior district. Current fractional subdistrict membership reproduces each city employment total. District moves include moves across provinces. These shares describe recent arrivals among current residents and do not measure departures, all gross moves, moves within districts, earlier migration, or individual sector switching. Source: 2010 population census. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

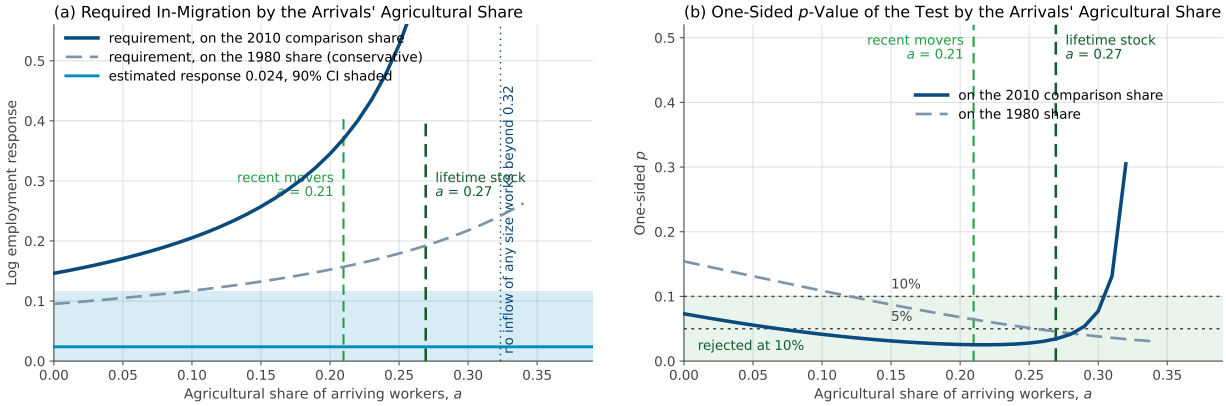
and a log employment response of $\ln(1+g(a))$. Equation 20 has two properties the test relies on. The requirement rises steeply in $a$, because arrivals who farm dilute the share less. And it is infinite once $a \geq s - \Delta$: an inflow with an agricultural share that high cannot pull the average down that far at any size.

The base $s$ is the comparison city's *2010* share, 0.374 on the main city sample. Both coefficients are cross-sectional 2010 differentials, so the share the extra arrivals dilute is the share the comparison city has when they arrive. Using the 1980 share instead would attribute the common 1980-2010 structural change to the inflow as well. The 1980 base is roughly 20 percentage points higher and lowers the requirement by more than a factor of three; I report it throughout as a conservative alternative.

**Measuring the arrivals' sector mix.** No census aggregate crosses migrant status with industry. I therefore build that cross from the individual records of the 1980 and 2010 censuses, aggregating person to village, then to year-2000 subdistrict through the harmonized crosswalk, then to the reclassified urban centers. Recomputing each city's overall agricultural share from these person records matches the census share for the median city (correlation 1.000).

**The test.** Both $\Delta$ and the employment response are estimated, so the threshold is itself a random variable and the test needs their joint covariance under the same spatial kernel. I therefore stack the two equations: each city contributes two rows, one for log employment and one for the agricultural share, every regressor is interacted with an equation indicator, and one Conley regression is run on the stack. The two rows of a city sit at distance zero, so the kernel gives the cross-equation covariance directly. The stacked fit is saturated and reproduces the two separate regressions exactly. The delta method then propagates the covariance through Equation 20. The resulting standard error on the gap is 0.351, against 0.057 on the employment response, so the vertical distance between the two curves in Figure E6 understates the uncertainty.

Figure E6: Required In-Migration and Its One-Sided Test by the Arrivals' Agricultural Share




*Notes:* Log employment and the agricultural employment share in the 2010 census are each regressed on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200 km; inference uses the joint covariance of the two coefficients. The horizontal axis is $a$, the agricultural share of the arriving workers; solid curves take the 2010 mean agricultural share $s$ as the base, and dashed curves the 1980 mean. Left panel: the curves are $\ln(1+g(a))$, the log employment response Equation 20 requires; the flat line and shaded band are the estimated response to a one-standard-deviation rise in agro-similarity and its 90% confidence interval; the dotted line marks the asymptote $a = s - \Delta$, where $\Delta$ is the estimated fall in the agricultural share per standard deviation of agro-similarity, and beyond it an inflow of any size fails to produce that fall. Right panel: the one-sided $p$-value at each $a$, with reference lines at 10% and 5%. In both panels the dashed vertical lines mark the 2010 agricultural shares of recent movers (moved district within five years) and of the lifetime stock (born outside the district). The one-sided test rejects at 10% for every $a$ up to 0.30 and at 5% from 0.07 to 0.28.

**The calculation at the average city.** A city with the sample's mean workforce, 91,751 workers with an industry code, and its mean agricultural share has 34,345 of them in agriculture. The high-exposure city has 2,208 more workers and 30,390 agricultural workers. If net employment growth equals arrivals, with the incumbent workforce fixed, and those arrivals farm at the measured 26.9%, they add 595 farmers, so agricultural employment would total about 34,939. It is at 30,390, a gap of roughly 4,549 workers relative to that calculation. Under those assumptions, 86,210 arrivals would be required, against net employment growth of 2,208, a shortfall of about 84,002 workers on a workforce of 91,751.

**The modern-job accounting.** Exposure adds modern-sector jobs to a city faster than it adds workers, and I test whether the workers a city gains could have filled those jobs with nobody already in the city changing sector. At the mean city, one standard deviation of exposure adds about 9,100 modern-sector jobs and 2,200 workers, the coefficients of Table 4 taken at the 2010 means. I count how many of the new jobs the workers added could fill, first if every one of them took a modern-sector job and then if they took such jobs at the rate the census measures for migrants, and I attach to each share a one-sided upper bound that uses the joint precision of the two coefficients, since both come from the same cities. The workers added fill about a quarter of the gain, 24%, and at most 82% at 95% confidence; at the migrants' rate they fill 16%, at most 54%. In the main specification, the test rejects that the net addition of workers fills the whole modern-job gain. This is an arrivals calculation with the incumbent workforce fixed and without offsetting departures or labor-force entry and exit. The checks below report how the conclusion varies with the variance estimator, job conversion, and design-based specification.

**The share of the modern-job gain.** Each log coefficient of Table 4 is approximately a proportional change, so at the mean city it is also a change in people: a coefficient $b$ on an outcome with 2010 mean $\bar{X}$ moves that outcome by $\bar{X}(e^b - 1)$. On modern-sector employment (mean 53,037) the coefficient gives 9,100 jobs; on employment (mean 90,259) it gives 2,200 workers. If nobody already working in the city changes sector, only the workers added can fill the new modern-sector jobs, and the share they fill is the workers added, times the fraction of them in modern-sector work, divided by the jobs gained: $\varrho = f\bar{E}(e^{b_E} - 1)/[\bar{M}(e^{b_M} - 1)]$, with $\bar{E}$ and $\bar{M}$ the two means, $b_E$ and $b_M$ the two coefficients and $f$ the fraction. With every worker added in a modern-sector job ($f = 1$) the share is 24%; at the rate at which workers born outside their district of residence hold modern-sector jobs in the 2010 census, 65%, it is 16%. The share is uncertain because the two coefficients are, and its numerator, the employment response, is the imprecise one, so a standard error of the ratio itself (50 percentage points by the delta method) is a poor guide. I therefore test each candidate share directly, asking whether the data reject that the workers added fill that much of the gain. The two coefficients move together across cities (correlation 0.82 in one regression that stacks the two outcomes, as in the test above), and the one-sided test uses their joint covariance. The largest share the test does not reject at 5% is the one-sided 95% upper bound: 82% with every worker added in a modern-sector job and 54% at the migrants' rate. A share of 100%, the workers added filling every new job, is rejected in both cases ($p = 0.012$ and $p < 0.001$). The net modern-job gain exceeds net employment growth by 6,900, with a 90% interval of 1,900 to 11,900. Repeated cross-sections do not identify individual moves or occupational transitions.

**Variance estimators and job conversions.** The strength of the rejection varies with the standard errors and the way jobs are counted. Under the paper's uniform Conley kernel, a Bartlett kernel and errors clustered by city, the headline row and every row of Panel A of Table E9 that counts workers reject a share of 100% at 5%. Province clusters, the loosest of the four, still reject in the headline row ($p = 0.033$, bound 93%); only the two rows that put the comparison city at the mean of the cities' shares do not reject at 5% ($p = 0.067$ and 0.071, bounds 106% and 106%), because larger cities are more modern, so the mean of the cities' modern shares is below the modern share of the mean city, and those two rows put fewer modern-sector jobs in the denominator. Counting residents aged 15 and over in place of workers weakens the comparison, because residents aged 15 and over grow by more than workers and only the working residents among them fill jobs. At the city's 0.563 workers per resident aged 15 and over, the residents added fill 52% of the gain, with a bound of 104%. The accounting counts workers because a job is filled by a worker.

**Two moments.** City size and the employment shares are two moments, and one inflow has to fit both. With net employment growth equated to arrivals and the incumbent workforce fixed, arrivals imply both the agricultural share and the modern share of the more exposed city. Each implied share is a restriction that holds when the estimate is consistent with it, and the one-sided Wald test of the two restrictions together, whose null distribution is a mixture of chi-squared distributions with weights set by the correlation of the two restrictions (Kodde & Palm 1986; Wolak 1989), rejects the hypothesis that arrivals with no switching produced both moments at 1% under each of the four variances, with $p$ at most 0.002 (under province clusters). Testing the two shares jointly removes the concern that a one-moment test picks the share that happens to reject.

Table E9: The Share of the Modern-Job Gain by Job Conversion, Variance Estimator and Specification

| | | One-sided 95% upper bound | | | | One-sided $p$, share of 100% or more | | | |
|---|---|---|---|---|---|---|---|---|---|
| Route or specification | Share | Uniform | Bartlett | City | Province | Uniform | Bartlett | City | Province |
| Workers, every one in a modern job (headline) | 24% | 82% | 79% | 86% | 93% | 0.012 | 0.008 | 0.020 | 0.033 |
| *Panel A: Other job conversions* | | | | | | | | | |
| Workers, modern rate of migrants, employment-weighted | 16% | 54% | 52% | 56% | 61% | <0.001 | <0.001 | 0.001 | 0.004 |
| Workers, modern rate of migrants, unweighted | 15% | 51% | 49% | 54% | 58% | <0.001 | <0.001 | 0.001 | 0.004 |
| All workers with an industry code, every one modern | 24% | 84% | 81% | 87% | 95% | 0.014 | 0.010 | 0.022 | 0.037 |
| Comparison city at the mean shares | 27% | 94% | 90% | 98% | 106% | 0.033 | 0.025 | 0.045 | 0.067 |
| Share route, at the mean shares | 30% | 95% | 92% | 98% | 106% | 0.033 | 0.026 | 0.043 | 0.071 |
| Residents aged 15 and over at the city rate of employment, every one modern | 52% | 104% | 104% | 109% | 118% | 0.067 | 0.066 | 0.087 | 0.122 |
| Residents aged 15 and over, every one a worker in a modern job | 92% | 184% | 185% | 194% | 210% | 0.458 | 0.456 | 0.460 | 0.467 |
| *Panel B: Design-based specifications* | | | | | | | | | |
| Every worker added takes a modern job | | | | | | | | | |
| Double-robust, whole-match expectation | 35% | 133% | 137% | none | none | 0.166 | 0.161 | 0.157 | 0.256 |
| Double-robust, origin expectation | *log modern employment falls (−0.100), so there is no gain to fill* | | | | | | | | |
| Recentered, whole-match | 51% | none | none | none | none | 0.358 | 0.358 | 0.352 | 0.410 |
| Recentered, origin | *log modern employment falls (−0.006), so there is no gain to fill* | | | | | | | | |
| Workers added take modern jobs as often as migrants do | | | | | | | | | |
| Double-robust, whole-match expectation | 23% | 87% | 89% | none | none | 0.021 | 0.029 | 0.031 | 0.088 |
| Double-robust, origin expectation | *log modern employment falls (−0.100), so there is no gain to fill* | | | | | | | | |
| Recentered, whole-match | 33% | none | none | none | none | 0.167 | 0.209 | 0.207 | 0.291 |
| Recentered, origin | *log modern employment falls (−0.006), so there is no gain to fill* | | | | | | | | |

*Notes:* The main city sample, in the main specification with the full set of controls. Each row is the share of the modern-job gain that the people added could fill if nobody already working in the city changed sector, its one-sided 95% upper bound from inverting a one-sided test on the joint covariance of the two coefficients, and the one-sided $p$-value of a share of 100% or more. The headline row takes columns 2 and 6 of Table 4 at their mean levels with every worker added in a modern job. In Panel A, the first two rows use the rate at which migrants, workers born outside their district of residence, work in manufacturing or services, the third counts all workers with an industry code, the fourth and fifth put the comparison city at the mean shares, the fourth with the coefficient on log modern employment and the fifth with the coefficient on the modern share, and the last two count residents. Panel B repeats the two routes in workers, the headline row and the first row of Panel A, under four design-based specifications: the two double-robust specifications of Appendix C.1, which add as a control the exposure a city would expect under counterfactual settler assignments, from the whole-match reshuffle or the origin redraw, and the two recentered ones, which subtract that expected exposure from the treatment. The mean levels of employment and modern employment are the main specification's in every row. An upper bound of none means the bound is not finite. Variances: Conley SE within 200 km with a uniform kernel (the paper's) and a Bartlett kernel, and errors clustered by city and by province, each on one stacked regression of the row's specification. Source: 1980 and 2010 population censuses.

**The design-based specifications.** The design-based specifications of Appendix C.1 change both estimates and precision. Panel B of Table E9 repeats the headline row and the first row of Panel A under the four design-based specifications. When every additional worker takes a modern-sector job, only the main specification rejects. With the whole-match expectation as a control the workers added could fill 35% of the gain and the bound passes 100% (133%,

$p = 0.166$); at the migrants' rate the test still rejects at 5% under three of the four variances ($p = 0.021$ under the uniform kernel, bound 87%). With the recentered whole-match treatment the share is 51% with no finite upper bound ($p = 0.358$). In the two origin specifications log modern-sector employment falls, so there is no gain to fill, and the table shows the coefficient in place of a share. The point estimate of the share stays below 100% of the gain in every specification that has one.

**Birthplace and age composition.** The identity treats the extra workers as people who moved in. Transmigrant placement ran through the early 1980s, so a child born to an incumbent in that decade is a prime-age worker by 2010, and a fertility response would place locally born workers in the inflow. Table E10 reports statistically insignificant responses. Splitting 2010 employment by birthplace, native-born employment does not rise detectably, and the point estimate is concentrated among workers born elsewhere, whose employment rises by 20.8% while the migrant share of the workforce rises by 3.6 percentage points; neither is distinguishable from zero. The responses of children and youths per native-born adult are also statistically insignificant. These proxies do not rule out fertility responses or labor-force entry and exit. Both ratios are computed inside the native-born population, so adult in-migrants do not inflate the numerator or the denominator.

Table E10: In-Migration and Births as Sources of the Extra Workers

| Outcome, 2010 | Coefficient | SE | $t$ |
|---|---|---|---|
| *Panel A: Where the employment response comes from* | | | |
| Log employment, all workers | 0.0248 | 0.0582 | 0.43 |
| Log employment, born in this district | -0.0369 | 0.0375 | -0.98 |
| Log employment, born elsewhere | 0.1889 | 0.1307 | 1.45 |
| Migrant share of employment | 0.0359 | 0.0260 | 1.38 |
| *Panel B: Fertility, measured inside the native-born population* | | | |
| Children aged 0 to 5 per native-born adult aged 16 to 50 | 0.0313 | 0.0253 | 1.24 |
| Youth aged 5 to 16 per native-born adult aged 16 to 50 | 0.0299 | 0.0282 | 1.06 |

*Notes:* Each row is one regression of a 2010 population census outcome on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200 km. The columns are the coefficient, its standard error and their ratio. Native-born means born in the district of residence and born elsewhere means born outside it; Panel A counts employed workers of the person-level census cross that measures the sector mix of arrivals, in logs (so its first row differs slightly from column 2 of Table 4), and the migrant share is the born-elsewhere fraction of employment. Panel B's ratios are computed within the native-born population. SE is the Conley standard error and $t$ the coefficient divided by it.

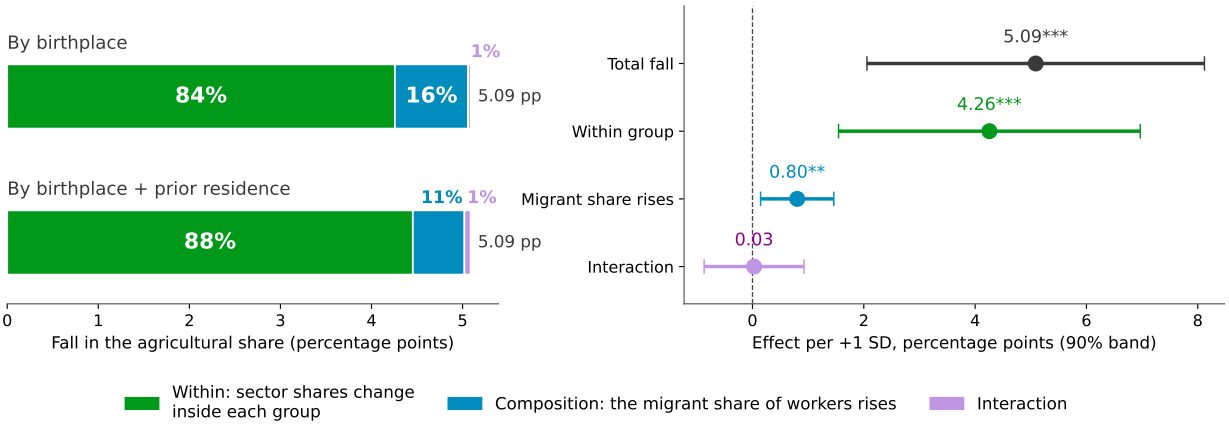
**Scope of the bound.** The bound asks whether the *arrivals'* composition could produce the fall, holding the incumbent workforce fixed. The birthplace decomposition of Appendix E.4 measures a wider object, because its composition component also allows the locally born share of the workforce to fall. The two quantities are therefore not nested, and both are small: the bound puts the arrivals-only channel at 4.9% of the fall using the lifetime stock and 7.6% using five-year movers, against 16% measured directly.

Two limitations remain. The requirement is computed on gross inflow with no offsetting outflow. A small net population response can conceal arrivals of modern-sector workers and departures of agricultural workers, producing sectoral reallocation without incumbent workers changing occupations. The native-employment coefficient in Table E10 does not identify those gross flows. And $s$ and $a$ enter as measured constants. They are means over the same 134 cities and are estimated far more precisely than the coefficients, but they contain sampling error that the reported standard errors omit.

## E.4 Decomposing the Agricultural-Share Fall: Construction

This appendix gives the construction behind Figure E7: the identity the decomposition relies on, the data it needs, the calculation it replaces, and the checks at each step. The exhibits label the two components by what they measure, the migrant share of the workforce and sector composition within origin groups; in the text I use the literature's terms, composition and within, for the same two objects.

Figure E7: Decomposition of the Fall in the Urban Agricultural Employment Share, by Origin




*Notes:* The figure uses the main city sample in the 2010 population census; each effect is estimated by a regression across urban centers on standardized agro-similarity in the main specification with the full set of controls and Conley SE within 200 km. Intervals are 90% confidence intervals from the same standard errors; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Each city's 2010 workforce is partitioned by birthplace (four groups) and, separately, by how recently a worker arrived (three groups). The stacked bars split the estimated fall in the agricultural employment share (percentage points per standard deviation, each piece labeled with its percentage of the fall) into a within component ("within birthplace group"), a composition component ("migrant share rises") and their interaction; the right half repeats the total and the three birthplace pieces with their intervals. Because the census is a repeated cross-section, the within component includes cohort replacement.

### The identity

Index cities by $c$ and partition each city's 2010 workforce into origin groups $g$. Let $\omega_{gc}$ be group $g$'s share of city $c$'s workers and $a_{gc}$ that group's own agricultural employment share. The city agricultural share is then an exact accounting identity:

$$s_c \;=\; \sum_g \omega_{gc}\, a_{gc}. \tag{21}$$

Write each factor as its sample mean plus a deviation, $\omega_{gc} = \bar{\omega}_g + u_{gc}$ and $a_{gc} = \bar{a}_g + v_{gc}$, and Equation 21 becomes

$$s_c \;=\; \underbrace{\sum_g \bar{\omega}_g \bar{a}_g}_{\text{constant}} \;+\; \underbrace{\sum_g \bar{a}_g\, u_{gc}}_{\text{migrant share}} \;+\; \underbrace{\sum_g \bar{\omega}_g\, v_{gc}}_{\text{within origin}} \;+\; \underbrace{\sum_g u_{gc} v_{gc}}_{\text{interaction}}. \tag{22}$$

The second term is the *composition* component: it moves only because the local-versus-migrant makeup of the workforce changes, holding what each group does fixed. The third is the *within* component. It moves only because groups change what they do, holding that makeup fixed. I construct the three right-hand terms as variables and regress each on the *same* main specification used for the total. Because Equation 22 is an identity and the regressions share a design matrix, the three coefficients sum to the coefficient on $s_c$ exactly, so no residual category absorbs the difference. Each component nonetheless has its own Conley standard error, so the split reports precision as well as magnitude.

Two partitions are used. The first is birthplace, in the four mutually exclusive groups of Figure 13 and Panel B of Table C1: born in the current district, born in a different district of the same province, born in a different outer-island

province, and born on Java or Bali. The second combines birthplace and prior residence in three groups: workers born in the current district, lifetime migrants living there five years before the census, and lifetime migrants living elsewhere five years earlier. This partition uses observed locations and does not establish a complete migration history.

The interpretation comes with one caveat. The census is a repeated cross-section, so "born in this district" in 2010 is not the same list of people as in 1980. The within term measures sector composition among workers of a given origin, including whoever replaced whoever left. The within term therefore holds both sector changes inside a cohort and the replacement of one cohort by another; Budí-Ors (2026) puts cohort replacement at 68% of Indonesia's aggregate agricultural-share fall.

### The data the identity needs

Evaluating Equation 21 requires each city's workers classified by origin *and* industry in the same data. The city-level census data hold the birthplace shares and, separately, the industry counts, but nothing crossing them, so the cross is built from the 2010 individual census records for the outer islands (93.5 million persons). Three definitional choices align it with the published outcome:

- The employment base is every person with a valid industry code (40.1 million workers), which is the denominator every sectoral share in this paper uses. Using the stricter economically active count as the base would produce shares that do not sum to those of the main results
- Agriculture is industry codes 1 through 6, as everywhere else in the paper
- No birth-cohort restriction is imposed. The 2010 birthplace shares of Figure 13 and Table C1 restrict to the 1955-1990 cohort in order to match 1980 birth-district characteristics; imposing that here would place the cross on a different population than the published agricultural share and break the reconciliation below. Birthplace is recorded for 100% of the workers in the base, so no group has a thin denominator

The person-level counts are collapsed to the 2010 village, crosswalked to the year-2000 subdistrict geography the rest of the paper uses, and rolled up to urban centers over the reclassified membership. Both partitions recover the total workforce exactly in every one of the 2,020 subdistricts, and Equation 21 holds in every city.

The remeasured share is compared with the agricultural share of Table 4 before anything is split. Its mean across the main city sample matches to three decimals (0.374 against 0.374), the two measures correlate at 1.000, and the main specification coefficient on the remeasured share, −0.0509 (standard error 0.0184), reproduces that table's −0.0509 (0.0184) on the same cities.

### Subdistricts claimed by two urban centers

Urban centers are satellite-drawn built-up polygons while the census reports by subdistrict, and five outer-island subdistricts are large enough to contain two separate built-up areas. The sample records both claims. Summing constituent subdistricts into city totals would give the whole subdistrict to each of the two claimants, so 95,728 workers across the main city sample, or 0.78% of the total, would be counted twice. Both claimants lie inside the sample in two of the five cases.

The city totals throughout the paper divide such a subdistrict. Each village is assigned to the claimant whose built-up polygon overlaps more of its area, or, for a village outside both polygons, to the nearer claimant. Each city's allocation weight is its assigned villages' share of observed 2010 employment in the subdistrict. Village polygons without matched employment records do not enter these weights. I apply the same weights to the available additive census counts in every year, including 1980 population, urban population, employment, and agricultural employment, then recalculate the city shares and logs from the allocated counts. This allocates the unobserved 1980 city portions in proportion to observed 2010 employment. The weights sum to one within each shared subdistrict and preserve its recorded totals;

conservation does not establish that the 2010 shares describe its 1980 distribution. With the subdistricts divided, the within share is 83.7%, and leaving out the 5 cities marked as having a shared subdistrict gives 83.7%. This omission comparison concerns the decomposition and does not separately validate the allocation of 1980 baselines.

## E.5 Public Provision and Other Alternative Channels

I argue that agricultural productivity reshaped the employment of nearby cities through the goods market. The Indonesian state settled these villages, and the same state built the roads, schools, health posts and electricity lines that serve them. If public provision followed the settlements into the places where they turned out productive, public provision could contribute to the urban response. This subsection examines road quality, village facilities, and the allocation of the INPRES program.

**Roads.** The trade channel runs on the road network, so the sharpest form of the objection is that exposed corridors received better roads. I test this with the Indonesian Road Management System panel assembled by Gertler et al. (2024), which gives road-quality-implied travel speeds on a fixed network in 1990 and 2007 for Sumatra and Sulawesi, the two outer islands the source covers. Each subdistrict takes the speed of its nearest 1990 district, and I average the speed over three areas of each city, weighting subdistricts by their 1990 population: the city itself, its countryside with settlements and its countryside without settlements (103, 103 and 97 cities). In 1990 road speed did not differ detectably with exposure in any of the three areas, at −0.46 km/h per standard deviation of exposure in the city (standard error 1.15), −0.94 km/h in the countryside with settlements (0.90) and −0.09 km/h in the countryside without settlements (0.96), against a mean of 60.6 km/h across rural subdistricts. The countryside with settlements sat farther out in the road network, by 39.4 km (standard error 22.5), marginally significant under both kernels and insignificant with province clusters. Holding the 1990 speed fixed, the 2007 speed differs by −0.92 km/h in the countryside with settlements (standard error 0.36) and by −0.73 km/h in the countryside without settlements (0.30), both statistically significant, and by −0.73 km/h in the city (0.50), statistically insignificant, against a mean of 62.3 km/h across rural subdistricts in 2007. Roads in the countryside of more exposed cities therefore improved less, and road quality on this network cannot account for the transformation of those cities. With controls for the 1983 baseline of road access in the village census, roads passable by a four-wheel vehicle, paved roads and distance to the subdistrict seat, which shows some imbalance in favor of the more exposed places, the two countryside estimates are −0.76 km/h (standard error 0.36) and −0.67 km/h (0.27). With the rural subdistrict as the unit, the same specification gives −0.44 km/h (standard error 0.44) over 583 subdistricts, statistically insignificant, also with standard errors clustered by 1990 district (86 districts), the level at which the speed is measured (Table E11).

The test has two limits. Network distance is identical to the last digit across 1990 and 2007 for every Sumatran and Sulawesi district in the source, so the travel-time calculation changes road speeds on a fixed road network and cannot detect newly built roads. It covers only Sumatra and Sulawesi.

**Facilities.** I track what the state puts inside a village, city amenities from the 1980 village census to the 2003 and 2011 waves. Column (2) regresses the 1980 level itself on exposure, so the table shows whether public provision already differed before the settlements matured, and columns (3) and (4) give the change over each window with the amenity's own 1980 level held fixed (Appendix Table E12). Facility counts are per 1,000 people at the population of each year.

Two facts come out of it. The 1980 levels differ in three of the seven government-provided amenities. Senior-secondary schools and polyclinics show a 1980 gradient, at 0.0149 (standard error 0.0042) and 0.0151 (standard error 0.0051) per 1,000 people and per standard deviation of exposure, against means of 0.072 and 0.073, and the population share on a road passable by a four-wheel vehicle is higher by 5.2 percentage points (standard error 3.0), marginally significant, against a mean of 80%. The exposed cities entered the sample period with slightly more of two facilities the state provides. With each amenity's own 1980 level held fixed, the changes are statistically insignificant over both windows in schooling, in health centers and in both road-surface measures.

Table E11: Road Speed and Exposure in the City and Its Countryside

| | City | Countryside with settlements | Countryside without settlements | Rural subdistricts |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Road speed 2007, 1990 speed held fixed | −0.73 | −0.92*** | −0.73** | −0.44 |
| | (0.50) | (0.36) | (0.30) | (0.44) |
| *p-value* | 0.143 | 0.009 | 0.015 | 0.319 |
| *N* | 103 | 103 | 97 | 583 |
| with the 1983 road measures | −0.67 | −0.76**† | −0.67** | −0.42 |
| | (0.53) | (0.36) | (0.27) | (0.43) |
| *p-value* | 0.205 | 0.036 | 0.012 | 0.326 |
| *N* | 103 | 103 | 97 | 576 |
| Road speed 1990 | −0.46 | −0.94 | −0.09 | 0.20 |
| | (1.15) | (0.90) | (0.96) | (0.79) |
| *p-value* | 0.688 | 0.301 | 0.929 | 0.796 |
| *N* | 103 | 103 | 97 | 583 |
| Network distance 1990 (km) | 47.1 | 39.4*† | 27.8 | 34.8 |
| | (33.5) | (22.5) | (28.0) | (29.3) |
| *p-value* | 0.160 | 0.080 | 0.321 | 0.235 |
| *N* | 103 | 103 | 97 | 583 |
| Change in road speed, 1990 to 2007 | −0.35 | −0.18 | −0.66 | −0.60 |
| | (1.06) | (0.77) | (0.74) | (0.87) |
| *p-value* | 0.738 | 0.820 | 0.371 | 0.490 |
| *N* | 103 | 103 | 97 | 583 |

*Notes:* Each cell is one regression of the outcome on standardized agro-similarity in the main specification with the full set of controls and Conley SE within 200 km; the first two blocks hold the area's 1990 speed fixed, and the second adds the three 1983 village-census road measures (roads passable by a four-wheel vehicle, paved roads, distance to the subdistrict seat). Speeds are km/h on a fixed network from the Indonesian Road Management System panel of Gertler et al. (2024), Sumatra and Sulawesi; each subdistrict takes the speed of its nearest 1990 district, and an area averages its subdistricts by 1990 population. Columns (1) to (3) are the cities of the main city sample on the two islands; column (4) is the rural subdistricts of the paper's rural tables, with the settler control of those tables and exposure standardized over them. Conley SE (uniform kernel) in parentheses and the uniform-kernel p-value beneath, shaded by band (0.10 or more, below 0.10, below 0.05, below 0.01). The stars use the uniform-kernel SE: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. † significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with province clusters or, in column (4), with 1990-district clusters.

Polyclinics are the one exception, positive and significant by 2011, at 0.0139 (standard error 0.0060) against a 1980 mean of 0.073, an estimate that stays significant with the Bartlett kernel and marginally significant with province clusters. A facility stock does not identify its financing.

Of the 21 coefficients in the government-provided block, only the three 1980 levels and the polyclinic change to 2011 cross a conventional threshold. Without the 1980 level, all 14 changes are statistically insignificant.

**Electricity.** Grid connection is where the change regressions do move when the 1980 level is left out, and the two margins of it separate cleanly once that level is held fixed. On the extensive margin, without the 1980 level, the share of a city's population living in a village on the state grid falls by 6.4 percentage points per standard deviation of exposure over 1980 to 2003 and by 7.2 percentage points over 1980 to 2011, both statistically significant. That is the less exposed cities catching up. Agriculturally favored places began 10.9 percentage points more connected, a statistically significant difference, so a change regression with no lagged outcome was measuring convergence, and holding the 1980 level fixed, as columns (3) and (4) do, removes the result in both windows, leaving −0.1 and −0.4 percentage points. The 1980 measure is the narrower of the two. The 1980 form records only a village's main source of electricity, so a 1980 village counts as on the grid only where the state utility is its main supplier, while a 2003 or 2011 village counts where any household takes grid electricity.

The intensive margin behaves differently and is the only electricity result that survives the 1980 level. The share of households connected also starts higher in favored places, at 5.7 percentage points per standard deviation with a

Table E12: City Amenities and Central Government Programs From the 1980 Baseline

| | 1980 baseline | | Change, 1980 level held fixed | |
|---|---|---|---|---|
| | Mean in 1980 (1) | 1980 level on exposure (2) | 1980 to 2003 (3) | 1980 to 2011 (4) |
| *Panel A: Government-provided: schooling and health, facilities per 1,000 people* | | | | |
| Primary schools | 1.159 | -0.0149 | -0.0156 | 0.0060 |
| | | (0.0786) | (0.0195) | . |
| | | [0.0726] | [0.0298] | [0.0193] |
| *p-value* | | 0.850 | 0.423 | . |
| Junior-secondary schools | 0.144 | 0.0053 | -0.0085 | -0.0069 |
| | | (0.0110) | (0.0087) | (0.0101) |
| | | [0.0107] | [0.0096] | [0.0088] |
| *p-value* | | 0.627 | 0.326 | 0.494 |
| Senior-secondary schools | 0.072 | 0.0149*** | 0.0009 | 0.0050 |
| | | (0.0042) | (0.0055) | (0.0062) |
| | | [0.0061] | [0.0071] | [0.0064] |
| *p-value* | | 0.000 | 0.876 | 0.422 |
| Health centers (puskesmas) | 0.081 | -0.0001 | 0.0019 | 0.0032 |
| | | (0.0111) | (0.0016) | (0.0024) |
| | | [0.0106] | [0.0023] | [0.0025] |
| *p-value* | | 0.994 | 0.219 | 0.178 |
| Polyclinics | 0.073 | 0.0151*** | 0.0076 | 0.0139** |
| | | (0.0051) | (0.0058) | (0.0060) |
| | | [0.0058] | [0.0063] | [0.0065] |
| *p-value* | | 0.003 | 0.190 | 0.020 |
| *Panel B: Government-provided: road surface, share of population* | | | | |
| Asphalt main road | 0.435 | -0.0244 | -0.0343 | 0.0141 |
| | | (0.0495) | (0.0278) | (0.0289) |
| | | [0.0494] | [0.0283] | [0.0266] |
| *p-value* | | 0.622 | 0.217 | 0.625 |
| Passable by four-wheel vehicle | 0.797 | 0.0516*† | 0.0010 | 0.0104 |
| | | (0.0300) | (0.0038) | (0.0096) |
| | | [0.0284] | [0.0054] | [0.0109] |
| *p-value* | | 0.085 | 0.800 | 0.279 |
| *Panel C: Mixed provision: grid electricity* | | | | |
| Villages on the grid, share of population | 0.353 | 0.1090*** | -0.0005 | -0.0040 |
| | | (0.0346) | (0.0128) | (0.0128) |
| | | [0.0377] | [0.0131] | [0.0124] |
| *p-value* | | 0.002 | 0.968 | 0.757 |
| Households connected, share of households | 0.175 | 0.0574*** | 0.0339 | 0.0449*† |
| | | (0.0181) | (0.0251) | (0.0233) |
| | | [0.0222] | [0.0259] | [0.0249] |
| *p-value* | | 0.002 | 0.177 | 0.054 |
| Observations | | 133 | 133 | 133 |

*Notes:* Each cell in columns (2) to (4) is one regression of the outcome, from the village census (PODES), on standardized agro-similarity on the cities of the main city sample the census covers in 1980 and in the endline year, in the main specification with the full set of controls and Conley SE within 200 km; column (2) leaves out the four 1980 baselines because its outcome is a 1980 value, and columns (3) and (4) add the outcome's own 1980 level. Conley SE (uniform kernel) in parentheses and Bartlett-kernel SE in brackets, with the uniform-kernel p-value beneath, shaded by band (0.10 or more, below 0.10, below 0.05, below 0.01); a dot marks a cell with no uniform-kernel standard error. The stars use the uniform-kernel SE: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. † significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with province clusters (17 provinces), or with both.

standard error of 1.8, and the plain changes are small and imprecise, 1.1 percentage points (standard error 2.4) by 2003 and 1.9 percentage points (standard error 2.1) by 2011. With the 1980 level held fixed the changes grow to 3.4 percentage points by 2003, statistically insignificant, and 4.5 percentage points by 2011, marginally significant under both kernels and insignificant with province clusters. So the pattern is a modest deepening of connections inside places that were already connected, on a base of about 18% of households connected in 1980.

Electrification reflects both the state's supply and household demand. The grid is a state utility, while the decision to connect a household responds to demand and ability to pay. The connection estimates do not distinguish those explanations. Table A1 adds the 1980 levels of senior-secondary schools and polyclinics and the two grid measures, then the road measures, to the controls of the main specification.

Table E12: City Amenities and Central Government Programs From the 1980 Baseline, Continued

| | 1980 baseline | | Change, 1980 level held fixed | |
|---|---|---|---|---|
| | Mean in 1980 (1) | 1980 level on exposure (2) | 1980 to 2003 (3) | 1980 to 2011 (4) |
| *Panel D: Central government program: INPRES, 1980 level* | | | | |
| INPRES schools, share of primary schools | 0.3355 | -0.0063 | | |
| | | (0.0129) | | |
| *p-value* | | 0.623 | | |
| INPRES schools per 1,000 people | 0.3831 | -0.0097 | | |
| | | (0.0243) | | |
| *p-value* | | 0.689 | | |
| INPRES teachers, share of primary-school teachers | 0.3228 | -0.0047 | | |
| | | (0.0115) | | |
| *p-value* | | 0.683 | | |
| INPRES buildings, share of school buildings | 0.3384 | -0.0040 | | |
| | | (0.0127) | | |
| *p-value* | | 0.753 | | |
| INPRES public latrines, share of latrines | 0.4010 | 0.0207 | | |
| | | (0.0592) | | |
| *p-value* | | 0.727 | | |
| Placebo: schools INPRES did not build, per 1,000 | 0.7778 | -0.0023 | | |
| | | (0.0657) | | |
| *p-value* | | 0.973 | | |

*Notes, continued:* Panel D: The presidential-instruction (INPRES) primary-school construction program built over 61,000 schools from 1973 to 1978, so the 1980 levels show where it had already built. The 1980 village census records INPRES items separately from the rest. The first, third, fourth and fifth rows of Panel D are INPRES shares of the total, the second is INPRES schools per 1,000 people, and the sixth, the placebo, is the schools the program did not build per 1,000 people. Each Panel D row regresses the 1980 level on exposure as column (2) does, on the 133 cities of the main city sample that the 1980 village census covers (the latrine row on the 120 cities with a public latrine), and shows the uniform-kernel SE only. The one main-sample city the 1980 village census does not cover is Pemangkat (West Kalimantan).

**INPRES school allocation.** The sharpest available test examines a central-government program directly. Indonesia's INPRES primary-school construction program built over 61,000 schools between 1973 and 1978, before the 1980 village census was taken (Duflo 2001). That census records INPRES-built schools, INPRES-paid teachers, INPRES-built school buildings and INPRES-built public latrines separately from everything else, so the program is identified by its own label, and the question becomes whether the central government's flagship rural program had already favored agriculturally advantaged places by the start of my sample period.

Panel D of Table E12 reports six versions of the test, and each estimate is not statistically significant. The INPRES share of a city's primary schools changes by −0.0063 per standard deviation of exposure (standard error 0.0129) against a mean of 0.336, the INPRES share of primary-school teachers by −0.0047 (standard error 0.0115) against a mean of 0.323, and the INPRES share of school buildings by −0.0040 (standard error 0.0127) against a mean of 0.338, three separate questionnaire items that agree both in their null and in placing about a third of primary schooling under the program. Public latrines give the cleanest form of the contrast, since INPRES and non-INPRES latrines are counted separately within one facility type in one year and one place, and the INPRES share there is 0.0207 (standard error 0.0592) against a mean of 0.401. The placebo is null, with non-INPRES schools per 1,000 people at −0.0023 (standard error 0.0657). One qualification: 13 of the 133 cities have no public latrine in 1980, so the latrine row covers 120 cities.

The road evidence covers road quality on a fixed network in Sumatra and Sulawesi, and the facility evidence covers stocks and connections. These tests do not identify their financing or rule out other public programs. Around the settlements, the state grid extended to more villages on the subdistrict's exposure. In the transmigration villages themselves, the population share with a main road passable by car rises with their own agro-similarity, with a marginally significant estimate; that road response becomes imprecise with Bartlett distance weights or province clustering. The electricity, school and health responses are imprecise (Table E13). Rural public investment remains a possible contributor through food production and trade.

Table E13: Public Provision in the Transmigration Villages, 2011

| | (1) Asphalt road | (2) Car road | (3) Grid reached | (4) Households on grid | (5) Primary | (6) Junior sec. | (7) Senior sec. | (8) Health centers |
|---|---|---|---|---|---|---|---|---|
| Agro-similarity of the village | -0.032 | 0.028* | 0.051 | 0.020 | 0.009 | 0.023 | -0.003 | 0.021 |
| | (0.034) | (0.016) | (0.037) | (0.023) | (0.033) | (0.018) | (0.010) | (0.025) |
| Outcome mean | 0.331 | 0.896 | 0.705 | 0.422 | 0.975 | 0.314 | 0.128 | 0.403 |
| Transmigration villages | 808 | 808 | 813 | 813 | 813 | 813 | 813 | 813 |

*Notes:* Each column regresses a transmigration village's 2011 outcome on the village's own agro-similarity (Bazzi et al. 2016), standardized over the 814 transmigration villages, with the village geography controls (ruggedness, distance to the coast, three slope classes, temperature, log altitude); Conley SE within 200 km. A village's 2011 records are the 2011 villages that descend from its year-2000 code. Columns 1 and 2: population share with an asphalt main road and with a main road passable by car. Column 3: population share in hamlet units the state grid has reached. Column 4: households on the state grid. Columns 5 to 7: schools per 1,000 people. Column 8: puskesmas and puskesmas pembantu per 1,000 people. The villages were settled from 1979 to 1988, so no 1980 value exists for them. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: village census (PODES) 2011.

This subsection also holds the tests behind Section 5.3: public provision in the transmigration villages and in the rural subdistricts with a settlement, the state electricity grid wave by wave, and the alternative channels in the city and its countryside (Table E14). Every regression is the main specification with the full set of controls and Conley SE within 200 km, with each outcome's own 1980 level held fixed where one exists. Table E14 shows the Bartlett-kernel standard error beside the uniform one and marks with a dagger an estimate statistically significant at the 10% level with the uniform kernel and not with the Bartlett kernel, with province clusters, or with both. Two standardized coefficient plots in the Online Supplement show every estimate of public provision by area and by rural subdistrict, of the road network, of banks, of new district seats and of the spending of rural households. The tests on the conflict provinces, the exposure of neighboring cities and the city boundary are panels of Table A1.

**Public provision in the transmigration villages.** The electricity, school and health responses to the village's own agro-similarity are imprecise with village geography controls: the population shares with an asphalt main road and in hamlets the state grid has reached, the share of households on the grid, schools per 1,000 people at the three levels and health centers per 1,000 people are statistically insignificant. The share with a main road passable by car is marginally significant under the uniform kernel (Table E13) and becomes imprecise with Bartlett distance weights or province clustering. Without the geography controls, the grid and car-road shares are statistically significant. The villages were settled from 1979 to 1988 and lack comparable pre-settlement provision baselines.

**The state electricity grid.** The state electricity grid was extended to more of the villages in the more exposed rural subdistricts with a settlement from 1990 on, and households in villages already on the grid did not connect at a higher rate. With the 1980 level measured on the state grid at both ends, the share of people in villages on the grid rises with exposure by 0.0362 (standard error 0.0149) in 1990 and by 0.1073 (0.0227) in 2011, while the household share within those villages changes by 0.0137 (0.0218) in 2011. Electricity from any supplier in 2011 does not move detectably with exposure (−0.0000, standard error 0.0010), so part of the grid's rise replaced other suppliers. A grid line can follow demand that is already there, so the test does not separate the state's supply from household demand (Figure E8).

**Provision by area, roads, banks, new district seats and rural spending.** At a given 1980 level, the city's roads do not respond detectably to exposure, while the countryside with settlements and the countryside without settlements both gain asphalt roads by 1990 and grid connections by 2011 (the grid gain is marginally significant where the settlements are), and senior secondary schools rise only in the countryside with settlements. With each rural subdistrict as a unit, asphalt roads and household electricity rose with exposure in subdistricts with and without a settlement alike, and the two slopes are statistically indistinguishable. In 1980 asphalt roads were already more common in the subdistricts with a settlement, by 0.0471 (0.0186). The road network around more exposed cities did not improve detectably more

Table E14: Alternative Channels in the City and Its Countryside

| | The city | | The countryside with settlements | | The countryside without settlements | |
|---|---|---|---|---|---|---|
| | Estimate | Cities | Estimate | Cities | Estimate | Cities |
| *Panel A: Finance* | | | | | | |
| Banks | 0.0110 | 133 | 0.0119*† | 131 | 0.0077*** | 124 |
| per 1,000 people, 1980 to 2011 | (0.0092) | [0.0093] | (0.0072) | [0.0060] | (0.0026) | [0.0034] |
| Bank presence | 0.0250 | 122 | -0.0139 | 123 | 0.0184*† | 117 |
| population share, 1980 to 2011 | (0.0248) | [0.0222] | (0.0137) | [0.0130] | (0.0112) | [0.0118] |
| Finance and insurance employment | 0.0009** | 134 | | | | |
| share of employment, 1980 to 2010 | (0.0004) | [0.0005] | | | | |
| *Panel B: Public provision* | | | | | | |
| Asphalt main road | 0.0271 | 131 | 0.0592*** | 129 | 0.0673*** | 123 |
| population share, 1980 to 1990 | (0.0172) | [0.0280] | (0.0207) | [0.0217] | (0.0162) | [0.0175] |
| Main road passable by car all year | 0.0193 | 122 | 0.0200* | 121 | 0.0284*** | 116 |
| population share, 1980 to 2011 | (0.0127) | [0.0136] | (0.0106) | [0.0091] | (0.0084) | [0.0083] |
| Road density around the city | -0.0757 | 131 | | | | |
| current road map, 1950 density held fixed | (0.0893) | [0.0990] | | | | |
| Households on the state grid | | | 0.0549*† | 131 | 0.0789*** | 125 |
| household share, 1980 to 2011 | | | (0.0286) | [0.0263] | (0.0194) | [0.0198] |
| Population in a village with grid power | | | 0.0712*** | 123 | 0.0552*** | 119 |
| population share, 1980 to 2011 | | | (0.0219) | [0.0217] | (0.0168) | [0.0168] |
| Senior secondary schools | | | 0.0421*** | 131 | 0.0030 | 125 |
| per 1,000 people, 1980 to 2011 | | | (0.0100) | [0.0121] | (0.0038) | [0.0043] |
| Health centers | 0.0095 | 128 | 0.0075 | 131 | -0.0067 | 125 |
| per 1,000 people, 1980 to 2011 | (0.0097) | [0.0106] | (0.0186) | [0.0174] | (0.0155) | [0.0146] |
| Seat of a district created 1999 to 2010 | -0.0064 | 134 | | | | |
| seat in 1998 held fixed | (0.0518) | [0.0558] | | | | |
| Log modern employment, 2010 | 0.1653**† | 134 | | | | |
| less education, health, social | (0.0711) | [0.0660] | | | | |
| and government services | | | | | | |
| *Panel C: Rural demand* | | | | | | |
| Log spending per person, 2011 | | | -0.0225 | 131 | -0.0307 | 128 |
| rural households | | | (0.0393) | [0.0353] | (0.0273) | [0.0264] |
| rightmost estimate: whole hinterland | | | | | | |
| Non-food share of spending, 2011 | | | -0.0023 | 131 | -0.0060 | 128 |
| rural households | | | (0.0078) | [0.0080] | (0.0054) | [0.0047] |
| rightmost estimate: whole hinterland | | | | | | |

*Notes:* Each cell regresses the row's outcome, measured over the column's area, on standardized agro-similarity on the main city sample, in the main specification with the full set of controls and Conley SE within 200 km plus the outcome's own 1980 value or the earlier value its label names; the modern-sector row, which counts workers with a recorded industry, holds the log of its 1980 count fixed, and the two spending rows do not add a 1980 value, since the household survey waves used here start in 1994. Conley SE within 200 km in parentheses (uniform kernel, which the stars use) and Bartlett-kernel SE in brackets; Cities counts the cities in each cell. The countryside with settlements is the rural subdistricts with a transmigration village within 300 travel minutes of the city, its own subdistricts left out; the countryside without settlements is the rural subdistricts within the city's 300-minute window (208.7 minutes on the subdistrict travel-time matrix, its calibrated equivalent) with no transmigration village, and the whole hinterland is every rural subdistrict within that window. Road density is measured over the subdistricts that hold the city's catchment villages, every subdistrict that belongs to a city left out; electricity and schools are tested in the countryside only. A presence share is the population share in villages with the item (for roads, of villages reached by land); health centers are puskesmas and auxiliary centers (pustu). The presence shares to 2011 leave out the 8 cities of West Kalimantan, whose villages were merged between the 1990 and 1993 village censuses; the car-access and bank-presence shares also leave out the West Sumatra subdistricts recorded as nagari in 2011, a larger unit than the 1980 village, which the grid-village share keeps, measured on their hamlets. Without those subdistricts these estimates of the countryside without settlements change their star level (uniform-kernel SE): senior secondary schools, per 1,000 people, 1980 to 2011: 0.0067* (0.0041) on 123 cities; banks, per 1,000 people, 1980 to 2011: 0.0060** (0.0028) on 122 cities. Sources: village censuses (PODES) 1980 to 2014, population censuses 1980 and 2010, Susenas 2011, the current national and provincial road map and the 1950 atlas, and the World Bank district proliferation crosswalk. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. † significant at 10% with the uniform kernel and not significant at 10% with the Bartlett kernel, with errors clustered by province (a least-squares test not shown), or with both.

than its earlier level predicts. Banks per person and bank presence do not move detectably with exposure in the city at a given 1980 level, and in the countryside with settlements bank presence does not move detectably while banks per person rise, a marginally significant gain. The village censuses count banks and do not record deposits or lending. Exposure does not detectably predict the seat of a new district. Log modern-sector employment without education, health, social and government services rises by 0.165 (0.071), statistically significant under both Conley kernels and not statistically

Table E15: Public Provision in Rural Subdistricts With a Settlement and Roads Around the City

*Panel A: Rural subdistricts with a settlement, 2011*

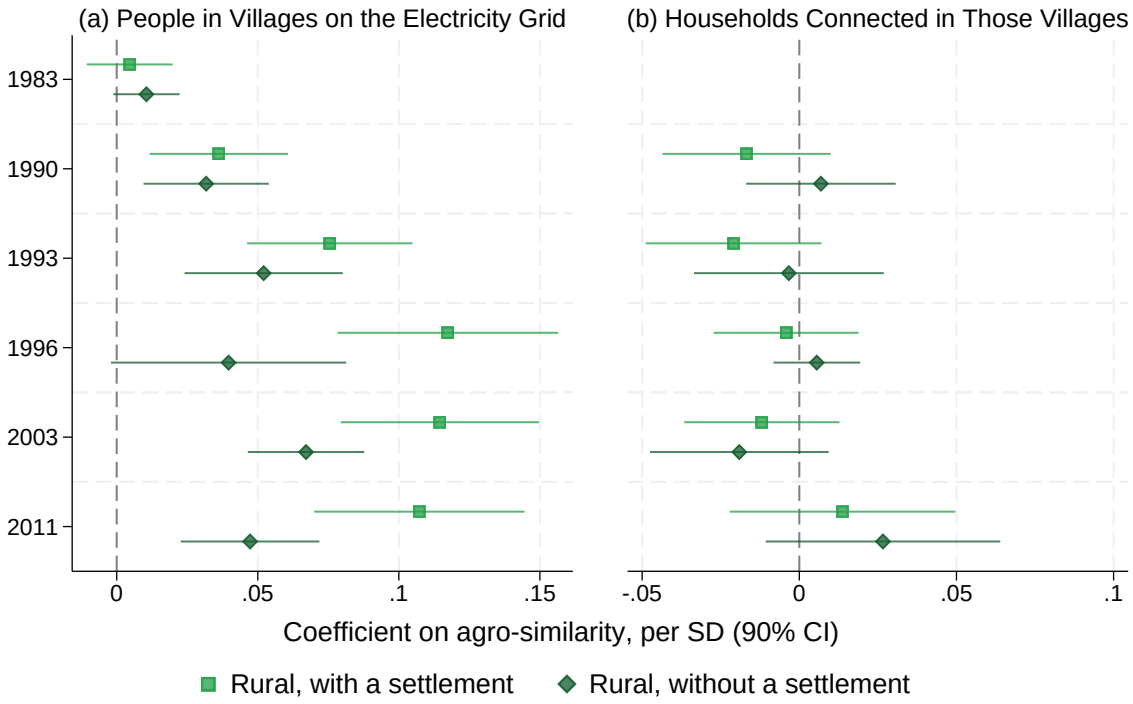
| | Asphalt main road | Households connected to the grid | Schools per 1,000 people | | | Health centers per 1,000 people |
|---|---|---|---|---|---|---|
| | | | Primary | Junior secondary | Senior secondary | |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Agro-similarity | 0.0116 | 0.0137 | 0.0187 | -0.0064 | 0.0082 | -0.0206 |
| | (0.0257) | (0.0218) | (0.0336) | (0.0086) | (0.0077) | (0.0158) |
| 1980 level of the outcome | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Subdistrict controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Outcome mean | 0.627 | 0.653 | 1.077 | 0.290 | 0.130 | 0.308 |
| Rural subdistricts | 651 | 665 | 764 | 787 | 787 | 771 |
| with a settlement | 168 | 172 | 185 | 191 | 191 | 187 |

*Panel B: The city and its catchment*

| | Log road density on the current map | | Road corridors, change 1990 to 2007 | |
|---|---|---|---|---|
| | Catchment | Within 5 km of a transmigration village | Roughness | Unpaved share |
| | (1) | (2) | (3) | (4) |
| Agro-similarity | -0.0757 | -0.1667 | 0.0586 | 0.0145 |
| | (0.0893) | (0.1992) | (0.3007) | (0.0203) |
| 1950 road density | ✓ | ✓ | | |
| 1990 level of the outcome | | | ✓ | ✓ |
| Main specification controls | ✓ | ✓ | ✓ | ✓ |
| Outcome mean | 1.665 | 1.802 | -0.991 | -0.466 |
| Cities | 131 | 131 | 132 | 125 |

*Notes:* Panel A: Each column regresses a rural subdistrict's 2011 outcome on agro-similarity (per SD over the 794 rural subdistricts of the main sample, 191 of them with a settlement) times an indicator for a transmigration village in the subdistrict and times its complement, the indicator, the outcome's 1980 level, the settler count, the four 1980 census baselines and the 16 agro-climatic covariates; the row is the coefficient for subdistricts with a settlement, with Conley SE within 200 km in parentheses. The settler count is the number of settlers placed in the subdistrict's own transmigration villages where it has one and, where it has none, the settlers of the villages within 300 travel minutes weighted by inverse distance, as for cities. Column 1: the population share living in a village whose main road is asphalt. Column 2: the share of households connected to the state electricity grid among the households of the villages the grid has reached, with the 1980 population and household shares with electricity from the grid and from another supplier held fixed. Columns 3 to 6: facilities per 1,000 people; health centers are community health centers and their branches. Columns 1 and 2 leave out the 71 subdistricts of West Kalimantan, which merged its villages after 1990, and the 51 West Sumatra subdistricts recorded as nagari in 2011, a larger unit; column 2 holds the subdistricts where the grid had reached a village by 2011. Panel B: Each column regresses a road measure on standardized agro-similarity on the main city sample (134 cities), in the main specification with the full set of controls and Conley SE within 200 km, adding the earlier level its check mark names. Columns 1 and 2: ln(1 + km of road per 100 square km) on the current national and provincial road map, over the rural subdistricts that hold the city's transmigration villages within 300 travel minutes (column 1) or the part of them within 5 km of a transmigration village (column 2), with the same measure on the 1950 road atlas held fixed. Columns 3 and 4: the change from 1990 to 2007 in roughness (International Roughness Index, meters per km) and in the unpaved share of the road-condition survey posts within 10 km of the straight lines from the city to its transmigration villages. Each regression excludes cities with missing outcome data: 3 cities have no rural subdistrict holding their transmigration villages (columns 1 and 2), 2 have no survey post with roughness measured in both years (column 3), and 9 have none with the surface recorded in both years (column 4). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Sources: village censuses (PODES) of 1980 and 2011; population census 1980; the 1950 road atlas, the current national and provincial road network, and the road-condition survey of 1990 and 2007.

Figure E8: Exposure and the State Electricity Grid by Village-Census Wave




*Notes:* Each marker is the coefficient on agro-similarity (per SD over the 794 in-sample rural subdistricts) for rural subdistricts with and without a settlement, with its 90% interval from Conley SE within 200 km. One regression per wave and panel: the outcome on agro-similarity times the indicators of a subdistrict with and without a settlement, the with-settlement indicator, the 1980 population shares in villages whose main source of electricity is the state grid (PLN) and another supplier (panel (b): also the 1980 household shares using each), and the controls of the main specification measured at the subdistrict, except that the settler count of a subdistrict with a settlement is the number of settlers placed in the transmigration villages that overlap it. Sample: the 794 in-sample rural subdistricts less the 51 West Sumatra subdistricts recorded as nagari in 2011, 743 in all (191 with a settlement, 552 without), less the 71 of West Kalimantan from 1993 on, when the province merged its villages; the 41 subdistricts in the window that belong to a reclassified city are already outside the 794. Panel (a): the share of the subdistrict's people in villages with a household on PLN. Panel (b): households on PLN over all households of those villages, from 1990. Electricity from any supplier in 2011, with a settlement: −0.0000 (standard error 0.0010). Source: village censuses (PODES) 1980 to 2011 on year-2000 subdistricts; population census 1980.

significant with province clusters. The industry codes left out hold public and private employers alike, so the cut does not separate public from private employment. Rural households around more exposed cities do not spend detectably more per person, and the non-food share of their spending does not move detectably. The household survey (Susenas) starts after 1980, so these regressions do not add a 1980 level.

**The conflict provinces.** The effect does not differ measurably in the provinces of the conflicts of around 2000 (Barter & Côté 2015; van Klinken 2007). With exposure interacted with an indicator of Aceh, West Kalimantan, Central Kalimantan and Central Sulawesi on the main city sample, the interaction term is −0.027 (standard error 0.032) on the agricultural employment share and −0.015 (0.084) on log modern-sector employment, and it is not statistically significant on any of the seven outcomes. The slope outside stays within a fifth of the headline, at −0.042 (0.021) and 0.170 (0.081). Leaving the 26 cities of the four provinces out instead gives −0.024 (0.022) on the agricultural employment share and 0.081 (0.092) on log modern-sector employment, about half their size on the main city sample, with every standard error larger. The conflict-province cities are about as exposed as the others; the halving comes from estimating the control coefficients on the 108 remaining cities, and the standard deviation of exposure left after the controls falls from 0.46

to 0.30 (Table A1). Where the sources record destinations, part of the displacement left the four provinces, as when Javanese residents fled Aceh for North Sumatra (Barter & Côté 2015), so the drop leaves some of the places that received the displaced in the sample.

**The exposure of neighboring cities.** The exposure of neighboring cities leaves the shift in sector shares in place, and its estimates on city size are mixed. With the exposure of the 137 other measured cities weighted by inverse great-circle distance in kilometers and added beside a city's own, the agricultural employment share and the modern share do not move detectably with it (0.006, standard error 0.075; −0.013, 0.075), and the own-exposure estimates stay at the headline, −0.052 against −0.051 on the agricultural employment share. Log population, log employment and log modern-sector employment fall with the exposure of neighbors, by −0.607 (0.281), −0.601 (0.272) and −0.613 (0.307); the weighted measure varies little once own exposure and the controls are known (its standard deviation is 0.29 against 1.00 for own exposure, and the two explain 0.87 of it), and without three pairs of twin cities within 12 km of each other, left out of the sample and still counted as neighbors of the other cities, or with province effects, the three size estimates remain negative but smaller, and of the six estimates only log employment without the twin cities and log modern-sector employment with province effects are marginally significant. The plain mean over the other measured cities within 100 km, on the 125 cities with such a neighbor, has the same negative sign on the three size outcomes and is statistically insignificant on every outcome with the uniform kernel and with province clusters, with log population at −0.133 (0.085) (Table A1).

# F The Model

This appendix gives the model's production, household demand, worker choice and equilibrium conditions, followed by the calibration and counterfactual definitions.

## F.1 Production and Land

Food and local services use labor and land, and traded non-agricultural output uses labor.

$$Y_{jA} = (A_{jA}T_{jA})^{1-\alpha}L_{jA}^{\alpha}, \tag{23}$$

$$Y_{jS} = \bar{A}_{jS}N_j^{\eta}T_{jH}^{1-\gamma_S}(L_{jS}^{\text{local}})^{\gamma_S}, \tag{24}$$

$$Y_{jM} = A_{jM}L_{jM}, \qquad L_{jS} = L_{jS}^{\text{local}} + L_{jM}. \tag{25}$$

The sector-choice decision is between agriculture and total non-agricultural employment. The latter is allocated between local and traded activities at their common wage $w_{jS}$.

Labor receives its marginal revenue product in agriculture and local services:

$$w_{jA} = p_{jA}\alpha(A_{jA}T_{jA})^{1-\alpha}L_{jA}^{\alpha-1}, \tag{26}$$

$$w_{jS} = p_{jS}\gamma_S\bar{A}_{jS}N_j^{\eta}T_{jH}^{1-\gamma_S}(L_{jS}^{\text{local}})^{\gamma_S-1}. \tag{27}$$

The corresponding land payments are $\Pi_{jA} = (1-\alpha)p_{jA}Y_{jA}$ and $\Pi_{jS} = (1-\gamma_S)p_{jS}Y_{jS}$. Traded production has unit cost $q_{jM} = w_{jS}/A_{jM}$ and does not pay land rent.

**Land allocation and income.** The location's endowment $\bar{T}_j$ is allocated between agriculture and local services. Equal rents across active uses imply

$$T_{jA} = \bar{T}_j\frac{\Pi_{jA}}{\Pi_{jA}+\Pi_{jS}}, \quad T_{jH} = \bar{T}_j - T_{jA}, \quad R_j = \frac{\Pi_{jA}+\Pi_{jS}}{N_j}. \tag{28}$$

The measured endowment is crop area plus the built-up footprint, $\bar{T}_j = T_j^{\text{crop}} + T_j^{\text{built}}$.

Table F1: Model Notation

| Symbol | Definition |
|---|---|
| $A, S, M$ | Food, local services, traded non-agricultural output |
| $N_j$ | Population and total model labor in location $j$ |
| $L_{jA}, L_{jS}$ | Agricultural and total non-agricultural employment |
| $L_{jS}^{\text{local}}, L_{jM}$ | Local and traded non-agricultural employment |
| $T_{jA}, T_{jH}$ | Agricultural and local-service land |
| $p_{jA}, p_{jS}, p_{jM}$ | Local consumption prices |
| $q_{jA}, q_{jM}$ | Origin supply-price index for food and unit labor cost for the traded good |
| $W_j, R_j$ | Sector-inclusive wage index and per-person land-rent rebate |
| $w_{jA}, w_{jS}, \bar{w}_j$ | Agricultural and non-agricultural wages and their employment-weighted mean |
| $\theta_S, \theta_L$ | Sector-choice and location-choice elasticities |

*Notes:* $L_{jS}$ includes employment in both non-agricultural activities. The destination price index $p_{jM}$ differs from the origin unit price $q_{jM}$.

**The non-agricultural wage.** Holding local-service productivity fixed, differentiating its wage condition gives

$$d\ln w_{jS} = d\ln p_{jS} + \eta\, d\ln N_j + (1-\gamma_S) d\ln T_{jH} + (\gamma_S - 1) d\ln L_{jS}^{\text{local}}. \tag{29}$$

The main specification sets $\eta = 0$. Local prices, land use and the division of non-agricultural employment between activities jointly determine the wage response.

## F.2 Household Demand and Worker Choice

Households have preferences $U = (c_A - \bar{a})^{\psi_A} c_S^{\psi_S} c_M^{\psi_M}$, with weights summing to one. For expenditure resources $y_j$, their budget is

$$p_{jA} c_A + p_{jS} c_S + p_{jM} c_M = y_j. \tag{30}$$

For $y_j > p_{jA}\bar{a}$, discretionary expenditure is $D_j = y_j - p_{jA}\bar{a} > 0$, and utility maximization gives

$$c_A = \bar{a} + \frac{\psi_A D_j}{p_{jA}}, \quad c_S = \frac{\psi_S D_j}{p_{jS}}, \quad c_M = \frac{\psi_M D_j}{p_{jM}}. \tag{31}$$

Indirect utility is proportional to $D_j/P_j$, where $P_j = p_{jA}^{\psi_A} p_{jS}^{\psi_S} p_{jM}^{\psi_M}$. The food expenditure share is $\psi_A + (1-\psi_A) p_{jA}\bar{a}/y_j$, and local and traded non-agricultural expenditure shares are $\psi_S D_j/y_j$ and $\psi_M D_j/y_j$.

Sectoral heterogeneity generates the agricultural employment share

$$\pi_{A|j} = \frac{w_{jA}^{\theta_S}}{w_{jA}^{\theta_S} + w_{jS}^{\theta_S}}, \qquad W_j = (w_{jA}^{\theta_S} + w_{jS}^{\theta_S})^{1/\theta_S}. \tag{32}$$

Households spend $y_j = W_j + R_j$, where $W_j$ summarizes sectoral earnings opportunities and $R_j$ is their share of local land rents. The index $W_j$ is distinct from the arithmetic mean wage. For Fréchet efficiency draws, the expected maximum is $\Gamma(1 - 1/\theta_S) W_j$ when $\theta_S > 1$. The equations use the index $W_j$ for all positive $\theta_S$.

With amenities $\Phi_L^j$, location choice is

$$V_j = \Phi_L^j \frac{D_j}{P_j}, \qquad \pi_j = \frac{V_j^{\theta_L}}{\sum_k V_k^{\theta_L}}, \qquad N_j = \bar{L}\pi_j. \tag{33}$$

Amenities are $\Phi_L^j = \exp(\kappa z_j^{\text{amen}})/\operatorname{median}_k \exp(\kappa z_k^{\text{amen}})$, where $z_j^{\text{amen}}$ is the standardized amenity index and $\kappa = 0.5$ is held fixed. With moving costs, an origin's workers choose destinations with probabilities proportional to $(V_j/\tau_{oj}^L)^{\theta_L}$. Staying has cost $\tau_{oo}^L = 1$; for $o \neq j$, $\tau_{oj}^L = \max\{d_{oj}, 30\}^\nu$, with travel time in minutes. The cost-source sensitivity uses 1980 origin populations. The main costly-migration comparison uses the reference model allocation as its origin population.

## F.3 Trade Prices and Equilibrium

Food and the traded non-agricultural good have trade costs $\tau_{kjg} = \max\{d_{kj}, 1\}^{\delta_g}$ between locations and unit cost within a location, for $g \in \{A, M\}$. The food supply-price index is inverse marginal labor productivity; the traded non-agricultural price is unit labor cost:

$$q_{kA} = \frac{L_{kA}^{1-\alpha}}{\alpha(A_{kA}T_{kA})^{1-\alpha}}, \qquad q_{kM} = \frac{w_{kS}}{A_{kM}}. \tag{34}$$

Destination consumption prices and origin expenditure shares are

$$p_{jg} = \left[\sum_k (\tau_{kjg} q_{kg})^{1-\sigma}\right]^{1/(1-\sigma)}, \qquad \lambda_{kj}^g = \frac{(\tau_{kjg} q_{kg})^{1-\sigma}}{\sum_\ell (\tau_{\ell jg} q_{\ell g})^{1-\sigma}}. \tag{35}$$

Traded non-agricultural employment satisfies origin sales at the common wage, and local-service prices equate expenditure with local revenue:

$$w_{kS} L_{kM} = \sum_j \lambda_{kj}^M N_j \psi_M D_j, \qquad p_{jS} Y_{jS} = N_j \psi_S D_j. \tag{36}$$

These equations are solved jointly with labor and land allocation, sectoral wages, rents and location choice. Local-service prices are retained at the levels that satisfy the expenditure equation. The model does not impose goods-market clearing by location.

Production revenue is allocated to factor payments. With $\bar{w}_j = \pi_{A|j} w_{jA} + (1 - \pi_{A|j}) w_{jS}$,

$$N_j \bar{w}_j + \Pi_{jA} + \Pi_{jS} = p_{jA} Y_{jA} + p_{jS} Y_{jS} + q_{jM} Y_{jM}. \tag{37}$$

The origin price $q_{jM}$ values traded output in this equation.

**Employment responses.** The food-price derivative in the main text holds production inputs and other origins' supply indices fixed. In the complete equilibrium, local-service prices and input allocations also adjust. The agricultural employment share falls if its sectoral wage falls relative to the common non-agricultural wage. Changes in total employment additionally reflect population movement. Model fit therefore uses both the population and agricultural-share exposure coefficients.

## F.4 Calibration and Model Fit

**Geography and local characteristics.** The model has 134 reporting cities, 42 additional cities and 166 rural district remainders. Rural locations are district remainders after the cities' subdistricts have been assigned. Locations in Maluku and North Maluku are excluded under the model's provincial coverage rule, and locations require measured land inputs. The additional cities outside the settlement exposure window participate in the economy but do not enter the exposure regressions. Their exposure is undefined. The direct productivity shock applies where a rural remainder contains at least one transmigration village: 77 rural locations satisfy this condition and 89 do not. The gain applies to the entire host remainder, including its constituent subdistricts without a settlement.

Crop area and the satellite built-up footprint determine the land endowment. Agricultural productivity is constructed from agricultural employment per unit of crop area, winsorized at the 1st and 99th percentiles, and normalized by its median. Local-service productivity uses non-agricultural employment and its land scale, with a common normalization. Traded-output productivity uses census employment in manufacturing, trade, transport and skilled private services, valued at the reference allocation's model wages and scaled to total traded-good expenditure. This employment basket is broader than the purchased-services expenditure items used to set $\psi_M$. Amenities use standardized satellite measures and village-census infrastructure on each location's own constituent subdistricts. Where components are missing, their observed-component mean is adjusted to the complete-index scale using the relationship estimated among fully measured locations. The underlying missing components remain missing.

**Parameter sources.** The expenditure weights are $\psi_A = 0.550$ for food, $\psi_M = 0.082$ for traded non-agricultural output, and $\psi_S = 1-\psi_A-\psi_M$ for local services, based on Susenas expenditure shares. The food input measures purchased food and excludes food produced at home or received as gifts. The subsistence quantity $\bar{a} = 0.108$ is the BPS food-poverty threshold for 2000, expressed in 2011 rupiah and divided by the reference non-agricultural wage. These weights allocate spending above subsistence. The model's total food expenditure share also includes subsistence spending. Agricultural and local-service labor shares are fixed at 0.5 and 0.7, and the agglomeration elasticity is set to zero. The substitution elasticity is $\sigma = 4$, informed by Broda & Weinstein (2006). The reference trade-cost elasticities to travel time are 0.5 for food and 1.0 for traded output. The food elasticity is informed by published estimates, and the traded-output elasticity by the project's IFLS service-trip estimates. The empirical food-price-gap gradients do not directly identify trade costs.

**Matching the empirical responses.** The location and sector elasticities are selected jointly to match the standardized exposure coefficients for log population and the agricultural employment share. The model regressions include the settler count, 16 agro-climatic controls, and the same 4 observed 1980 controls: log population, log total employment, urban population share and log agricultural employment. The quantity and expenditure-share comparisons use 134 cities, and the displayed food-price comparison uses 112 cities with an observed 1996 price. Non-agricultural employment, agricultural employment, the 2011 food expenditure share and the 1996 raw-food price index are untargeted.

Let $\widehat{b}$ collect the targeted empirical coefficients and $b(\theta)$ the model coefficients. The minimum-distance objective is

$$J(\theta) = [b(\theta) - \widehat{b}]'\Omega[b(\theta) - \widehat{b}], \qquad \Omega = \operatorname{diag}\left(\widehat{\operatorname{se}}(\widehat{b}_k)^{-2}\right). \tag{38}$$

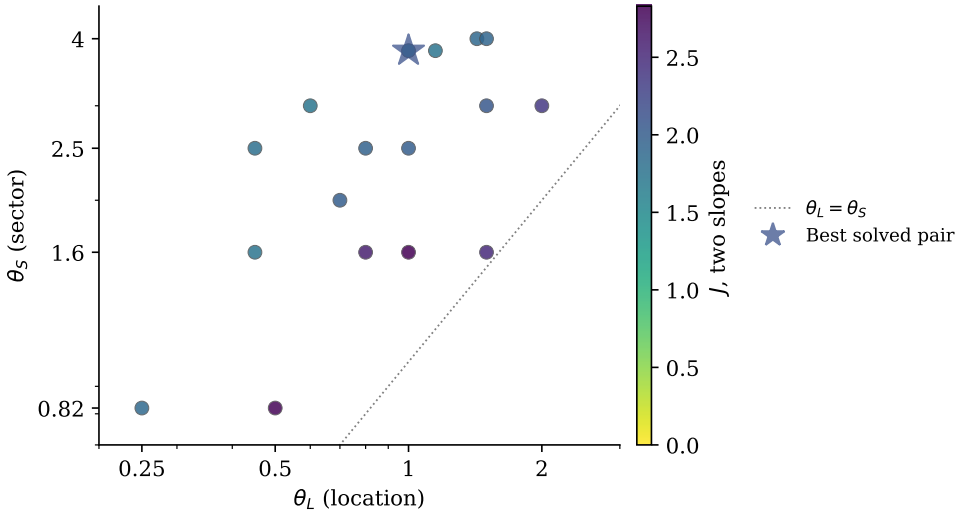
The empirical standard errors allow spatial correlation within 200 kilometers. The objective summarizes fit using the individual coefficient variances; it does not define a joint test or a parameter confidence set. Matching auxiliary regression coefficients is an indirect-inference approach (Hansen 1982; Gourieroux et al. 1993; Smith 1993).

The calculations attempted 29 parameter pairs; 7 did not solve within the time limit. Of the completed solutions, 17 satisfy $\theta_L < \theta_S$ and 5 have the reverse ordering. The best fit among the ordered pairs is at $(1.0, 3.8)$. Along $\theta_L = 1$, fit improves as $\theta_S$ rises toward the unsolved parameter combinations. The best completed calibration is therefore at the boundary of the available solutions. These calculations do not establish a unique global optimum or parameter confidence intervals.

## F.5 Counterfactual Restrictions and Outcome Units

Every comparison raises agricultural productivity at rural settlement hosts by 0.05 log units. The main figure compares higher trade costs with higher moving costs, using a new unshocked allocation at each cost. Higher trade costs multiply all between-location costs for both food and traded non-agricultural output. The chosen point reduces the economy-wide share of food expenditure on purchases from other locations to 10% of its reference value. Higher moving costs use origin-destination choices starting from the reference allocation; the chosen point leaves 10% of reference shock-induced population reallocation, measured as half the sum of absolute population changes across all model locations. The unrestricted economy provides the denominator for both comparisons.

Figure F1: Model Fit Across Location and Sector Elasticities




*Notes:* Each point is a completed calibration with $\theta_L < \theta_S$. Fit is measured by the sum of squared model-data coefficient differences in empirical standard-error units for log population and the agricultural employment share. Empirical coefficients use 134 cities, the main controls, and spatial standard errors at a 200-kilometer cutoff. The star indicates the best-fitting completed calibration. Parameter combinations without a solution are omitted. The display does not provide confidence intervals for the parameters.

For an outcome $y_j$, let $\Delta y_j^r$ be the change under restriction $r$ and $z_j$ standardized exposure. The common-unit regression is

$$\Delta y_j^r = a^r + b^r z_j + X_j' c^r + u_j^r. \tag{39}$$

Share changes are in percentage points, and population and employment changes are in logs. The same outcome units and exposure standardization apply across restrictions. The signed bar is $100 b^r / |b^{\text{full}}|$. A negative bar denotes a negative exposure gradient, and its magnitude relative to the full-response marker gives the restricted response as a share of the full response.

The separate fixed-price comparison holds food prices, traded non-agricultural prices and traded non-agricultural employment at their unshocked levels while allowing population to move. Its trade comparison fixes each location's population. Fixed prices and autarky are different restrictions: under autarky, local prices can respond to the productivity gain. These fixed-price comparisons are used in the cost-sensitivity figure below. They are distinct from the main figure's costly-but-open economies.

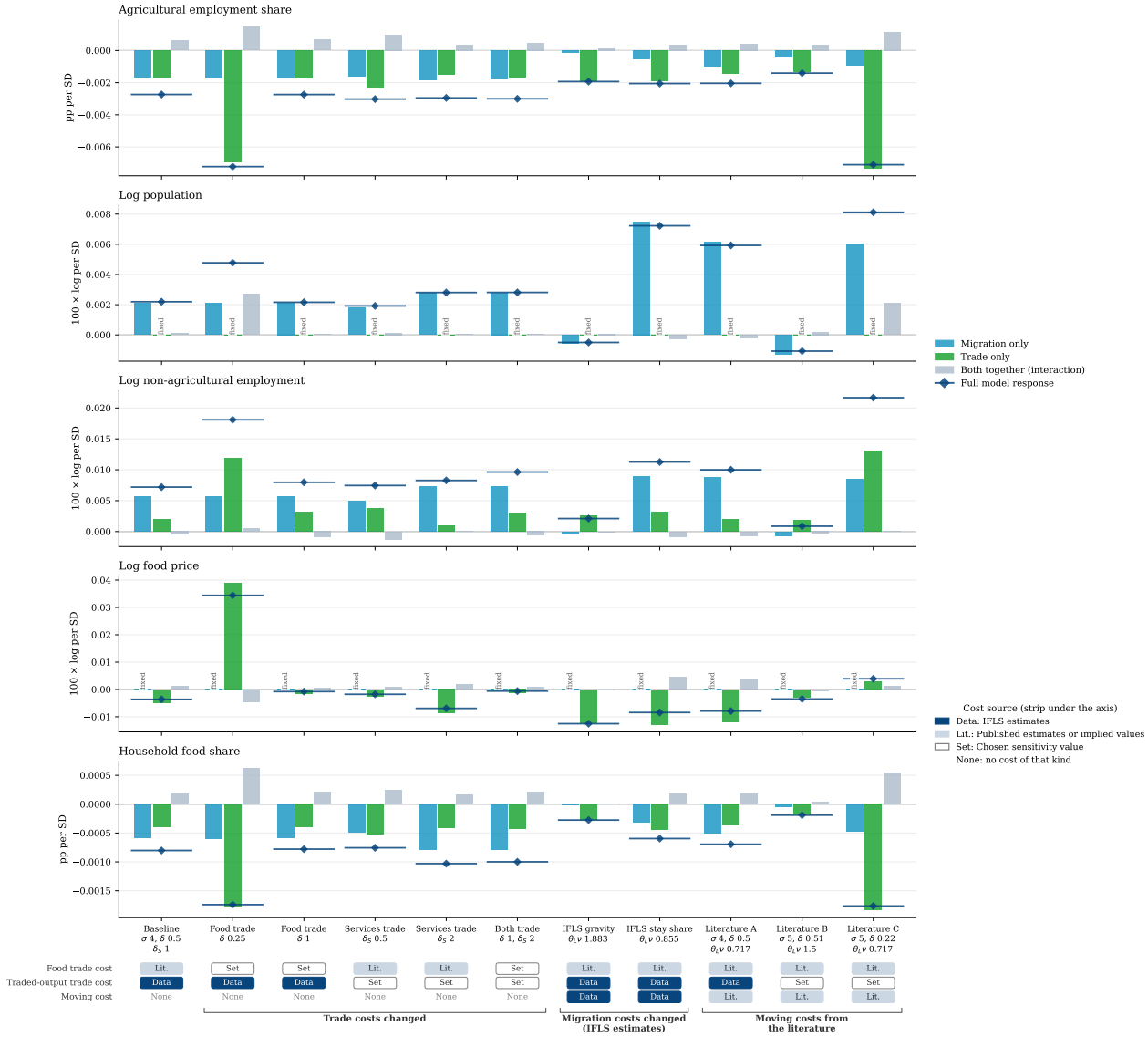
The pooled agricultural employment share is $\sum_j L_{jA} / \sum_j N_j$, calculated over the reporting cities before and after the shock. It includes effects common to those cities. The exposure coefficient $b^r$ measures differences across cities and does not identify that common response. The restricted percentages can overlap, since changing one channel also changes the prices and wages determining the other. Historical attribution requires a historical productivity change and the corresponding counterfactual economy; the uniform productivity experiment does not provide that attribution.

## F.6 Sensitivity to Trade and Moving Costs

The reference calculations set the substitution elasticity to $\sigma = 4$ and the travel-time elasticities of trade costs to $\delta_A = 0.5$ and $\delta_M = 1.0$. The food trade cost is informed by published estimates, and the traded-output cost by the project's IFLS service-trip estimates. The project's IFLS migration regression estimates the composite distance coefficient governing destination shares; a separate scaling matches the share who remain in their origin. These are alternative moving-cost calibrations. The literature comparisons use moving costs from Bryan & Morten (2019) and Tombe & Zhu (2019), together

with their stated trade-cost alternatives. Moving-cost comparisons start from the 1980 origin populations, while the main figure's costly-migration experiment starts from the reference model allocation.

Figure F2: Channel Comparisons Under Alternative Trade and Moving Costs




*Notes:* Model responses on 134 reporting cities, at $(\theta_L, \theta_S) = (0.45, 1.6)$. The 5 outcomes are cross-city responses after a 0.05 increase in log agricultural productivity at rural settlement hosts: agricultural and food expenditure shares in percentage points; population, non-agricultural employment and the food price in 100 times log changes. Migration only fixes food and traded non-agricultural prices and traded non-agricultural employment; trade only fixes population. The interaction is the full response less these two responses, and the line shows the full response. The regressions use standardized exposure and the main empirical controls. The source rows separately classify food, traded-output and moving costs as published estimates or implied values, IFLS estimates, or chosen sensitivity values. The substitution elasticity follows published estimates. These fixed-price and fixed-population restrictions differ from the costly-but-open restrictions in the main figure. The displayed model comparisons do not report sampling uncertainty.

Trade contributes more to the agricultural-share exposure response when food trade costs are lower. Higher costs of trading non-agricultural output can reverse the ranking of trade and migration. Moving costs estimated from IFLS reduce the migration comparison and worsen the fit to the two targeted slopes. The main figure's agricultural-share ranking holds along its two comparisons with costly but open channels. The wider cost comparison shows where that ranking changes.

## F.7 Values Tried in the Sensitivity Runs

Table F2: Values Tried in the Sensitivity Runs

| Symbol | Parameter | Baseline | Values tried | Solved at |
|---|---|---|---|---|
| $\theta_L$, $\theta_S$ | Choice elasticities | $(1, 3.8)$ | All 17 solved ordered pairs ($\theta_L < \theta_S$): $\theta_L$ 0.25 to 2, $\theta_S$ 0.82 to 4; $J$ 1.66 to 2.83 | Each pair |
| $\sigma$ | Substitution across origins | 4 | Lit. 5 (Literature B, C) | $(0.45, 1.6)$; Lit. 11-pair grid |
| $\delta_A$ | Food trade-cost elasticity | 0.5 | Set 0.25, 1.0; Lit. 0.51 (B), 0.2225 (C) | $(0.45, 1.6)$; Lit. 11-pair grid |
| $\delta_M$ | Traded-output trade-cost elasticity | 1.0 | Set 0.5, 2.0; Set 0.75 (B, C), holding $(\sigma - 1)\delta_M$ at 3 | $(0.45, 1.6)$; Lit. 11-pair grid |
| $\delta_A$, $\delta_M$ | Both trade-cost elasticities | 0.5, 1.0 | Set 1.0 and 2.0 | $(0.45, 1.6)$ |
| $\theta_L\nu$ | Moving-cost elasticity | None | Data 1.883 (IFLS gravity), 0.855 (IFLS share who stay); Lit. 0.717 (A, C), 1.50 (B) | $(0.45, 1.6)$; Lit. 11-pair grid |
| $\times m$ | Every trade cost raised in steps | $m = 1$ | $m$ = 1.34, 1.75, 2.45 and 10,000 | $(1, 3.8)$ |
| $\theta_L\nu \times s$ | Moving cost raised in steps | None | $\theta_L\nu = 1.883 \times s$, $s$ = 0.535, 0.626, 0.725 and 8 | $(1, 3.8)$ |

*Notes:* Values tried in the sensitivity runs; the baseline is the value in Table 7. Set, half or double the baseline, chosen to bracket it; Lit., a published value; Data, measured in the IFLS. Literature A changes only the moving cost, to the value of Bryan & Morten (2019). Literature B takes $\sigma$, the food decay and the moving cost from Tombe & Zhu (2019) and Tombe & Zhu (2015), with the services decay set so the distance elasticity of services flows stays at $-3$. Literature C sets the food decay so the distance elasticity of food flows is $-0.89$, the median in Head & Mayer (2014), with $\sigma$ and $\delta_M$ as in B and the moving cost as in A. The cost settings use $(\theta_L, \theta_S) = (0.45, 1.6)$, with fit distance $J = 1.74$ at the reference costs; seven of the ten settings other than the baseline were run only there, and Literatures A to C at all 11 pairs of the grid, the estimate among them. The costs raised in steps were run at the estimate. $\theta_L\nu$: the distance elasticity of moves. $\times m$: every between-location trade cost times $m$, raised until locations buy 50%, 25%, 10% and 0% of the food they buy elsewhere at the estimate. $\times s$: the IFLS moving cost 1.883 times $s$, raised until the shock moves 50%, 25%, 10% and 0% of the workers it moves at the estimate.

Table F2 reports the parameter ranges and cost settings used in the sensitivity runs, with the choice elasticities used for each comparison.